Quantitative density decay for Pythagorean-pair-free prime-power boxes

By Rongzhong Xiao

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1. Introduction

1.1. **History and motivations.** The Pythagorean equation $x^2+y^2=z^2$ is a central nonlinear question in partition regularity. Erdős and Graham [16, Section 3, Problem 6] asked whether every finite colouring of $\mathbb{N}$ contains a monochromatic solution.

The finite two-colour problem was first approached computationally: Cooper and Overstreet [4] found a two-colouring of $\{1,\dots,7664\}$ with no monochromatic triple by a SAT computation. Heule, Kullmann, and Marek [18, Theorem 1] then gave a certified cube-and-conquer

proof (with a formally checkable certificate of roughly 200 terabytes) that $\{1,\dots,7824\}$ admits such a colouring, whereas every two-colouring of $\{1,\dots,7825\}$ contains a monochromatic Pythagorean triple. Two independent 2026 preprints give purely theoretic proofs of the same two-colour theorem: Leng [22] uses an analytic and ergodic argument, while Wesley [27] gives an elementary argument based on linear algebra and counting. Thus the finite two-colour problem is settled, whereas the corresponding question for three or more colours remains open.

A second line of progress uses multiplicative harmonic analysis. Frantzikinakis and Host [8, Corollary 2.8 and Problem 3] developed higher-order Fourier tools for bounded multiplicative functions and obtained monochromatic solutions for several quadratic equations, including $16x^2+9y^2=z^2$ and $x^2-xy+y^2=z^2$; they also isolated the two-leg Pythagorean question, where x,y have the same colour and $x^2+y^2=z^2$ for some z. Frantzikinakis, Klurman, and Moreira [9, Theorems 1.1–1.2] subsequently proved that every finite colouring of $\mathbb{N}$ contains such distinct same-coloured legs and established the analogous statements for the other two pairs of entries. Their density theorem gives the same conclusion inside every set of positive multiplicative density.[1]

In a sequel, Frantzikinakis, Klurman, and Moreira [10, Theorems 1.2–1.4 and 1.9–1.10] extended this framework to generalized Pythagorean equations $ax^2+by^2=cz^2$. They proved unconditional partition and density regularity for the relevant pairs when ac or bc is a square; in the complementary $(a+b)c$-square case they obtained corresponding results under an Elliott-type correlation conjecture, together with an unconditional theorem for pretentious multiplicative level sets. These results place the Pythagorean pair theorem within a broader theory of homogeneous quadratic forms.

Recent work broadens this multiplicative picture. Azevedo and Moreira [11] showed that finitely many completely multiplicative functions can be simultaneously close to 1 on the three entries of a Pythagorean triple, and treated square coefficients satisfying Rado's condition. Frantzikinakis and Mountakis [13] proved a corresponding recurrence theorem for generalized triples $ax^2+by^2=cz^2$ in pretentious systems. Leng, McNamara, and Mountakis [12] obtained almost-Pythagorean quadruples $x^2+y^2w^2=z^2$ in multiplicatively dense sets, with w allowed to lie in a second dense set.

Here, our aim is quantitative and finite. We replace an arbitrary multiplicative Følner sequence by the explicit prime-power boxes below and obtain a decay rate for the largest subset containing no two distinct legs of an integer right triangle.

1.2. **Prime boxes and the main result.** For a positive integer N, let $p_1<\cdots<p_N$ be the first N primes and define

$$\Phi_N=\left\{\prod_{i=1}^{N}p_i^{a_i}:a_i\in\{0,1,\dots,N-1\}\right\}.$$

Unique factorization gives $|\Phi_N|=N^N$. These sets are products of finite geometric progressions whose exponent vectors form the integer box $\{0,\dots,N-1\}^N$.

The boxes form a multiplicative Følner sequence: multiplication by any fixed positive integer changes only a boundary proportion tending to zero. For a multiplicative Følner sequence $\Psi=(\Psi_K)$, define

$$\overline{d}_\Psi(A)=\limsup_{K\to\infty}|A\cap\Psi_K|/|\Psi_K|.$$

[1] Ordinary natural density is insufficient: the odd integers have density 1/2, but the sum of two odd squares is 2 (mod 4) and hence is not a square. Multiplicative Følner sequences retain the local information lost by ordinary density.

The density theorem of Frantzikinakis, Klurman and Moreira [9, Theorem 1.2(1), with $a = b = c = 1$] states that positive multiplicative density forces distinct x, y with $x^2 + y^2$ a square.

We study the following finite extremal formulation of the two-leg density problem, with both the prime support and the exponent range specified by N. Call $A \subseteq \Phi_N$ admissible if there are no distinct $a, b \in A$ and $c \in \mathbb{N}$ with $a^2 + b^2 = c^2$, and put

$$f(N) = \max\{|A| : A \subseteq \Phi_N \text{ is admissible}\}.$$

The integer c is not required to belong to A or to Φ_N. The fact that the sum of two odd squares is 2 (mod 4) implies that $f(N) \geq N^{N-1}$. The theorem below bounds the density of every admissible subset of a sufficiently large box, with the threshold uniform over the choice of that subset.

Theorem 1.1. *There are absolute constants $C_{\mathrm{main}}, C_1, C_2, N_0 \geq 1$ with the following properties. For $0 < \delta < 1/10$ and every integer $N \geq 1$,*

$$N \geq \exp\left(\exp\left(C_{\mathrm{main}}\delta^{-1295}\log^2(2/\delta)\right)\right) \quad \Longrightarrow \quad f(N) \leq 2\delta N^N. \tag{1.1}$$

For every integer $N \geq N_0$,

$$\frac{f(N)}{N^N} \leq C_1\left(\frac{(\log\log\log N)^2}{\log\log N}\right)^{1/1295} \leq C_2(\log\log N)^{-1/1296}. \tag{1.2}$$

For each $0 < b < 1/1295$, there are $C_b, N_b > 0$ such that

$$f(N) \leq C_b N^N (\log\log N)^{-b} \qquad \text{for every integer } N \geq N_b. \tag{1.3}$$

In particular, $b = 1/1296$ is an admissible numerical exponent. The first inequality in (1.2) records the more precise logarithmic correction. The constants in the theorem are absolute where stated, but their numerical values are not evaluated.

1.3. **Notation.** For $x \geq 1$, write $[x] = \{1, \dots, \lfloor x \rfloor\}$, $e(t) = e^{2\pi it}$, and $\mathbb{T} = \mathbb{R}/\mathbb{Z}$. The distance to the nearest integer is $\|t\|_{\mathbb{R}/\mathbb{Z}}$. A finite-set average is normalized:

$$\mathbb{E}_{x\in S}F(x) = |S|^{-1}\sum_{x\in S}F(x).$$

We write $\mathbf{1}_S$ for an indicator and $A \ll B$ when $A \leq CB$ with the stated dependence of C; all logarithms are natural. The iterated logarithms are $\log_2 x = \log\log x$, $\log_3 x = \log\log\log x$, and $\mathrm{Exp}_2(x) = \exp(\exp x)$.

Let $\mathcal{M}$ be the set of all completely multiplicative functions $g : \mathbb{N} \to \mathbb{C}$ with $|g(n)| = 1$. Their unique extension to $\mathbb{Q}_{>0}$ is

$$g(a/b) = g(a)\overline{g(b)}. \tag{1.4}$$

For $r \in \mathbb{Q}_{>0}$, $v_p(r)$ denotes its (possibly negative) prime valuation. For a primitive Dirichlet character χ, denote its conductor by q_χ. For real $X \geq 2$, set

$$\mathbb{D}_X(g, \chi n^{it})^2 = \sum_{\substack{p\leq X\\ p \text{ prime}}} \frac{1 - \Re(g(p)\overline{\chi(p)}p^{-it})}{p}.$$

2. Proof outline

We explain the proof of Theorem 1.1 before developing its ingredients. Fix $0 < \delta < 1/10$. The construction must work simultaneously for every admissible set and every multiplicative character. Its quantitative objective is a finite measure whose support has bounded rational height. We describe this objective first, then the arithmetic averages, the analytic comparison, and the choice of scales.

2.1. **Fourier transfer.** A positive rational r is called a Pythagorean ratio if $1+r^2$ is the square of a rational number. Write $r = u/v$ in the lowest terms, with $u, v \in \mathbb{N}$, and define its height by $\mathrm{ht}(r) = \max(u, v)$. For a finitely supported probability measure μ on these ratios, we seek the uniform lower bound

$$\Re \sum_r \mu(r) g(r) \geq -\delta \qquad \text{for every } g \in \mathcal{M}. \tag{2.1}$$

This estimate controls $f(N)$ through translation of exponent vectors. A ratio r acts on exponent vectors by translation through $d(r) = (v_{p_1}(r), \ldots, v_{p_N}(r))$. If $\mathrm{ht}(r) \leq H$ and $N \geq H$, every prime in r occurs among the first N primes, and

$$\sum_{i=1}^{N} |v_{p_i}(r)| \leq \frac{2 \log H}{\log 2}.$$

The proportion of exponent vectors for which translation leaves the box is at most this last quantity divided by N. On the other vectors, two points of an admissible set cannot differ by $d(r)$: their corresponding integers would be distinct legs of an integer right triangle.

Finite Fourier analysis on $(\mathbb{Z}/N\mathbb{Z})^N$ supplies the other side of the estimate. If an admissible set has density α, its zero Fourier coefficient contributes α^2. The other frequencies contribute at least $-\delta\alpha$ by (2.1). Thus Theorem 3.3 proves

$$\alpha^2 - \delta\alpha \leq \frac{2 \log H}{N \log 2}.$$

For $N \geq 2 \log H/(\delta^2 \log 2)$, the right side is at most δ^2, and the positive root of the resulting quadratic is less than 2δ. The remaining task is therefore to construct μ satisfying (2.1) with a suitable H.

2.2. **Local averages.** Euclid's identity

$$(m^2 - n^2)^2 + (2mn)^2 = (m^2 + n^2)^2 \qquad (m > n > 0)$$

produces the ratios

$$R(m, n) = \frac{m^2 - n^2}{2mn}.$$

The measure will be an average of weighted distributions of these ratios. The two weights have different purposes. A fixed nonnegative function w of two real variables controls the factor r^{it}. A nonnegative periodic function W_0 controls values at small primes. The function w has integral one and compact support in $\{(s, t) : s > t > 0,\ s + t < C_0 - 1\}$, where $C_0 \geq 2$ is a fixed integer. The period D_0 of W_0 depends only on δ, and the mean of W_0 modulo D_0 is one. Define

$$Z_X = X^{-2} \sum_{m,n \in \mathbb{Z}} w(m/X, n/X) W_0(m, n),$$

$$K_X(g) = \frac{X^{-2}}{Z_X} \sum_{m > n > 0} w(m/X, n/X) W_0(m, n) g(R(m, n)).$$

These expressions are used only when $Z_X > 0$. The normalization estimate in (4.46) gives $Z_X \geq 1/2$ for every sufficiently large X. Consequently $K_X(1) = 1$, so each K_X is the transform of a probability measure.

The construction of W_0 in Section 4 uses

$$B = (200/\delta)^2, \qquad T_0 = \lceil 100/\delta \rceil, \qquad D_0 = \prod_{\substack{p \leq B \\ p \text{ prime}}} p^{2T_0 + 5}.$$

At each prime $p \leq B$, a congruence condition fixes the valuation of R to be $2j$ and constrains its unit part to be one modulo p^3. The same integer j is used at all these primes. Its probabilities

are proportional to $T_0+1-|j|$ for $0<|j|\le T_0$. To describe the relevant identity, let z be any complex number of modulus one. Then

$$\frac{1}{T_0}\left\{\frac{1}{T_0+1}\left|\sum_{j=0}^{T_0} z^j\right|^2-1\right\}\ge -\frac{1}{T_0}.$$

In the application, $z=\prod_{\substack{p\le B\\ p \text{ prime}}} z_p^2$, where the corrected prime phases z_p are defined in (4.123). That definition separates the factor r^{it} and the local unit characters as in (4.124). The identity explains the use of one common integer at all small primes. The conditional unit distribution also eliminates every local character that is nontrivial on the prescribed unit subgroup.

For the remaining primes, the local principal-character factors are nonnegative, while a nonprincipal factor at a prime $p>B$ has modulus at most $2/\sqrt{p}$. The real weight is chosen so that

$$\int_{s>t>0} w(s,t)R(s,t)^{i\tau}\,ds\,dt=\left(\frac{\sin(\tau/2)}{\tau/2}\right)^4\ge 0,$$

with value one at $\tau=0$. Theorem 4.9 compares the finite integer average with the product of these local and real averages, including all unit-character data. It yields Theorem 4.10: there is an absolute constant C and an integer U, depending only on δ and satisfying $2\le U\ll\delta^{-3}$, such that every $M_1\ge\exp(C\delta^{-3})$ satisfies

$$\Re K_X(\widetilde{g})\ge -\delta/50 \quad\text{if}\quad y\ge M_1,\quad X\ge (M_1y)^U, \tag{2.2}$$

whenever $\widetilde{g}\in\mathcal{M}$ agrees with $\chi(p)p^{it}$ at all primes $p>y$, with $q_\chi,|t|\le M_1$. The values at primes at most y remain arbitrary.

2.3. **The inverse theorem.** For $g\in\mathcal{M}$, complete multiplicativity factors the kernel integrand into four functions of linear forms:

$$g(R(m,n))=\overline{g(2)}\,g(m-n)g(m+n)\overline{g(m)}\,\overline{g(n)}. \tag{2.3}$$

Three applications of Cauchy–Schwarz control this average by the U^3 norm. A finite Fourier expansion of W_0 and a subdivision of the support of w give Theorem 8.1. In particular, there exist $\epsilon_\delta>0$ and $X_\delta\ge 1$ with $\log(2X_\delta)\ll\delta^{-3}$ such that, for every integer $X\ge X_\delta$,

$$\log(2/\epsilon_\delta)\ll\delta^{-3},\qquad \|g\|_{U^3[C_0X]}<\epsilon_\delta\implies |K_X(g)|\le\delta/50. \tag{2.4}$$

Thus only functions with a larger uniformity norm require a comparison with the models in (2.2).

The quantitative inverse theorem proved in Section 7 has the following form. Put $L=2+\log(2/\epsilon)$. If

$$\|g\|_{U^3[Z]}\ge\epsilon\quad\text{and}\quad Z\ge\operatorname{Exp}_2(CL^{431}),$$

then some primitive character χ and real number t satisfy

$$q_\chi,|t|\le\exp(CL^{431}),\qquad \mathbb{D}_Z(g,\chi n^{it})^2\le CL^{431}. \tag{2.5}$$

The proof is divided into two parts. In Section 6, a correlation with a scalar quadratic phase is converted into rational approximations to its coefficients. After these coefficients are controlled, a mean-value theorem for multiplicative functions gives the character and real parameter. In Section 7, the U^3 inverse theorem first supplies a quadratic nilsequence. The group and orbit are constructed explicitly. Further correlations reduce this nilsequence to the scalar case.

The power 431 is obtained from two bounds in the initial inverse theorem and the subsequent scalar reduction. If d is the dimension of the constructed group, M is the Lipschitz bound, and η is the correlation, then

$$d\ll L^{43},\qquad \log(10dM/\eta)\ll L^{87},\qquad (d+1)^8\log(10dM/\eta)\ll L^{431}.$$

The interval thresholds are checked in Theorem 7.10; they have the same numerical power.

2.4. **Concentration.** Fix ranges $q_\chi \le Q$ and $|t| \le T_1$ that include all parameters produced by (2.5) at $\epsilon = \epsilon_\delta$. For each $g \in \mathcal{M}$ and real $z \ge 2$, define

$$E_g(z) = \min_{\substack{\chi \text{ primitive},\ q_\chi \le Q \\ |t| \le T_1}} \mathbb{D}_z(g, \chi n^{it})^2.$$

This is a nonnegative, nondecreasing function of z. A minimizing pair exists because the character set is finite and the parameter interval is compact. If (χ, t) minimizes the sum at z and $2 \le y \le z$, its contribution from primes in $(y, z]$ is at most $E_g(z) - E_g(y)$; see Theorem 8.2.

For $y \ge Q$, define $\widetilde{g}(p) = g(p)$ for $p \le y$ and $\widetilde{g}(p) = \chi(p)p^{it}$ for $p > y$. Both choices have modulus one. Their completely multiplicative extension is therefore in $\mathcal{M}$. The quotient $h = g/\widetilde{g}$ is one at every prime at most y. Section 5 applies the Turán–Kubilius inequality to the additive function whose value at a prime is the argument of $h(p)$. It proves that the values of h are close, in mean square, to a single complex number of modulus one on all four relevant linear forms. In (2.3), that number occurs twice and its conjugate occurs twice, so their product is one.

The congruence weight is invariant under multiplying both coordinates by a unit modulo D_0. This invariance gives the comparison coefficient b_δ with

$$|K_X(g) - K_X(\widetilde{g})| \le b_\delta \sqrt{E_g(C_0X) - E_g(y)}, \qquad b_\delta^2 \ll \log(2/\delta), \tag{2.6}$$

Here the pair (χ, t) minimizes the sum at $z = C_0X$, and the scale assumptions are those in Theorem 5.5. The estimate follows by averaging over the weighted residue classes.

2.5. **Scales and height.** Set $\ell = \log(2/\delta)$. Substitution of (2.4) into (2.5) allows a bound $C_1 \ll \delta^{-1293}$ for the minimized prime sum, because $1293 = 3 \cdot 431$. Choose a sufficiently large initial parameter M_1 with

$$\log\log M_1 \ll \delta^{-1293}. \tag{2.7}$$

Take M_1 to be an integer dominating Q, T_1, D_0, the uniformity threshold X_δ, and the inverse-theorem threshold at ϵ_δ. The verification that these requirements are consistent with (2.7) is given in (8.14)–(8.16). Starting from $Z_0 = M_1^{a_0}$, with a fixed integer $a_0 \ge 2$, define integer scales by

$$X_j = \left\lceil (M_1 Z_{j-1})^U \right\rceil, \qquad Z_j = C_0 X_j \qquad (1 \le j \le k). \tag{2.8}$$

The scales are chosen independently of g. If $E_g(Z_j) > C_1$, the inverse theorem implies the small-kernel estimate (2.4). If $E_g(Z_j) \le C_1$, use (2.2) and (2.6). The result in the second case is

$$\Re K_{X_j}(g) \ge -\delta/50 - b_\delta \sqrt{E_g(Z_j) - E_g(Z_{j-1})}.$$

To sum the errors, put $F_j = \min(C_1, E_g(Z_j))$. This sequence is nondecreasing and lies in $[0, C_1]$. Cauchy–Schwarz gives

$$\frac{1}{k} \sum_{j=1}^{k} \sqrt{F_j - F_{j-1}} \le \left(\frac{F_k - F_0}{k} \right)^{1/2} \le \sqrt{C_1/k}.$$

The two cases at each scale are combined in Theorem 8.3. Taking k to be a sufficiently large constant times $b_\delta^2 C_1 \delta^{-2}$, rounded up, yields

$$k \ll \delta^{-1295}\ell, \qquad \Re\left(\frac{1}{k} \sum_{j=1}^{k} K_{X_j}(g) \right) \ge -\delta.$$

The average of the corresponding probability measures is the required measure μ_δ.

Every ratio in its support has height at most Z_k^2. On taking logarithms in (2.8), one obtains a linear recurrence for $\log Z_j$, with multiplier U. Its solution is written out in (8.19). Since $\log U \ll \ell$, it gives

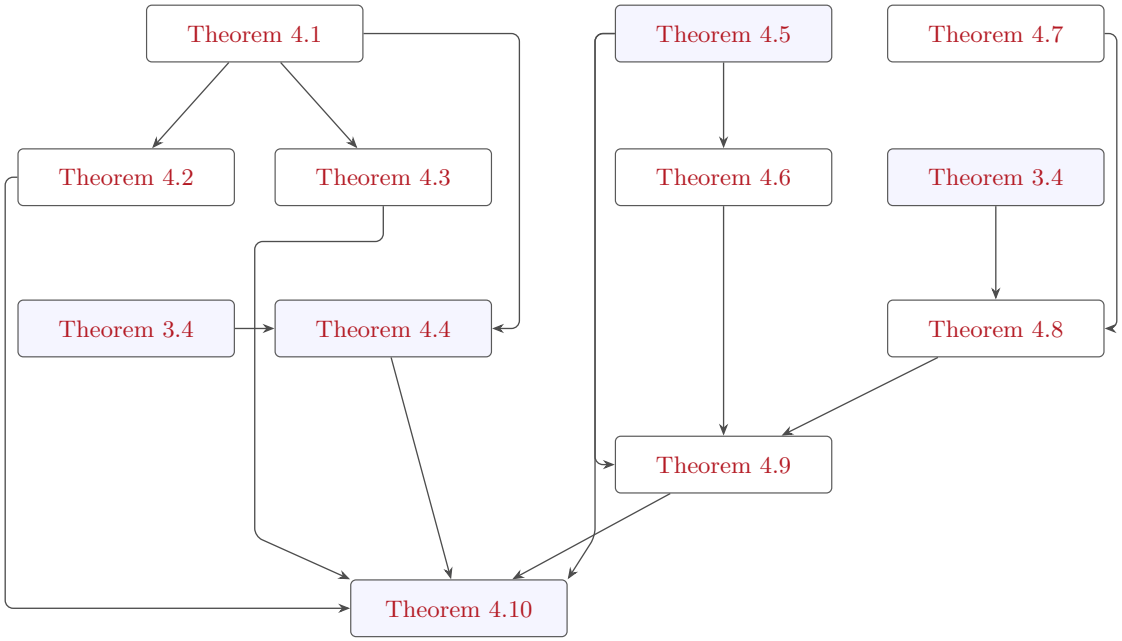
$$\log\log \mathrm{ht}(r) \le k \log U + O(\delta^{-1293}) \ll \delta^{-1295}\ell^2 \qquad (r \in \operatorname{supp} \mu_\delta).$$

The two factors of ℓ have separate sources: one comes from b_δ^2, and the other comes from $\log U$. Theorem 8.4 verifies all thresholds in this calculation. The finite Fourier estimate then proves (1.1), and its elementary inversion proves (1.2) and (1.3).

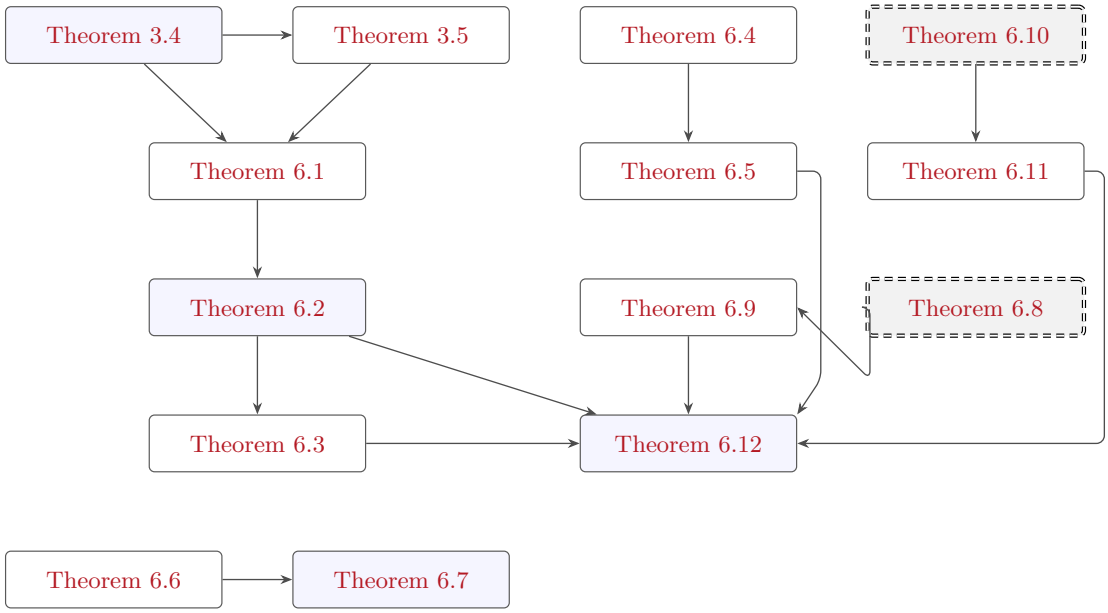
2.6. **Relations between the results.** The four panels below summarize the principal proof dependencies of the results in the paper. An arrow $A \to B$ means that the proof of B uses A, together with the hypotheses and other inputs required in that proof. It does not assert that A alone implies B, or that the two statements are equivalent. Directed paths record successive uses of results.

Repeated boxes represent the same result and connect the panels. Their pale shading distinguishes these repeated nodes. The double dashed boxes mark Theorem 6.10 and Theorem 6.8 as external inputs; other external inputs and elementary estimates are cited in the proofs rather than drawn as additional nodes. An edge can also record an estimate established within the indicated result's proof, such as the prime or normalization estimates. Crossing arrows have no junction unless they meet a result box.

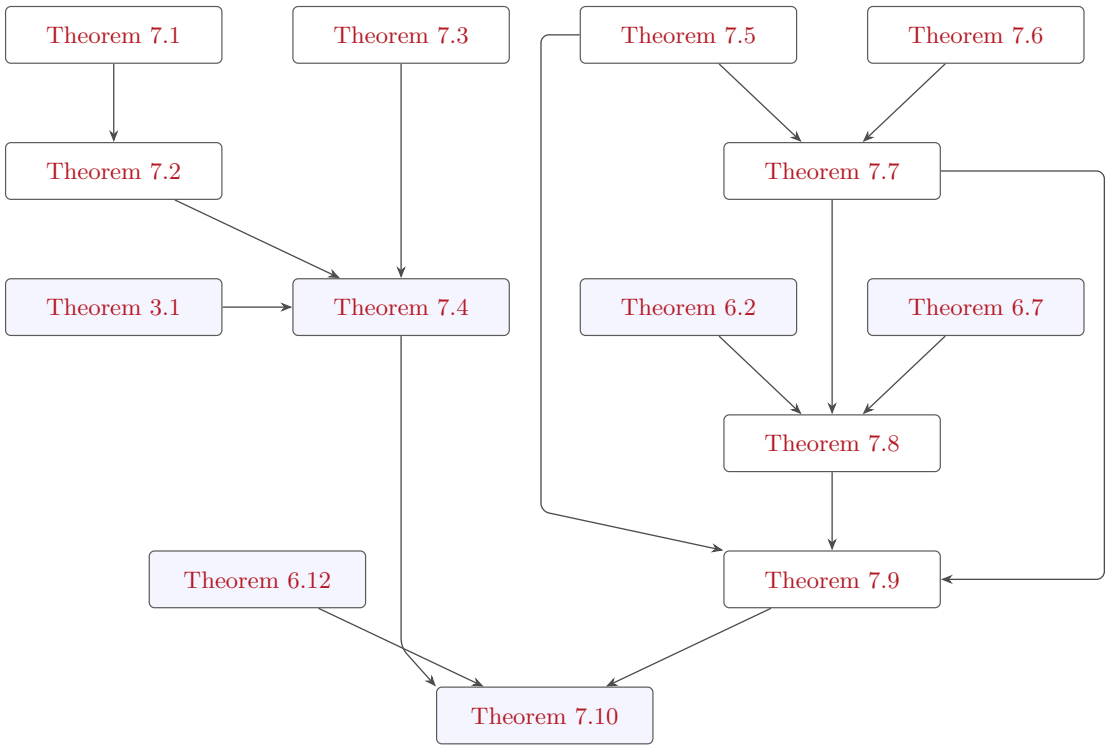
A. Sections 3 and 4



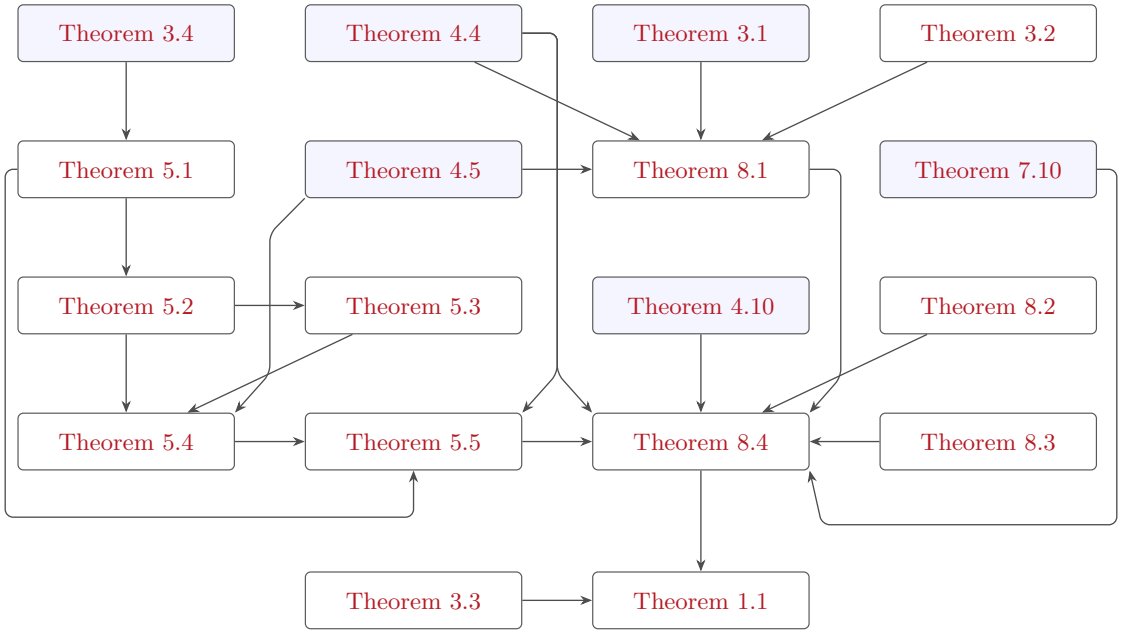
B. Sections 3 and 6



C. Sections 3, 6 and 7



D. Sections 1, 3, 4, 5, 7 and 8



2.7. **Reading guide.** Section 3 contains the finite Fourier reduction and the elementary estimates for primes. Its four-form inequality explains the occurrence of U^3. Its transfer proposition is the final step from a measure on ratios to the extremal bound.

Section 4 develops the local arithmetic in the order in which the finite average is constructed. It begins with valuations and unit parts, defines the weights, proves the required quadrature estimates, and then establishes the joint distribution comparison. The final result of that section is the lower bound for finite character models. Section 5 proves the mean-square comparison needed to replace a general function by such a model.

Sections 6 and 7 contain the inverse-theorem argument. The first treats scalar quadratic exponentials, and the second reduces the general U^3 correlation to that case. A reader checking the numerical exponent should follow the rank estimates, the explicit group construction, and the scalar reduction in Section 7, ending with Theorem 7.10.

Section 8 makes all final parameters explicit in their order of choice. It first establishes the periodic uniformity estimate, then proves the monotone-sum averaging inequality, constructs the measure, calculates its height, and completes the proof of Theorem 1.1. A reader interested first in how the estimates fit together may read this section after the present outline, referring back to the precise results cited there.

3. Fourier transfer

Recall that $\mathcal{M}$ is the class of completely multiplicative functions with values in $\mathbb{S}^1$, defined in Section 1.3. We first describe their restrictions to finitely generated subgroups of $\mathbb{Q}_{>0}$. This identifies the precise finite Fourier transform that will be used in the transfer argument. If $p_1, \dots, p_s$ are distinct primes, put

$$v(r) = (v_{p_1}(r), \dots, v_{p_s}(r)) \in \mathbb{Z}^s$$

for rational numbers whose numerator and denominator have no prime divisors outside this list. Every $\theta = (\theta_1, \dots, \theta_s) \in \mathbb{T}^s$ determines an element $g_\theta \in \mathcal{M}$ by

$$g_\theta(p_i) = e(\theta_i), \qquad g_\theta(p) = 1 \ (p \notin \{p_1, \dots, p_s\}), \qquad g_\theta(r) = e(\theta \cdot v(r)).$$

Conversely, the restriction of any $g \in \mathcal{M}$ to this subgroup of $\mathbb{Q}_{>0}$ has this form. Thus a finitely supported probability measure μ on positive rationals has a finite-dimensional Fourier transform

$$\widehat{\mu}(\theta) = \sum_r \mu(r) e(\theta \cdot v(r)) = \sum_r \mu(r) g_\theta(r).$$

3.1. **Interval uniformity.** For a finite abelian group G, all averages over G are normalized counting averages. For $\omega \in \{0,1\}^3$, write $|\omega| = \omega_1 + \omega_2 + \omega_3$ and $\omega \cdot h = \omega_1 h_1 + \omega_2 h_2 + \omega_3 h_3$. If $F : G \to \mathbb{C}$, define its Gowers norm $\| \cdot \|_{U^3(G)}$:

$$\|F\|_{U^3(G)}^8 = \mathbb{E}_{x,h_1,h_2,h_3 \in G} \prod_{\omega \in \{0,1\}^3} \mathcal{C}^{|\omega|} F(x + \omega \cdot h),$$

where $\mathcal{C}z = \overline{z}$. To verify that the right side is nonnegative, put $\Delta_h F(x) = F(x)\overline{F(x+h)}$. For fixed h_1, h_2, let $H = \Delta_{h_1}\Delta_{h_2} F$. The average over x, h_3 is

$$\mathbb{E}_{x,h_3} H(x)\overline{H(x+h_3)} = |\mathbb{E}_x H(x)|^2,$$

because $(x, x+h_3)$ is uniform on G^2. Therefore

$$\|F\|_{U^3(G)}^8 = \mathbb{E}_{h_1,h_2 \in G} \left|\mathbb{E}_{x \in G} \Delta_{h_1}\Delta_{h_2} F(x)\right|^2. \tag{3.1}$$

For a positive integer Z, write $[Z] = \{1, \dots, Z\}$ and let

$$\mathcal{Q}_Z = \{(x, h_1, h_2, h_3) \in \mathbb{Z}^4 : x + \omega \cdot h \in [Z] \text{ for every } \omega \in \{0,1\}^3\}.$$

The normalized interval convention used throughout the paper is

$$\|f\|_{U^3[Z]}^8 = \frac{1}{|\mathcal{Q}_Z|} \sum_{(x,h) \in \mathcal{Q}_Z} \prod_{\omega \in \{0,1\}^3} \mathcal{C}^{|\omega|} f(x + \omega \cdot h). \tag{3.2}$$

The following identity verifies its positivity and its compatibility with the cyclic-group formulation.

Lemma 3.1. *Let $Z \geq 1$ and $P > 8Z$ be integers. For $f : [Z] \to \mathbb{C}$, let f_Z denote its zero extension to $\mathbb{Z}/P\mathbb{Z}$. Then*

$$\|f_Z\|_{U^3(\mathbb{Z}/P\mathbb{Z})} = \|f\|_{U^3[Z]} \|\mathbf{1}_{[Z]}\|_{U^3(\mathbb{Z}/P\mathbb{Z})}. \tag{3.3}$$

In particular, if $|f| \leq 1$, then $0 \leq \|f\|_{U^3[Z]} \leq 1$ and $\|f_Z\|_{U^3(\mathbb{Z}/P\mathbb{Z})} \leq \|f\|_{U^3[Z]}$.

Proof. We divide the proof into three steps.

Step 1. Lift supported cubes to the interval. Put $G = \mathbb{Z}/P\mathbb{Z}$, and identify $[Z]$ with its image in G. Call a cyclic cube supported if all eight vertices belong to $[Z]$. Cubes are counted by their parameter tuples; repeated vertices and zero increments are allowed.

Let $(\bar{x}, \bar{h}_1, \bar{h}_2, \bar{h}_3) \in G^4$ be supported. Choose the unique representatives $x, x_i \in [Z]$ of $\bar{x}$ and $\bar{x} + \bar{h}_i$, respectively, and put

$$h_i = x_i - x, \qquad |h_i| \leq Z - 1 \qquad (1 \leq i \leq 3).$$

For each $\omega \in \{0,1\}^3$, let $y_\omega \in [Z]$ represent $\bar{x} + \omega \cdot \bar{h}$. Then

$$x + \omega \cdot h \equiv y_\omega \pmod{P}, \qquad |x + \omega \cdot h - y_\omega| \leq |x - y_\omega| + \sum_{i=1}^3 \omega_i |h_i| \leq 4(Z-1) < P.$$

The difference is a multiple of P with absolute value less than P, so it is zero. Thus every integer vertex $x + \omega \cdot h$ belongs to $[Z]$, and the lifted tuple lies in $\mathcal{Q}_Z$. Conversely, reduction modulo P sends every tuple of $\mathcal{Q}_Z$ to a supported cyclic cube. Lifting recovers its initial vertex x and each adjacent vertex $x + h_i$, hence each increment h_i. Lifting and then reducing also recovers the original cyclic tuple. These maps are therefore inverse bijections.

Step 2. Count the lifted cubes and identify the normalized average. Write

$$N_Z = |\mathcal{Q}_Z|, \qquad S_f = \sum_{(x,h)\in\mathcal{Q}_Z} \prod_{\omega\in\{0,1\}^3} \mathcal{C}^{|\omega|} f(x+\omega\cdot h).$$

Any cyclic cube with a vertex outside $[Z]$ contributes zero to the average for f_Z. On supported cubes, the bijection preserves every factor. For $\mathbf{1}_{[Z]}$, each supported cube contributes one. Since a normalized average over G^4 has denominator P^4,

$$\|f_Z\|_{U^3(G)}^8 = \frac{S_f}{P^4}, \qquad \|\mathbf{1}_{[Z]}\|_{U^3(G)}^8 = \frac{N_Z}{P^4}. \tag{3.4}$$

The first quantity is real and nonnegative by (3.1); hence $S_f \geq 0$, although individual summands may be complex. The constant cubes $(x,0,0,0)$, with $x \in [Z]$, give $N_Z \geq Z > 0$. Thus the interval average S_f/N_Z is nonnegative, and its nonnegative eighth root is well-defined. By (3.2) and (3.4),

$$\|f_Z\|_{U^3(G)}^8 = \frac{S_f}{N_Z}\frac{N_Z}{P^4} = \|f\|_{U^3[Z]}^8 \|\mathbf{1}_{[Z]}\|_{U^3(G)}^8.$$

Taking nonnegative eighth roots proves (3.3).

Step 3. Establish the norm bounds. Finally, if $|f| \leq 1$, the triangle inequality gives

$$0 \leq S_f = |S_f| \leq \sum_{(x,h)\in\mathcal{Q}_Z} \left| \prod_{\omega\in\{0,1\}^3} \mathcal{C}^{|\omega|} f(x+\omega\cdot h) \right| \leq N_Z.$$

Hence $0 \leq \|f\|_{U^3[Z]} \leq 1$. The bijection also gives $N_Z \leq |G|^4 = P^4$, so (3.4) implies $\|\mathbf{1}_{[Z]}\|_{U^3(G)} \leq 1$. The remaining inequality follows from (3.3). □

3.2. **A four-form inequality.** The factorization of $g(R(m,n))$, where $R(m,n) = (m^2 - n^2)/2mn$, involves the four forms $m-n, m+n, m, n$. We prove the finite-group inequality for these forms here. The proof introduces one averaging variable for each of the three factors that must be eliminated by Cauchy–Schwarz.

Lemma 3.2. *Let $P \geq 3$ be odd. For functions $a,b,c,d : \mathbb{Z}/P\mathbb{Z} \to \mathbb{C}$ of modulus at most one,*

$$\left|\mathbb{E}_{m,n\in\mathbb{Z}/P\mathbb{Z}} a(m-n)b(m+n)c(m)d(n)\right| \leq \|a\|_{U^3(\mathbb{Z}/P\mathbb{Z})}. \tag{3.5}$$

Proof. We divide the proof into four steps.

Step 1. Factor the four-form average. Put $G = \mathbb{Z}/P\mathbb{Z}$ and $q = 2^{-1} \in G$, which exists because P is odd. All averages below are normalized finite sums: for $F : G^k \to \mathbb{C}$,

$$\mathbb{E}_{v_1,\dots,v_k\in G} F(v_1,\dots,v_k) = \frac{1}{P^k} \sum_{v_1,\dots,v_k\in G} F(v_1,\dots,v_k).$$

Let Λ denote the average inside the absolute value in (3.5). For each fixed $u_1 \in G$, the substitution

$$m = qu_1 + u_3, \qquad n = -qu_1 - u_2$$

is a bijection from $(u_2,u_3) \in G^2$ to $(m,n) \in G^2$. Indeed, its inverse is $u_2 = -n - qu_1$, $u_3 = m - qu_1$. Since $2q = 1$, averaging also over u_1 gives

$$\Lambda = \mathbb{E}_{u_1,u_2,u_3} A(u_1,u_2,u_3)B(u_2,u_3)C(u_1,u_3)D(u_1,u_2), \tag{3.6}$$

where

$$\begin{aligned} A(u_1,u_2,u_3) &= a(u_1+u_2+u_3),\\ B(u_2,u_3) &= b(-u_2+u_3),\\ C(u_1,u_3) &= c(qu_1+u_3),\\ D(u_1,u_2) &= d(-qu_1-u_2). \end{aligned}$$

Thus B, C, D do not depend on u_1, u_2, u_3, respectively, and have modulus at most one.

Step 2. Introduce the finite differences. To record the resulting products, write

$$\Delta_i F = F(\dots, u_i^0, \dots)\overline{F(\dots, u_i^1, \dots)},$$

with the other variables unchanged. The superscripts $0, 1$ label two summation variables, not powers. For example,

$$\Delta_2\Delta_1 A = A(u_1^0, u_2^0, u_3)\overline{A(u_1^1, u_2^0, u_3)A(u_1^0, u_2^1, u_3)}A(u_1^1, u_2^1, u_3).$$

Step 3. Apply three Cauchy–Schwarz inequalities. First write (3.6) as

$$\Lambda = \mathbb{E}_{u_2,u_3} B\, \mathbb{E}_{u_1}(ACD).$$

Cauchy–Schwarz over (u_2, u_3) and $|B| \le 1$ give

$$|\Lambda|^2 \le \left(\mathbb{E}_{u_2,u_3}|B|^2\right) \mathbb{E}_{u_2,u_3} |\mathbb{E}_{u_1} ACD|^2 \le \mathbb{E}_{u_2,u_3} |\mathbb{E}_{u_1} ACD|^2 = \mathbb{E}_{u_1^0,u_1^1,u_2,u_3} \Delta_1 A\, \Delta_1 C\, \Delta_1 D =: S_1.$$

The last equality expands the squared modulus into a double sum over u_1^0, u_1^1. In particular, $S_1 \ge 0$.

Next, $\Delta_1 C$ does not depend on u_2 and has modulus at most one. Taking the average over u_2 first, Cauchy–Schwarz over (u_1^0, u_1^1, u_3) gives

$$S_1^2 \le \mathbb{E}_{u_1^0,u_1^1,u_3} |\mathbb{E}_{u_2} \Delta_1 A\, \Delta_1 D|^2 = \mathbb{E}_{u_1^0,u_1^1,u_2^0,u_2^1,u_3} \Delta_2\Delta_1 A\, \Delta_2\Delta_1 D =: S_2.$$

Again $S_2 \ge 0$, since it is an average of squared moduli. Finally, $\Delta_2\Delta_1 D$ does not depend on u_3 and has modulus at most one. Taking the average over u_3 first, Cauchy–Schwarz over $(u_1^0, u_1^1, u_2^0, u_2^1)$ yields

$$S_2^2 \le \mathbb{E}_{u_1^0,u_1^1,u_2^0,u_2^1} |\mathbb{E}_{u_3} \Delta_2\Delta_1 A|^2 = \mathbb{E}_{u_1^0,u_1^1,u_2^0,u_2^1,u_3^0,u_3^1} \prod_{\omega\in\{0,1\}^3} \mathcal{C}^{|\omega|} a\left(\sum_{i=1}^3 u_i^{\omega_i}\right) =: S_3.$$

Each index equal to 1 contributes a conjugation, producing the exponent $|\omega|$.

Step 4. Identify the final U^3 average. To identify this last sum, set

$$x = u_1^0 + u_2^0 + u_3^0, \qquad h_i = u_i^1 - u_i^0 \quad (1 \le i \le 3).$$

For each prescribed $(x, h_1, h_2, h_3) \in G^4$, choose u_1^0, u_2^0 arbitrarily. The remaining variables are then uniquely determined by

$$u_3^0 = x - u_1^0 - u_2^0, \qquad u_i^1 = u_i^0 + h_i \quad (1 \le i \le 3).$$

Thus each tuple in G^4 has exactly P^2 preimages in G^6. Moreover,

$$\sum_{i=1}^3 u_i^{\omega_i} = \sum_{i=1}^3 (u_i^0 + \omega_i h_i) = x + \omega \cdot h.$$

Consequently, the definition of the group U^3 norm gives

$$S_3 = \frac{P^2}{P^6} \sum_{x,h_1,h_2,h_3\in G} \prod_{\omega\in\{0,1\}^3} \mathcal{C}^{|\omega|} a(x + \omega \cdot h) = \|a\|_{U^3(G)}^8.$$

Since $S_1, S_2 \ge 0$, combining the three Cauchy–Schwarz inequalities and taking nonnegative eighth roots proves (3.5):

$$|\Lambda|^8 \le S_1^4 \le S_2^2 \le S_3 = \|a\|_{U^3(G)}^8. \qquad \square$$

3.3. **Transfer to the exponent box.** Call $r \in \mathbb{Q}_{>0}$ a Pythagorean ratio if $1+r^2$ is the square of a rational number. Such a ratio is different from one, because $\sqrt{2}$ is irrational. For a reduced fraction $r = u/v$ with $u, v \in \mathbb{N}$, its height is $\max(u, v)$.

Proposition 3.3. *Let $0 < \delta < 1/10$ and $H \geq 2$. If a finitely supported probability measure μ is supported on Pythagorean ratios of height at most H and satisfies*

$$\Re \sum_r \mu(r) g(r) \geq -\delta \qquad \text{for every } g \in \mathcal{M}. \tag{3.7}$$

then $f(N) \leq 2\delta N^N$ for every integer

$$N \geq \max\left\{H, \frac{2\log H}{\delta^2 \log 2}\right\}. \tag{3.8}$$

Proof. We divide the proof into four steps.

Step 1. Encode the box and control translations. Fix an integer N satisfying (3.8), and let $A \subseteq \Phi_N$ be admissible. Put

$$B_N = \{0, \dots, N-1\}^N, \qquad G = (\mathbb{Z}/N\mathbb{Z})^N, \qquad \pi(x) = \prod_{i=1}^N p_i^{x_i}.$$

Unique factorization makes $\pi : B_N \to \Phi_N$ a bijection. Identify B_N with the standard representatives of G, and set

$$\mathcal{A} = \pi^{-1}(A), \qquad F = \mathbf{1}_{\mathcal{A}}, \qquad \alpha = \frac{|A|}{N^N} = \mathbb{E}_{x \in G} F(x).$$

Arguments of F are evaluated modulo N. We will bound the same weighted correlation from above by admissibility and from below by (3.7).

For $r = u/v \in \operatorname{supp} \mu$ in lowest terms, every prime dividing u or v is at most $H \leq N$, hence occurs among the first N primes. Put $d(r) = (v_{p_i}(r))_{i=1}^N \in \mathbb{Z}^N$. Since u and v are coprime, unique factorization gives

$$\sum_{i=1}^N |d_i(r)| \log p_i = \sum_{i=1}^N \big(v_{p_i}(u) + v_{p_i}(v)\big) \log p_i = \log(uv),$$

$$\|d(r)\|_1 \leq \frac{\log(uv)}{\log 2} \leq \frac{2\log H}{\log 2}.$$

For $d \in \mathbb{Z}^N$, define the boundary set

$$E_d = \{x \in B_N : x - d \notin B_N\},$$

where subtraction is in $\mathbb{Z}^N$. In coordinate i, the condition $0 \leq x_i - d_i < N$ fails when $x_i < d_i$ if $d_i \geq 0$, and when $x_i \geq N + d_i$ if $d_i < 0$. Among $0, \dots, N-1$, exactly $\min(N, |d_i|)$ choices fail. Counting these failures over the coordinates yields

$$|E_d| \leq N^{N-1} \sum_{i=1}^N \min(N, |d_i|) \leq N^{N-1} \|d\|_1.$$

Step 2. Use admissibility to bound the correlation. If $x \notin E_{d(r)}$, then the cyclic translate $x - d(r)$ is the ordinary integer vector in B_N. If both $F(x)$ and $F(x - d(r))$ were one, the labels $a = \pi(x)$ and $b = \pi(x - d(r))$ would lie in A and satisfy

$$\frac{a}{b} = \prod_{i=1}^N p_i^{d_i(r)} = r, \qquad a^2 + b^2 = b^2(1 + r^2) = (bs)^2,$$

where $s \in \mathbb{Q}_{>0}$ satisfies $1 + r^2 = s^2$. The labels are distinct because $r \neq 1$. Moreover, bs is an integer: writing $bs = k/\ell$ in lowest terms, the integrality of $(bs)^2$ gives $\ell^2 \mid k^2$, hence $\ell = 1$.

This contradicts admissibility. Thus $F(x)F(x-d(r))$ vanishes outside $E_{d(r)}$ and is at most one on that set. Since $\mu(r) \geq 0$ and $\sum_r \mu(r) = 1$,

(3.9)
$$0 \leq \mathcal{K} := \sum_r \mu(r)\mathbb{E}_{x\in G}F(x)F(x-d(r)) \leq \sum_r \mu(r)\frac{|E_{d(r)}|}{N^N} \leq \frac{1}{N}\sum_r \mu(r)\|d(r)\|_1 \leq \frac{2\log H}{N\log 2} \leq \delta^2.$$

The last inequality uses (3.8).

Step 3. Compute the Fourier lower bound. For the lower bound, define

$$\widehat{F}(\xi) = \mathbb{E}_{x\in G}F(x)e(-\xi\cdot x/N) = \frac{1}{N^N}\sum_{x\in G}F(x)e(-\xi\cdot x/N).$$

The finite geometric-series identity, applied in each coordinate, gives

$$\mathbb{E}_{x\in G}e((\xi-\eta)\cdot x/N) = \mathbf{1}_{\xi=\eta}, \qquad \sum_{\xi\in G} e(\xi\cdot z/N) = N^N\mathbf{1}_{z=0}. \tag{3.10}$$

Indeed, $\sum_{t=0}^{N-1} e(jt/N)$ equals N if $N \mid j$ and zero otherwise. Since F is real-valued, expanding the two Fourier coefficients and using (3.10) gives, for every $d \in \mathbb{Z}^N$,

$$\begin{aligned}\sum_{\xi\in G}|\widehat{F}(\xi)|^2 e(\xi\cdot d/N) &= \mathbb{E}_{x,y\in G}F(x)F(y)\sum_{\xi\in G}e(\xi\cdot(y-x+d)/N)\\ &= N^N\mathbb{E}_{x,y\in G}F(x)F(y)\mathbf{1}_{y=x-d}\\ &= \mathbb{E}_{x\in G}F(x)F(x-d).\end{aligned}$$

Taking $d=0$, using $F^2=F$, and evaluating $\widehat{F}(0)$ directly gives

$$\sum_{\xi\in G}|\widehat{F}(\xi)|^2 = \alpha, \qquad \widehat{F}(0) = \alpha, \qquad \sum_{\xi\neq 0}|\widehat{F}(\xi)|^2 = \alpha-\alpha^2. \tag{3.11}$$

For each $\xi \in G$, prescribe

$$g_\xi(p_i) = e(\xi_i/N) \quad (1\leq i\leq N), \qquad g_\xi(p) = 1 \quad \big(p\notin\{p_1,\dots,p_N\}\big),$$

and extend by prime factorization. This defines a completely multiplicative function of modulus one, so $g_\xi \in \mathcal{M}$. For every $r \in \operatorname{supp}\mu$,

$$g_\xi(r) = \prod_{i=1}^N g_\xi(p_i)^{d_i(r)} = e(\xi\cdot d(r)/N).$$

At $\xi=0$, $g_0=1$ and $\sum_r \mu(r)g_0(r)=1$, so the zero frequency contributes exactly α^2. Since $\mathcal{K}$ is real, interchanging the finite sums and taking real parts gives

$$\mathcal{K} = \sum_{\xi\in G}|\widehat{F}(\xi)|^2\Re\sum_r \mu(r)g_\xi(r) \geq \alpha^2 - \delta\sum_{\xi\neq 0}|\widehat{F}(\xi)|^2 = \alpha^2-\delta(\alpha-\alpha^2) \geq \alpha^2-\delta\alpha. \tag{3.12}$$

The inequality uses (3.7), the nonnegative coefficients $|\widehat{F}(\xi)|^2$ and (3.11).

Step 4. Solve the resulting quadratic. Combining (3.9) and (3.12), and using $\alpha \geq 0$, we obtain

$$\alpha^2-\delta\alpha-\delta^2 \leq 0, \qquad \alpha \leq \frac{1+\sqrt{5}}{2}\delta < 2\delta.$$

Hence $|A| \leq 2\delta N^N$. Maximizing over admissible subsets $A \subseteq \Phi_N$ proves the proposition. □

3.4. **Prime estimates.** We first prove the following upper-bound sieve estimate. Let $z \geq 2$, and let $h : \mathbb{N} \to [0,1]$ be multiplicative with $h(p) < 1$ for $p \leq z$. Suppose that, for fixed positive constants κ and B,

$$\prod_{\substack{a \leq p \leq b \\ p \text{ prime}}} (1-h(p))^{-1} \leq \left(\frac{\log b}{\log a}\right)^{\kappa} \exp(B/\log a) \qquad (2 \leq a \leq b \leq z). \tag{3.13}$$

Let $\mathcal{S}$ be a set of primes, assume that $h(d) = 0$ when d has a prime divisor outside $\mathcal{S}$, and let R be a random positive integer. Then

$$\mathbb{P}\left(p \nmid R \text{ for all } p \leq z,\ p \in \mathcal{S}\right) \leq C_{\kappa,B} \prod_{\substack{p \leq z \\ p \text{ prime}}} (1-h(p)) + \sum_{\substack{d \leq z \\ p \text{ prime},\ p \mid d \Rightarrow p \in \mathcal{S}}} \mu^2(d)\, |\mathbb{P}(d \mid R) - h(d)|\,. \tag{3.14}$$

Here $\mu(d)$ is the Möbius function: it is zero if a square of a prime divides d, and it equals $(-1)^k$ if d is a product of k distinct primes. The constant depends only on the two displayed dimension parameters. We verify those parameters in the application.

Proof of (3.14). We divide the proof into four steps.

Step 1. Remove the degenerate case and form the Euler product. Multiplicativity gives $h(1) = h(1)^2$. If $h(1) = 0$, then $h(n) = h(1)h(n) = 0$ for every $n \geq 1$. The $d = 1$ term in the error sum in (3.14) then equals $|\mathbb{P}(1 \mid R) - h(1)| = 1$. Since the left side is a probability, the estimate follows. We may therefore assume $h(1) = 1$.

Step 2. Invoke the upper-bound sieve weights. Put

$$Q = \prod_{\substack{p \leq z \\ p \in \mathcal{S}}} p, \qquad V(z) = \prod_{\substack{p \leq z \\ p \text{ prime}}} (1-h(p)).$$

The event on the left of (3.14) is $\gcd(R, Q) = 1$. Since $h(p) = 0$ outside $\mathcal{S}$, multiplicativity and expansion of the finite product give

$$V(z) = \prod_{\substack{p \mid Q \\ p \text{ prime}}} (1-h(p)) = \sum_{d \mid Q} \mu(d) h(d).$$

This explains the Euler product, but its divisor sum need not be supported on $d \leq z$.

Step 3. Restrict the weights and obtain a majorant. We apply the construction in [7, Lemma 2.2, p. 16, and proof of Theorem 2.4, pp. 18–19] with $g(p) = h(p)$ for $p \leq z$ and $g(p) = 0$ for $p > z$. Condition (Ω) in [7, p. 18] is then exactly (3.13). This construction gives real weights $(c_d)_{d \geq 1}$ satisfying

$$c_1 = 1, \qquad |c_d| \leq 1 \quad (d \geq 1),$$
$$c_d = 0 \quad \text{unless } d \leq z \text{ is squarefree},$$
$$\sum_{d \mid m} c_d \geq 0 \quad (m \geq 1).$$

Since $g(d) = h(d)$ whenever $c_d \neq 0$, [7, (2.8), p. 17, and (2.11), p. 19] also gives

$$\sum_{d \geq 1} c_d h(d) = \sum_{d \geq 1} c_d g(d) \leq C_{\kappa,B} V(z), \tag{3.15}$$

where $C_{\kappa,B} > 0$ depends only on κ, B.

Restrict these weights to divisors of Q by setting $\lambda_d = c_d \mathbf{1}_{d \mid Q}$. Thus

$$\lambda_1 = 1, \qquad |\lambda_d| \leq 1, \qquad \lambda_d = 0 \quad \text{unless } d \mid Q \text{ and } d \leq z. \tag{3.16}$$

For $m = \gcd(n, Q)$, the divisor sum of c_d equals one if $m = 1$ and is nonnegative otherwise. Hence

$$\mathbf{1}_{\gcd(n,Q)=1} \le \sum_{d \mid \gcd(n,Q)} c_d = \sum_{d \mid n} \lambda_d \qquad (n \ge 1). \tag{3.17}$$

If $c_d \neq 0$ and $d \nmid Q$, then d has a prime factor outside $\mathcal{S}$, so $h(d) = 0$. Thus (3.15) gives

$$\sum_{d} \lambda_d h(d) = \sum_{d} c_d h(d) \le C_{\kappa,B} V(z). \tag{3.18}$$

Step 4. Take expectations and bound the error. Define the divisibility errors by

$$\varepsilon_d = \mathbb{P}(d \mid R) - h(d).$$

Taking expectations in (3.17) and interchanging the finite divisor sum with expectation yields

$$\mathbb{P}(\gcd(R, Q) = 1) \le \sum_{d} \lambda_d \mathbb{P}(d \mid R) = \sum_{d} \lambda_d h(d) + \sum_{d} \lambda_d \varepsilon_d \le C_{\kappa,B} V(z) + \sum_{\substack{d \le z \\ d \mid Q}} |\varepsilon_d|. \tag{3.19}$$

The last step uses (3.18), the support in (3.16), and $|\lambda_d \varepsilon_d| \le |\varepsilon_d|$. For $d \le z$, the condition $d \mid Q$ is equivalent to $\mu^2(d) = 1$ and p prime, $p \mid d \Rightarrow p \in \mathcal{S}$. Consequently,

$$\sum_{\substack{d \le z \\ d \mid Q}} |\varepsilon_d| = \sum_{\substack{d \le z \\ p \text{ prime},\ p \mid d \Rightarrow p \in \mathcal{S}}} \mu^2(d) \left| \mathbb{P}(d \mid R) - h(d) \right|.$$

Substitution into (3.19) proves (3.14). In particular, the coefficient one on the error sum comes from $|\lambda_d| \le 1$. □

Lemma 3.4. *There is an absolute C such that, for every real $x \ge 2$,*

$$\log x \le E(x) := \prod_{\substack{p \le x \\ p \text{ prime}}} (1 - p^{-1})^{-1} \le C \log x. \tag{3.20}$$

For every set of primes $\mathcal{S}$, let h be the multiplicative function defined by $h(1) = 1$ and, for each prime p and integer $k \ge 1$, $h(p^k) = p^{-k}$ if $p \in \mathcal{S}$ and $h(p^k) = 0$ otherwise. Then h satisfies (3.13) with absolute constants κ, B independent of $\mathcal{S}$.

Proof. We divide the proof into three steps.

Step 1. Establish the Chebyshev and Euler-product bounds. We first prove the Chebyshev bound

$$\vartheta(x) := \sum_{\substack{p \le x \\ p \text{ prime}}} \log p \le C_0 x \qquad (x \ge 2), \qquad C_0 = 4 \log 2. \tag{3.21}$$

For each integer $n \ge 1$, every prime $n < p \le 2n$ divides $(2n)!$ but not $n!$, and hence divides $\binom{2n}{n}$. The product of these distinct primes therefore divides the binomial coefficient. The binomial theorem gives $\binom{2n}{n} \le 2^{2n}$, so

$$\vartheta(2n) - \vartheta(n) \le \log \binom{2n}{n} \le 2n \log 2.$$

Taking $n = 2^{j-1}$ and summing over $1 \le j \le k$ yields

$$\vartheta(2^k) \le \sum_{j=1}^{k} 2^j \log 2 = (2^{k+1} - 2) \log 2 \le 2^{k+1} \log 2.$$

For $2^{k-1} < x \le 2^k$, monotonicity gives $\vartheta(x) \le \vartheta(2^k) \le 4x \log 2$, proving (3.21).

Expanding the Euler factors as geometric series and using unique factorization gives

$$E(x) = \sum_{\substack{n\ge 1\\ p \text{ prime},\ p\mid n\Rightarrow p\le x}} \frac{1}{n} \ge \sum_{n=1}^{\lfloor x\rfloor} \frac{1}{n} \ge \int_1^{\lfloor x\rfloor+1} \frac{dt}{t} = \log(\lfloor x\rfloor+1) \ge \log x.$$

Every $n \le x$ occurs in the first sum, and $1/t \le 1/n$ on $[n, n+1]$.

For the upper bound, put

$$\sigma = 1 + \frac{1}{\log x}, \qquad E_\sigma(x) = \prod_{\substack{p\le x\\ p \text{ prime}}} (1-p^{-\sigma})^{-1}.$$

Since $\sigma > 1$, geometric expansion and comparison with an integral give

$$E_\sigma(x) \le \sum_{n=1}^{\infty} n^{-\sigma} \le 1 + \int_1^{\infty} t^{-\sigma}\, dt = 1 + \frac{1}{\sigma-1} = 1 + \log x. \tag{3.22}$$

Using $-\log(1-u) = \sum_{k\ge1} u^k/k$ and $1 - e^{-v} \le v$ for $v \ge 0$, we obtain

$$\begin{aligned} \log E(x) - \log E_\sigma(x) &= \sum_{\substack{p\le x\\ p \text{ prime}}} \sum_{k\ge1} \frac{p^{-k}(1-e^{-k(\sigma-1)\log p})}{k} \\ &\le (\sigma-1) \sum_{\substack{p\le x\\ p \text{ prime}}} (\log p) \sum_{k\ge1} p^{-k} \\ &= \frac{1}{\log x} \sum_{\substack{p\le x\\ p \text{ prime}}} \frac{\log p}{p-1}. \end{aligned} \tag{3.23}$$

All series here have nonnegative terms and converge. Partial summation and (3.21) give

$$\sum_{\substack{p\le x\\ p \text{ prime}}} \frac{\log p}{p} = \frac{\vartheta(x)}{x} + \int_2^x \frac{\vartheta(t)}{t^2}\, dt \le C_0 \left(1 + \log\frac{x}{2}\right).$$

The identity follows by multiplying

$$1/p = 1/x + \int_p^x t^{-2}\, dt$$

by $\log p$ and summing. Since $1/(p-1) \le 2/p$, (3.23) implies

$$\log E(x) - \log E_\sigma(x) \le 2C_0 \frac{1+\log(x/2)}{\log x} \le \frac{2C_0}{\log 2} = 8.$$

The last inequality uses

$$(1+\log(x/2))/\log x = 1 + (1-\log 2)/\log x \le 1/\log 2.$$

Together with (3.22), this proves

$$E(x) \le e^8(1+\log x) \le C\log x, \qquad C = e^8\left(1 + \frac{1}{\log 2}\right),$$

and completes (3.20).

Step 2. Verify the sieve-dimension estimate. Now fix $2 \le a \le b \le z$. Since $h(p)$ is either $1/p$ or zero, the logarithmic series gives

$$\log \prod_{\substack{a\le p\le b\\ p \text{ prime}}} (1-h(p))^{-1} \le \sum_{\substack{a\le p\le b\\ p \text{ prime}}} \sum_{k\ge1} \frac{1}{kp^k} = \sum_{\substack{a\le p\le b\\ p \text{ prime}}} \frac{1}{p} + \sum_{\substack{a\le p\le b\\ p \text{ prime}}} \sum_{k\ge2} \frac{1}{kp^k}. \tag{3.24}$$

For the first sum, apply partial summation with $f(t) = 1/(t\log t)$, whose derivative is

$$f'(t) = -(\log t + 1)/(t^2(\log t)^2).$$

This yields

$$\sum_{\substack{a\le p\le b\\ p \text{ prime}}} \frac{1}{p} = \frac{\vartheta(b)}{b\log b} - \frac{\vartheta(a)}{a\log a} + \int_a^b \frac{\vartheta(t)(\log t+1)}{t^2(\log t)^2}\,dt + \frac{\mathbf{1}_{\{a \text{ is prime}\}}}{a}. \tag{3.25}$$

The final term restores the lower endpoint when a is prime. The integral needed after applying (3.21) is

$$\int_a^b \frac{\log t+1}{t(\log t)^2}\,dt = \int_{\log a}^{\log b}\left(\frac{1}{u}+\frac{1}{u^2}\right)du = \log\frac{\log b}{\log a} + \frac{1}{\log a} - \frac{1}{\log b}. \tag{3.26}$$

Discarding the nonpositive lower-endpoint term in (3.25) and using (3.21) and (3.26), we get

$$\sum_{\substack{a\le p\le b\\ p \text{ prime}}} \frac{1}{p} \le C_0\log\frac{\log b}{\log a} + \frac{C_0}{\log a} + \frac{1}{a} \le C_0\log\frac{\log b}{\log a} + \frac{C_0+1}{\log a}. \tag{3.27}$$

Here the terms $C_0/\log b$ cancel, and $1/a \le 1/\log a$.

For the terms with $k \ge 2$, use $1/k \le 1$, sum the geometric series, and then sum over all integers at least a:

$$\sum_{\substack{p\ge a\\ p \text{ prime}}}\sum_{k\ge 2}\frac{1}{kp^k} \le \sum_{n\ge\lceil a\rceil}\frac{1}{n(n-1)} = \sum_{n\ge\lceil a\rceil}\left(\frac{1}{n-1}-\frac{1}{n}\right) = \frac{1}{\lceil a\rceil - 1} \le \frac{2}{a}. \tag{3.28}$$

The last inequality follows from $\lceil a\rceil - 1 \ge a - 1 \ge a/2$. Combining (3.24), the first bound in (3.27), and (3.28) gives

$$\log\prod_{\substack{a\le p\le b\\ p \text{ prime}}}(1-h(p))^{-1} \le C_0\log\frac{\log b}{\log a} + \frac{C_0}{\log a} + \frac{3}{a} \le C_0\log\frac{\log b}{\log a} + \frac{C_0+3}{\log a}.$$

Step 3. Conclude the lemma. Exponentiation proves (3.13) with $\kappa = 4\log 2$ and $B = 4\log 2 + 3$. These constants are independent of $\mathcal{S}$. □

Lemma 3.5. *There is an absolute constant $C > 0$, independent of A, B, N, with the following property. Let A, B be real numbers with $2 \le A < B$, let $N \ge 1$ be an integer, and put $z = \lceil B\rceil$. Then*

$$\frac{1}{N}\#\{1 \le n \le N : p \nmid n \text{ for every prime } A \le p < B\} \le C\frac{\log A}{\log B} + \frac{z}{N}. \tag{3.29}$$

Proof. We divide the proof into three steps.

Step 1. Set up the sieve probability. Let $\mathcal{S} = \{p \text{ prime} : A \le p < B\}$, and let R take each value in $\{1,\dots,N\}$ with probability $1/N$. Thus

$$\rho := \frac{1}{N}\#\{1 \le n \le N : p \nmid n \text{ for all } p \in \mathcal{S}\} = \mathbb{P}(p \nmid R \text{ for all } p \in \mathcal{S}). \tag{3.30}$$

Define the multiplicative function

$$h(d) = \begin{cases} d^{-1}, & \text{if every prime divisor of } d \text{ belongs to } \mathcal{S},\\ 0, & \text{otherwise.}\end{cases}$$

In particular, $h(1) = 1$ and $0 \le h(p) \le 1/2 < 1$. By Theorem 3.4, h satisfies (3.13) with fixed absolute parameters κ_0, β_0, independent of $\mathcal{S}$. Every prime in $\mathcal{S}$ is less than $B \le z$, so (3.14) applies and gives

$$\rho \le C_{\text{sieve}} \prod_{\substack{A \le p < B \\ p \text{ prime}}} (1 - p^{-1}) + \sum_{\substack{d \le z \\ p \text{ prime},\ p \mid d \Rightarrow p \in \mathcal{S}}} \mu^2(d) \left| \mathbb{P}(d \mid R) - \frac{1}{d} \right|. \tag{3.31}$$

Here $C_{\text{sieve}} = C_{\kappa_0, \beta_0}$ is absolute and independent of A, B, N. The product contains only primes in $\mathcal{S}$ because $h(p) = 0$ outside that set.

Step 2. Bound the divisibility error. There are exactly $\lfloor N/d \rfloor$ multiples of d in $\{1, \dots, N\}$. Hence, for every positive integer d,

$$\mathbb{P}(d \mid R) = \frac{\lfloor N/d \rfloor}{N},$$
$$\left| \mathbb{P}(d \mid R) - \frac{1}{d} \right| = \frac{1}{N} \left| \lfloor N/d \rfloor - \frac{N}{d} \right| \le \frac{1}{N}.$$

This uses $0 \le t - \lfloor t \rfloor < 1$ and also holds when $d > N$. Since $\mu^2(d) \le 1$ and there are at most z positive integers $d \le z$, the error in (3.31) is bounded by

$$\sum_{\substack{d \le z \\ p \text{ prime},\ p \mid d \Rightarrow p \in \mathcal{S}}} \mu^2(d) \left| \mathbb{P}(d \mid R) - \frac{1}{d} \right| \le \frac{1}{N} \sum_{d \le z} 1 = \frac{z}{N}. \tag{3.32}$$

Step 3. Estimate the Euler product and conclude. To estimate the product, write $E(t^-) = \prod_{\substack{p < t \\ p \text{ prime}}} (1 - p^{-1})^{-1}$, with a strict upper endpoint. Factoring the product over primes below B at A gives

$$E(B^-) = E(A^-) \prod_{\substack{A \le p < B \\ p \text{ prime}}} (1 - p^{-1})^{-1}, \qquad \prod_{\substack{A \le p < B \\ p \text{ prime}}} (1 - p^{-1}) = \frac{E(A^-)}{E(B^-)}. \tag{3.33}$$

Let C_E be the absolute constant in (3.20). Its upper bound gives

$$E(A^-) \le E(A) \le C_E \log A.$$

If B is not prime, then $E(B^-) = E(B)$. If B is prime, then

$$E(B^-) = (1 - B^{-1}) E(B) \ge E(B)/2.$$

Thus in both cases,

$$E(B^-) \ge \frac{1}{2} E(B) \ge \frac{1}{2} \log B. \tag{3.34}$$

Combining (3.33)–(3.34), we obtain

$$\prod_{\substack{A \le p < B \\ p \text{ prime}}} (1 - p^{-1}) \le 2 C_E \frac{\log A}{\log B}. \tag{3.35}$$

Substituting (3.32) and (3.35) into (3.31) yields

$$\rho \le 2 C_{\text{sieve}} C_E \frac{\log A}{\log B} + \frac{z}{N}.$$

By (3.30), this proves (3.29) with $C = 2 C_{\text{sieve}} C_E$. Both factors are absolute, so C is independent of A, B, N. □

4. Weighted Pythagorean ratios

Throughout this section,

$$R(m,n)=\frac{m^2-n^2}{2mn},\qquad m>n>0.$$

We average these ratios using a congruence weight at small primes and a fixed density in the two real variables. For $y\ge 1$, call a unit-modulus completely multiplicative function g a finite character model if $g(p)=\chi(p)p^{it}$ for every prime $p>y$, where χ is a Dirichlet character of conductor at most y and $t\in\mathbb{R}$. We approximate the integer average by its local factors, with an explicit error, and bound its real part below for every such model.

4.1. **Local distributions.** Write $\mathbb{S}^1=\{z\in\mathbb{C}:|z|=1\}$. For p-adic fields and topology, see [15, Sections 3.1–3.2 and 4.1–4.4]; for Haar measure, see [3, Section 6, Proposition I.6.3]. All subsets of p-adic spaces considered below are Borel.

p-adic numbers and residue classes. Fix a prime p. Write $r\in\mathbb{Q}^\times$ as $r=p^k a/b$, where $k\in\mathbb{Z}$, $a,b\in\mathbb{Z}\setminus\{0\}$, and $p\nmid ab$. Set $v_p(r)=k$ and $|r|_p=p^{-k}$, with $|0|_p=0$. The field $\mathbb{Q}_p$ is the completion of $\mathbb{Q}$ for this absolute value. Extend v_p to $\mathbb{Q}_p$ by $v_p(x)=-\log_p|x|_p$ for $x\neq 0$, and set $v_p(0)=+\infty$. Then

$$\mathbb{Z}_p=\{x\in\mathbb{Q}_p:v_p(x)\ge 0\},\qquad \mathbb{Z}_p^\times=\{x\in\mathbb{Q}_p:v_p(x)=0\}=\mathbb{Z}_p\setminus p\mathbb{Z}_p.$$

Every $x\in\mathbb{Z}_p$ has a unique expansion $x=\sum_{j\ge 0}x_jp^j$ with $x_j\in\{0,\dots,p-1\}$. For each integer $a\ge 1$, its residue class modulo p^a is determined by the first a digits, giving $\mathbb{Z}_p/p^a\mathbb{Z}_p\simeq\mathbb{Z}/p^a\mathbb{Z}$. The ring $\mathbb{Z}_p$ is compact. The residue classes $b+p^a\mathbb{Z}_p$ form a basis of open and closed sets and generate its Borel σ-algebra.

Haar measure and random variables on $\mathbb{Z}_p$. Let λ_p be the translation-invariant Borel probability measure on $\mathbb{Z}_p$. The p^a residue classes modulo p^a have equal measure, so

$$\lambda_p(\mathbb{Z}_p)=1,\qquad \lambda_p(b+p^a\mathbb{Z}_p)=p^{-a}\quad(a\ge 1,\ b\in\mathbb{Z}_p). \tag{4.1}$$

These values determine λ_p. A $\mathbb{Z}_p$-valued random variable X is Haar uniform on $\mathbb{Z}_p$ if it has law λ_p, or equivalently,

$$\mathbb{P}(X\equiv b\pmod{p^a})=p^{-a}\quad(a\ge 1,\ b\in\mathbb{Z}/p^a\mathbb{Z}).$$

One construction is $X=\sum_{j\ge 0}D_jp^j$ with independent digits D_j uniform on $\{0,\dots,p-1\}$: a prescribed residue fixes a digits and has probability p^{-a}. Also $\mathbb{P}(X=x)\le p^{-a}$ for every a, so each singleton has probability zero.

A Haar pair (M,N) consists of independent $\mathbb{Z}_p$-valued random variables M,N, each with law λ_p. Their joint law is $\lambda_p\otimes\lambda_p$. For $A,B\subseteq\mathbb{Z}_p$, $a\ge 1$, and $b,c\in\mathbb{Z}/p^a\mathbb{Z}$,

$$\mathbb{P}(M\in A,\ N\in B)=\lambda_p(A)\lambda_p(B),\qquad \mathbb{P}(M\equiv b,\ N\equiv c\pmod{p^a})=p^{-2a}.$$

Similarly, a Haar vector in $\mathbb{Z}_p^d$ has law $\lambda_p^{\otimes d}$. Haar scalar and unrestricted Haar pair refer to these laws before conditioning; independence between different primes is assumed only when stated.

For $\mathbb{P}(E)>0$, conditional probability is $\mathbb{P}(F\mid E)=\mathbb{P}(F\cap E)/\mathbb{P}(E)$. For X with law λ_p and $B\subseteq\mathbb{Z}_p$ with $\lambda_p(B)>0$, its conditional law given $X\in B$ assigns $A\subseteq\mathbb{Z}_p$ the probability $\lambda_p(A\cap B)/\lambda_p(B)$. In particular, $b+p^aX$ has the normalized restriction of λ_p to $b+p^a\mathbb{Z}_p$:

$$\mathbb{P}(b+p^aX\in A)=p^a\lambda_p\big(A\cap(b+p^a\mathbb{Z}_p)\big)\quad(a\ge 0,\ b\in\mathbb{Z}_p,\ A\subseteq\mathbb{Z}_p). \tag{4.2}$$

Indeed, for $j \geq 0$ and $c \in \mathbb{Z}_p$, the preimage of $b + p^a c + p^{a+j}\mathbb{Z}_p$ is $c + p^j\mathbb{Z}_p$. Both sides of (4.2) assign this target class probability $p^{-j} = p^a p^{-(a+j)}$; equality on residue classes proves the formula.

Haar measure on the unit group. Since $\lambda_p(\mathbb{Z}_p^\times) = 1 - p^{-1}$, define

$$\lambda_p^\times(A) = \frac{\lambda_p(A)}{1 - p^{-1}} \quad (A \subseteq \mathbb{Z}_p^\times). \tag{4.3}$$

For X with law λ_p, this is its conditional law given $X \in \mathbb{Z}_p^\times$. Multiplication by a unit permutes residue classes, hence preserves λ_p and $\lambda_p^\times$. Thus $\lambda_p^\times$ is normalized multiplicative Haar measure on $\mathbb{Z}_p^\times$. A $\mathbb{Z}_p^\times$-valued random variable U is Haar uniform if it has law $\lambda_p^\times$, equivalently if

$$\mathbb{P}(U \equiv u \pmod{p^a}) = \frac{p^{-a}}{1 - p^{-1}} = \frac{1}{\varphi(p^a)} \quad (a \geq 1,\ u \in (\mathbb{Z}/p^a\mathbb{Z})^\times).$$

Here $\varphi(p^a) = (p-1)p^{a-1}$. These residue probabilities also prove uniqueness, since the unit classes generate the Borel sets. If U has this law, then for $s \geq 1$ its conditional law on $1 + p^s\mathbb{Z}_p$ assigns $A \subseteq 1 + p^s\mathbb{Z}_p$ mass $\lambda_p^\times(A)/(p^{-s}/(1-p^{-1})) = p^s\lambda_p(A)$. It is the normalized multiplicative Haar measure on this subgroup, since multiplication by its elements preserves the subgroup and λ_p.

Primitive pairs. A pair $(m, n) \in \mathbb{Z}_p^2$ is primitive if at least one coordinate is a unit, equivalently if $\min(v_p(m), v_p(n)) = 0$. Put $C_p = \mathbb{Z}_p^2 \setminus (p\mathbb{Z}_p)^2$. For a Haar pair (M, N) under $\mathbb{P}$, let $E_p = \{(M, N) \in C_p\}$. Independence and conditional probability give

$$\mathbb{P}(E_p) = 1 - p^{-2}, \qquad \mathbb{P}((M, N) \in A \mid E_p) = \frac{(\lambda_p \otimes \lambda_p)(A \cap C_p)}{1 - p^{-2}} \quad (A \subseteq \mathbb{Z}_p^2). \tag{4.4}$$

The induced measure on the projective line. The projective line $\mathbb{P}^1(\mathbb{Q}_p)$ consists of nonzero pairs $(m, n) \in \mathbb{Q}_p^2$ modulo common nonzero scaling, with the quotient topology. Write $[m : n] = m/n$ when $n \neq 0$, and $[1 : 0] = \infty$. The map $\iota : p\mathbb{Z}_p \to \mathbb{P}^1(\mathbb{Q}_p)$, $u \mapsto [1 : u]$, is a homeomorphism onto $(\mathbb{Q}_p \setminus \mathbb{Z}_p) \cup \{\infty\}$; thus $\iota(u) = 1/u$ for $u \neq 0$, $\iota(0) = \infty$, and $\mathbb{P}^1(\mathbb{Q}_p) = \mathbb{Z}_p \sqcup \iota(p\mathbb{Z}_p)$.

Let (M, N) be a Haar pair under $\mathbb{P}$, with primitive event E_p. Since the pair is nonzero on E_p, define μ_p as the law of $[M : N]$ under $\mathbb{P}(\,\cdot \mid E_p)$. For $A \subseteq \mathbb{Z}_p$ and $B \subseteq p\mathbb{Z}_p$, the events $[M : N] \in A$ and $[M : N] \in \iota(B)$, restricted to E_p, are respectively $\{N \in \mathbb{Z}_p^\times,\ M \in NA\}$ and $\{M \in \mathbb{Z}_p^\times,\ N \in MB\}$. Multiplication by a unit preserves λ_p. Independence and division by $\mathbb{P}(E_p) = 1 - p^{-2}$ therefore give

$$\begin{aligned} \mu_p(A) &= \frac{(1 - p^{-1})\lambda_p(A)}{1 - p^{-2}} = \frac{p}{p+1}\lambda_p(A), \\ \mu_p(\iota(B)) &= \frac{(1 - p^{-1})\lambda_p(B)}{1 - p^{-2}} = \frac{p}{p+1}\lambda_p(B). \end{aligned} \tag{4.5}$$

In particular, $\mu_p(\mathbb{Z}_p) = p/(p+1)$ and $\mu_p(\iota(p\mathbb{Z}_p)) = 1/(p+1)$. Given $M \in \mathbb{Z}_p^\times$ and $N \in p\mathbb{Z}_p$, the random variable $u = N/M$ has law $p\,\lambda_p|_{p\mathbb{Z}_p}$. The unnormalized restriction $\lambda_p|_{p\mathbb{Z}_p}$ has mass $1/p$.

For $m, n \in \mathbb{Q}_p$ with $mn \neq 0$, put $R(m, n) = (m^2 - n^2)/(2mn)$ and $R(t) = R(t, 1) = (t^2 - 1)/(2t)$ for $t \in \mathbb{Q}_p^\times$. Then $R(m, n) = R(m/n)$ and $R(p^j m, p^j n) = R(m, n)$. Assign $R(M, N)$ the value zero on the null event $MN = 0$. Its law is the same under $\mathbb{P}$ and $\mathbb{P}(\,\cdot \mid E_p)$. Indeed, for $J = \min(v_p(M), v_p(N))$, $\mathbb{P}(J = j) = p^{-2j} - p^{-2j-2} = p^{-2j}(1 - p^{-2})$ for $j \geq 0$. Given $M, N \in p^j\mathbb{Z}_p$, the scaled pair $(p^{-j}M, p^{-j}N)$ is a Haar pair by (4.2) and independence; $J = j$ means that this scaled pair is primitive. Hence its law given $J = j$ is (4.4). Homogeneity makes

the conditional law of $R(M,N)$ independent of j, hence equal to its law given $E_p = \{J = 0\}$. Averaging over j, using $\sum_{j\ge 0} p^{-2j}(1-p^{-2}) = 1$, proves the assertion.

For t with law μ_p, define, outside the null set $\{0, 1, -1, \infty\}$,

$$V_p = v_p(R(t)), \qquad U_p = p^{-V_p} R(t) \in \mathbb{Z}_p^\times.$$

The unit part U_p need not be an ordinary integer.

Lemma 4.1. *With the notation above, for every odd prime p and every integer $k \ge 1$,*

$$\mu_p(V_p = k) = \mu_p(V_p = -k) = \frac{2(p-1)}{p+1} p^{-k}, \qquad \mu_p(V_p = 0) = \frac{p-3}{p+1}. \tag{4.6}$$

For $p = 2$, one has $|V_2| \ge 2$ almost surely, and for every integer $h \ge 2$,

$$\mu_2(V_2 = h) = \frac{2^{1-h}}{3}, \qquad \mu_2(V_2 = -h) = \frac{2^{2-h}}{3}. \tag{4.7}$$

For every nonzero integer j with $\mu_p(V_p = j) > 0$, the conditional distribution of U_p given $V_p = j$ is the normalized Haar measure $\lambda_p^\times$ on $\mathbb{Z}_p^\times$. Explicitly, for every integer $a \ge 1$ and every $b \in (\mathbb{Z}/p^a\mathbb{Z})^\times$,

$$\mu_p\big(U_p \equiv b \pmod{p^a} \mid V_p = j\big) = \frac{1}{\varphi(p^a)} = \frac{1}{(p-1)p^{a-1}}. \tag{4.8}$$

Proof. We divide the proof into three steps.

Step 1. Compute the odd-prime shells. We use the residue-class measures in (4.1), the unit measure in (4.3), the projective measure (4.5), and compactness of $\mathbb{Z}_p$. We omit the four null points throughout the valuation calculations. For $c \in \mathbb{Z}_p$ and every integer $k \ge 0$,

$$\lambda_p\{x \in \mathbb{Z}_p : v_p(x-c) = k\} = \lambda_p(c+p^k\mathbb{Z}_p) - \lambda_p(c+p^{k+1}\mathbb{Z}_p) = p^{-k} - p^{-k-1} = (1-p^{-1})p^{-k}. \tag{4.9}$$

For $t \in \mathbb{Q}_p \setminus \{0, 1, -1\}$, factoring the numerator of R gives

$$v_p(R(t)) = v_p(t-1) + v_p(t+1) - v_p(2) - v_p(t), \qquad R(1/t) = -R(t). \tag{4.10}$$

Suppose first that p is odd. If $t \in p\mathbb{Z}_p \setminus \{0\}$, then $t-1$ and $t+1$ are units, so $V_p = -v_p(t)$. For $t = \iota(u)$ with $u \in p\mathbb{Z}_p \setminus \{0\}$, the second identity in (4.10) likewise gives $V_p = -v_p(u)$. For $k \ge 1$, each of these two disjoint regions therefore contributes

$$\frac{p}{p+1}(p^{-k} - p^{-k-1}) = \frac{p-1}{p+1} p^{-k} \tag{4.11}$$

to $\mu_p(V_p = -k)$, by (4.5) and (4.9).

If $t \in 1 + p\mathbb{Z}_p$, then t and $t+1$ are units, so $V_p = v_p(t-1)$. If $t \in -1 + p\mathbb{Z}_p$, then t and $t-1$ are units, so $V_p = v_p(t+1)$. These balls are disjoint, and each contributes (4.11) to $\mu_p(V_p = k)$. This proves both nonzero-valuation formulas in (4.6). The remaining points lie in $\mathbb{Z}_p$ and satisfy $t \bmod p \notin \{0, 1, -1\}$. Both the numerator and denominator of $R(t)$ are units there. The $p-3$ remaining residue classes each have μ_p-measure $1/(p+1)$. Thus

$$\mu_p(V_p = 0) = \frac{p-3}{p+1}, \qquad \sum_{k\ge 1} \mu_p(V_p = k) = \sum_{k\ge 1} \mu_p(V_p = -k) = \frac{2}{p+1}. \tag{4.12}$$

The three probabilities in (4.12) sum to one.

Step 2. Compute the dyadic shells. Now let $p = 2$. If $t = 2^k a$, where $k \ge 1$ and $a \in \mathbb{Z}_2^\times$, then $t^2 - 1$ is odd and $V_2 = -k - 1$. For $h \ge 2$, the set giving $V_2 = -h$ in $2\mathbb{Z}_2$ is therefore $\{t : v_2(t) = h - 1\}$. The region parametrized by $\iota(u)$ contributes the same measure, since $R(1/u) = -R(u)$. Hence

$$\mu_2(V_2 = -h) = 2 \cdot \frac{2}{3}\big(2^{-(h-1)} - 2^{-h}\big) = \frac{2^{2-h}}{3} \qquad (h \ge 2).$$

If t is odd and $t \neq \pm 1$, exactly one of $t-1, t+1$ has valuation one, while the other has valuation at least two. Consequently,

$$V_2 = \begin{cases} v_2(t-1), & t \in 1+4\mathbb{Z}_2, \quad t \neq 1, \\ v_2(t+1), & t \in -1+4\mathbb{Z}_2, \quad t \neq -1. \end{cases} \tag{4.13}$$

The two balls in (4.13) partition $\mathbb{Z}_2^\times$. Each contributes $(2/3)(2^{-h} - 2^{-h-1})$ to $\mu_2(V_2 = h)$, giving

$$\mu_2(V_2 = h) = 2 \cdot \frac{2}{3}\big(2^{-h} - 2^{-h-1}\big) = \frac{2^{1-h}}{3} \qquad (h \geq 2).$$

These cases exhaust the projective line apart from the four null points. In particular, $|V_2| \geq 2$ almost surely, and the total probabilities are

$$\sum_{h \geq 2} \mu_2(V_2 = h) = \frac{1}{3}, \qquad \sum_{h \geq 2} \mu_2(V_2 = -h) = \frac{2}{3}.$$

Step 3. Identify the conditional unit law. To prove (4.8), we first justify the measure-preservation fact needed below. Let $B = c + p^r\mathbb{Z}_p$ and $C = d + p^r\mathbb{Z}_p$, where $r \geq 0$ is an integer and $c, d \in \mathbb{Z}_p$. Suppose that $F : B \to C$ satisfies

$$v_p(F(x) - F(y)) = v_p(x - y) \qquad (x, y \in B). \tag{4.14}$$

For each $a \geq r$, congruence of x, y modulo p^a is equivalent to congruence of $F(x), F(y)$ modulo p^a. Thus F induces an injection between the residue classes of B and C modulo p^a. Both sets have p^{a-r} classes, so the induced map is a bijection. Each target residue class has exactly one source residue class as its preimage, and their normalized Haar probabilities are both p^{r-a}. For a fixed $z \in C$, the preimages of $z + p^a\mathbb{Z}_p$ are nested nonempty compact balls. Their intersection contains a point x, and $F(x) \equiv z \pmod{p^a}$ for all $a \geq r$ implies $F(x) = z$. Hence F is onto and carries normalized Haar measure on B to normalized Haar measure on C, because residue classes generate the Borel sets. The same argument applies to $F : \mathbb{Z}_p^\times \to \mathbb{Z}_p^\times$ satisfying (4.14): for $a \geq 1$ there are $\varphi(p^a)$ residue classes, each of probability $1/\varphi(p^a)$ under $\lambda_p^\times$.

For positive valuations, put $r = 1$ when p is odd and $r = 2$ when $p = 2$. For $\epsilon \in \{1, -1\}$, let $B_\epsilon = \epsilon + p^r\mathbb{Z}_p$. The two balls are disjoint. For $t, s \in B_\epsilon$, direct subtraction gives

$$R(t) - R(s) = \frac{t - s}{2}\left(1 + \frac{1}{ts}\right).$$

For odd p, the factor multiplying $t - s$ is congruent to 1 modulo p, since $ts \equiv 1 \pmod{p}$. For $p = 2$, one has $ts \equiv 1 \pmod 4$, so $v_2(ts + 1) = 1$ and $(ts + 1)/(2ts)$ is odd. The factor is therefore a unit in both cases. Since $R(\epsilon) = 0$, the map R is an isometry $B_\epsilon \to p^r\mathbb{Z}_p$. The preceding argument proves that it is onto and preserves normalized Haar measure. By (4.5), the conditional distribution of t given $t \in B_\epsilon$ is normalized additive Haar measure on that ball. Its image under R is therefore normalized additive Haar measure on $p^r\mathbb{Z}_p$. For every $j \geq r$, $a \geq 1$, and $b \in (\mathbb{Z}/p^a\mathbb{Z})^\times$, it follows that

$$\mu_p\big(U_p \equiv b \pmod{p^a} \mid V_p = j,\ t \in B_\epsilon\big) = \frac{p^{-(j+a)}}{p^{-j} - p^{-j-1}} = \frac{1}{(p-1)p^{a-1}}.$$

Here the numerator is the additive Haar measure of $p^j b + p^{j+a}\mathbb{Z}_p$, using any integer representative of b; the coset is independent of that choice. The denominator is the measure of $p^j\mathbb{Z}_p^\times$. The common normalizing factor for $p^r\mathbb{Z}_p$ cancels in the quotient. Both balls give the same conditional unit law. Since they exhaust the event $V_p = j$, the law of total probability gives (4.8) for positive j.

For negative valuations, set $\delta = v_p(2)$ and $d = 2/p^\delta \in \mathbb{Z}_p^\times$; thus $(\delta, d) = (0, 2)$ for odd p and $(\delta, d) = (1, 1)$ for $p = 2$. Write $t = p^k a$ with $k \geq 1$ and $a \in \mathbb{Z}_p^\times$. Given $v_p(t) = k$ in the

region $p\mathbb{Z}_p$, the variable $a = p^{-k}t$ has distribution $\lambda_p^\times$, by (4.5) and the residue-class measures. Since $V_p = -k - \delta$, its unit part is

$$U_p = p^{k+\delta}R(p^k a) = \frac{p^{2k}a - a^{-1}}{d} =: F_{p,k}(a).$$

The numerator is a unit, so $F_{p,k}$ maps $\mathbb{Z}_p^\times$ into itself. For units a, b,

$$F_{p,k}(a) - F_{p,k}(b) = (a-b)\frac{p^{2k} + (ab)^{-1}}{d}.$$

The second factor is a unit. Thus $F_{p,k}$ satisfies (4.14) on $\mathbb{Z}_p^\times$ and preserves $\lambda_p^\times$. In the other region, parametrized by $t = \iota(u)$ with $v_p(u) = k$, the identity $R(1/u) = -R(u)$ changes only the sign of the unit part. Multiplication by -1 permutes unit residue classes modulo every p^a, so this region gives the same unit law. The two regions exhaust the event $V_p = -k-\delta$. The law of total probability therefore proves (4.8) for every negative valuation of positive probability. □

For $z \in \mathbb{S}^1$ and a continuous character $\eta : \mathbb{Z}_p^\times \to \mathbb{S}^1$, put

$$K_p(z, \eta) = \int z^{V_p}\eta(U_p)\, d\mu_p.$$

The conductor exponent of a nontrivial η is the least integer $a \geq 1$ such that η is trivial on $1 + p^a\mathbb{Z}_p$. For the trivial character the conductor exponent is zero. Such an exponent exists for every continuous character: continuity puts the image of $1 + p^a\mathbb{Z}_p$, for sufficiently large a, in the arc $\{e^{i\theta} : |\theta| < \pi/3\}$. This image is a subgroup, and the only subgroup of $\mathbb{S}^1$ contained in that arc is $\{1\}$, since the powers of any nonidentity element leave the arc.

Lemma 4.2. *Let p be an odd prime, $z \in \mathbb{S}^1$, and $\eta : \mathbb{Z}_p^\times \to \mathbb{S}^1$ a continuous character. If η is trivial, then $K_p(z, 1)$ is real and*

$$K_p(z,1) = \frac{p-3}{p+1} + \frac{4(p-1)}{p+1}\sum_{k\geq 1} p^{-k}\Re(z^k), \qquad K_p(z,1) \geq \frac{p^2 - 6p + 1}{(p+1)^2}. \tag{4.15}$$

In particular, $K_p(z, 1) \in [0, 1]$ for $p \geq 7$. If η is nontrivial with conductor exponent $a \geq 1$, then

$$|K_p(z,\eta)| \leq \begin{cases} 2\sqrt{p}/(p+1), & a = 1, \\ 2/(p+1), & a \geq 2. \end{cases} \tag{4.16}$$

Proof. We divide the proof into three steps.

Step 1. Treat the principal character. For $\eta = 1$, decompose the integral according to V_p:

$$K_p(z,1) = \mu_p(V_p = 0) + \sum_{k\geq 1}\big(z^k\mu_p(V_p = k) + z^{-k}\mu_p(V_p = -k)\big).$$

The series converges absolutely. Substituting (4.6) and using $z^{-k} = \overline{z^k}$ proves the equality in (4.15) and its reality. For $c = \Re z \in [-1, 1]$, the geometric series gives

$$f(c) := \sum_{k\geq 1} p^{-k}\Re(z^k) = \Re\frac{z}{p - z} = \frac{pc - 1}{p^2 - 2pc + 1}, \qquad f'(c) = \frac{p(p^2-1)}{(p^2 - 2pc + 1)^2} > 0.$$

Thus $f(c) \geq f(-1) = -1/(p+1)$, and

$$K_p(z,1) \geq \frac{p-3}{p+1} - \frac{4(p-1)}{(p+1)^2} = \frac{p^2 - 6p + 1}{(p+1)^2}. \tag{4.17}$$

Since $|z^{V_p}\eta(U_p)| = 1$ almost surely and μ_p is a probability measure, $|K_p(z, \eta)| \leq 1$. For $p \geq 7$, the numerator in (4.17) is positive, which proves the asserted interval bound.

Step 2. Reduce a nonprincipal factor to a finite character sum. Now suppose that η is nontrivial. Put $I = \int_{\mathbb{Z}_p^\times} \eta(u)\, d\lambda_p^\times(u)$ and choose $u_0 \in \mathbb{Z}_p^\times$ with $\eta(u_0) \neq 1$. Multiplicative invariance of Haar measure gives

$$I = \int_{\mathbb{Z}_p^\times} \eta(u_0 u)\, d\lambda_p^\times(u) = \eta(u_0) I, \qquad I = 0.$$

By (4.8), for every $j \neq 0$,

$$\int_{\{V_p = j\}} z^{V_p} \eta(U_p)\, d\mu_p = z^j \mu_p(V_p = j) I = 0.$$

Consequently only $V_p = 0$ contributes. Up to a null set, this event is $A = \{t \in \mathbb{Z}_p : t \bmod p \notin \{0, 1, -1\}\}$, where $U_p = R(t)$, so

$$K_p(z, \eta) = \int_A \eta(R(t))\, d\mu_p(t).$$

In particular, this expression is independent of z.

Let $a \geq 1$ be the conductor exponent and let η_a be the induced character of $(\mathbb{Z}/p^a\mathbb{Z})^\times$. By (4.5), every class $r + p^a\mathbb{Z}_p \subseteq \mathbb{Z}_p$ has measure $p^{1-a}/(p+1)$. On A, the value of $R(t)$ modulo p^a depends only on t modulo p^a, since $2t$ is a unit. Hence

$$K_p(z, \eta) = \frac{1}{(p+1)p^{a-1}} \sum_{\substack{t \bmod p^a \\ p \nmid t(t^2-1)}} \eta_a\big((t^2-1)(2t)^{-1}\big), \tag{4.18}$$

where the inverse is taken modulo p^a.

If $a = 1$, write $\chi = \eta_1$, of order $d > 1$,[2] and extend it by $\chi(0) = 0$. Then (4.18) becomes

$$K_p(z, \eta) = \frac{1}{p+1} \sum_{t \in \mathbb{F}_p \setminus \{0,1,-1\}} \chi\left(\frac{t^2-1}{2t}\right). \tag{4.19}$$

We use the following form of Weil's character-sum bound [24, Theorem 5.41]. Let q be a prime power, let $\xi : \mathbb{F}_q^\times \to \mathbb{S}^1$ be a multiplicative character of order $m > 1$, and set $\xi(0) = 0$. If a monic polynomial $F \in \mathbb{F}_q[X]$ of positive degree is not an mth power and has r distinct roots in its splitting field, then, for every $\alpha \in \mathbb{F}_q^\times$,

$$\left| \sum_{x \in \mathbb{F}_q} \xi(\alpha F(x)) \right| \leq (r-1)\sqrt{q}. \tag{4.20}$$

For the application to (4.19), put $P(T) = T^{d-1}(T-1)(T+1)$ and $\gamma = 2^{d-1} \in \mathbb{F}_p^\times$. For $t \notin \{0, 1, -1\}$,

$$\chi(\gamma P(t)) = \chi(2)^{-1}\chi(t)^{-1}\chi(t-1)\chi(t+1) = \chi\left(\frac{t^2-1}{2t}\right).$$

because $\chi(u)^{d-1} = \chi(u)^{-1}$ for every unit u. The polynomial P is monic and has exactly three distinct roots $0, 1, -1$, since p is odd. Its root at 1 has multiplicity one, so P is not a dth power. The terms at $0, 1, -1$ vanish under the extension of χ by zero. Apply (4.20) with $q = p$, $\xi = \chi$, $m = d$, $F = P$, $r = 3$, and $\alpha = \gamma$. This bounds the sum in (4.19) by $2\sqrt{p}$, proving the first line of (4.16).

Step 3. Handle higher conductor. Suppose finally that $a \geq 2$. Define $\psi(b) = \eta(1 + bp^{a-1})$ for $b \in \mathbb{F}_p$, using any integer representative of b. This is well-defined because η depends only on a unit modulo p^a. Moreover,

$$(1 + bp^{a-1})(1 + cp^{a-1}) \equiv 1 + (b+c)p^{a-1} \pmod{p^a},$$

[2]The order of χ is the least positive integer d such that $\chi(u)^d = 1$ for every $u \in \mathbb{F}_p^\times$; equivalently, χ^d is the trivial character. It divides $p-1$ and equals 1 exactly when χ is trivial.

since $2a-2 \geq a$. Hence ψ is an additive character of $\mathbb{F}_p$. It is nontrivial: otherwise η would be trivial on $1+p^{a-1}\mathbb{Z}_p$, contrary to the minimality of a.

Fix $t \in A$. For $h \in p^{a-1}\mathbb{Z}_p$, direct subtraction gives

$$R(t+h) = R(t) + hR'(t) - \frac{h^2}{2t^2(t+h)}, \qquad R'(t) = \frac{1+t^{-2}}{2}.$$

The quantities $2, t, t+h, R(t)$ are units. After division by $R(t)$, the remainder belongs to $p^{2a-2}\mathbb{Z}_p \subseteq p^a\mathbb{Z}_p$. With $h = bp^{a-1}$ this gives

$$\frac{R(t+bp^{a-1})}{R(t)} \equiv 1 + bp^{a-1}\frac{R'(t)}{R(t)} \pmod{p^a}. \tag{4.21}$$

Group the sum in (4.18) by residues modulo p^{a-1}. The p lifts of each such residue have equal measure. For a representative $t \in A$, put $\beta_t = R'(t)/R(t) \bmod p \in \mathbb{F}_p$. (4.21) gives their average as

$$\frac{1}{p}\sum_{b\in\mathbb{F}_p} \eta(R(t+bp^{a-1})) = \frac{\eta(R(t))}{p}\sum_{b\in\mathbb{F}_p}\psi(b\beta_t) = \begin{cases} \eta(R(t)), & \beta_t = 0, \\ 0, & \beta_t \neq 0. \end{cases}$$

Indeed, when $\beta_t \neq 0$, multiplication by β_t permutes $\mathbb{F}_p$. The sum of ψ is zero: translating its argument by b_0 with $\psi(b_0) \neq 1$ multiplies the sum by $\psi(b_0)$ without changing it. Since $R(t)$ is a unit,

$$\beta_t = 0 \iff R'(t) \equiv 0 \pmod{p} \iff t^2 \equiv -1 \pmod{p}.$$

Thus only the residue classes $r + p\mathbb{Z}_p$ with $r^2 = -1$ in $\mathbb{F}_p$ can contribute. There are at most two, and each has measure $1/(p+1)$. Since $|\eta(R(t))| = 1$ on A,

$$|K_p(z,\eta)| \leq \sum_{\substack{r\in\mathbb{F}_p \\ r^2=-1}} \mu_p(r+p\mathbb{Z}_p) \leq \frac{2}{p+1},$$

where an integer representative of r is understood. This proves the second line of (4.16). □

4.2. **Weights.** For a prime p, an integer $j \neq 0$, and an integer $s \geq 1$, let

$$E_{p,j,s} = \{V_p = 2j,\ U_p \in 1 + p^s\mathbb{Z}_p\}.$$

Its probability is

$$b_{p,j,s} := \mu_p(E_{p,j,s}) = \begin{cases} \dfrac{2}{p+1}p^{1-s-2|j|}, & p \text{ odd}, \\ 2^{2-s-2j}/3, & p = 2,\ j > 0, \\ 2^{3-s-2|j|}/3, & p = 2,\ j < 0. \end{cases} \tag{4.22}$$

Indeed, $U_p \in 1 + p^s\mathbb{Z}_p$ means $U_p \equiv 1 \pmod{p^s}$. The conditional unit law (4.8) therefore gives

$$b_{p,j,s} = \mu_p(V_p = 2j)\,\mu_p(U_p \equiv 1 \pmod{p^s} \mid V_p = 2j) = \frac{\mu_p(V_p = 2j)}{(p-1)p^{s-1}}.$$

Substituting (4.6) and (4.7) proves (4.22). These values satisfy $b_{p,j,s} \geq p^{-s-2|j|} > 0$. Hence, for integers $T \geq 1$ and $0 < |j| \leq T$,

$$\left\| \frac{\mathbf{1}_{E_{p,j,s}}}{b_{p,j,s}} \right\|_\infty = b_{p,j,s}^{-1} \leq p^{s+2|j|} \leq p^{s+2T}. \tag{4.23}$$

Fix integers $T \geq 1$ and $s \geq 1$, and a finite set of primes $\mathcal{B}$. Let J be a random variable taking values in $\{\pm 1, \dots, \pm T\}$ with law

$$\mathbb{P}(J = j) = \frac{T+1-|j|}{T(T+1)} \quad (0 < |j| \leq T). \tag{4.24}$$

Given $J = j$, let $t_p \in \mathbb{P}^1(\mathbb{Q}_p)$, $p \in \mathcal{B}$, be independent random variables with respective laws $\mu_p(\cdot \mid E_{p,j,s})$. Here V_p, U_p are evaluated at t_p; in particular, $V_p = 2J$ for every $p \in \mathcal{B}$ almost surely.

Lemma 4.3. *Let J and t_p, $p \in \mathcal{B}$, be the random variables defined above. For each $p \in \mathcal{B}$, let $z_p \in \mathbb{S}^1$ and let $\eta_p : \mathbb{Z}_p^\times \to \mathbb{S}^1$ be a continuous character. Then $\mathbb{E}[\prod_{p\in\mathcal{B}} z_p^{V_p}\eta_p(U_p)]$ is zero if some η_p has conductor exponent greater than s. Otherwise this expectation is real and belongs to $[-1/T, 1]$.*

Proof. We divide the proof into three steps.

Step 1. Normalize the valuation labels and condition the unit variables. The probabilities in (4.24) sum to one, since

$$\sum_{0<|j|\le T} \frac{T+1-|j|}{T(T+1)} = \frac{2}{T(T+1)} \sum_{j=1}^{T} (T+1-j) = 1. \tag{4.25}$$

Put $H_{p,s} = 1 + p^s\mathbb{Z}_p$. For $0 < |j| \le T$ and a Borel set $D \subseteq \mathbb{Z}_p^\times$, the definition of the conditional law and (4.8) give

$$\mathbb{P}(U_p \in D \mid J = j) = \frac{\mu_p(U_p \in D \cap H_{p,s} \mid V_p = 2j)}{\mu_p(U_p \in H_{p,s} \mid V_p = 2j)} = \frac{\lambda_p^\times(D \cap H_{p,s})}{\lambda_p^\times(H_{p,s})} =: \nu_{p,s}(D).$$

Here $\lambda_p^\times(H_{p,s}) = 1/((p-1)p^{s-1}) > 0$. The normalized restriction $\nu_{p,s}$ is invariant under multiplication by elements of $H_{p,s}$, so it is Haar probability on this subgroup. In particular, this conditional unit law does not depend on j.

Step 2. Evaluate the local character integral. Let a_p be the conductor exponent of η_p, with $a_p = 0$ for the trivial character. By definition, η_p is trivial on $H_{p,s}$ exactly when $a_p \le s$. Thus

$$I_{p,s} := \int_{H_{p,s}} \eta_p(u)\, d\nu_{p,s}(u) = \begin{cases} 1, & a_p \le s, \\ 0, & a_p > s. \end{cases}$$

Indeed, the integral is one in the first case. In the second, choose $h_0 \in H_{p,s}$ with $\eta_p(h_0) \ne 1$. Haar invariance gives $I_{p,s} = \eta_p(h_0)I_{p,s}$, forcing $I_{p,s} = 0$.

Step 3. Average over the labels and apply the Fej'er identity. Write

$$Y = \prod_{p\in\mathcal{B}} z_p^{V_p}\eta_p(U_p) \quad \text{and} \quad z = \prod_{p\in\mathcal{B}} z_p^2 \in \mathbb{S}^1.$$

Given $J = j$, the variables t_p are independent and $V_p = 2j$ almost surely. Hence

$$\mathbb{E}[Y \mid J = j] = \prod_{p\in\mathcal{B}} z_p^{2j}\mathbb{E}[\eta_p(U_p) \mid J = j] = z^j \prod_{p\in\mathcal{B}} I_{p,s}. \tag{4.26}$$

If some $a_p > s$, then $\mathbb{E}[Y \mid J = j] = 0$ for every $0 < |j| \le T$. Averaging with the probabilities in (4.24) gives $\mathbb{E}[Y] = 0$.

Suppose now that $a_p \le s$ for every $p \in \mathcal{B}$. Then all $I_{p,s} = 1$. Averaging (4.26) over J and pairing j with $-j$ gives

$$\mathbb{E}[Y] = \frac{1}{T(T+1)} \sum_{0<|j|\le T} (T+1-|j|)z^j = \frac{2}{T(T+1)} \sum_{j=1}^{T} (T+1-j)\Re(z^j), \tag{4.27}$$

since $z^{-j} = \overline{z^j}$. This expression is real. Expanding a squared absolute value, we obtain

$$\left|\sum_{k=0}^{T} z^k\right|^2 = \sum_{k,\ell=0}^{T} z^{k-\ell} = (T+1) + 2\sum_{j=1}^{T} (T+1-j)\Re(z^j). \tag{4.28}$$

For each $-T \le j \le T$, there are $T+1-|j|$ pairs $(k,\ell) \in \{0,\dots,T\}^2$ with $k-\ell=j$, which proves the second equality. Combining (4.27) and (4.28) yields

$$\mathbb{E}[Y] = \frac{1}{T}\left\{ \frac{1}{T+1}\left|\sum_{k=0}^{T} z^k\right|^2 - 1 \right\} \ge -\frac{1}{T}. \tag{4.29}$$

Finally, $|Y|=1$ almost surely, so $|\mathbb{E}[Y]| \le \mathbb{E}[|Y|] = 1$. Together with reality and (4.29), this proves the interval bound. □

Fix $0<\delta<1/10$ and define

$$T_0 = \lceil 100/\delta \rceil, \qquad B = (200/\delta)^2, \qquad D_0 = \prod_{\substack{p \le B \\ p \text{ prime}}} p^{2T_0+5}. \tag{4.30}$$

For $0<|j|\le T_0$, define a function on $\mathbb{Z}_p^2$ by

$$W_{p,j}(m,n) = \frac{\mathbf{1}_{\min(v_p(m),v_p(n))=0}\mathbf{1}_{E_{p,j,3}}(m,n)}{(1-p^{-2})b_{p,j,3}}. \tag{4.31}$$

Set $W_{p,j}(m,n)=0$ whenever $mn(m^2-n^2)=0$. This union of four lines has product Haar measure zero. The denominator normalizes the weight to have mean one on $\mathbb{Z}_p^2$. Define the global congruence weight $W_0 : \mathbb{Z}^2 \to [0,\infty)$ by

$$W_0(m,n) = \sum_{0<|j|\le T_0} \frac{T_0+1-|j|}{T_0(T_0+1)} \prod_{\substack{p \le B \\ p \text{ prime}}} W_{p,j}(m,n). \tag{4.32}$$

Lemma 4.4. *The function W_0 is nonnegative, is periodic modulo D_0 in each variable, and satisfies*

$$D_0^{-2} \sum_{a,b \bmod D_0} W_0(a,b) = 1.$$

For all integers m,n,c with $\gcd(c,D_0)=1$, we have $W_0(cm,cn)=W_0(m,n)$. Moreover,

$$\log(2D_0) + \log(2\|W_0\|_\infty) \le C\delta^{-3}. \tag{4.33}$$

Here $C>0$ is an absolute constant.

Proof. We divide the proof into four steps.

Step 1. Prove local periodicity. Nonnegativity follows from (4.31) and (4.32). Fix $p \le B$ and $0<|j|\le T_0$, and put

$$F = m^2-n^2, \qquad G = 2mn, \qquad \ell = 2|j|+1, \qquad k = \ell+4.$$

We show that $W_{p,j}$ depends only on $(m,n) \bmod p^k$. Primitivity is determined modulo p; non-primitive pairs have weight zero. For a primitive pair, write $\alpha = v_p(F)$ and $\beta = v_p(G)$, with $v_p(0)=\infty$. If p is odd and $p \mid G$, exactly one coordinate is divisible by p, so F is a unit. If $p=2$, opposite parity makes F odd, whereas two odd coordinates give $\beta=1$. Thus

$$\min(\alpha,\beta)=0 \quad (p \text{ odd}), \qquad \min(\alpha,\beta)\le 1 \quad (p=2). \tag{4.34}$$

The residues of m,n modulo p^k determine those of F,G. If either F or G is divisible by $p^{\ell+1}$, the pair has weight zero. Indeed, when $FG\neq 0$, (4.34) gives $|\alpha-\beta| \ge \ell > 2|j|$; when $FG=0$, the defining zero convention applies. Otherwise $\alpha,\beta \le \ell$ are determined exactly. The units $F_0 = p^{-\alpha}F$ and $G_0 = p^{-\beta}G$ are determined modulo $p^{k-\alpha}$ and $p^{k-\beta}$, respectively. Since both exponents are at least four, these units are known modulo p^3. As $R = p^{\alpha-\beta}F_0/G_0$, we have

$$[m:n] \in E_{p,j,3} \iff \alpha-\beta = 2j \quad \text{and} \quad F_0 \equiv G_0 \pmod{p^3}.$$

Both conditions are determined by the prescribed residues. This also proves that the zero convention at $FG = 0$ is compatible with periodicity. Since $k \le 2T_0 + 5$, each $W_{p,j}$ depends only on the pair modulo p^{2T_0+5}. Their products and sum in (4.32) are therefore periodic modulo D_0 in each variable.

Step 2. Compute the mean by Haar measure and the Chinese remainder theorem. Put $L = 2T_0 + 5$. A Haar pair in $\mathbb{Z}_p^2$ is primitive with probability $1 - p^{-2}$, and its projective point has law μ_p given primitivity. Every residue pair modulo p^L has product Haar measure p^{-2L}. Thus the preceding periodicity and (4.31) give

$$p^{-2L} \sum_{r,s \bmod p^L} W_{p,j}(r,s) = \int_{\mathbb{Z}_p^2} W_{p,j}\, d(\lambda_p \otimes \lambda_p) = \frac{(1-p^{-2})\mu_p(E_{p,j,3})}{(1-p^{-2})b_{p,j,3}} = 1. \tag{4.35}$$

For each fixed j, the Chinese remainder theorem[3] yields

$$D_0^{-2} \sum_{a,b \bmod D_0} \prod_{\substack{p \le B \\ p \text{ prime}}} W_{p,j}(a,b) = \prod_{\substack{p \le B \\ p \text{ prime}}} \left(p^{-2L} \sum_{r,s \bmod p^L} W_{p,j}(r,s) \right) = 1. \tag{4.36}$$

The coefficients $\omega_j = (T_0 + 1 - |j|)/(T_0(T_0+1))$ sum to one by (4.25). Averaging (4.36) with these coefficients proves the asserted mean of W_0.

Step 3. Prove unit-scaling invariance. For $u \in \mathbb{Z}_p^\times$, simultaneous multiplication of m, n by u preserves primitivity and multiplies both F and G by u^2. Hence it preserves their valuations, their unit quotient, and the exceptional set $FG = 0$. Therefore

$$W_{p,j}(um, un) = W_{p,j}(m,n) \qquad (m,n \in \mathbb{Z}_p,\ u \in \mathbb{Z}_p^\times). \tag{4.37}$$

If $\gcd(c, D_0) = 1$, then $c \in \mathbb{Z}_p^\times$ for every $p \le B$. Applying (4.37) to each factor in (4.32) proves the stated identity for W_0.

Step 4. Bound the supremum norm. By (4.23) with $s = 3$,

$$\|W_{p,j}\|_\infty \le (1-p^{-2})^{-1} b_{p,j,3}^{-1} \le (1-p^{-2})^{-1} p^{3+2T_0}.$$

Since $\omega_j \ge 0$ and $\sum_j \omega_j = 1$, it follows that

$$\|W_0\|_\infty \le \sum_{0<|j|\le T_0} \omega_j \prod_{\substack{p \le B \\ p \text{ prime}}} \|W_{p,j}\|_\infty \le \prod_{\substack{p \le B \\ p \text{ prime}}} (1-p^{-2})^{-1} p^{3+2T_0} \le 2 \prod_{\substack{p \le B \\ p \text{ prime}}} p^{3+2T_0}.$$

For the last inequality, each factor exceeds one and

$$\prod_{n=2}^{N} (1-n^{-2})^{-1} = \left(\prod_{n=2}^{N} \frac{n}{n-1} \right) \left(\prod_{n=2}^{N} \frac{n}{n+1} \right) = \frac{2N}{N+1} \longrightarrow 2.$$

Thus the product over $p \le B$ is at most two. Taking logarithms and using the Chebyshev estimate $\sum_{\substack{p \le B \\ p \text{ prime}}} \log p \le 4B$ from (3.21), we obtain

$$\log D_0 \le 4(2T_0+5)B, \qquad \log \|W_0\|_\infty \le \log 2 + 4(2T_0+3)B.$$

Finally, $T_0 \le 101\delta^{-1}$ and $B = 40000\delta^{-2}$, so

$$\log(2D_0) + \log(2\|W_0\|_\infty) \le 3\log 2 + (16T_0 + 32)B \le C\delta^{-3}.$$

This proves (4.33). □

[3]For pairwise coprime $d_1, \dots, d_k \ge 2$ and $D = \prod_{\ell=1}^k d_\ell$, the map $\mathbb{Z}/D\mathbb{Z} \to \prod_{\ell=1}^k \mathbb{Z}/d_\ell\mathbb{Z}$, $a \mapsto (a \bmod d_\ell)_\ell$, is a ring isomorphism.

We next construct a fixed real probability density, independently of δ. Let $b = \mathbf{1}_{[-1/2,1/2]}$ and $h = b * b * b * b$. Then

$$h(s) = \frac{1}{6}\sum_{k=0}^{4}(-1)^k\binom{4}{k}(s+2-k)_+^3, \qquad a_+ = \max(a,0). \tag{4.38}$$

Put $c = \mathbf{1}_{[0,1]}$, and write c^{*r} for its r-fold convolution. Since $b(s) = c(s+1/2)$, we have $h(s) = c^{*4}(s+2)$. The value $(c*c)(x)$ is the length of $[0,1]\cap[x-1,x]$, so

$$(c*c)(x) = x_+ - 2(x-1)_+ + (x-2)_+.$$

Convolving once more means integrating over $0 \le a \le 1$. For $q = 1, 2$, the antiderivative of a positive-part power gives

$$\int_0^1 (x-a-k)_+^q\,da = \frac{(x-k)_+^{q+1} - (x-k-1)_+^{q+1}}{q+1}.$$

Applying this identity to each term, first with $q = 1$ and then with $q = 2$, yields

$$c^{*3}(x) = \frac{1}{2}\big[x_+^2 - 3(x-1)_+^2 + 3(x-2)_+^2 - (x-3)_+^2\big],$$
$$c^{*4}(x) = \frac{1}{6}\big[x_+^3 - 4(x-1)_+^3 + 6(x-2)_+^3 - 4(x-3)_+^3 + (x-4)_+^3\big].$$

Replacing x by $s+2$ proves (4.38). The convolution definition gives $h \ge 0$ and $h(s) = 0$ for $|s| > 2$. Integrating (4.38) over $[-2,2]$ gives

$$\int_{\mathbb{R}} h(s)\,ds = \frac{4^4 - 4\cdot 3^4 + 6\cdot 2^4 - 4}{24} = 1.$$

Moreover, for every real x,

$$\frac{d}{dx}x_+^3 = 3x_+^2, \qquad \frac{d^2}{dx^2}x_+^3 = 6x_+.$$

Both derivatives are continuous, including at zero. (4.38) and the compact support therefore give $h \in C_c^2(\mathbb{R})$. In fact, $\operatorname{supp} h = [-2,2]$.

For $t > 1$, put $r(t) = (t^2-1)/(2t)$ and $\phi(t) = \log r(t)$. Direct differentiation and the solution of $r(t) = e^s$ give

$$\phi'(t) = \frac{r'(t)}{r(t)} = \frac{t^2+1}{t(t^2-1)} > 0, \qquad \phi^{-1}(s) = e^s + \sqrt{1+e^{2s}} \quad (s \in \mathbb{R}).$$

Thus $\phi : (1,\infty) \to \mathbb{R}$ is an increasing smooth bijection. Fix a nonnegative $C_c^2((1,2))$ probability density ψ_0. Define

$$\rho(t) = h(\log r(t))\frac{t^2+1}{t(t^2-1)} \quad (t>1), \qquad w(u,v) = \frac{\psi_0(v)}{v}\rho(u/v) \quad (u>v>0), \tag{4.39}$$

and set $\rho(t) = 0$ for $t \le 1$ and $w = 0$ outside $u > v > 0$. Since $\phi' > 0$ and $\operatorname{supp} h = [-2,2]$, the support of ρ is exactly

$$\operatorname{supp}\rho = [t_-, t_+], \qquad t_- = e^{-2} + \sqrt{1+e^{-4}}, \qquad t_+ = e^2 + \sqrt{1+e^4}.$$

In particular, $1 < t_- < t_+ < \infty$, and $\rho \in C_c^2(\mathbb{R})$. The support of w is contained in

$$\{(u,v) : v \in \operatorname{supp}\psi_0,\ t_- v \le u \le t_+ v\},$$

a compact subset of $u > v > 0$. The formula in (4.39) is C^2 on this open set, and it vanishes near its boundary. Thus its zero extension satisfies $w \in C_c^2(\mathbb{R}^2)$.

The factor multiplying $h(\log r(t))$ in $\rho(t)$ is ds/dt for $s = \log r(t)$. Consequently this substitution gives

$$\rho(t)\,dt = h(s)\,ds, \qquad \int_1^\infty \rho(t)\,dt = \int_{\mathbb{R}} h(s)\,ds = 1.$$

For fixed $v > 0$, substitute $u = vt$, so that $du = v\,dt$. It follows that

$$\int_{\mathbb{R}^2} w(u,v)\,du\,dv = \int_0^\infty \psi_0(v)\left(\int_1^\infty \rho(t)\,dt\right)dv = 1.$$

For $\tau \in \mathbb{R}$, use $R(vt, v) = r(t)$ and the same two substitutions to obtain

$$\int_{u>v>0} w(u,v)R(u,v)^{i\tau}\,du\,dv = \int_0^\infty \psi_0(v)\,dv \int_1^\infty \rho(t)r(t)^{i\tau}\,dt = \int_{\mathbb{R}} h(s)e^{i\tau s}\,ds. \tag{4.40}$$

These integrals are absolutely convergent, since the complex factors have modulus one and both densities have integral one. Expanding the fourfold convolution defining h gives

$$\int_{\mathbb{R}} h(s)e^{i\tau s}\,ds = \int_{[-1/2,1/2]^4} e^{i\tau(a_1+a_2+a_3+a_4)}\,da_1\,da_2\,da_3\,da_4 = \left(\int_{-1/2}^{1/2} e^{i\tau a}\,da\right)^4.$$

Here Fubini's theorem applies to the real and imaginary parts; see [1, Theorem 5.32, p. 132]. The absolute value of the four-variable integrand has integral one. Direct integration gives

$$\int_{-1/2}^{1/2} e^{i\tau a}\,da = \begin{cases} \dfrac{e^{i\tau/2} - e^{-i\tau/2}}{i\tau} = \dfrac{\sin(\tau/2)}{\tau/2}, & \tau \neq 0, \\ 1, & \tau = 0. \end{cases} \tag{4.41}$$

Combining (4.40)–(4.41), we obtain

$$\int_{u>v>0} w(u,v)R(u,v)^{i\tau}\,du\,dv = \left(\frac{\sin(\tau/2)}{\tau/2}\right)^4 \in [0,1], \tag{4.42}$$

where the value at $\tau = 0$ is one. The interval bound follows from $|\sin x| \le |x|$.

Choose an integer $C_0 \ge 2$ large enough that

$$u + v < C_0 - 1 \quad \text{on } \operatorname{supp} w. \tag{4.43}$$

Every axis-parallel rectangle with both side lengths at most $1/2$ that meets $\operatorname{supp} w$ is contained in $\{u + v < C_0\}$. Indeed, if (u_0, v_0) is a point of that intersection, then every (u, v) in the rectangle satisfies

$$u + v \le u_0 + v_0 + |u - u_0| + |v - v_0| \le u_0 + v_0 + 1 < C_0.$$

This is the purpose of the margin in (4.43).

Let $g : \mathbb{Q}_{>0} \to \mathbb{S}^1$ be completely multiplicative: $g(q_1q_2) = g(q_1)g(q_2)$ for all $q_1, q_2 \in \mathbb{Q}_{>0}$. To form the finite kernel average, fix an integer $X \ge 1$ and sum over the integer pairs (m, n) with $m > n > 0$. The continuous weight is sampled at $(m/X, n/X)$, while $W_0(m, n)$ imposes the prescribed congruence weight. For this reason we first define the unnormalized weighted sum $I_X(g)$, and then normalize it by its value at the constant function 1:

$$\begin{aligned} I_X(g) &= X^{-2} \sum_{\substack{m,n\in\mathbb{Z}\\ m>n>0}} w(m/X, n/X)W_0(m,n)g(R(m,n)), \\ Z_X &= I_X(1), \qquad K_X(g) = I_X(g)/Z_X \quad \text{when } Z_X > 0. \end{aligned} \tag{4.44}$$

The sum is finite because w has compact support. Every contributing pair satisfies $m > n > 0$, so $R(m, n) = (m^2 - n^2)/(2mn) \in \mathbb{Q}_{>0}$ and $g(R(m, n))$ is defined.

The positivity of Z_X follows from the quadrature estimate in Theorem 4.5 below. By Theorem 4.4, W_0 is nonnegative, periodic modulo D_0, and has average one on residue pairs. Applying (4.46) with $D = D_0$ and $V = W_0$ therefore gives, for a constant $A_w > 0$ depending only on the fixed density w,

$$|Z_X - 1| \le A_w\left(\frac{D_0}{X} + \frac{D_0^2}{X^2}\right).$$

Put $C_w = \max\{1, 4A_w\}$. If $X \ge C_w D_0$, then $D_0/X \le 1/C_w \le 1$, and hence

$$|Z_X - 1| \le \frac{2A_w}{C_w} \le \frac{1}{2}, \qquad \frac{1}{2} \le Z_X \le \frac{3}{2}.$$

Thus $K_X(g)$ is defined in this range. Its coefficients are nonnegative and sum to one, so it is an average of the unit-modulus values $g(R(m,n))$; in particular, $K_X(1) = 1$ and $|K_X(g)| \le 1$.

4.3. **Quadrature.** In the quadrature estimates, $\|\cdot\|_1$ denotes the Lebesgue integral of the absolute value on $\mathbb{R}^2$.

Lemma 4.5. *Let $F \in C_c^2(\mathbb{R}^2)$. For every integer $d \ge 1$ and every coset $\mathbf{a} + d\mathbb{Z}^2$,*

$$\left| \sum_{n \in \mathbf{a} + d\mathbb{Z}^2} F(n) - d^{-2} \int_{\mathbb{R}^2} F \right| \le d^{-1}\big(\|\partial_1 F\|_1 + \|\partial_2 F\|_1\big) + \|\partial_1 \partial_2 F\|_1. \tag{4.45}$$

In particular, let $D \ge 1$ be an integer, $X > 0$, and $V : \mathbb{Z}^2 \to [0, \infty)$ be periodic modulo D in each variable, with $D^{-2} \sum_{a,b \bmod D} V(a,b) = 1$. Then

$$\left| X^{-2} \sum_{m,n \in \mathbb{Z}} w(m/X, n/X) V(m,n) - 1 \right| \le C_w(D/X + D^2/X^2), \tag{4.46}$$

where w is the fixed density in (4.39) and C_w depends only on w.

Proof. We divide the proof into two steps.

Step 1. Prove the residue-class quadrature estimate.

For a compactly supported absolutely continuous function $f : \mathbb{R} \to \mathbb{C}$, we first prove

$$\left| \sum_{n \in a + d\mathbb{Z}} f(n) - d^{-1} \int_{\mathbb{R}} f \right| \le \int_{\mathbb{R}} |f'|. \tag{4.47}$$

Put $t_i = a + di$ and $J_i = [t_i, t_i + d)$. The fundamental theorem of calculus gives

$$f(t) - f(t_i) = \int_{t_i}^{t} f'(s)\, ds$$

for $t \in J_i$. Hence

$$\left| f(t_i) - d^{-1} \int_{J_i} f(t)\, dt \right| \le d^{-1} \int_{J_i} \int_{t_i}^{t} |f'(s)|\, ds\, dt \le \int_{J_i} |f'(s)|\, ds. \tag{4.48}$$

Summing over the intervals, which partition $\mathbb{R}$, proves (4.47). Write $x_i = a_1 + di$, $y_j = a_2 + dj$, and define

$$G(x) = \int_{\mathbb{R}} F(x,y)\, dy, \qquad A(x) = \int_{\mathbb{R}} |\partial_2 F(x,y)|\, dy. \tag{4.49}$$

The function F and its derivatives vanish outside a common compact rectangle. Differentiation under the integral gives

$$G'(x) = \int_{\mathbb{R}} \partial_1 F(x,y)\, dy, \qquad \int_{\mathbb{R}} |G'| \le \|\partial_1 F\|_1. \tag{4.50}$$

To control A, put $B(s) = \int_{\mathbb{R}} |\partial_1 \partial_2 F(s,y)|\, dy$. For $x < x'$, the reverse triangle inequality and the fundamental theorem of calculus give

$$\begin{aligned} |A(x') - A(x)| &\le \int_{\mathbb{R}} |\partial_2 F(x',y) - \partial_2 F(x,y)|\, dy \\ &\le \int_{\mathbb{R}} \int_{x}^{x'} |\partial_1 \partial_2 F(s,y)|\, ds\, dy = \int_{x}^{x'} B(s)\, ds. \end{aligned} \tag{4.51}$$

Since B is continuous, bounded, and compactly supported, this makes A Lipschitz, hence absolutely continuous. Dividing by $|x'-x|$ and taking the limit at points of differentiability yields $|A'(x)| \le B(x)$ almost everywhere. Consequently,

$$\int_{\mathbb{R}} A = \|\partial_2 F\|_1, \qquad \int_{\mathbb{R}} |A'| \le \int_{\mathbb{R}} B = \|\partial_1\partial_2 F\|_1. \tag{4.52}$$

Both G and A are compactly supported. Applying (4.47) to $y \mapsto F(x_i, y)$, to G, and to A, respectively, gives

$$\begin{aligned} \left|\sum_j F(x_i, y_j) - d^{-1}G(x_i)\right| &\le A(x_i), \\ \left|\sum_i G(x_i) - d^{-1}\int_{\mathbb{R}} G\right| &\le \|\partial_1 F\|_1, \\ \sum_i A(x_i) &\le d^{-1}\|\partial_2 F\|_1 + \|\partial_1\partial_2 F\|_1. \end{aligned} \tag{4.53}$$

Since $\int_{\mathbb{R}} G = \int_{\mathbb{R}^2} F$, the triangle inequality now yields

$$\begin{aligned} \left|\sum_{i,j} F(x_i, y_j) - d^{-2}\int_{\mathbb{R}^2} F\right| &\le \sum_i A(x_i) + d^{-1}\left|\sum_i G(x_i) - d^{-1}\int_{\mathbb{R}} G\right| \\ &\le d^{-1}\big(\|\partial_1 F\|_1 + \|\partial_2 F\|_1\big) + \|\partial_1\partial_2 F\|_1. \end{aligned} \tag{4.54}$$

All sums are finite, and all integrals are absolutely convergent, by compact support. This proves (4.45).

Step 2. Apply the estimate to the periodic normalization. For the second assertion, recall that $w \in C_c^2(\mathbb{R}^2)$ and $\int_{\mathbb{R}^2} w = 1$. Set $F_X(x,y) = w(x/X, y/X)$. The substitution $(x,y) = (Xu, Xv)$ gives

$$\begin{aligned} \int_{\mathbb{R}^2} F_X &= X^2, \\ \|\partial_i F_X\|_1 &= X\|\partial_i w\|_1 \quad (i = 1, 2), \\ \|\partial_1\partial_2 F_X\|_1 &= \|\partial_1\partial_2 w\|_1. \end{aligned} \tag{4.55}$$

Put $L_w = \|\partial_1 w\|_1 + \|\partial_2 w\|_1$ and $M_w = \|\partial_1\partial_2 w\|_1$. For integers $0 \le a, b < D$, let $S_{a,b}$ be the sum of F_X over $(a,b) + D\mathbb{Z}^2$. Applying (4.45) to this coset gives

$$\left|S_{a,b} - \frac{X^2}{D^2}\right| \le L_w\frac{X}{D} + M_w. \tag{4.56}$$

Periodicity and $\sum_{a,b \bmod D} V(a,b) = D^2$ give

$$\sum_{m,n\in\mathbb{Z}} F_X(m,n)V(m,n) - X^2 = \sum_{a,b \bmod D} V(a,b)\left(S_{a,b} - \frac{X^2}{D^2}\right). \tag{4.57}$$

Since $V \ge 0$, it follows that

$$\begin{aligned} \left|X^{-2}\sum_{m,n\in\mathbb{Z}} F_X(m,n)V(m,n) - 1\right| &\le X^{-2}\sum_{a,b \bmod D} V(a,b)\left|S_{a,b} - \frac{X^2}{D^2}\right| \\ &\le L_w\frac{D}{X} + M_w\frac{D^2}{X^2}. \end{aligned} \tag{4.58}$$

This proves (4.46) with a constant depending only on w. The bound uses the total residue mass D^2 of V, so it has no factor $\|V\|_\infty$. □

For the next lemma, fix $X > 0$ and an integer $K \geq 1$. Partition $\mathbb{R}^2$ into a grid of axis-parallel half-open squares of side $1/K$. Let Q_γ be the finitely many squares that meet $\operatorname{supp} w$. For each γ, define

$$C_\gamma = \{(m,n) \in \mathbb{R}^2 : (m/X, n/X) \in Q_\gamma\}.$$

Thus C_γ is the dilation of Q_γ by the factor X. Define $F : \mathbb{R}^2 \to [0,\infty)$ by $F(m,n) = w(m/X, n/X)$, and set

$$I = \int_{\mathbb{R}^2} F = X^2, \qquad I_\gamma = \int_{C_\gamma} F, \qquad \pi_\gamma = I_\gamma / I.$$

For each integer $d \geq 1$ and each $\mathbf{a} \in \mathbb{Z}^2$, let $N_{\mathbf{a}}$ be the sum of F over $\mathbf{a} + d\mathbb{Z}^2$. When $N_{\mathbf{a}} > 0$, define a random vector $\mathbf{Z}$ taking values in this coset by

$$\mathbb{P}(\mathbf{Z} = \mathbf{z}) = \frac{F(\mathbf{z})}{N_{\mathbf{a}}} \qquad (\mathbf{z} \in \mathbf{a} + d\mathbb{Z}^2).$$

Almost surely, $\mathbf{Z}$ lies in a unique cell C_γ. Let Γ be the index of this cell.

Lemma 4.6. *For every integer $d \geq 1$ and every $\mathbf{a} \in \mathbb{Z}^2$,*

$$\sum_\gamma \left| \sum_{\mathbf{n} \in (\mathbf{a} + d\mathbb{Z}^2) \cap C_\gamma} F(\mathbf{n}) - \frac{I_\gamma}{d^2} \right| \leq C_w I \left(\frac{K}{Xd} + \frac{K^2}{X^2} \right). \tag{4.59}$$

If $d \leq c_w X$, then $N_{\mathbf{a}} \geq I/(2d^2) > 0$. Thus $\mathbf{Z}$ and Γ are defined. In this case, for every integer $Q \geq 1$ coprime to d and every $\mathbf{b} \in (\mathbb{Z}/Q\mathbb{Z})^2$,

$$\sum_\gamma \left| \mathbb{P}(\Gamma = \gamma, \mathbf{Z} \equiv \mathbf{b} \pmod{Q}) - \frac{\pi_\gamma}{Q^2} \right| \leq C_w \left(\frac{dK}{XQ} + \frac{d^2K^2}{X^2} \right). \tag{4.60}$$

Here $C_w > 0$ and the sufficiently small constant $c_w > 0$ depend only on w.

Proof. We divide the proof into four steps.

Step 1. Establish bounded-variation quadrature on one rectangle. We first extend (4.47) to compactly supported functions $h : \mathbb{R} \to \mathbb{R}$ of bounded pointwise total variation, denoted by $\operatorname{Var}(h)$. The values at jump points are included in this variation. For $t_j = u + jd$, comparison on $[t_j, t_j + d]$ gives

$$\begin{aligned} \left| h(t_j) - \frac{1}{d} \int_{t_j}^{t_j+d} h(t)\, dt \right| &\leq \operatorname{Var}_{[t_j, t_j+d]}(h), \\ \left| \sum_{j \in \mathbb{Z}} h(u + jd) - \frac{1}{d} \int_{\mathbb{R}} h \right| &\leq \operatorname{Var}(h). \end{aligned} \tag{4.61}$$

The second inequality follows by summing the first; variation is additive on adjacent intervals. If h is absolutely continuous on $[\alpha, \beta]$, its half-open restriction satisfies

$$\operatorname{Var}(\mathbf{1}_{[\alpha,\beta)} h) \leq |h(\alpha)| + \int_\alpha^\beta |h'(x)|\, dx + |h(\beta)|. \tag{4.62}$$

The jumps at α and β have absolute values $|h(\alpha)|$ and $|h(\beta)|$, respectively. For a rectangle $C = [\alpha, \beta) \times [\eta, \zeta)$, define

$$\begin{aligned} V_1(C) &= \int_C |\partial_1 F| + \int_\eta^\zeta \big(|F(\alpha, y)| + |F(\beta, y)|\big)\, dy, \\ V_2(C) &= \int_C |\partial_2 F| + \int_\alpha^\beta \big(|F(x, \eta)| + |F(x, \zeta)|\big)\, dx, \\ V_{12}(C) &= \int_C |\partial_1 \partial_2 F| + \int_\eta^\zeta \big(|\partial_2 F(\alpha, y)| + |\partial_2 F(\beta, y)|\big)\, dy \\ &\quad + \int_\alpha^\beta \big(|\partial_1 F(x, \eta)| + |\partial_1 F(x, \zeta)|\big)\, dx + \sum_{\substack{\xi \in \{\alpha, \beta\} \\ \nu \in \{\eta, \zeta\}}} |F(\xi, \nu)|. \end{aligned} \tag{4.63}$$

To apply (4.61) in both coordinates, set

$$\begin{aligned} g(x) &= \int_\eta^\zeta F(x, y)\, dy, \qquad G_C(x) = \mathbf{1}_{[\alpha,\beta)}(x) g(x), \\ k(x) &= \int_\eta^\zeta |\partial_2 F(x, y)|\, dy + |F(x, \eta)| + |F(x, \zeta)|, \\ A_C(x) &= \mathbf{1}_{[\alpha,\beta)}(x) k(x) \qquad (x \in \mathbb{R}). \end{aligned} \tag{4.64}$$

The function g is continuously differentiable. For $\alpha \le x < x' \le \beta$, the fundamental theorem of calculus and $\big||r| - |s|\big| \le |r - s|$ give

$$|k(x') - k(x)| \le \int_x^{x'} \left\{ \int_\eta^\zeta |\partial_1 \partial_2 F(t, y)|\, dy + |\partial_1 F(t, \eta)| + |\partial_1 F(t, \zeta)| \right\} dt. \tag{4.65}$$

Thus k is absolutely continuous, and $|k'|$ is bounded almost everywhere by the integrand in braces. Applying (4.62) to g and k therefore yields

$$\mathrm{Var}(G_C) \le V_1(C), \qquad \int_{\mathbb{R}} A_C = V_2(C), \qquad \mathrm{Var}(A_C) \le V_{12}(C). \tag{4.66}$$

The jumps of A_C at α and β have absolute values $k(\alpha)$ and $k(\beta)$, respectively. Their sum is included in $V_{12}(C)$, by (4.63) and (4.64). Write $x_i = a_1 + id$ and $y_j = a_2 + jd$. By (4.61) and (4.62),

$$\left| \sum_{j \in \mathbb{Z}} \mathbf{1}_C(x_i, y_j) F(x_i, y_j) - \frac{G_C(x_i)}{d} \right| \le A_C(x_i) \qquad (i \in \mathbb{Z}). \tag{4.67}$$

Since $\int_{\mathbb{R}} G_C = \int_C F$, a second application in x gives

$$\begin{aligned} \left| \sum_{\mathbf{n} \in (\mathbf{a} + d\mathbb{Z}^2) \cap C} F(\mathbf{n}) - \frac{1}{d^2} \int_C F \right| &\le \sum_{i \in \mathbb{Z}} A_C(x_i) + \frac{1}{d} \left| \sum_{i \in \mathbb{Z}} G_C(x_i) - \frac{1}{d} \int_{\mathbb{R}} G_C \right| \\ &\le \frac{V_1(C) + V_2(C)}{d} + V_{12}(C). \end{aligned} \tag{4.68}$$

Each point of $\operatorname{supp} F$ belongs to exactly one C_γ. The variation bounds use all four edges of C.

Step 2. Sum the cell errors. Choose $R_0 \ge 1$ such that $\operatorname{supp} w \subset [-R_0, R_0]^2$. Let M_w bound the supremum norms of w, its first derivatives, and its mixed second derivative. The scaling $F(x, y) = w(x/X, y/X)$ gives, on all of $\mathbb{R}^2$,

$$|F| \le M_w, \qquad |\partial_i F| \le M_w / X \quad (i = 1, 2), \qquad |\partial_1 \partial_2 F| \le M_w / X^2. \tag{4.69}$$

Each cell has side length $h = X/K$, and at most $(2R_0K+3)^2 \le (2R_0+3)^2K^2$ cells meet the support. Each cell has area h^2 and perimeter $4h$. At each of the four vertices, $|F| \le M_w$. Substituting (4.69) into (4.63) gives

$$V_1(C_\gamma) + V_2(C_\gamma) \le M_w(2h^2/X + 4h), \qquad V_{12}(C_\gamma) \le M_w(h^2/X^2 + 4h/X + 4). \tag{4.70}$$

Summing these inequalities and using $K \ge 1$ and $I = X^2$, we obtain

$$\begin{aligned} \sum_\gamma \big(V_1(C_\gamma) + V_2(C_\gamma)\big) &\le C_w IK/X, \\ \sum_\gamma V_{12}(C_\gamma) &\le C_w IK^2/X^2. \end{aligned} \tag{4.71}$$

Together with (4.68), this proves (4.59) for every $d \ge 1$.

Step 3. Normalize the original coset. To prove the second assertion, apply (4.45) to F on the original coset. The derivative integrals are $O_w(X)$ in first order and $O_w(1)$ in mixed second order. Hence

$$N_{\mathbf{a}} = \frac{I}{d^2}(1+\varepsilon_{\mathbf{a}}), \qquad |\varepsilon_{\mathbf{a}}| \le C_w\left(\frac{d}{X} + \frac{d^2}{X^2}\right). \tag{4.72}$$

Choose $c_w > 0$ so small that the last bound is at most $1/2$ whenever $d \le c_wX$. Assume $d \le c_wX$ from now on. Then $|\varepsilon_{\mathbf{a}}| \le 1/2$ and $N_{\mathbf{a}} \ge I/(2d^2) > 0$, as asserted.

Step 4. Impose the extra congruence. Let $Q \ge 1$ be coprime to d, and let $\mathbf{b} \in (\mathbb{Z}/Q\mathbb{Z})^2$. The Chinese remainder theorem, applied in each coordinate, gives $\mathbf{c} \in \mathbb{Z}^2$ such that

$$\{\mathbf{n} \in \mathbf{a} + d\mathbb{Z}^2 : \mathbf{n} \equiv \mathbf{b} \pmod{Q}\} = \mathbf{c} + dQ\mathbb{Z}^2. \tag{4.73}$$

Define $N_{\gamma,\mathbf{b}} = \sum_{\mathbf{n} \in (\mathbf{c}+dQ\mathbb{Z}^2)\cap C_\gamma} F(\mathbf{n})$. Applying (4.59) with modulus dQ yields

$$\sum_\gamma \left| N_{\gamma,\mathbf{b}} - \frac{I_\gamma}{d^2Q^2} \right| \le C_w I\left(\frac{K}{XdQ} + \frac{K^2}{X^2}\right). \tag{4.74}$$

This step places no size restriction on dQ. By the definition of $\mathbf{Z}$, the joint probability in (4.60) is $N_{\gamma,\mathbf{b}}/N_{\mathbf{a}}$. Since $I_\gamma \ge 0$ and $\sum_\gamma I_\gamma = I$, the triangle inequality gives

$$\begin{aligned} \sum_\gamma \left| \frac{N_{\gamma,\mathbf{b}}}{N_{\mathbf{a}}} - \frac{\pi_\gamma}{Q^2} \right| &\le \frac{1}{N_{\mathbf{a}}} \sum_\gamma \left| N_{\gamma,\mathbf{b}} - \frac{I_\gamma}{d^2Q^2} \right| + \frac{1}{Q^2}\left| \frac{I}{d^2N_{\mathbf{a}}} - 1 \right| \\ &\le C_w\left\{ \frac{dK}{XQ} + \frac{d^2K^2}{X^2} + \frac{d/X + d^2/X^2}{Q^2} \right\}. \end{aligned} \tag{4.75}$$

Here we used $N_{\mathbf{a}}^{-1} \le 2d^2/I$ and $|I/(d^2N_{\mathbf{a}}) - 1| = |\varepsilon_{\mathbf{a}}|/|1+\varepsilon_{\mathbf{a}}| \le 2|\varepsilon_{\mathbf{a}}|$. Finally, $K, Q \ge 1$ imply

$$\frac{d}{XQ^2} \le \frac{dK}{XQ}, \qquad \frac{d^2}{X^2Q^2} \le \frac{d^2K^2}{X^2}. \tag{4.76}$$

Absorbing these two terms proves (4.60).

□

4.4. **Joint valuation laws.** Write $\mathbf{Z} = (M, N)$ for the random lattice point defined before Theorem 4.6. Recall that $\Gamma = \gamma$ when $\mathbf{Z} \in C_\gamma$. For a finite set $\mathcal{P}$ of primes, our application concerns the joint distribution of

$$\Big(\Gamma, \big(v_p(M-N), v_p(M+N), v_p(M), v_p(N)\big)_{p\in\mathcal{P}}\Big).$$

Let μ, ν be probability measures on a countable set S. Define

$$d_{\mathrm{TV}}(\mu,\nu) = \frac{1}{2}\sum_{x\in S} |\mu(x) - \nu(x)|. \tag{4.77}$$

For $f : S \to \mathbb{C}$ with $|f(x)| \le 1$ for every $x \in S$, the triangle inequality gives

$$|\mathbb{E}_\mu f - \mathbb{E}_\nu f| = \left|\sum_{x\in S} f(x)\big(\mu(x)-\nu(x)\big)\right| \le \sum_{x\in S}|\mu(x)-\nu(x)| = 2d_{\mathrm{TV}}(\mu,\nu). \tag{4.78}$$

Fix $\kappa, B_1 > 0$. Let z, D be integers with $2 \le z \le D^{1/2}$, and put $s = \log D/\log z$. Assume $s \ge 100$. We use divisor sums to bound from below the indicator that an integer has no prime divisor at most z. The fundamental lemma of sieve theory, in its lower-bound form [7, Theorem 3.6(a)], provides a function $\lambda^- : \mathbb{N} \to [-1, 1]$ depending only on z, D, κ, B_1. Its values $\lambda^-(e)$ are the sieve coefficients. They satisfy $\lambda^-(1) = 1$ and vanish unless $e \le D$ is squarefree and every prime divisor of e is at most z. Their divisor sums satisfy

$$\sum_{e\mid n}\lambda^-(e) \le \mathbf{1}_{n=1} \quad \text{for every } n \ge 1 \text{ with prime divisors at most } z. \tag{4.79}$$

For $n = 1$, the sum equals $\lambda^-(1) = 1$. For $n > 1$ whose prime divisors are all at most z, [7, (3.1)] gives $\sum_{e\mid n}\lambda^-(e) \le 0$. More generally, put $P_z = \prod_{\substack{p\le z\\ p \text{ prime}}} p$. Since a nonzero coefficient requires e to be a squarefree divisor of P_z, for every $n \ge 1$ we have

$$\sum_{e\mid n}\lambda^-(e) = \sum_{e\mid \gcd(n,P_z)}\lambda^-(e) \le \mathbf{1}_{\gcd(n,P_z)=1}. \tag{4.80}$$

The inequality follows by applying (4.79) to $\gcd(n, P_z)$. Let $\mathcal{S}_{\mathrm{sf}}$ be the set of squarefree positive integers. Choose a number $h(p) \in [0, 1)$ for each prime p. Define $h : \mathcal{S}_{\mathrm{sf}} \to [0, 1]$ by

$$h(1) = 1, \qquad h(e) = \prod_{\substack{p\mid e\\ p \text{ prime}}} h(p) \quad (e \in \mathcal{S}_{\mathrm{sf}},\ e > 1). \tag{4.81}$$

Thus $h(ab) = h(a)h(b)$ whenever $a, b \in \mathcal{S}_{\mathrm{sf}}$ are coprime.

Assume

$$\prod_{\substack{a\le p\le b\\ p \text{ prime}}}(1-h(p))^{-1} \le \left(\frac{\log b}{\log a}\right)^{\kappa}\exp(B_1/\log a) \quad (2 \le a \le b \le z). \tag{4.82}$$

Then

$$\sum_{e}\lambda^-(e)h(e) = \prod_{\substack{p\le z\\ p \text{ prime}}}(1-h(p))\big\{1 + O_{\kappa,B_1}\big(e^{-s\log s + s\log\log\log s + C_{\kappa,B_1}s}\big)\big\}. \tag{4.83}$$

Here the sum is over the support of λ^-, so only squarefree values of h occur. The O-term is a relative error; its uniform bound is given below. To verify (4.83), extend h to a multiplicative function $\widetilde{h} : \mathbb{N} \to [0, 1]$ by setting

$$\widetilde{h}(1) = 1, \qquad \widetilde{h}(p) = h(p), \qquad \widetilde{h}(p^j) = 0 \quad (p \text{ prime},\ j \ge 2). \tag{4.84}$$

For squarefree e, multiplicativity gives $\widetilde{h}(e) = \prod_{\substack{p\mid e\\ p \text{ prime}}} h(p) = h(e)$. Consequently,

$$\sum_{e\ge 1}\lambda^-(e)\widetilde{h}(e) = \sum_{e}\lambda^-(e)h(e), \qquad \prod_{\substack{p\le z\\ p \text{ prime}}}(1-\widetilde{h}(p)) = \prod_{\substack{p\le z\\ p \text{ prime}}}(1-h(p)). \tag{4.85}$$

We check the remaining hypotheses of [7, Theorem 3.6(a)]. The function $\widetilde{h}$ satisfies $0 \le \widetilde{h}(p) < 1$. Condition (4.82) is precisely the condition (Ω) in that theorem, with parameters κ, B_1. The

integers z, D satisfy $2 \le z \le D^{1/2}$. The theorem uses $\max\{100, \log D/\log z\}$, which equals s under our assumption $s \ge 100$. Hence [7, Theorem 3.6(a), p. 38] gives

(4.86)
$$\sum_{e\ge 1} \lambda^-(e)\widetilde{h}(e) = \prod_{\substack{p\le z\\ p \text{ prime}}} (1-\widetilde{h}(p))\left\{1 + O_{\kappa,B_1}(\exp[-s\log s + s\log\log\log s + O_{\kappa,B_1}(s)])\right\}.$$

Choose $A_{\kappa,B_1} > 0$ to bound the implied constant in the outer O-term of (4.86). Choose $C_{\kappa,B_1} > 0$ so that the inner $O_{\kappa,B_1}(s)$-term in (4.86) has absolute value at most $C_{\kappa,B_1}s$. Both constants depend only on κ, B_1. The identities (4.85) then give

$$\left|\frac{\sum_e \lambda^-(e)h(e)}{\prod_{\substack{p\le z\\ p \text{ prime}}}(1-h(p))} - 1\right| \le A_{\kappa,B_1}\exp(-s\log s + s\log\log\log s + C_{\kappa,B_1}s). \tag{4.87}$$

The denominator is positive because $h(p) < 1$ for every prime p. This bound holds uniformly for all z, D, h satisfying the stated assumptions. It gives the precise meaning of the O-term in (4.83) and proves that estimate. See also [14, Proposition 3.1]. We record the weighted lower bound in the construction of [7, Theorem 3.6(a), pp. 38–39]. Let $\mathcal{P}_1, \dots, \mathcal{P}_t$ be its partition of all primes $p \le z$, with products $P_j = \prod_{p\in\mathcal{P}_j} p$ and even cutoffs $k_j \ge 0$. For $1 \le j \le t$, put

$$\begin{aligned} U_j &= \sum_{\substack{e\mid P_j\\ \omega(e)\le k_j}} \mu(e)h(e), \quad V_j = \prod_{\substack{p\mid P_j\\ p \text{ prime}}} (1-h(p)),\\ W_j &= \sum_{\substack{e\mid P_j\\ \omega(e)=k_j+1}} h(e), \qquad L_j = \log(V_j^{-1}). \end{aligned} \tag{4.88}$$

Here μ is the Möbius function and $\omega(e)$ counts distinct prime divisors. By [7, p. 17], $U_j \ge V_j > 0$. Define

$$\begin{aligned} U &= \prod_{j=1}^t U_j, \qquad V = \prod_{j=1}^t V_j, \qquad \eta = \sum_{j=1}^t \frac{W_j}{U_j},\\ \mathfrak{E} &= \sum_{j=1}^t \exp(L_j\mathbf{1}_{k_j>0})\frac{L_j^{k_j+1}}{(k_j+1)!}, \qquad \Sigma = \sum_e \lambda^-(e)h(e) = U(1-\eta). \end{aligned} \tag{4.89}$$

The identity for Σ follows from [7, (3.2), p. 30]. The bounds in [7, (2.5)–(2.6), p. 17, and p. 31] give $V \le U \le Ve^{\mathfrak{E}}$ and $0 \le \eta \le \mathfrak{E}$. To bound Σ from above, take independent Bernoulli variables B_p with $\mathbb{P}(B_p = 1) = h(p)$ for $p \le z$, and apply (4.79) to $N = \prod_{\substack{p\le z\\ p \text{ prime}}} p^{B_p}$. Taking expectations gives $\Sigma \le V$. Hence

$$\begin{aligned} 0 \le V - \Sigma = V - U + U\eta \le U\mathfrak{E} \le V\mathfrak{E}e^{\mathfrak{E}},\\ \Sigma \ge U(1-\mathfrak{E}) \ge V(1-\mathfrak{E}) \qquad (\mathfrak{E} \le 1). \end{aligned} \tag{4.90}$$

Thus the lower bound in [7, (3.4), p. 31] is used with $\mathfrak{E} \le 1$. The parameter choices in [7, (3.11), pp. 38–39] give

$$\mathfrak{E} \ll_{\kappa,B_1} \exp(-s\log s + s\log\log\log s + C_{\kappa,B_1}s) \qquad (s \ge s_0(\kappa,B_1)). \tag{4.91}$$

In particular, $\mathfrak{E} \le 1/2$ for sufficiently large s. For $100 \le s \le s_0(\kappa, B_1)$, those choices give $\mathfrak{E} = O_{\kappa,B_1}(1)$. The first bound in (4.90) then supplies this remaining range after enlarging the constants in (4.87).

Fix from now on the four primitive forms

$$L_1(m,n) = m-n, \quad L_2(m,n) = m+n, \quad L_3(m,n) = m, \quad L_4(m,n) = n. \tag{4.92}$$

For distinct $i, j \in \{1, 2, 3, 4\}$, the coefficient matrix of (L_i, L_j) has determinant ± 1 or ± 2. For $p \geq 5$, this determinant is a unit in $\mathbb{Z}_p$. Thus $(m, n) \mapsto (L_i(m, n), L_j(m, n))$ is a bijection of $\mathbb{Z}_p^2$.

Lemma 4.7. *Let $p \geq 5$ be prime, and let $\mathbf{Z}_p = (X, Y)$ be a Haar-distributed random vector on $\mathbb{Z}_p^2$. Put $\nu_r = v_p(L_r(\mathbf{Z}_p))$ for $1 \leq r \leq 4$, and write $\boldsymbol{\nu} = (\nu_1, \ldots, \nu_4)$. Almost surely, $\boldsymbol{\nu}$ has one of the forms*

$$(j, j, j, j), \qquad (j, \ldots, j, \underset{i}{j + k}, j, \ldots, j), \quad j \geq 0,\ k \geq 1,\ 1 \leq i \leq 4, \tag{4.93}$$

where j, k, i are integers. For each vector $\mathbf{a} = (a_1, \ldots, a_4)$ in (4.93), define

$$\begin{aligned} E_p(\mathbf{a}) &= \{\nu_r \geq a_r \text{ for every } 1 \leq r \leq 4\}, \\ g_p(\mathbf{a}) &= \mathbb{P}(E_p(\mathbf{a})), \\ h_{\mathbf{a}}(p) &= \mathbb{P}(\boldsymbol{\nu} \neq \mathbf{a} \mid E_p(\mathbf{a})). \end{aligned}$$

Then

$\mathbf{a}$	$g_p(\mathbf{a})$	$h_{\mathbf{a}}(p)$
(j, j, j, j)	p^{-2j}	$4/p - 3/p^2$
$a_i = j + k, \quad a_r = j\ (r \neq i)$	p^{-2j-k}	$2/p - 1/p^2$.

(4.94)

Proof. We divide the proof into four steps.

Step 1. Establish the independent Haar coordinates. For $r \neq s$, the coefficient matrix of (L_r, L_s) belongs to $\mathrm{GL}_2(\mathbb{Z}_p)$ by (4.92). It permutes the residue pairs modulo every power of p, so it preserves normalized Haar measure. Thus $L_r(\mathbf{Z}_p)$ and $L_s(\mathbf{Z}_p)$ are independent Haar-distributed random variables on $\mathbb{Z}_p$. In particular,

$$\mathbb{P}(L_r(\mathbf{Z}_p) = 0) \leq \mathbb{P}(p^t \mid L_r(\mathbf{Z}_p)) = p^{-t} \quad (t \geq 1).$$

Thus $\mathbb{P}(L_r(\mathbf{Z}_p) = 0) = 0$ for $1 \leq r \leq 4$. All four valuations are therefore finite with probability one.

Step 2. Classify the possible valuation vectors. At a point where all four valuations are finite, set $j = \min\{v_p(X), v_p(Y)\}$. The inequality $v_p(X \pm Y) \geq j$ gives $\min_r \nu_r = j$. Writing $\mathbf{Z}_p = p^j \mathbf{W}$, we obtain

$$\mathbf{W} \in \mathbb{Z}_p^2 \setminus p\mathbb{Z}_p^2, \qquad \nu_r = j + v_p(L_r(\mathbf{W})) \quad (1 \leq r \leq 4).$$

If two distinct forms vanished modulo p at $\mathbf{W}$, their invertible coefficient matrix would imply $\mathbf{W} \in p\mathbb{Z}_p^2$. Hence at most one of the four values $L_r(\mathbf{W})$ is divisible by p. If none is divisible by p, then $\nu_r = j$ for every r. Otherwise, let i be the unique index for which $p \mid L_i(\mathbf{W})$, and put $k = v_p(L_i(\mathbf{W})) \geq 1$. Then $\nu_i = j + k$ and $\nu_r = j$ for $r \neq i$. These are the two possibilities in (4.93).

Step 3. Compute the equal-valuation row. For the first row of (4.94), fix $j \geq 0$ and let $\mathbf{a} = (j, j, j, j)$. The conditions on L_3, L_4 require both X, Y to be divisible by p^j, and these conditions also suffice for L_1, L_2. Therefore

$$E_p(\mathbf{a}) = \{X, Y \in p^j\mathbb{Z}_p\}, \qquad g_p(\mathbf{a}) = p^{-j}p^{-j} = p^{-2j}.$$

Conditional on this event, $\mathbf{W} = p^{-j}\mathbf{Z}_p$ has normalized Haar distribution on all of $\mathbb{Z}_p^2$. Consequently, each of the p^2 residue pairs modulo p has conditional probability p^{-2}. The event $\boldsymbol{\nu} \neq \mathbf{a}$ occurs exactly when at least one $L_r(\mathbf{W})$ vanishes modulo p. Writing $\mathbf{W} = (x, y)$, the four zero sets are the lines

$$x - y = 0, \qquad x + y = 0, \qquad x = 0, \qquad y = 0 \quad \text{in } \mathbb{F}_p^2.$$

Each line contains p points, and two distinct lines meet only at the origin. Their union has $1+4(p-1)$ points, so

$$h_{\mathbf{a}}(p)=\frac{1+4(p-1)}{p^2}=\frac{4}{p}-\frac{3}{p^2}.$$

Step 4. Compute the exceptional row and conclude. For the second row, let $a_i=j+k$ and $a_r=j$ for $r\neq i$, where $j\geq 0$ and $k\geq 1$. Choose $\ell\neq i$ and put $(S,T)=(L_i(\mathbf{Z}_p),L_\ell(\mathbf{Z}_p))$. These are independent Haar-distributed random variables. The inverse coordinate map shows that $S,T\in p^j\mathbb{Z}_p$ is equivalent to $\mathbf{Z}_p\in p^j\mathbb{Z}_p^2$. Consequently,

$$E_p(\mathbf{a})=\{S\in p^{j+k}\mathbb{Z}_p,\ T\in p^j\mathbb{Z}_p\},\qquad g_p(\mathbf{a})=p^{-(j+k)}p^{-j}=p^{-2j-k}.$$

The event restricts the independent coordinates S,T separately. Thus, conditional on $E_p(\mathbf{a})$, we can write

$$S=p^{j+k}U,\qquad T=p^jV, \tag{4.95}$$

where U,V are independent and have normalized Haar distribution on $\mathbb{Z}_p$.

Since L_i,L_ℓ form a coordinate system, for every $r\neq i$ we can write

$$L_r=\alpha_rL_i+\beta_rL_\ell,\qquad \alpha_r\in\mathbb{Z}_p,\quad \beta_r\in\mathbb{Z}_p^\times. \tag{4.96}$$

To verify that β_r is a unit, take determinants of coefficient matrices. The determinant for (L_i,L_r) is β_r times the determinant for (L_i,L_ℓ), and both determinants are units. The identities (4.95) and (4.96) give

$$L_r(\mathbf{Z}_p)=p^j(\alpha_rp^kU+\beta_rV),\qquad \alpha_rp^kU+\beta_rV\equiv\beta_rV\pmod p\quad(r\neq i).$$

where the congruence uses $k\geq 1$. Since β_r is a unit, under this conditional law

$$\nu_i>a_i\iff p\mid U,\qquad \nu_r>a_r\iff p\mid V\quad(r\neq i).$$

Thus $\boldsymbol{\nu}\neq\mathbf{a}$ exactly when $p\mid U$ or $p\mid V$. Independence and inclusion–exclusion yield

$$h_{\mathbf{a}}(p)=\frac{1}{p}+\frac{1}{p}-\frac{1}{p^2}=\frac{2}{p}-\frac{1}{p^2}.$$

This proves the second row of (4.94). In particular,

$$\begin{aligned}\mathbb{P}(\boldsymbol{\nu}=\mathbf{a})&=g_p(\mathbf{a})(1-h_{\mathbf{a}}(p))\\&=\begin{cases}p^{-2j}(p-1)(p-3)/p^2, & \mathbf{a}=(j,j,j,j),\\ p^{-2j-k}(1-1/p)^2, & a_i=j+k,\ a_r=j\ (r\neq i).\end{cases}\end{aligned}$$

Both expressions are positive for $p\geq 5$, so every vector in (4.93) occurs with positive probability. □

For the next theorem, let Y,y be real numbers with $Y\geq 3$ and $5\leq y\leq Y$, and let $\mathcal{P}$ be a set of primes in $[5,y]$. Let $\mathcal{C}$ be a nonempty finite set. Let $\mathbf{Z}$ be a $\mathbb{Z}^2$-valued random vector and C a $\mathcal{C}$-valued random variable, defined on the same probability space. Assume that $\mathbf{Z}$ has finite support and that $\mathbb{P}(L_i(\mathbf{Z})=0)=0$ for $1\leq i\leq 4$.

Let $\pi=(\pi_c)_{c\in\mathcal{C}}$ be a prescribed probability vector: $\pi_c\geq 0$ for every $c\in\mathcal{C}$ and $\sum_{c\in\mathcal{C}}\pi_c=1$. For a real constant $A_0\geq 0$, consider the joint residue-discrepancy condition[4]

$$\sum_{c\in\mathcal{C}}\left|\mathbb{P}(C=c,\mathbf{Z}\equiv\mathbf{b}\pmod Q)-\frac{\pi_c}{Q^2}\right|\leq A_0\left(\frac{1}{YQ}+\frac{1}{Y^2}\right), \tag{4.97}$$

for every positive integer Q supported on $\mathcal{P}$ and every $\mathbf{b}\in(\mathbb{Z}/Q\mathbb{Z})^2$.

[4]We use this phrase for (4.97). Its antecedents are estimates for divisibility probabilities in the Kubilius model and Elliott's generalization; see [14, Sections 1.1–1.2, pp. 1–2].

For each $p \in \mathcal{P}$, let $\mathbf{Z}_p$ be a Haar-distributed random vector on $\mathbb{Z}_p^2$. Assume that these vectors are independent. Let C^* be a $\mathcal{C}$-valued random variable, independent of $(\mathbf{Z}_p)_{p\in\mathcal{P}}$, with $\mathbb{P}(C^* = c) = \pi_c$ for every $c \in \mathcal{C}$. Write

$$\mathbf{V} = (v_p(L_i(\mathbf{Z})))_{p,i}, \qquad \mathbf{A} = (v_p(L_i(\mathbf{Z}_p)))_{p,i}, \qquad u = \frac{\log Y}{\log y}.$$

The indices in these vectors range over $p \in \mathcal{P}$ and $1 \le i \le 4$. Their common state space is the countable set $(\mathbb{N}_0^4)^{\mathcal{P}}$, where $\mathbb{N}_0 = \{0, 1, 2, \dots\}$. If $\mathcal{P}$ is empty, this space consists of one empty vector. All products over an empty set below are assigned the value one.

Theorem 4.8. *There exist absolute constants $K \ge 1$ and $0 < c \le 1$ such that, for every choice of the preceding data satisfying (4.97),*

$$d_{\mathrm{TV}}((C, \mathbf{V}), (C^*, \mathbf{A})) \le K\left(e^{-cu} + \frac{A_0(1 + \log Y)^{16}}{\sqrt{Y}}\right).$$

Proof. Throughout the proof, $B > 0$ denotes an absolute constant that may increase from one occurrence to the next. We divide the rest proof into six steps.

Step 1. Verify the sieve condition. By (3.21), $\vartheta(t) \le 4t$. Partial summation gives

$$\begin{aligned} \sum_{\substack{p \le t \\ p \text{ prime}}} \frac{\log p}{p} &\le B \log t && (t \ge 2), \\ \sum_{\substack{a \le p \le b \\ p \text{ prime}}} \frac{1}{p} &\le 4 \log \frac{\log b}{\log a} + \frac{4}{\log a} && (2 \le a \le b). \end{aligned} \tag{4.98}$$

Writing $\vartheta(a^-) = \sum_{\substack{p<a \\ p \text{ prime}}} \log p$, the identities are

$$\begin{aligned} \sum_{\substack{p \le t \\ p \text{ prime}}} \frac{\log p}{p} &= \frac{\vartheta(t)}{t} + \int_2^t \frac{\vartheta(v)}{v^2}\, dv, \\ \sum_{\substack{a \le p \le b \\ p \text{ prime}}} \frac{1}{p} &= \frac{\vartheta(b)}{b \log b} - \frac{\vartheta(a^-)}{a \log a} + \int_a^b \frac{\vartheta(v)(\log v + 1)}{v^2 (\log v)^2}\, dv. \end{aligned}$$

The first is at most $4 + 4\log(t/2)$. The second is at most $4\log(\log b/\log a) + 4/\log a$. Suppose $h(p) = 0$ for $p < 5$ and $0 \le h(p) \le 4/p$ for $p \ge 5$. The inequality $-\log(1-x) \le x + x^2/(2(1-x))$ for $0 \le x < 1$ yields

$$\begin{aligned} -\log(1 - h(p)) &\le \frac{4}{p} + \frac{40}{p^2} && (p \ge 5), \\ \sum_{n \ge a} \frac{1}{n^2} &\le \int_{\lceil a \rceil - 1}^{\infty} \frac{dt}{t^2} \le \frac{1}{a-1} \le \frac{1}{\log a} && (a \ge 2). \end{aligned}$$

Summing the first bound and using (4.98) gives

$$\prod_{\substack{a \le p \le b \\ p \text{ prime}}} (1 - h(p))^{-1} \le \left(\frac{\log b}{\log a}\right)^{16} \exp(56/\log a) \qquad (2 \le a \le b).$$

Thus (4.82) holds with the fixed parameters $\kappa = 16$ and $B_1 = 56$.

Step 2. Express the model probabilities and total variation. Let $\mathscr{D} \subset \mathbb{N}^4$ be the set of tuples obtained from the local vectors in (4.93) by

$$d_i = \prod_{p \in \mathcal{P}} p^{a_{i,p}} \quad (1 \le i \le 4), \qquad q(\mathbf{d}) = \operatorname{lcm}(d_1, d_2, d_3, d_4).$$

Unique factorization identifies each valuation vector with its tuple. We use this identification in events such as $\mathbf{A} = \mathbf{d}$. For $\mathbf{d} \in \mathscr{D}$, put

$$\mathbf{a}_p = (v_p(d_1), \dots, v_p(d_4)), \qquad g_{\mathbf{d}} = \prod_{p \in \mathcal{P}} g_p(\mathbf{a}_p),$$

$$h_{\mathbf{d}}(p) = \begin{cases} h_{\mathbf{a}_p}(p), & p \in \mathcal{P}, \\ 0, & p \notin \mathcal{P}, \end{cases} \qquad h_{\mathbf{d}}(e) = \prod_{\substack{p \mid e \\ p \text{ prime}}} h_{\mathbf{d}}(p) \quad (e \in \mathcal{S}_{\mathrm{sf}}).$$

Here $h_{\mathbf{d}}(1) = 1$. By Theorem 4.7, the exact probability at p is $g_p(\mathbf{a}_p)(1 - h_{\mathbf{d}}(p))$. The vectors $\mathbf{Z}_p$ are independent as p varies over $\mathcal{P}$. Also, C^* is independent of $\mathbf{A}$. Therefore

$$V_{\mathbf{d}} = \prod_{p \in \mathcal{P}} (1 - h_{\mathbf{d}}(p)), \qquad \mathbb{P}(\mathbf{A} = \mathbf{d}) = g_{\mathbf{d}} V_{\mathbf{d}},$$

$$m_{c,\mathbf{d}} := \mathbb{P}(C^* = c, \mathbf{A} = \mathbf{d}) = \pi_c g_{\mathbf{d}} V_{\mathbf{d}}, \qquad \sum_{c \in \mathcal{C}} \sum_{\mathbf{d} \in \mathscr{D}} m_{c,\mathbf{d}} = 1. \tag{4.99}$$

Both joint distributions have total probability one. Using $|t| = 2t_+ - t$, where $t_+ = \max\{t, 0\}$ for $t \in \mathbb{R}$, in (4.77) gives

$$\Delta := d_{\mathrm{TV}}((C, \mathbf{V}), (C^*, \mathbf{A})) = \sum_{c \in \mathcal{C}} \sum_{\mathbf{d} \in \mathscr{D}} \big(m_{c,\mathbf{d}} - \mathbb{P}(C = c, \mathbf{V} = \mathbf{d})\big)_+.$$

The sum is restricted to $\mathscr{D}$ because $\mathbb{P}(\mathbf{A} = \mathbf{d}) = 0$ for $\mathbf{d} \notin \mathscr{D}$. Set $S = Y^{1/4}$ and $H = Y^{1/8}$. The terms with $q(\mathbf{d}) > H$ sum to at most $\mathbb{P}(q(\mathbf{A}) > H)$.

Step 3. Bound the probability that $q(\mathbf{A}) > H$. Put $M_p = \max_{1 \le i \le 4} v_p(L_i(\mathbf{Z}_p))$. Each $L_i(\mathbf{Z}_p)$ has normalized Haar distribution on $\mathbb{Z}_p$. The union bound therefore gives

$$\mathbb{P}(M_p \ge k) \le 4p^{-k} \qquad (k \ge 1).$$

Choose $\alpha = \min\{1/4, 1/\log y\}$. Telescoping and Tonelli's theorem yield

$$\mathbb{E}p^{\alpha M_p} = 1 + \sum_{k \ge 1} (p^{\alpha k} - p^{\alpha(k-1)}) \mathbb{P}(M_p \ge k) \le 1 + \frac{4(p^\alpha - 1)}{p - p^\alpha} \le 1 + B\alpha \frac{\log p}{p}.$$

For the last inequality, $\alpha \log p \le 1$ implies $p^\alpha - 1 \le e\alpha \log p$, while $p - p^\alpha \ge p(1 - 5^{-3/4})$. Since $q(\mathbf{A}) = \prod_{p \in \mathcal{P}} p^{M_p}$, independence and (4.98) give

$$\mathbb{E}q(\mathbf{A})^\alpha \le \exp\left(B\alpha \sum_{\substack{p \le y \\ p \text{ prime}}} \frac{\log p}{p} \right) \le B, \qquad \alpha \log H \ge \frac{\log 5}{32} u.$$

The last inequality uses $\alpha \log y = \min\{\log y/4, 1\} \ge \log 5/4$. Markov's inequality now gives

$$\mathbb{P}(q(\mathbf{A}) > H) \le BH^{-\alpha} \le Be^{-\beta u}, \qquad \beta = \frac{\log 5}{32}.$$

Step 4. Apply the residue-discrepancy condition. Let $\mathbf{d} \in \mathscr{D}$ satisfy $q = q(\mathbf{d}) \le H$. Write $\mathcal{E}$ for the squarefree positive integers supported on $\mathcal{P}$, including 1. Define

$$E_{\mathbf{d}} = \{d_i \mid L_i(\mathbf{Z}) \text{ for } 1 \le i \le 4\},$$

$$E_{\mathbf{d},e} = E_{\mathbf{d}} \cap \bigcap_{p \mid e} \bigcup_{i=1}^{4} \{v_p(L_i(\mathbf{Z})) \ge a_{i,p} + 1\} \qquad (e \in \mathcal{E}).$$

This event depends only on $\mathbf{Z}$ modulo qe: at $p \mid e$ the required exponents are at most $v_p(q)+1$, and at other primes they are at most $v_p(q)$. Let $\mathcal{R}_{\mathbf{d},e} \subset (\mathbb{Z}/qe\mathbb{Z})^2$ be its residue classes. The Chinese remainder theorem and Theorem 4.7 give

$$\frac{|\mathcal{R}_{\mathbf{d},e}|}{(qe)^2} = g_{\mathbf{d}} h_{\mathbf{d}}(e).$$

By the triangle inequality, summing (4.97) over these classes yields

$$\begin{aligned} \sum_{c\in\mathcal{C}} \left|\mathbb{P}(C=c, E_{\mathbf{d},e}) - \pi_c g_{\mathbf{d}} h_{\mathbf{d}}(e)\right| &\le |\mathcal{R}_{\mathbf{d},e}| A_0 \left(\frac{1}{Yqe} + \frac{1}{Y^2}\right) \\ &\le A_0 \left(\frac{qe}{Y} + \frac{(qe)^2}{Y^2}\right). \end{aligned} \tag{4.100}$$

Step 5. Obtain a lower bound for each exact probability. Assume $u \ge u_0$, where $u_0 \ge 1600$ is a sufficiently large absolute constant. Set $z = \lfloor y \rfloor$ and $D = \lfloor S/q \rfloor$. Since $S/q \ge Y^{1/8}$ and $D \ge S/(2q)$,

$$s = \frac{\log D}{\log z} \ge \frac{\log Y/8 - \log 2}{\log y} \ge u/8 - 1 \ge u/16 \ge 100. \tag{4.101}$$

Thus $2 \le z \le D^{1/2}$. By (4.94), $0 \le h_{\mathbf{d}}(p) \le 4/p < 1$ for $p \in \mathcal{P}$. Step 1 verifies (4.82) with the same parameters $(\kappa, B_1) = (16, 56)$ for every $\mathbf{d}$. Use the coefficients constructed in [7, Theorem 3.6(a), pp. 38–39], with these parameters. Increase u_0 so that (4.91) gives $\mathfrak{E} \le 1/2$ whenever $s \ge u_0/16$. This choice is absolute, and (4.101) gives $s \ge u/16 \ge u_0/16$, so $\mathfrak{E} \le 1/2$. Thus (4.90) holds for every $\mathbf{d} \in \mathscr{D}$ with $q(\mathbf{d}) \le H$.

Define the squarefree integer

$$T = \prod_{\substack{p\in\mathcal{P} \\ \exists i:\ v_p(L_i(\mathbf{Z}))>a_{i,p}}} p. \qquad \{\mathbf{V}=\mathbf{d}\} = E_{\mathbf{d}} \cap \{T=1\}.$$

Every prime divisor of T belongs to $\mathcal{P}$, so it is at most $\lfloor y \rfloor = z$. Thus T satisfies the prime-support restriction in (4.79); see also [7, (3.1), p. 30]. For $e \in \mathcal{E}$, $E_{\mathbf{d}} \cap \{e \mid T\} = E_{\mathbf{d},e}$. Apply (4.79) to T, multiply by $\mathbf{1}_{C=c}\mathbf{1}_{E_{\mathbf{d}}}$, and take expectations:

$$\mathbb{P}(C=c, \mathbf{V}=\mathbf{d}) \ge \sum_{\substack{e\in\mathcal{E} \\ e\le D}} \lambda^-(e)\mathbb{P}(C=c, E_{\mathbf{d},e}). \tag{4.102}$$

Because $h_{\mathbf{d}}(p) = 0$ outside $\mathcal{P}$, (4.83) takes the form

$$\sum_{\substack{e\in\mathcal{E} \\ e\le D}} \lambda^-(e) h_{\mathbf{d}}(e) = V_{\mathbf{d}}(1+\delta_{\mathbf{d}}), \qquad |\delta_{\mathbf{d}}| \le Be^{-u\log u/64}.$$

Indeed, after increasing u_0, the exponent in (4.87) satisfies

$$-s\log s + s\log\log\log s + Bs \le -\tfrac{1}{2}s\log s \le -\tfrac{1}{64}u\log u.$$

Set $r_{c,\mathbf{d},e} = \mathbb{P}(C=c, E_{\mathbf{d},e}) - \pi_c g_{\mathbf{d}} h_{\mathbf{d}}(e)$. Substitution in (4.102) gives

$$\begin{aligned} \mathbb{P}(C=c, \mathbf{V}=\mathbf{d}) &\ge m_{c,\mathbf{d}}(1+\delta_{\mathbf{d}}) + \sum_{\substack{e\in\mathcal{E} \\ e\le D}} \lambda^-(e) r_{c,\mathbf{d},e}, \\ \big(m_{c,\mathbf{d}} - \mathbb{P}(C=c, \mathbf{V}=\mathbf{d})\big)_+ &\le Be^{-u\log u/64} m_{c,\mathbf{d}} + \sum_{\substack{e\in\mathcal{E} \\ e\le D}} |r_{c,\mathbf{d},e}|. \end{aligned} \tag{4.103}$$

For the second inequality, take the absolute value of the signed remainder and use $|\lambda^-(e)| \le 1$. Summing its first term over $c \in \mathcal{C}$ and $q(\mathbf{d}) \le H$ gives at most $Be^{-u\log u/64}$ by (4.99).

Step 6. Sum the residue errors and conclude. Each d_i divides $q(\mathbf{d})$, so at most $\tau(q)^4$ tuples have least common multiple q. Let $d_{16}(q)$ count ordered factorizations of q into sixteen positive integers. Then

$$\tau(q)^4 \le d_{16}(q), \qquad \sum_{q\le H}\frac{d_{16}(q)}{q} \le \left(\sum_{n\le H}\frac{1}{n}\right)^{16} \le (1+\log H)^{16}. \tag{4.104}$$

For the first inequality at $q=p^a$, assign to $(a_1,\dots,a_4)\in\{0,\dots,a\}^4$ the integers

$$n_{\boldsymbol{\varepsilon}} = \#\{1\le r\le a : (\mathbf{1}_{r\le a_i})_{i=1}^4 = \boldsymbol{\varepsilon}\}, \qquad \sum_{\boldsymbol{\varepsilon}} n_{\boldsymbol{\varepsilon}} = a, \qquad a_i = \sum_{\boldsymbol{\varepsilon}} \varepsilon_i n_{\boldsymbol{\varepsilon}},$$

where $\boldsymbol{\varepsilon}\in\{0,1\}^4$. The last identity proves injectivity into the sixteen-part weak compositions of a, counted by $d_{16}(p^a)$. Multiplication over primes proves $\tau(q)^4\le d_{16}(q)$. Expanding $d_{16}(q)$ and allowing each factor to range independently up to H proves the remaining inequalities in (4.104).

Sum (4.100) over the terms in (4.103). In the resulting numerical bound, enlarge the sum over e to all positive integers $e\le S/q$. Using $\sum_{e\le R} e\le R^2$ and $\sum_{e\le R} e^2\le R^3$ for $R\ge 1$, we obtain

$$\begin{aligned} &A_0\sum_{q\le H} d_{16}(q)\sum_{e\le S/q}\left(\frac{qe}{Y}+\frac{(qe)^2}{Y^2}\right)\\ &\quad\le A_0\left(\frac{S^2}{Y}+\frac{S^3}{Y^2}\right)\sum_{q\le H}\frac{d_{16}(q)}{q}\\ &\quad\le A_0(Y^{-1/2}+Y^{-5/4})(1+\log H)^{16} \le 2A_0Y^{-1/2}(1+\log Y)^{16}. \end{aligned}$$

Combining Steps 2, 3, and 5 with this bound gives

$$\Delta \le Be^{-\beta u} + Be^{-u\log u/64} + 2A_0Y^{-1/2}(1+\log Y)^{16} \qquad (u\ge u_0). \tag{4.105}$$

Choose one absolute $B\ge 1$ valid for both exponential terms in (4.105), and set

$$c=\frac{1}{64}, \qquad K=\max\{2B, e^{u_0/64}\}.$$

Since $\beta>1/64$ and $\log u\ge 1$ for $u\ge u_0$, these constants give the stated estimate in that range. For $1\le u<u_0$, we have $\Delta\le 1\le Ke^{-u/64}$. If $\mathcal{P}=\varnothing$, both valuation vectors are empty. Taking $Q=1$ in (4.97) gives directly

$$\Delta = \frac{1}{2}\sum_{c\in\mathcal{C}}|\mathbb{P}(C=c)-\pi_c| \le \frac{A_0}{2}(Y^{-1}+Y^{-2}) \le A_0Y^{-1/2},$$

which also satisfies the claimed bound. □

4.5. **Local approximation.** We compare the joint distribution of the valuations and the character values of the unit parts of the four linear forms. Residues modulo D determine these quantities at primes dividing D_0q, except on an event of small probability.

Fix $0<\delta<1/10$, and let $D_0=D_0(\delta)$ be defined by (4.30). Let $q\ge 1$ be an integer and $0<\eta\le 1$. For each prime $p\mid q$, put $a_p=v_p(q)$ and let χ_p be a character of $(\mathbb{Z}/p^{a_p}\mathbb{Z})^\times$. Set $a_p=0$ for $p\nmid q$. Let $\mathcal{B}=\{p \text{ prime} : p\mid D_0q\}$ and $J=|\mathcal{B}|$. Since $2\mid D_0$, we have $J\ge 1$. For $p\in\mathcal{B}$, define

$$A=8J/\eta, \qquad b_p=v_p(D_0), \qquad k_p=\left\lceil\frac{\log A}{\log p}\right\rceil, \qquad K_p=b_p+k_p-1.$$

Set

$$D=\prod_{p\in\mathcal{B}} p^{K_p+\max(a_p,1)}. \tag{4.106}$$

Then

$$D_0 \mid D, \qquad D \le D_0 q A^J \prod_{\substack{p\in\mathcal{B}\\ p\nmid q}} p \le D_0^2 q A^J. \tag{4.107}$$

Indeed, $A \ge 8$ gives $k_p \ge 1$, so each exponent in D is at least b_p. Moreover,

$$D = D_0 q \left(\prod_{p\in\mathcal{B}} p^{k_p-1}\right) \prod_{\substack{p\in\mathcal{B}\\ p\nmid q}} p, \qquad p^{k_p-1} < A \quad (p\in\mathcal{B}).$$

This gives the first upper bound in (4.107). The last product divides D_0, which gives the second.

Let $V : (\mathbb{Z}/D_0\mathbb{Z})^2 \to [0,\infty)$ satisfy

$$D_0^{-2} \sum_{\mathbf{a}\in(\mathbb{Z}/D_0\mathbb{Z})^2} V(\mathbf{a}) = 1,$$

and extend V periodically to $\mathbb{Z}^2$. Retain the density w from (4.39). Let $\psi : \mathbb{R}^2 \to \mathbb{C}$ be Borel measurable, with $|\psi| \le 1$ on $\operatorname{supp} w$. Assume that $T \ge 0$ and that, for every integer $k \ge 1$,

$$|\psi(\mathbf{x}) - \psi(\mathbf{x}')| \le 2T/k$$

whenever $\mathbf{x}, \mathbf{x}' \in \operatorname{supp} w$ lie in the same square of side $1/k$. Let $X \ge 1$ be an integer, and choose

$$K = \max\{1, \lceil 8T/\eta \rceil\}, \qquad X_* = \frac{X}{DK}, \qquad 5 \le y \le X_*.$$

Assume $D \le c_w X$, where $c_w > 0$ is sufficiently small and depends only on w. Let $\mathcal{P}$ be any set of primes at most y disjoint from $\mathcal{B}$.

Define

$$N_{X,V} = \sum_{m,n\in\mathbb{Z}} w(m/X, n/X) V(m,n).$$

By (4.46) and $D_0 \le D \le c_w X$,

$$\left|\frac{N_{X,V}}{X^2} - 1\right| \le C_w(D_0/X + D_0^2/X^2) \le \frac{1}{2} \tag{4.108}$$

after decreasing c_w if necessary. Thus $N_{X,V} > 0$. Let $\mathbf{Z}$ be the $\mathbb{Z}^2$-valued random vector with distribution

$$\mathbb{P}(\mathbf{Z} = (m,n)) = \frac{w(m/X, n/X) V(m,n)}{N_{X,V}} \qquad ((m,n)\in\mathbb{Z}^2).$$

For the comparison distribution, let $\mathbf{A}$ take values in $\{0,\dots,D_0-1\}^2$, with

$$\mathbb{P}(\mathbf{A} = \mathbf{a}) = \frac{V(\mathbf{a})}{D_0^2} \qquad \big(\mathbf{a}\in\{0,\dots,D_0-1\}^2\big).$$

Let $(\mathbf{U}_p)_{p\in\mathcal{B}\cup\mathcal{P}}$ be independent Haar vectors in $\mathbb{Z}_p^2$, independent of $\mathbf{A}$. Put

$$\mathbf{Z}_p = \begin{cases} \mathbf{A} + p^{b_p}\mathbf{U}_p, & p\in\mathcal{B},\\ \mathbf{U}_p, & p\in\mathcal{P}. \end{cases}$$

Conditional on $\mathbf{A} = \mathbf{a}$, these vectors are independent. For $p \in \mathcal{B}$, the distribution is normalized Haar measure on $\mathbf{a} + p^{b_p}\mathbb{Z}_p^2$; when $b_p = 0$, this is all of $\mathbb{Z}_p^2$. For $p \in \mathcal{P}$, the distribution is normalized Haar measure on $\mathbb{Z}_p^2$.

For $z \in \mathbb{Q}_p^\times$, write $u_p(z) = p^{-v_p(z)} z \in \mathbb{Z}_p^\times$. For $p \mid q$, evaluate $\chi_p(u)$, $u \in \mathbb{Z}_p^\times$, by reducing u modulo p^{a_p}. Using the forms in (4.92), define the two random tuples

$$\begin{aligned} \mathcal{D}_{\mathrm{int}} &= \Big((v_p(L_i(\mathbf{Z})))_{p\in\mathcal{B}\cup\mathcal{P},\,1\le i\le 4}, (\chi_p(u_p(L_i(\mathbf{Z}))))_{p\mid q,\,1\le i\le 4} \Big), \\ \mathcal{D}_{\mathrm{loc}} &= \Big((v_p(L_i(\mathbf{Z}_p)))_{p\in\mathcal{B}\cup\mathcal{P},\,1\le i\le 4}, (\chi_p(u_p(L_i(\mathbf{Z}_p))))_{p\mid q,\,1\le i\le 4} \Big). \end{aligned} \tag{4.109}$$

All four $L_i(\mathbf{Z})$ are positive almost surely because of the support of w. Each L_i has a coefficient that is a unit in $\mathbb{Z}_p$, so $L_i(\mathbf{Z}_p)$ has no atoms, even conditional on $\mathbf{A}$. In particular, $L_i(\mathbf{Z}_p) \neq 0$ almost surely. Thus both tuples in (4.109) are defined almost surely. They take values in the countable set

$$\mathscr{S} = (\mathbb{Z}_{\ge 0}^4)^{\mathcal{B}\cup\mathcal{P}} \times \prod_{\substack{p\mid q \\ p \text{ prime}}} \{\chi_p(u) : u \in (\mathbb{Z}/p^{a_p}\mathbb{Z})^\times\}^4. \tag{4.110}$$

Write $\mathbb{E}_{\mathrm{loc}}$ for expectation with respect to $\mathbf{A}$ and the vectors $\mathbf{U}_p$.

Theorem 4.9. *Under the preceding hypotheses, every function $F : \mathscr{S} \to \mathbb{C}$ with $|F| \le 1$ satisfies*

$$\begin{aligned} \Big| \mathbb{E}[F(\mathcal{D}_{\mathrm{int}})\psi(\mathbf{Z}/X)] - \mathbb{E}_{\mathrm{loc}}[F(\mathcal{D}_{\mathrm{loc}})] \int w\psi \Big| &\le C_w \big\{ \eta + e^{-c\log X_*/\log y} \\ &\quad + X_*^{-1/2}(1+\log X_*)^{16} + D/X \big\}. \end{aligned} \tag{4.111}$$

Here C_w depends only on w, and $c > 0$ is absolute.

Proof. Let $0 < c_w \le 1$ be no larger than the constant in Theorem 4.6. With C_w as in (4.46), decrease c_w so that $C_w(c_w + c_w^2) \le 1/4$. We divide the rest proof into seven steps.

Step 1. Condition on a residue modulo D_0. Let $\mathbf{a} \in \{0,\dots,D_0-1\}^2$ satisfy $V(\mathbf{a}) > 0$. Write $\mathbb{P}_{\mathbf{a}}, \mathbb{E}_{\mathbf{a}}$ for the integer distribution conditioned on $\mathbf{Z} \equiv \mathbf{a} \pmod{D_0}$, and $\mathbb{P}_{\mathbf{a}}^{\mathrm{loc}}, \mathbb{E}_{\mathbf{a}}^{\mathrm{loc}}$ for the local distribution conditioned on $\mathbf{A} = \mathbf{a}$. Since V is constant on $\mathbf{a} + D_0\mathbb{Z}^2$,

$$N_{\mathbf{a}} = \sum_{\mathbf{z}\in\mathbf{a}+D_0\mathbb{Z}^2} w(\mathbf{z}/X), \qquad N_{\mathbf{a}} \ge \frac{X^2}{2D_0^2} > 0,$$

$$\mathbb{P}_{\mathbf{a}}(\mathbf{Z} = \mathbf{z}) = \frac{w(\mathbf{z}/X)}{N_{\mathbf{a}}} \qquad (\mathbf{z} \in \mathbf{a} + D_0\mathbb{Z}^2).$$

The lower bound follows from Theorem 4.5 and $D_0 \le D \le c_w X$. We first prove the comparison with a bound independent of $\mathbf{a}$.

Step 2. Bound large valuations at primes in $\mathcal{B}$. For $p \in \mathcal{B}$, the conditional local vector is $\mathbf{Z}_p = \mathbf{a} + p^{b_p}\mathbf{U}_p$. Each L_i has a unit coefficient in $\mathbb{Z}_p$, so $L_i(\mathbf{U}_p)$ has Haar distribution on $\mathbb{Z}_p$. If $v_p(L_i(\mathbf{a})) < b_p$, this valuation is unchanged by adding $p^{b_p}L_i(\mathbf{U}_p)$. Otherwise $L_i(\mathbf{Z}_p)/p^{b_p}$ has Haar distribution on $\mathbb{Z}_p$. Since $K_p + 1 = b_p + k_p$ and $p^{k_p} \ge A$,

$$\mathbb{P}_{\mathbf{a}}^{\mathrm{loc}}\big(v_p(L_i(\mathbf{Z}_p)) > K_p\big) = \begin{cases} 0, & v_p(L_i(\mathbf{a})) < b_p, \\ p^{-k_p}, & v_p(L_i(\mathbf{a})) \ge b_p \end{cases} \le A^{-1}. \tag{4.112}$$

The second case uses the Haar probability p^{-k_p} of $p^{k_p}\mathbb{Z}_p$.

Let $\mathcal{R}_{\mathrm{exc}}$ be the following set of residues:

$$\mathcal{R}_{\mathrm{exc}} = \Big\{ \mathbf{c} \in (\mathbb{Z}/D\mathbb{Z})^2 : L_i(\mathbf{c}) \equiv 0 \pmod{p^{K_p+1}} \text{ for some } p \in \mathcal{B},\ 1 \le i \le 4 \Big\}.$$

This is well defined because $v_p(D) \ge K_p + 1$. For $\mathbf{c} \notin \mathcal{R}_{\mathrm{exc}}$, the residue determines each valuation $r = v_p(L_i)$ at $p \in \mathcal{B}$, and $r \le K_p$. For $p \mid q$, it also determines $p^{-r}L_i$ modulo p^{a_p}, since

$$v_p(D) - r = K_p + a_p - r \ge a_p.$$

Thus every component of (4.109) belonging to $\mathcal{B}$ is fixed on each residue outside $\mathcal{R}_{\mathrm{exc}}$.

Step 3. Compare the residue distributions modulo D. Put

$$\mathcal{R}_{\mathbf{a}} = \{\mathbf{c} \in (\mathbb{Z}/D\mathbb{Z})^2 : \mathbf{c} \equiv \mathbf{a} \pmod{D_0}\}, \qquad r_0 = |\mathcal{R}_{\mathbf{a}}| = (D/D_0)^2.$$

The Chinese remainder theorem defines a unique $\mathbf{C} \in (\mathbb{Z}/D\mathbb{Z})^2$ by the congruences $\mathbf{C} \equiv \mathbf{Z}_p \pmod{p^{v_p(D)}}$ for $p \in \mathcal{B}$. Under $\mathbb{P}^{\mathrm{loc}}_{\mathbf{a}}$, the residues $\mathbf{Z}_p \bmod p^{v_p(D)}$ are independent. Each is uniformly distributed among the residues congruent to $\mathbf{a} \pmod{p^{b_p}}$. Therefore $\mathbf{C}$ is uniform on $\mathcal{R}_{\mathbf{a}}$. The union bound and (4.112) give

$$\mathbb{P}^{\mathrm{loc}}_{\mathbf{a}}(\mathbf{C} \in \mathcal{R}_{\mathrm{exc}}) \le \frac{4J}{A} = \frac{\eta}{2}. \tag{4.113}$$

For $\mathbf{c} \in \mathcal{R}_{\mathbf{a}}$, define $N_{\mathbf{c}} = \sum_{\mathbf{z} \in \mathbf{c} + D\mathbb{Z}^2} w(\mathbf{z}/X)$. Theorem 4.5 gives

$$\begin{aligned} \left| N_{\mathbf{c}} - \frac{X^2}{D^2} \right| &\le C_w(X/D + 1), \quad N_{\mathbf{c}} \ge \frac{X^2}{2D^2} > 0, \\ \left| N_{\mathbf{a}} - \frac{X^2}{D_0^2} \right| &\le C_w(X/D_0 + 1), \quad N_{\mathbf{a}} \ge \frac{X^2}{2D_0^2} > 0. \end{aligned} \tag{4.114}$$

Consequently the integer residue probabilities are $\theta_{\mathbf{c}} = N_{\mathbf{c}}/N_{\mathbf{a}}$, and $\sum_{\mathbf{c} \in \mathcal{R}_{\mathbf{a}}} \theta_{\mathbf{c}} = 1$. Inserting $X^2/(D^2 N_{\mathbf{a}})$ and using $r_0 = (D/D_0)^2$ yields

$$\begin{aligned} \sum_{\mathbf{c} \in \mathcal{R}_{\mathbf{a}}} \left| \theta_{\mathbf{c}} - \frac{1}{r_0} \right| &\le \frac{r_0 C_w(X/D + 1)}{N_{\mathbf{a}}} + \frac{|X^2/D_0^2 - N_{\mathbf{a}}|}{N_{\mathbf{a}}} \\ &\le C_w \left(\frac{D}{X} + \frac{D^2}{X^2} + \frac{D_0}{X} + \frac{D_0^2}{X^2} \right). \end{aligned}$$

Since $D_0 \le D \le c_w X$, we obtain

$$\ell_{\mathbf{a}} := \sum_{\mathbf{c} \in \mathcal{R}_{\mathbf{a}}} \left| \theta_{\mathbf{c}} - \frac{1}{r_0} \right| \le C_w(D/X + D^2/X^2) \le C_w' D/X.$$

The total-variation distance of the two residue distributions is $\ell_{\mathbf{a}}/2$. Thus (4.113) implies

$$\mathbb{P}_{\mathbf{a}}(\mathbf{Z} \bmod D \in \mathcal{R}_{\mathrm{exc}}) \le \frac{\eta}{2} + \frac{\ell_{\mathbf{a}}}{2}. \tag{4.115}$$

Step 4. Approximate ψ on the spatial squares. Use the squares Q_γ of side $1/K$ from Theorem 4.6. Choose $\mathbf{x}_\gamma \in Q_\gamma \cap \operatorname{supp} w$, and put $\psi_\gamma = \psi(\mathbf{x}_\gamma)$ and $\psi^{(K)} = \sum_\gamma \psi_\gamma \mathbf{1}_{Q_\gamma}$. The oscillation assumption and the choice of K give

$$|\psi(\mathbf{x}) - \psi^{(K)}(\mathbf{x})| \le 2T/K \le \eta/4 \qquad (\mathbf{x} \in \operatorname{supp} w). \tag{4.116}$$

This includes $T = 0$. Set

$$\begin{gathered} \pi_\gamma = \int_{Q_\gamma} w = I_\gamma/X^2, \qquad I_\psi = \int w\psi, \qquad I_\psi^{(K)} = \sum_\gamma \pi_\gamma \psi_\gamma = \int w\psi^{(K)}, \\ \sum_\gamma \pi_\gamma = 1, \qquad |I_\psi^{(K)} - I_\psi| \le \eta/4. \end{gathered} \tag{4.117}$$

Both I_ψ and $I_\psi^{(K)}$ have modulus at most one. Replacing $\psi(\mathbf{Z}/X)$ by $\psi^{(K)}(\mathbf{Z}/X)$ in an expectation multiplied by F also costs at most $\eta/4$.

Step 5. Apply Theorem 4.8 at primes in $\mathcal{P}$. For $\mathbf{c} \in \mathcal{R}_{\mathbf{a}}$, let $\mathbb{P}_{\mathbf{c}}$ be the integer distribution conditioned on $\mathbf{Z} \equiv \mathbf{c} \pmod{D}$. It assigns probability $w(\mathbf{z}/X)/N_{\mathbf{c}}$ to $\mathbf{z} \in \mathbf{c} + D\mathbb{Z}^2$. This is defined by (4.114); it has finite support, on which all four forms are positive. Let Γ be the

index of the square containing $\mathbf{Z}/X$. Every positive integer Q supported on $\mathcal{P}$ is coprime to D. Applying (4.60) with $d = D$ gives, for every $\mathbf{b} \in (\mathbb{Z}/Q\mathbb{Z})^2$,

$$\sum_{\gamma} \left| \mathbb{P}_{\mathbf{c}}(\Gamma = \gamma, \mathbf{Z} \equiv \mathbf{b} \pmod{Q}) - \frac{\pi_\gamma}{Q^2} \right| \le C_w \left(\frac{DK}{XQ} + \frac{D^2K^2}{X^2} \right) = C_w \left(\frac{1}{X_* Q} + \frac{1}{X_*^2} \right).$$

This is (4.97) with $Y = X_*$ and $A_0 = C_w$. The cell index set is finite and nonempty, and (π_γ) is a probability vector. Also, $2, 3 \mid D_0$, so $\mathcal{P} \subset [5, y]$, while $5 \le y \le X_*$. Thus all hypotheses of Theorem 4.8 hold.

Write $\mathbf{V}_{\mathcal{P}}$ and $\mathbf{V}^*_{\mathcal{P}}$ for the valuation vectors at primes in $\mathcal{P}$ under the integer and local distributions, respectively. Let $\mu_{\mathbf{c}}$ be the law of $(\Gamma, \mathbf{V}_{\mathcal{P}})$ under $\mathbb{P}_{\mathbf{c}}$, and let ν be the product of (π_γ) and the law of $\mathbf{V}^*_{\mathcal{P}}$. Theorem 4.8 gives

$$d_{\mathrm{TV}}(\mu_{\mathbf{c}}, \nu) \le \varepsilon_* := C_w \left\{ e^{-c \log X_* / \log y} + X_*^{-1/2} (1 + \log X_*)^{16} \right\}. \tag{4.118}$$

The vectors $\mathbf{Z}_p$, $p \in \mathcal{P}$, are independent of $\mathbf{A}$ and $(\mathbf{U}_p)_{p \in \mathcal{B}}$. Since $\mathbf{C}$ is determined by these latter variables, conditioning on $\mathbf{A} = \mathbf{a}$ and $\mathbf{C} = \mathbf{c}$ does not change the law of $\mathbf{V}^*_{\mathcal{P}}$. Consequently ν is the same for every $\mathbf{a}, \mathbf{c}$.

Step 6. Complete the comparison on the initial coset. For $\mathbf{c} \in \mathcal{R}_{\mathbf{a}} \setminus \mathcal{R}_{\mathrm{exc}}$, define $F_{\mathbf{c}} : (\mathbb{Z}^4_{\ge 0})^{\mathcal{P}} \to \mathbb{C}$ by inserting the fixed data at $\mathcal{B}$ and the valuation vector $\mathbf{v}$ at $\mathcal{P}$ into F. Set $F_{\mathbf{c}} = 0$ for $\mathbf{c} \in \mathcal{R}_{\mathbf{a}} \cap \mathcal{R}_{\mathrm{exc}}$. Put $\Phi_{\mathbf{c}}(\gamma, \mathbf{v}) = F_{\mathbf{c}}(\mathbf{v}) \psi_\gamma$. Then $|\Phi_{\mathbf{c}}| \le 1$ and

$$\begin{aligned} \sum_{\mathbf{c} \in \mathcal{R}_{\mathbf{a}}} \theta_{\mathbf{c}} \int \Phi_{\mathbf{c}} \, d\mu_{\mathbf{c}} &= \mathbb{E}_{\mathbf{a}} \left[\mathbf{1}_{\{\mathbf{Z} \bmod D \notin \mathcal{R}_{\mathrm{exc}}\}} F(\mathcal{D}_{\mathrm{int}}) \psi^{(K)}(\mathbf{Z}/X) \right], \\ \frac{1}{r_0} \sum_{\mathbf{c} \in \mathcal{R}_{\mathbf{a}}} \int \Phi_{\mathbf{c}} \, d\nu &= \mathbb{E}^{\mathrm{loc}}_{\mathbf{a}} \left[\mathbf{1}_{\{\mathbf{C} \notin \mathcal{R}_{\mathrm{exc}}\}} F(\mathcal{D}_{\mathrm{loc}}) \right] I^{(K)}_\psi. \end{aligned} \tag{4.119}$$

The first identity follows by conditioning on $\mathbf{Z} \bmod D$. For the second identity, work under $\mathbb{P}^{\mathrm{loc}}_{\mathbf{a}}$. The vector $\mathbf{C}$ is uniform on $\mathcal{R}_{\mathbf{a}}$ and independent of $\mathbf{V}^*_{\mathcal{P}}$. Since ν is a product measure, $\int \Phi_{\mathbf{c}} \, d\nu = I^{(K)}_\psi \mathbb{E}^{\mathrm{loc}}_{\mathbf{a}}[F_{\mathbf{c}}(\mathbf{V}^*_{\mathcal{P}})]$. Averaging over $\mathbf{c} \in \mathcal{R}_{\mathbf{a}}$ gives the second identity. By (4.78), (4.118), and $\sum \theta_{\mathbf{c}} = 1$,

$$\left| \sum_{\mathbf{c} \in \mathcal{R}_{\mathbf{a}}} \theta_{\mathbf{c}} \int \Phi_{\mathbf{c}} \, d\mu_{\mathbf{c}} - \frac{1}{r_0} \sum_{\mathbf{c} \in \mathcal{R}_{\mathbf{a}}} \int \Phi_{\mathbf{c}} \, d\nu \right| \le \sum_{\mathbf{c} \in \mathcal{R}_{\mathbf{a}}} 2\theta_{\mathbf{c}} \varepsilon_* + \ell_{\mathbf{a}} = 2\varepsilon_* + \ell_{\mathbf{a}}. \tag{4.120}$$

Define

$$B_{\mathbf{a}} = \mathbb{E}_{\mathbf{a}}[F(\mathcal{D}_{\mathrm{int}}) \psi(\mathbf{Z}/X)], \qquad M_{\mathbf{a}} = \mathbb{E}^{\mathrm{loc}}_{\mathbf{a}}[F(\mathcal{D}_{\mathrm{loc}})].$$

By (4.115), the contribution from $\mathbf{Z} \bmod D \in \mathcal{R}_{\mathrm{exc}}$ to the integer expectation has modulus at most $\eta/2 + \ell_{\mathbf{a}}/2$. By (4.113), the contribution from $\mathbf{C} \in \mathcal{R}_{\mathrm{exc}}$ to the local expectation, multiplied by $I^{(K)}_\psi$, has modulus at most $\eta/2$. These bounds use $|F|, |\psi^{(K)}|, |I^{(K)}_\psi| \le 1$.

For the integer approximation, (4.116) gives $|\mathbb{E}_{\mathbf{a}}[F(\mathcal{D}_{\mathrm{int}})(\psi - \psi^{(K)})(\mathbf{Z}/X)]| \le \eta/4$. For the spatial integral, (4.117) gives $|(I_\psi - I^{(K)}_\psi) M_{\mathbf{a}}| \le \eta/4$. Combining these four bounds with (4.119) and (4.120), we obtain

$$|B_{\mathbf{a}} - I_\psi M_{\mathbf{a}}| \le \frac{3\eta}{2} + 2\varepsilon_* + \frac{3\ell_{\mathbf{a}}}{2} \le C_w \left(\eta + \varepsilon_* + \frac{D}{X} \right). \tag{4.121}$$

This bound is independent of $\mathbf{a}$.

Step 7. Average over the residues modulo D_0. For every $\mathbf{a} \bmod D_0$, quadrature gives

$$\begin{aligned} N_{\mathbf{a}} &= \frac{X^2}{D_0^2}(1 + e_{\mathbf{a}}), \qquad |e_{\mathbf{a}}| \le E_0 := C_w(D_0/X + D_0^2/X^2) \le \frac{1}{4}, \\ \sigma_{\mathbf{a}} &= \frac{V(\mathbf{a})}{D_0^2}, \qquad \bar{e} = \sum_{\mathbf{a}} \sigma_{\mathbf{a}} e_{\mathbf{a}}, \qquad N_{X,V} = X^2(1 + \bar{e}). \end{aligned}$$

Here $\sum \sigma_{\mathbf{a}} = 1$ and $|\bar{e}| \le E_0$. The integer residue probabilities therefore satisfy

$$\begin{aligned} \rho_{\mathbf{a}} &:= \mathbb{P}(\mathbf{Z} \equiv \mathbf{a} \pmod{D_0}) = \sigma_{\mathbf{a}} \frac{1+e_{\mathbf{a}}}{1+\bar{e}}, \\ \sum_{\mathbf{a}} |\rho_{\mathbf{a}} - \sigma_{\mathbf{a}}| &= \frac{\sum_{\mathbf{a}} \sigma_{\mathbf{a}} |e_{\mathbf{a}} - \bar{e}|}{1+\bar{e}} \le \frac{2E_0}{1-E_0} \le C_w D_0/X. \end{aligned} \tag{4.122}$$

Set $B_{\mathbf{a}} = M_{\mathbf{a}} = 0$ when $V(\mathbf{a}) = 0$; then $\rho_{\mathbf{a}} = \sigma_{\mathbf{a}} = 0$ as well. Since $|I_\psi|, |M_{\mathbf{a}}| \le 1$,

$$\begin{aligned} &|\mathbb{E}[F(\mathcal{D}_{\mathrm{int}})\psi(\mathbf{Z}/X)] - I_\psi \mathbb{E}_{\mathrm{loc}}[F(\mathcal{D}_{\mathrm{loc}})]| \\ &\quad \le \sum_{\mathbf{a}} \rho_{\mathbf{a}} |B_{\mathbf{a}} - I_\psi M_{\mathbf{a}}| + \sum_{\mathbf{a}} |\rho_{\mathbf{a}} - \sigma_{\mathbf{a}}| \\ &\quad \le C_w \left(\eta + \varepsilon_* + \frac{D}{X} \right). \end{aligned}$$

The last inequality uses (4.121), (4.122), and $D_0 \le D$. Substituting the value of ε_* from (4.118) proves (4.111). The bound in (4.122) uses V only through the probability vector $(\sigma_{\mathbf{a}})$. Thus the constants are independent of $\|V\|_\infty$. □

4.6. **Model positivity.** Retain $0 < \delta < 1/10$ and the parameters T_0, B, D_0 in (4.30). Let χ be a primitive Dirichlet character of conductor $q = q_\chi$. For each prime $p \mid q$, the Chinese remainder theorem determines a character $\chi_p : (\mathbb{Z}/p^{v_p(q)}\mathbb{Z})^\times \to \mathbb{S}^1$ such that $\chi(n) = \prod_{\substack{p \mid q \\ p \text{ prime}}} \chi_p(n)$ whenever $\gcd(n, q) = 1$. For $u \in \mathbb{Z}_p^\times$, evaluate $\chi_p(u)$ at the residue of u modulo $p^{v_p(q)}$. For $p \nmid q$, let χ_p be the trivial character on $\mathbb{Z}_p^\times$.

Let $g : \mathbb{N} \to \mathbb{S}^1$ be completely multiplicative, and let $t \in \mathbb{R}$. Extend g to $\mathbb{Q}_{>0}$ by $g(a/b) = g(a)/g(b)$ for $a, b \in \mathbb{N}$. For every prime p, put

$$z_p = \frac{g(p) p^{-it}}{\prod_{\substack{\ell \mid q,\ \ell \ne p \\ \ell \text{ prime}}} \chi_\ell(p)}. \tag{4.123}$$

Each factor in the denominator has modulus one, so $z_p \in \mathbb{S}^1$. For $r \in \mathbb{Q}_p^\times$, write $u_p(r) = p^{-v_p(r)} r \in \mathbb{Z}_p^\times$. For every $r \in \mathbb{Q}_{>0}$,

$$g(r) = r^{it} \prod_{p \text{ prime}} z_p^{v_p(r)} \prod_{\substack{\ell \mid q \\ \ell \text{ prime}}} \chi_\ell(u_\ell(r)). \tag{4.124}$$

Both sides are homomorphisms from $\mathbb{Q}_{>0}$ to $\mathbb{S}^1$. For $r = p$, we have $u_p(p) = 1$ and $u_\ell(p) = p$ for $\ell \ne p$, so (4.124) follows from (4.123). Unique factorization then proves the identity for every positive rational r. Only finitely many prime exponents $v_p(r)$ are nonzero.

Assume $y \ge \max(B, q)$ and $g(p) = \chi(p) p^{it}$ for every $p > y$. Then (4.123) gives $z_p = 1$ for $p > y$. Take $\mathcal{B} = \{p \text{ prime} : p \mid D_0 q\}$ and $\mathcal{P} = \{p \text{ prime} : p \le y,\ p \notin \mathcal{B}\}$ in the local distribution preceding Theorem 4.9, with $V = W_0$. Since each prime divisor of D_0 is at most B, the disjoint union $\mathcal{B} \cup \mathcal{P}$ is the set of all primes at most y. The normalization of W_0 follows from Theorem 4.4.

We first express $g(R)$ in terms of the data in (4.109). For $p \le y$, set $c_p = 2p^{-v_p(2)} \in \mathbb{Z}_p^\times$. For $\mathbf{z} \in \mathbb{Z}_p^2$ with $\prod_{i=1}^4 L_i(\mathbf{z}) \ne 0$, the identity $R = L_1 L_2/(2L_3 L_4)$ gives

$$\begin{aligned} v_p(R(\mathbf{z})) &= v_p(L_1(\mathbf{z})) + v_p(L_2(\mathbf{z})) - v_p(L_3(\mathbf{z})) - v_p(L_4(\mathbf{z})) - v_p(2), \\ u_p(R(\mathbf{z})) &= c_p^{-1} \frac{u_p(L_1(\mathbf{z})) u_p(L_2(\mathbf{z}))}{u_p(L_3(\mathbf{z})) u_p(L_4(\mathbf{z}))}. \end{aligned} \tag{4.125}$$

On the data space $\mathscr{S}$ in (4.110), define $F : \mathscr{S} \to \mathbb{S}^1$ by

$$F\big((a_{p,i}),(\xi_{p,i})\big) = \prod_{\substack{p\le y\\ p \text{ prime}}} z_p^{a_{p,1}+a_{p,2}-a_{p,3}-a_{p,4}-v_p(2)} \prod_{\substack{p\mid q\\ p \text{ prime}}} \frac{\xi_{p,1}\xi_{p,2}}{\chi_p(c_p)\xi_{p,3}\xi_{p,4}}. \tag{4.126}$$

Here $(a_{p,i})$ are the valuation coordinates and $(\xi_{p,i})$ are the character-value coordinates. All factors have modulus one. When $q = 1$, the character product is one. By (4.124) and (4.125),

$$\begin{aligned} F(\mathcal{D}_{\mathrm{int}}) &= g(R(\mathbf{Z}))R(\mathbf{Z})^{-it},\\ F(\mathcal{D}_{\mathrm{loc}}) &= \prod_{\substack{p\le y\\ p \text{ prime}}} z_p^{v_p(r_p)}\chi_p(u_p(r_p)), \qquad r_p = R(\mathbf{Z}_p). \end{aligned} \tag{4.127}$$

The local quantities are defined almost surely, as noted after (4.109). In particular, (4.126) includes the factor 2 in the denominator of R.

Define $\psi : \mathbb{R}^2 \to \mathbb{C}$ by $\psi(u,v) = R(u,v)^{it}$ for $u > v > 0$, and by zero elsewhere. Since $R(\mathbf{Z}/X) = R(\mathbf{Z})$, the integer integrand in Theorem 4.9 is $g(R(\mathbf{Z}))$. Moreover, (4.42) gives $\int w\psi = K_\infty(t) \in [0,1]$. We now compute the corresponding local expectation. The size conditions needed to apply Theorem 4.9 are verified in the proof of Theorem 4.10.

Put $\omega_j = (T_0 + 1 - |j|)/(T_0(T_0+1))$ for $0 < |j| \le T_0$. The distribution at the primes $p \le B$ has the following description. Choose an integer J_0 with $\mathbb{P}(J_0 = j) = \omega_j$. Conditional on $J_0 = j$, the vectors $\mathbf{Z}_p$, $p \le B$, are independent with densities $W_{p,j}$ relative to product Haar measure on $\mathbb{Z}_p^2$. This follows from (4.31), (4.32), and (4.35). Their projective points therefore have distributions $\mu_p(\cdot \mid E_{p,j,3})$. Indeed, the two factors in the denominator of $W_{p,j}$ are the probability $1-p^{-2}$ of primitivity and the probability $b_{p,j,3}$ of $E_{p,j,3}$ under μ_p. Thus the contribution of these primes is

$$K_{\le B} := \sum_{0<|j|\le T_0} \omega_j \prod_{\substack{p\le B\\ p \text{ prime}}} \left(\frac{1}{b_{p,j,3}} \int_{E_{p,j,3}} z_p^{V_p}\chi_p(U_p)\, d\mu_p \right).$$

Here V_p, U_p are the valuation and unit part of R under μ_p. Applying Theorem 4.3 with $T = T_0$ and $s = 3$ gives $K_{\le B} \in [-1/T_0, 1]$.

For $B < p \le y$, the vector $\mathbf{Z}_p$ has Haar distribution on $\mathbb{Z}_p^2$, independently of the other such vectors and of the vectors at primes at most B. For $p \mid q$, this follows because $v_p(D_0) = 0$: conditional on $\mathbf{A}$, the vector $\mathbf{A} + \mathbf{U}_p$ is Haar, and its distribution does not depend on $\mathbf{A}$. For $p \nmid q$, it is part of the definition of the local distribution. The homogeneity of R and (4.5) show that the expectation of $z_p^{v_p(r_p)}\chi_p(u_p(r_p))$ is $K_p(z_p,\chi_p)$. Combining these expectations with (4.127) gives

$$\mathbb{E}_{\mathrm{loc}}[F(\mathcal{D}_{\mathrm{loc}})] \int w\psi = K_\infty(t)\, K_{\le B} \prod_{\substack{B<p\le y\\ p \text{ prime}}} K_p(z_p,\chi_p). \tag{4.128}$$

If every χ_p with $B < p \le y$ is trivial, Theorem 4.2 gives $K_p(z_p,\chi_p) \in [0,1]$. Together with $K_\infty(t) \in [0,1]$ and $K_{\le B} \ge -1/T_0$, this bounds (4.128) below by $-1/T_0$. If some χ_{p_0} with $B < p_0 \le y$ is nontrivial, Theorem 4.2 gives $|K_{p_0}(z_{p_0},\chi_{p_0})| \le 2/\sqrt{p_0} \le 2/\sqrt{B}$. Every other factor has modulus at most one, so the real part of (4.128) is at least $-2/\sqrt{B}$. The bounds from Theorem 4.2 apply here because $B > 5$. In either case,

$$\Re\left(K_\infty(t) K_{\le B} \prod_{\substack{B<p\le y\\ p \text{ prime}}} K_p(z_p,\chi_p) \right) \ge -\max(T_0^{-1}, 2B^{-1/2}) \ge -\delta/100. \tag{4.129}$$

The last inequality follows from $T_0 \ge 100/\delta$ and $B = (200/\delta)^2$.

In Theorem 4.10, $M_1 > 0$ is a real parameter and $C_* > 0$ is an absolute constant. The parameter bounds are

$$M_1 \geq \exp(C_*\delta^{-3}), \qquad q_\chi \leq M_1, \qquad |t| \leq M_1. \tag{4.130}$$

Theorem 4.10. *There are absolute constants $C_*, C_U > 0$ such that, for each $0 < \delta < 1/10$, an integer $2 \leq U \leq C_U\delta^{-3}$, depending only on δ, can be chosen so that, whenever (4.130) holds, $y \geq M_1$, and $X \geq (M_1 y)^U$ is an integer, every finite character model g with $g(p) = \chi(p)p^{it}$ for $p > y$ satisfies*

$$\Re K_X(g) \geq -\delta/50. \tag{4.131}$$

Proof. All constants depending on the fixed density w are absolute. Write $q = q_\chi$ and $W = \|W_0\|_\infty$. We divide the rest proof into six steps.

Step 1. Choose the approximation tolerance. Let C_{match} be the constant C_w in (4.111). Fix $C_\eta \geq \max\{1, 400C_{\mathrm{match}}\}$ and put

$$\eta = \frac{\delta}{C_\eta(1+W)} \in (0,1).$$

By Theorem 4.4, there is an absolute $A_\eta \geq 1$ such that

$$\log(1/\eta) = \log C_\eta + \log(1+W) + \log(1/\delta) \leq A_\eta\delta^{-3}. \tag{4.132}$$

Theorem 4.4 also gives $\log D_0 \leq C\delta^{-3}$. Choose $C_* \geq 1$ large enough that (4.130) implies $M_1 \geq \max\{D_0, B, 5\}$. In particular, $y \geq \max\{B, q, 5\}$.

Step 2. Bound the modulus. Use D from (4.106), and let J be the number of distinct prime divisors of $D_0 q$. Order them as $p_1 < \cdots < p_J$. Since $p_j \geq j+1$,

$$(J+1)! \leq \prod_{j=1}^{J} p_j \leq D_0 q, \qquad 2^J \leq D_0 q.$$

For $J \geq 4$, at least $J/2$ factors in $(J+1)!$ are at least $J/2$. Thus $\log((J+1)!) \geq (J/2)\log(J/2)$. Including $J < 4$ by enlarging an absolute constant gives

$$J\log(2J) \leq C\log(D_0 q), \qquad J \leq \frac{\log(D_0 q)}{\log 2}.$$

By (4.107),

$$\begin{aligned} \log D &\leq 2\log D_0 + \log q + J\log(8J) + J\log(1/\eta) \\ &\leq C(1+\log(1/\eta))\log(D_0 q) \\ &\leq C_D\delta^{-3}\log M_1. \end{aligned} \tag{4.133}$$

The last line uses (4.132) and $D_0, q \leq M_1$.

Step 3. Bound the spatial grid size. Let L_w bound $|\nabla \log R|$ on a fixed compact convex subset of $u > v > 0$ containing $\operatorname{supp} w$. For $\mathbf{x}, \mathbf{x}' \in \operatorname{supp} w$ in the same square of side $1/k$, the mean value inequality gives

$$\begin{aligned} |\psi(\mathbf{x}) - \psi(\mathbf{x}')| &\leq |t|\,|\log R(\mathbf{x}) - \log R(\mathbf{x}')| \\ &\leq \sqrt{2}L_w|t|/k \leq 2T/k, \qquad T = \max\{1, L_w\}(1+|t|). \end{aligned} \tag{4.134}$$

Choose $K = \max\{1, \lceil 8T/\eta \rceil\}$ as in Theorem 4.9. Then

$$\log K \leq C_w + \log(1+|t|) + \log(1/\eta) \leq C_K \log M_1. \tag{4.135}$$

Here we used $|t| \leq M_1$, (4.132), and $\log M_1 \geq C_*\delta^{-3}$.

Step 4. Choose the exponent U. Let $s_0 \ge 1$ be an absolute constant such that $16\log(1+s) \le s/4$ for $s \ge s_0$. Let $c > 0$ be the constant in (4.111). Let $0 < c_w \le 1$ be the constant in the condition $D \le c_w X$ imposed before Theorem 4.9. Choose an absolute $C' \ge 2$ satisfying

$$C' \ge 2(C_D + C_K), \qquad cC' \ge A_\eta, \qquad \frac{3}{2}C'C_* \ge \max\{s_0, 4A_\eta, \log(1/c_w)\}. \tag{4.136}$$

Set $U = \lceil C'\delta^{-3} \rceil$. The bounds (4.133) and (4.135) give

$$\log(DK) \le (C_D + C_K)\delta^{-3}\log M_1 \le \frac{U}{2}\log M_1.$$

Put $X_* = X/(DK)$, $s = \log X_*$, and $\sigma = (3U/2)\log M_1$. Since $X \ge (M_1 y)^U$ and $y \ge M_1$,

$$\begin{aligned} X_* &\ge M_1^{U/2} y^U, \\ s &\ge U\log y + (U/2)\log M_1 \ge \sigma, \\ \sigma &\ge \frac{3}{2}C'C_*\delta^{-6} \ge \max\{s_0, 4\log(1/\eta), \log(1/c_w)\}. \end{aligned} \tag{4.137}$$

The last inequality follows from (4.132), (4.136), and $\delta^{-3} \ge 1$. In particular, $5 \le y \le X_*$.

Step 5. Verify the error bounds and normalization. The choices above give

$$\begin{aligned} e^{-c\log X_*/\log y} &\le e^{-cU} \le e^{-A_\eta \delta^{-3}} \le \eta, \\ X_*^{-1/2}(1+\log X_*)^{16} &= e^{-s/2+16\log(1+s)} \le e^{-s/4} \le \eta, \\ D/X &\le M_1^{-U/2} y^{-U} \le e^{-\sigma} \le \min\{\eta, c_w\}. \end{aligned} \tag{4.138}$$

For the second line, use $s \ge s_0$ and $s/4 \ge \log(1/\eta)$ from (4.137). For the third, use $D \le DK \le M_1^{U/2}$ and $X \ge (M_1 y)^U$. Thus $D \le c_w X$, as required in Theorem 4.9. Since $D_0 \mid D$, Theorem 4.4 and (4.46) also give

$$|Z_X - 1| \le C_w(D_0/X + D_0^2/X^2) \le \frac{1}{2}.$$

The last inequality is (4.108) with $V = W_0$. Hence $Z_X \ge 1/2$.

Step 6. Apply the local approximation. Take $V = W_0$ and F as in (4.126). Define $\psi : \mathbb{R}^2 \to \mathbb{C}$ by $\psi(u,v) = R(u,v)^{it}$ for $u > v > 0$, and by zero elsewhere. By Theorem 4.4, V is nonnegative, is periodic modulo D_0, and has residue average one. The definition of F gives $|F| = 1$. By (4.134), $|\psi(\mathbf{x}) - \psi(\mathbf{x}')| \le 2T/k$ for every integer $k \ge 1$ and all $\mathbf{x}, \mathbf{x}' \in \operatorname{supp} w$ lying in the same square of side $1/k$. The bounds (4.137) and (4.138) give $5 \le y \le X_*$ and $D \le c_w X$. We may therefore apply Theorem 4.9. Moreover, $N_{X,W_0} = X^2 Z_X$ and (4.127) give the exact identity

$$\mathbb{E}[F(\mathcal{D}_{\mathrm{int}})\psi(\mathbf{Z}/X)] = \frac{\displaystyle\sum_{\substack{m,n\in\mathbb{Z}\\ m>n>0}} w(m/X, n/X)W_0(m,n)g(R(m,n))}{X^2 Z_X} = K_X(g).$$

Using (4.128), (4.111), and (4.138), we obtain

$$\left| K_X(g) - K_\infty(t)K_{\le B}\prod_{\substack{B<p\le y\\ p \text{ prime}}} K_p(z_p, \chi_p) \right| \le 4C_{\mathrm{match}}\eta \le \delta/100.$$

(4.129) therefore yields

$$\Re K_X(g) \ge -\delta/100 - \delta/100 = -\delta/50,$$

which is (4.131). Finally, $2 \le U \le C'\delta^{-3} + 1 \le (C'+1)\delta^{-3}$. Thus $C_U = C' + 1$ is admissible, and U depends only on δ. □

5. Weighted concentration

We compare multiplicative functions that agree at small primes. Although $\|W_0\|_\infty$ depends on δ, the comparison constant depends only on the primes dividing its period. The key property is invariance under simultaneous multiplication by units.

We first state the additive-function estimate that we shall use. For a real additive function $H : \mathbb{N} \to \mathbb{R}$ and a real number $Y \geq 2$, define

$$A_H(Y) = \sum_{\substack{p \text{ prime}, j\in\mathbb{N} \\ p^j \leq Y}} \frac{H(p^j)}{p^j}, \qquad B_H(Y) = \sum_{\substack{p \text{ prime}, j\in\mathbb{N} \\ p^j \leq Y}} \frac{|H(p^j)|^2}{p^j}.$$

Both sums range over primes p and integers $j \geq 1$. The Turán–Kubilius inequality[5] states that there is an absolute constant C_{TK} such that, for every such H and integer $M \geq 2$,

$$\frac{1}{M} \sum_{n\leq M} |H(n) - A_H(M)|^2 \leq C_{\mathrm{TK}} B_H(M). \tag{5.1}$$

Lemma 5.1. *Let $C_E \geq 1$ be the absolute constant in (3.20). For $Z \geq 4$ and $M \geq \sqrt{Z}$,*

$$\sum_{\substack{p \text{ prime}, j\in\mathbb{N} \\ p^j \leq Z,\ p^j > M}} p^{-j} \leq \log(2C_E) + 1. \tag{5.2}$$

If every prime factor of a positive integer D is at most $B \geq 2$, then

$$\frac{D}{\varphi(D)} \leq C_E \log(2B),$$

where φ is Euler's totient[6] and $\varphi(1) = 1$.

Proof. Write $E(x)$ for the product in the proof of Theorem 3.4; its constant is the C_E fixed above. The terms with $j = 1$ in (5.2) are at most

$$\sum_{\substack{\sqrt{Z}<p\leq Z \\ p \text{ prime}}} \frac{1}{p} \leq \log \frac{E(Z)}{E(\sqrt{Z})} \leq \log(2C_E).$$

For the terms with $j \geq 2$, the sum over all primes and powers is bounded by

$$\sum_{p \text{ prime}} \sum_{j\geq 2} p^{-j} = \sum_{p \text{ prime}} \frac{1}{p(p-1)} \leq \sum_{n\geq 2} \frac{1}{n(n-1)} = 1.$$

This proves (5.2). Finally,

$$D/\varphi(D) = \prod_{\substack{p|D \\ p \text{ prime}}} (1 - p^{-1})^{-1} \leq E(B)$$

when $D > 1$, and the asserted bound also holds for $D = 1$. □

[5]There is an absolute constant $C > 0$ such that every additive function $f : \mathbb{N} \to \mathbb{C}$, meaning $f(ab) = f(a)+f(b)$ whenever $(a, b) = 1$, satisfies

$$\sum_{n\leq x} \left| f(n) - \sum_{\substack{p \text{ prime}, j\in\mathbb{N} \\ p^j \leq x}} \frac{f(p^j)}{p^j} \right|^2 \leq Cx \sum_{\substack{p \text{ prime}, j\in\mathbb{N} \\ p^j \leq x}} \frac{|f(p^j)|^2}{p^j} \qquad (x \geq 2).$$

Here x is real, and the prime-power sums range over primes p and integers $j \geq 1$. See [5, p. 893, equation (1)].

[6]For each positive integer n, $\varphi(n) = \#\{a \in \mathbb{Z} : 1 \leq a \leq n,\ \gcd(a, n) = 1\}$, the number of integers from 1 to n that are coprime to n.

For the next two lemmas, let $h : \mathbb{N} \to \mathbb{S}^1$ be completely multiplicative, let $Z \geq 4$ be an integer, and suppose $2 \leq y \leq Z$ and $h(p) = 1$ for every prime $p \leq y$. Put

$$\Delta = \sum_{\substack{y<p\leq Z \\ p \text{ prime}}} \frac{1 - \Re h(p)}{p}.$$

Lemma 5.2. *There exist an absolute constant $C \geq 1$ and a number $\zeta \in \mathbb{S}^1$ such that, for every integer M with $\sqrt{Z} \leq M \leq Z$,*

$$\frac{1}{M} \sum_{n\leq M} |h(n) - \zeta|^2 \leq C\Delta. \tag{5.3}$$

Proof. We divide the proof into three steps.

Step 1. Represent the multiplicative function by an additive phase. For each prime p, choose $\theta_p = \operatorname{Arg} h(p) \in (-\pi, \pi]$ and define

$$H(n) = \sum_{p \text{ prime}} v_p(n)\theta_p.$$

Since $v_p(ab) = v_p(a) + v_p(b)$, the function H is real and completely additive, with $H(p^j) = j\theta_p$. Unique prime factorization and complete multiplicativity give

$$h(n) = \prod_{p \text{ prime}} e^{iv_p(n)\theta_p} = e^{iH(n)}.$$

Moreover, $\theta_p = 0$ for $p \leq y$.

Step 2. Bound the phase variance. We first bound $B_H(Z)$ in terms of Δ. Concavity of sine gives $\sin u \geq 2u/\pi$ on $[0, \pi/2]$. Hence, for $|\theta| \leq \pi$,

$$1 - \cos\theta = 2\sin^2(|\theta|/2) \geq \frac{2\theta^2}{\pi^2}, \qquad \theta^2 \leq \frac{\pi^2}{2}(1 - \cos\theta).$$

Differentiating the geometric series for $0 < x < 1$ gives

$$\sum_{j\geq 1} j^2 x^j = x\frac{d}{dx}\left(\frac{x}{(1-x)^2}\right) = \frac{x(1+x)}{(1-x)^3}.$$

Taking $x = 1/p \leq 1/2$, we obtain

$$\sum_{j\geq 1} \frac{j^2}{p^j} = \frac{1}{p}\frac{1 + 1/p}{(1 - 1/p)^3} \leq \frac{1}{p}\frac{3/2}{(1/2)^3} = \frac{12}{p}.$$

Since the summands are nonnegative and $\cos\theta_p = \Re h(p)$,

$$B_H(Z) = \sum_{\substack{y<p\leq Z \\ p \text{ prime}}} \theta_p^2 \sum_{\substack{j\geq 1 \\ p^j \leq Z}} \frac{j^2}{p^j} \leq 12 \sum_{\substack{y<p\leq Z \\ p \text{ prime}}} \frac{\theta_p^2}{p} \leq 6\pi^2 \sum_{\substack{y<p\leq Z \\ p \text{ prime}}} \frac{1 - \Re h(p)}{p} = 6\pi^2\Delta. \tag{5.4}$$

Step 3. Transfer the variance bound to the multiplicative function. Fix an integer M with $\sqrt{Z} \leq M \leq Z$. We compare $A_H(M)$ with the fixed center $A_H(Z)$. The upper bound

$M \le Z$ and Cauchy–Schwarz give

$$\begin{aligned}|A_H(Z) - A_H(M)|^2 &= \left| \sum_{\substack{p \text{ prime}, j\in\mathbb{N} \\ M<p^j\le Z}} \frac{H(p^j)}{p^j} \right|^2 \\ &\le \left(\sum_{\substack{p \text{ prime}, j\in\mathbb{N} \\ M<p^j\le Z}} \frac{|H(p^j)|^2}{p^j} \right) \left(\sum_{\substack{p \text{ prime}, j\in\mathbb{N} \\ M<p^j\le Z}} \frac{1}{p^j} \right) \\ &\le LB_H(Z), \qquad L := \log(2C_E) + 1.\end{aligned}$$

The last line uses (5.2), whose hypotheses $Z \ge 4$ and $M \ge \sqrt{Z}$ hold here. Since H is additive and $M \ge \sqrt{Z} \ge 2$, the Turán–Kubilius inequality (5.1) applies at M. Also $B_H(M) \le B_H(Z)$ because $M \le Z$. Writing $H(n) - A_H(Z) = (H(n) - A_H(M)) + (A_H(M) - A_H(Z))$ and using $|a+b|^2 \le 2|a|^2 + 2|b|^2$, we obtain

$$\begin{aligned}\mathbb{E}_{n\in[M]}|H(n) - A_H(Z)|^2 &\le 2\mathbb{E}_{n\in[M]}|H(n) - A_H(M)|^2 + 2|A_H(M) - A_H(Z)|^2 \\ &\le 2C_{\mathrm{TK}}B_H(M) + 2LB_H(Z) \\ &\le 2(C_{\mathrm{TK}} + L)B_H(Z) \le C\Delta,\end{aligned}$$

where $C = 12\pi^2(C_{\mathrm{TK}} + L)$ is absolute, by (5.4).

Set $\zeta = e^{iA_H(Z)} \in \mathbb{S}^1$. For real a, b, the inequality $|e^{ia} - e^{ib}| = 2|\sin((a-b)/2)| \le |a-b|$ gives

$$\mathbb{E}_{n\in[M]}|h(n) - \zeta|^2 \le \mathbb{E}_{n\in[M]}|H(n) - A_H(Z)|^2 \le C\Delta.$$

This is (5.3). The scalar ζ depends only on h and Z, so it works for every integer M in the stated range. □

Let $D \ge 1$ be an integer. A weight $v : \mathbb{Z}/D\mathbb{Z} \to [0,\infty)$ has mean one and is unit-invariant when

$$D^{-1} \sum_{a \bmod D} v(a) = 1, \qquad v(ua) = v(a) \quad (u \in (\mathbb{Z}/D\mathbb{Z})^\times). \tag{5.5}$$

Lemma 5.3. *There is an absolute constant $C \ge 1$ such that, whenever (5.5) holds, $Z \ge 4D^2$, and $y \ge D$, the scalar ζ in Theorem 5.2 satisfies*

$$\frac{1}{Z} \sum_{a\le Z} v(a)|h(a) - \zeta|^2 \le C\frac{D}{\varphi(D)}\Delta. \tag{5.6}$$

Proof. We divide the proof into four steps.

Step 1. Decompose the unit-invariant weight into gcd-orbits. Regard v as a D-periodic function on the integers. For each divisor $d \mid D$, put

$$Q_d = \frac{D}{d}, \qquad \mathcal{O}_d = \{a \bmod D : \gcd(a, D) = d\}.$$

We first show that the sets $\mathcal{O}_d$ are precisely the orbits of multiplication by units modulo D. Multiplication by a unit preserves the gcd with D. Conversely, suppose $d < D$ and write two residues in $\mathcal{O}_d$ as da_1 and da_2, with a_1, a_2 units modulo Q_d. Set $u_0 \equiv a_2a_1^{-1} \pmod{Q_d}$. For each $p^e \| D$, put $f = v_p(Q_d)$. If $f \ge 1$, choose a lift of $u_0 \bmod p^f$ to p^e; it remains nonzero modulo p. If $f = 0$, choose the residue $1 \bmod p^e$. The Chinese remainder theorem, applied to these pairwise coprime prime powers, gives a unit $u \bmod D$ with $u \equiv u_0 \pmod{Q_d}$. Thus $u(da_1) \equiv da_2 \pmod D$. For $d = D$, the set $\mathcal{O}_D$ is the singleton zero residue.

The map $t \bmod Q_d \mapsto dt \bmod D$ is a bijection from the units modulo Q_d to $\mathcal{O}_d$, since

$$\gcd(dt, D) = d\gcd(t, Q_d), \qquad dt_1 \equiv dt_2 \pmod{D} \iff t_1 \equiv t_2 \pmod{Q_d}. \tag{5.7}$$

Hence $|\mathcal{O}_d| = \varphi(Q_d)$, including $d = D$ by $\varphi(1) = 1$. Unit invariance makes v constant on each $\mathcal{O}_d$. Define its normalized mass by

$$M_d = \frac{1}{D}\sum_{a\in\mathcal{O}_d} v(a) \geq 0, \qquad \sum_{d|D} M_d = 1, \qquad v(a) = \frac{DM_d}{\varphi(Q_d)} \quad (a \bmod D \in \mathcal{O}_d). \tag{5.8}$$

The equality $\sum_{d|D} M_d = 1$ follows from (5.5), since the orbits partition $\mathbb{Z}/D\mathbb{Z}$. The formula for $v(a)$ uses $|\mathcal{O}_d| = \varphi(Q_d)$.

Step 2. Reduce the weighted sum to long prefixes. For each $d \mid D$, set $Y_d = Z/d$ and $m_d = \lfloor Y_d \rfloor$. Since $Z \geq 4D^2$ and $d \leq D$, we have

$$m_d \geq Z/d - 1 \geq Z/D - 1 \geq 2\sqrt{Z} - 1 \geq \sqrt{Z}, \qquad m_d \leq Z/d \leq Z. \tag{5.9}$$

Every prime factor of d is at most $D \leq y$, so $h(d) = 1$. Complete multiplicativity gives $h(dt) = h(t)$. Put $F(t) = |h(t) - \zeta|^2 \geq 0$. By (5.7), the integers $a \leq Z$ with $\gcd(a, D) = d$ are exactly $a = dt$ with $t \leq Y_d$ and $(t, Q_d) = 1$. Thus (5.8) and $D/Z = Q_d/Y_d$ give

$$\frac{1}{Z}\sum_{a\leq Z} v(a)F(a) = \sum_{d|D} \frac{DM_d}{Z\varphi(Q_d)} \sum_{\substack{t\leq Y_d \\ (t,Q_d)=1}} F(t) = \sum_{d|D} M_d \frac{Q_d}{\varphi(Q_d)} \frac{1}{Y_d} \sum_{\substack{t\leq Y_d \\ (t,Q_d)=1}} F(t). \tag{5.10}$$

Step 3. Apply prefix concentration. Since $F(t) \geq 0$, dropping the restriction $(t, Q_d) = 1$ can only increase the sum. The integers $t \leq Y_d$ are exactly those $t \leq m_d$, and $m_d \leq Y_d$. Therefore

$$\frac{1}{Y_d} \sum_{\substack{t\leq Y_d \\ (t,Q_d)=1}} F(t) \leq \frac{1}{Y_d} \sum_{t\leq m_d} F(t) \leq \mathbb{E}_{t\in[m_d]} F(t) \leq C\Delta. \tag{5.11}$$

The last inequality follows from Theorem 5.2, since (5.9) places every m_d in $[\sqrt{Z}, Z]$. That lemma supplies the same ζ for all these interval lengths.

Step 4. Use the totient comparison and conclude. Finally, $Q_d \mid D$, so the totient product formula gives

$$\frac{Q_d}{\varphi(Q_d)} = \prod_{\substack{p|Q_d \\ p \text{ prime}}} (1 - p^{-1})^{-1} \leq \prod_{\substack{p|D \\ p \text{ prime}}} (1 - p^{-1})^{-1} = \frac{D}{\varphi(D)}, \tag{5.12}$$

with an empty product equal to 1. Substituting (5.11) and (5.12) into (5.10), and using $\sum_{d|D} M_d = 1$, yields

$$\frac{1}{Z}\sum_{a\leq Z} v(a)|h(a) - \zeta|^2 \leq C\Delta \sum_{d|D} M_d \frac{Q_d}{\varphi(Q_d)} \leq C\frac{D}{\varphi(D)}\Delta.$$

This proves (5.6) with the constant from Theorem 5.2. □

For the two-variable comparison, let $D \geq 1$ be an integer and let $V : (\mathbb{Z}/D\mathbb{Z})^2 \to [0, \infty)$ have mean one:

$$D^{-2} \sum_{a,b \bmod D} V(a, b) = 1.$$

Assume $V(ua, ub) = V(a, b)$ for every unit u modulo D. Extend V periodically to $\mathbb{Z}^2$. Using the fixed w, C_0 of (4.39) and (4.43), for each integer $X \geq 1$ define $I_{X,V}$ and $Z_{X,V} = I_{X,V}(1)$

by (4.44) with V in place of W_0. Set $K_{X,V} = I_{X,V}/Z_{X,V}$ whenever $Z_{X,V} > 0$. The quadrature estimate (4.46) gives

$$|Z_{X,V} - 1| \le A_w \left(\frac{D}{X} + \frac{D^2}{X^2} \right),$$

where $A_w > 0$ depends only on w, uniformly in D and V. Put $B_w = \max\{1, 4A_w\}$. If $X \ge B_w D$, then $D/X \le 1/B_w \le 1$, so

$$|Z_{X,V} - 1| \le \frac{2A_w}{B_w} \le \frac{1}{2}, \qquad \frac{1}{2} \le Z_{X,V} \le \frac{3}{2}. \tag{5.13}$$

Thus $K_{X,V}$ is defined for every integer $X \ge B_w D$. This sufficient lower bound for X depends on V only through its chosen modulus D. The proposition below assumes the stronger condition $X \ge C_w D^2$, with $C_w \ge B_w$.

Proposition 5.4. *There is a constant C_w, depending only on w, such that whenever X is an integer and*

$$X \ge C_w D^2, \qquad Z = C_0 X, \qquad \max(2, D) \le y \le Z, \tag{5.14}$$

any two unit-modulus completely multiplicative functions f, g agreeing at every prime at most y satisfy

$$|K_{X,V}(f) - K_{X,V}(g)| \le C_w \sqrt{\frac{D}{\varphi(D)}} \left(\sum_{\substack{y < p \le Z \\ p \text{ prime}}} \frac{1 - \Re(f(p)\overline{g(p)})}{p} \right)^{1/2}. \tag{5.15}$$

Proof. We divide the proof into four steps.

Step 1. Fix the constants and define the probability measure. Let C_* be the absolute constant in Theorem 5.3. Set

$$A(w) = 2C_* \|w\|_\infty C_0(C_0 + 1), \qquad C_w \ge \max\{B_w, 4/C_0, 4\sqrt{A(w)}\}. \tag{5.16}$$

These constants depend only on the fixed w and its chosen C_0. Since $D \ge 1$, (5.14) gives $X \ge B_w D \ge D$ and $Z = C_0 X \ge 4D^2$. Also $Z_{X,V} \ge 1/2$ by (5.13). Set $h = f\overline{g}$ and let Δ denote the prime sum in (5.15). Then h is completely multiplicative, $|h| = 1$, and $h(p) = 1$ for every prime $p \le y$. Choose the scalar $\zeta \in \mathbb{S}^1$ from Theorem 5.2.

Step 2. Obtain marginal concentration for the four linear forms. For integer pairs $m > n > 0$, define

$$\nu_{X,V}(m,n) = \frac{w(m/X, n/X)V(m,n)}{X^2 Z_{X,V}}, \qquad \sum_{m>n>0} \nu_{X,V}(m,n) = 1.$$

The sum equals one by the definition of $Z_{X,V}$. Thus $\nu_{X,V}$ is a probability measure, depending only on w, V, X, and $K_{X,V}(\psi) = \mathbb{E}_{\nu_{X,V}} \psi(R(m,n))$ for $\psi = f, g$.

Put $\mathcal{L} = \{m-n, m+n, m, n\}$. For each $L \in \mathcal{L}$, use the coordinate map T_L and its inverse in the following table:

L	$T_L(m,n) = (a,b)$	$T_L^{-1}(a,b) = (m,n)$
$m-n$	$(m-n, n)$	$(a+b, b)$
$m+n$	$(m+n, n)$	$(a-b, b)$
m	(m, n)	(a, b)
n	(n, m)	(b, a).

Each map is linear with determinant ± 1, so it is a bijection of $\mathbb{Z}^2$ and of $(\mathbb{Z}/D\mathbb{Z})^2$. Define

$$V_L(a,b) = V(T_L^{-1}(a,b)), \qquad v_L(a) = \frac{1}{D} \sum_{b \bmod D} V_L(a,b).$$

The weight v_L is nonnegative and D-periodic. Linearity and the unit invariance of V give $V_L(ua, ub) = V_L(a, b)$ for every unit u mod D. The bijection modulo D and the substitution $b = uc$ therefore give

$$\frac{1}{D}\sum_{a \bmod D} v_L(a) = \frac{1}{D^2}\sum_{a,b \bmod D} V_L(a,b) = \frac{1}{D^2}\sum_{m,n \bmod D} V(m,n) = 1,$$
$$v_L(ua) = \frac{1}{D}\sum_{c \bmod D} V_L(ua, uc) = \frac{1}{D}\sum_{c \bmod D} V_L(a,c) = v_L(a).$$

Since $Z \geq 4D^2$ and $y \geq D$, Theorem 5.3 applies to every v_L and yields

$$\frac{1}{Z}\sum_{a\in[Z]} v_L(a)|h(a) - \zeta|^2 \leq C_* \frac{D}{\varphi(D)}\Delta \qquad (L \in \mathcal{L}). \tag{5.17}$$

The same ζ works for all four forms because it depends only on h and Z.

Step 3. Convert marginal bounds into second moments. On the support of $w(m/X, n/X)$, we have $m > n > 0$ and $m + n < Z$ by (4.43). Hence both $a = L(m, n)$ and its second coordinate b lie in $[Z]$. For fixed a, each b determines a unique integer pair, and each residue modulo D occurs at most $Z/D + 1$ times in $[Z]$. Thus, for $a \in [Z]$, nonnegativity and periodicity give

$$\begin{aligned} S_L(a) &:= \sum_{\substack{m,n\in\mathbb{Z}\\ m>n>0,\ L(m,n)=a}} w(m/X, n/X)V(m,n) \\ &\leq \|w\|_\infty \sum_{b\in[Z]} V_L(a,b) \\ &\leq \|w\|_\infty (Z/D+1)\sum_{b \bmod D} V_L(a,b) = \|w\|_\infty (Z+D)v_L(a). \end{aligned}$$

Grouping by $a = L(m, n)$ and applying (5.17), we obtain

$$\begin{aligned} \mathbb{E}_{\nu_{X,V}}|h(L) - \zeta|^2 &= \frac{1}{X^2 Z_{X,V}}\sum_{a\in[Z]} S_L(a)|h(a) - \zeta|^2 \\ &\leq \frac{\|w\|_\infty (Z+D)}{X^2 Z_{X,V}}\sum_{a\in[Z]} v_L(a)|h(a) - \zeta|^2 \\ &\leq C_* \frac{\|w\|_\infty (Z+D)Z}{X^2 Z_{X,V}}\frac{D}{\varphi(D)}\Delta \leq A(w)\frac{D}{\varphi(D)}\Delta. \end{aligned} \tag{5.18}$$

Here $h(L)$ denotes $h(L(m, n))$. The last inequality uses $Z = C_0 X$, $D \leq X$, and $Z_{X,V} \geq 1/2$:

$$\frac{(Z+D)Z}{X^2 Z_{X,V}} \leq \frac{2(C_0X + X)C_0X}{X^2} = 2C_0(C_0 + 1).$$

Step 4. Compare the rational phases and conclude. Since $y \geq 2$, we have $h(2) = 1$. Complete multiplicativity and the rational extension (1.4) give

$$h(R(m,n)) = \frac{h(m-n)h(m+n)}{h(2)h(m)h(n)} = h(m-n)h(m+n)\overline{h(m)}\,\overline{h(n)}. \tag{5.19}$$

For unit complex numbers α_j, β_j, we have

$$\prod_{j=1}^{4}\alpha_j - \prod_{j=1}^{4}\beta_j = \sum_{j=1}^{4}\left(\prod_{i<j}\beta_i\right)(\alpha_j - \beta_j)\left(\prod_{i>j}\alpha_i\right)$$

Taking absolute values bounds the left-hand side by $\sum_{j=1}^{4}|\alpha_j-\beta_j|$. Apply it to the four factors in (5.19) and to $(\beta_1,\beta_2,\beta_3,\beta_4)=(\zeta,\zeta,\overline{\zeta},\overline{\zeta})$. The latter four numbers have product $\zeta^2\overline{\zeta}^2=1$. Since $f(R)=g(R)h(R)$, $|g(R)|=1$, and conjugation preserves absolute differences, this gives

$$|f(R)-g(R)|=|h(R)-1|\le\sum_{L\in\mathcal{L}}|h(L)-\zeta|.$$

Taking expectations under the common measure $\nu_{X,V}$, then using Cauchy–Schwarz and (5.18), we conclude that

$$\begin{aligned}|K_{X,V}(f)-K_{X,V}(g)|&\le\mathbb{E}_{\nu_{X,V}}|f(R)-g(R)|\\&\le\sum_{L\in\mathcal{L}}\mathbb{E}_{\nu_{X,V}}|h(L)-\zeta|\\&\le\sum_{L\in\mathcal{L}}\left(\mathbb{E}_{\nu_{X,V}}|h(L)-\zeta|^2\right)^{1/2}\\&\le 4\sqrt{A(w)}\sqrt{\frac{D}{\varphi(D)}}\sqrt{\Delta}.\end{aligned}$$

The choice (5.16) proves (5.15). □

We now return to D_0,W_0,K_X from (4.30) and (4.44). Let χ be a primitive character of conductor q, let $t\in\mathbb{R}$, and let $g:\mathbb{N}\to\mathbb{S}^1$ be completely multiplicative. For $y\ge q$, define its finite model $\widetilde{g}:\mathbb{N}\to\mathbb{S}^1$ by complete multiplicativity and

$$\widetilde{g}(p)=\begin{cases}g(p), & p\le y,\\ \chi(p)p^{it}, & p>y.\end{cases}$$

Both functions extend uniquely to $\mathbb{Q}_{>0}$ by (1.4). We use these extensions, denoted by the same symbols, in $K_X(g)$ and $K_X(\widetilde{g})$.

Corollary 5.5. *There is a constant $C_w\ge 1$, depending only on w, with the following property. For each $0<\delta<1/10$, let D_0,W_0,K_X be defined by (4.30), (4.32), and (4.44), respectively. Here $C_0\ge 2$ is the fixed integer chosen in (4.43). One can choose $b_\delta>0$, depending only on δ and w, so that for all g,χ,t as above, every integer X, and every real y satisfying*

$$X\ge C_wD_0^2,\qquad Z=C_0X,\qquad \max(D_0,q,2)\le y\le Z,$$

we have

$$|K_X(g)-K_X(\widetilde{g})|\le b_\delta\left(\sum_{\substack{y<p\le Z\\ p\text{ prime}}}\frac{1-\Re(g(p)\overline{\chi(p)}p^{-it})}{p}\right)^{1/2},\tag{5.20}$$

$$b_\delta^2\le C_w\log(2/\delta).$$

Proof. We divide the proof into three steps.

Step 1. Choose the model constants. Let $P_w\ge 1$ be a fixed constant valid in Theorem 5.4, and let C_E be the constant in Theorem 5.1. Choose

$$C_w=\max\{1,P_w,6C_EP_w^2\},\qquad b_\delta=P_w\sqrt{\frac{D_0}{\varphi(D_0)}}.\tag{5.21}$$

Since D_0 depends only on δ, these choices have the stated dependence.

Step 2. Verify the model and apply the comparison proposition. For $p>y\ge q$, we have $p\nmid q$, so $|\chi(p)p^{it}|=1$. For $p\le y$, the prescribed value is $\widetilde{g}(p)=g(p)$. Thus complete multiplicativity defines $\widetilde{g}$ with modulus one on $\mathbb{N}$, and $g,\widetilde{g}$ agree at every prime at most y. This includes primes dividing q, where the definition uses $g(p)$.

By Theorem 4.4, W_0 is nonnegative and periodic modulo D_0 in each variable, with

$$\frac{1}{D_0^2}\sum_{a,b \bmod D_0} W_0(a,b) = 1, \qquad W_0(ua,ub) = W_0(a,b) \quad \big(u \in (\mathbb{Z}/D_0\mathbb{Z})^\times\big).$$

Since $C_w \ge P_w$, we have $X \ge C_w D_0^2 \ge P_w D_0^2$. Together with $\max(D_0,2) \le y \le Z$, this verifies the size assumptions of Theorem 5.4. We apply that proposition with $D = D_0$ and $V = W_0$ to $g, \widetilde{g}$. For the prime sum, we have

$$g(p)\overline{\widetilde{g}(p)} = g(p)\overline{\chi(p)p^{it}} = g(p)\overline{\chi(p)}p^{-it} \qquad (p > y),$$

$$\Delta := \sum_{\substack{y<p\le Z\\ p \text{ prime}}} \frac{1 - \Re(g(p)\overline{\chi(p)}p^{-it})}{p}.$$

The proposition therefore yields

$$|K_X(g) - K_X(\widetilde{g})| \le P_w\sqrt{\frac{D_0}{\varphi(D_0)}}\sqrt{\Delta} = b_\delta\sqrt{\Delta}. \tag{5.22}$$

Step 3. Estimate the totient factor. It remains to bound b_δ. By (4.30), $D_0 = \prod_{\substack{p\le B\\ p \text{ prime}}} p^{2T_0+5}$ with $B = 40000\delta^{-2}$. Euler's totient formula and Theorem 5.1 give

$$\frac{D_0}{\varphi(D_0)} = \prod_{\substack{p\le B\\ p \text{ prime}}} \frac{p^{2T_0+5}}{p^{2T_0+4}(p-1)} = \prod_{\substack{p\le B\\ p \text{ prime}}} (1-p^{-1})^{-1} \le C_E\log(2B). \tag{5.23}$$

Since $\delta < 1/10$ and $20000 < 20^4$, we have

$$\log(2B) = 2\log(2/\delta) + \log 20000 \le 2\log(2/\delta) + 4\log 20 \le 6\log(2/\delta). \tag{5.24}$$

Combining (5.21), (5.23), and (5.24), we obtain

$$b_\delta^2 = P_w^2\frac{D_0}{\varphi(D_0)} \le 6C_E P_w^2\log(2/\delta) \le C_w\log(2/\delta).$$

Together with (5.22), this proves (5.20). □

6. Quadratic correlations

The main result is Theorem 6.12: a unit-modulus completely multiplicative function with a large quadratic correlation on an arithmetic progression has small prime distance from a Dirichlet character times n^{it}. A prime-block decomposition produces correlated prime pairs, whose difference phases yield simultaneous rational approximations to the phase coefficients. Freezing the residual phase gives a large mean on a residue class, and a multiplicative mean-value theorem completes the argument.

6.1. Prime-block decomposition.

The following decomposition is based on the argument of Bourgain, Sarnak and Ziegler [2, Section 2]. We give its finite estimates, since their dependence on the correlation is used later. For $0 < \alpha < 1/10$, define

$$\begin{aligned} j_0 &= \left\lceil \frac{3\log(1/\alpha)}{\log(1+\alpha)} \right\rceil, \quad & j_1 &= \lceil j_0/\alpha \rceil, \\ x_j &= (1+\alpha)^j, & K &= \lceil x_{j_1+1} \rceil. \end{aligned} \tag{6.1}$$

Let P_j be the primes in $[x_j, x_{j+1})$ for $j_0 \le j \le j_1$. We use the abbreviations

$$s_j = |P_j|/x_j, \qquad S = \sum_{j=j_0}^{j_1} s_j, \qquad J = j_1 - j_0 + 1. \tag{6.2}$$

For a positive integer N, set

$$Q_j = \left\{ m \in \mathbb{N} : m \le N/x_{j+1},\ p \nmid m \text{ for every } p \in \bigcup_{i=j_0}^{j} P_i \right\}. \tag{6.3}$$

Lemma 6.1. *Let $0 < \alpha < 1/10$ and $N \in \mathbb{N}$, and use (6.1)–(6.3). The sets $P_jQ_j = \{pm : p \in P_j, m \in Q_j\}$ are pairwise disjoint subsets of $[N]$. Every element of P_jQ_j has a unique such representation. There is an absolute constant $C \ge 1$, independent of α, N, and the resulting parameters j_0, j_1, x_j, K, such that*

$$\sum_j |P_j||Q_j| \le N, \tag{6.4}$$

$$\left| [N] \setminus \bigcup_j P_jQ_j \right| \le C\alpha(1 + \log(1/\alpha))N + 2K, \tag{6.5}$$

$$s_j \le 2\alpha, \qquad S \le C(1 + \log(1/\alpha)),$$

$$J \le C\alpha^{-2}\log(1/\alpha), \qquad \log K \le C\alpha^{-1}\log(1/\alpha). \tag{6.6}$$

Proof. We use Theorems 3.4 and 3.5. All sums and unions indexed by j below range over $j_0 \le j \le j_1$. Put

$$L = \log(1/\alpha), \qquad A = x_{j_0}, \qquad B = x_{j_1+1}, \qquad K = \lceil B \rceil.$$

Then $L > 1$ and $2 < A < B$. We divide the rest proof into six steps.

Step 1: Bounds for s_j and S. The definition of j_0 gives $\log A = j_0 \log(1+\alpha) \ge 3L$, hence $A \ge \alpha^{-3}$. An interval of length h contains at most $h+1$ integers. Since $x_{j+1} - x_j = \alpha x_j$ and $x_j \ge A$, we obtain

$$|P_j| \le \alpha x_j + 1, \qquad s_j \le \alpha + x_j^{-1} \le \alpha + \alpha^3 \le 2\alpha. \tag{6.7}$$

Write $E(t) = \prod_{\substack{p \le t \\ p \text{ prime}}} (1 - p^{-1})^{-1}$ and $E(t^-) = \prod_{\substack{p < t \\ p \text{ prime}}} (1 - p^{-1})^{-1}$. By Theorem 3.4, there is an absolute $C_E \ge 1$ such that $\log t \le E(t) \le C_E \log t$ for $t \ge 2$. If t is prime, then $E(t^-) = (1 - t^{-1})E(t)$; otherwise $E(t^-) = E(t)$. Therefore

$$\tfrac{1}{2}\log t \le \tfrac{1}{2}E(t) \le E(t^-) \le C_E \log t, \qquad \prod_{\substack{A \le p < B \\ p \text{ prime}}} (1 - p^{-1})^{-1} = \frac{E(B^-)}{E(A^-)} \le 2C_E \frac{\log B}{\log A}.$$

For $0 < u < 1$, the identity $-\log(1-u) = \int_0^u (1-v)^{-1}\,dv$ gives $u \le -\log(1-u)$. Taking logarithms of the product above yields

$$\sum_{\substack{A \le p < B \\ p \text{ prime}}} \frac{1}{p} \le \log \prod_{\substack{A \le p < B \\ p \text{ prime}}} (1 - p^{-1})^{-1} \le \log(2C_E) + \log \frac{\log B}{\log A}.$$

Since $j_0/\alpha \le j_1 < j_0/\alpha + 1$,

$$\frac{\log A}{\log B} = \frac{j_0}{j_1 + 1} \le \alpha, \qquad \frac{\log B}{\log A} = \frac{j_1 + 1}{j_0} \le \frac{1}{\alpha} + \frac{2}{j_0} \le \frac{2}{\alpha}. \tag{6.8}$$

The last inequality uses $j_0 \ge 1$ and $2\alpha < 1$.

To estimate S, write $|P_j|/x_j$ as a sum of $1/x_j$ over $p \in P_j$. Each such prime satisfies $p < x_{j+1} = (1+\alpha)x_j$, so $1/x_j < (1+\alpha)/p$. Moreover, the intervals $[x_j, x_{j+1})$ partition $[A, B)$.

Thus the sets P_j are disjoint and contain each prime in $[A, B)$ exactly once. Summing the preceding comparison gives

$$\begin{aligned} S &= \sum_j \frac{|P_j|}{x_j} = \sum_j \sum_{p \in P_j} \frac{1}{x_j} \\ &\le (1+\alpha) \sum_j \sum_{p \in P_j} \frac{1}{p} = (1+\alpha) \sum_{\substack{A \le p < B \\ p \text{ prime}}} \frac{1}{p} \\ &\le 2 \log(2C_E) + 2 \log \frac{\log B}{\log A} \le C_S(1+L), \qquad C_S := 2\log(4C_E) + 2. \end{aligned} \tag{6.9}$$

The non-strict inequality includes empty sums. The last line uses $1 + \alpha < 2$ and (6.8).

Step 2: Disjoint product sets. Let $p \in P_j$, $m \in Q_j$, and $n = pm$. No prime in $\bigcup_{i=j_0}^{j} P_i$ divides m. Hence $v_p(n) = 1$, no other prime in P_j divides n, and no prime in P_i divides n when $j_0 \le i < j$. Also

$$n = pm < x_{j+1} m \le x_{j+1} \frac{N}{x_{j+1}} = N.$$

Thus $n \in [N]$. The integer n determines j as the least index i with $j_0 \le i \le j_1$ for which a prime in P_i divides n. It then determines p as its unique prime divisor in P_j, and $m = n/p$. Hence the representations are unique, the product sets are pairwise disjoint, and

$$\left| \bigcup_j P_j Q_j \right| = \sum_j |P_j Q_j| = \sum_j |P_j||Q_j| \le N.$$

This proves (6.4).

Step 3: The exceptional sets. For $n \in [N]$, define

$$T_j(n) = \sum_{p \in P_j} v_p(n).$$

The sum counts prime factors with multiplicity; for example, $p^2 \mid n$ with $p \in P_j$ implies $T_j(n) \ge 2$. Let $\mathcal{E}_0$ consist of the $n \in [N]$ with $T_j(n) = 0$ for every j. For $n \in [N] \setminus \mathcal{E}_0$, let $j(n)$ be the least index with $T_{j(n)}(n) > 0$, and let $\mathcal{E}_1$ consist of those n with $T_{j(n)}(n) \ge 2$.

For $n \in [N] \setminus (\mathcal{E}_0 \cup \mathcal{E}_1)$, there is a unique prime $p \in P_{j(n)}$ dividing n, and $v_p(n) = 1$. Set $m = n/p$. No prime in $\bigcup_{i=j_0}^{j(n)} P_i$ divides m, so $m \in Q_{j(n)}$ if and only if $m \le N/x_{j(n)+1}$. Step 2 shows that any representation of n in a product set must use this same index and prime. Let $\mathcal{E}_2$ consist of those $n \in [N] \setminus (\mathcal{E}_0 \cup \mathcal{E}_1)$ for which $m = n/p > N/x_{j(n)+1}$. It follows that $\mathcal{E}_0, \mathcal{E}_1, \mathcal{E}_2$ are disjoint and their union is $[N] \setminus \bigcup_j P_j Q_j$. In particular, if $N \notin \mathcal{E}_0$ and $T_{j(N)}(N) = 1$, then $m = N/p > N/x_{j(N)+1}$, so $N \in \mathcal{E}_2$.

Step 4: The first two exceptional sets. The integers in $\mathcal{E}_0$ have no prime divisor in $[A, B)$. Apply Theorem 3.5: its hypotheses $2 \le A < B$ and $N \ge 1$ hold, and its integer parameter is $\lceil B \rceil = K$. Writing C_R for its absolute constant and using (6.8), we obtain

$$|\mathcal{E}_0| \le C_R N \frac{\log A}{\log B} + K \le C_R \alpha N + K.$$

For $n \in \mathcal{E}_1$, the inequality $T_{j(n)}(n) \ge 2$ gives primes $p, q \in P_{j(n)}$ with $pq \mid n$. Here $p = q$ is allowed and means $p^2 \mid n$. For fixed p, q, there are $\lfloor N/(pq) \rfloor$ such multiples. Summing over all j and all ordered pairs $p, q \in P_j$ therefore gives

$$|\mathcal{E}_1| \le \sum_j \sum_{p,q \in P_j} \left\lfloor \frac{N}{pq} \right\rfloor \le N \sum_j \left(\sum_{p \in P_j} \frac{1}{p} \right)^2 \le N \sum_j s_j^2 \le 2\alpha N \sum_j s_j = 2\alpha N S.$$

Repeated counting only increases this upper bound. The last two inequalities use $p \geq x_j$ for $p \in P_j$ and $s_j \leq 2\alpha$ from (6.7).

Step 5: The remaining exceptional set. For $n \in \mathcal{E}_2$, put $j = j(n)$ and write $n = pm$ as in Step 3. Then $N/x_{j+1} < m \leq N/p \leq N/x_j$. For each fixed $p \in P_j$, the number of possible integers m is at most the number of all integers in $(N/x_{j+1}, N/x_j]$. The latter is exactly the floor difference below:

$$\left\lfloor \frac{N}{x_j} \right\rfloor - \left\lfloor \frac{N}{x_{j+1}} \right\rfloor \leq \frac{N}{x_j} - \frac{N}{x_{j+1}} + 1 = \frac{\alpha N}{(1+\alpha)x_j} + 1 \leq \frac{\alpha N}{x_j} + 1.$$

We have dropped the divisibility conditions on m and enlarged the upper endpoint from N/p to N/x_j. The sets P_j are disjoint and all their elements are less than $B \leq K$, so

$$\sum_j |P_j| = \#\{p \text{ prime} : A \leq p < B\} \leq K.$$

Summing over p and j gives

$$|\mathcal{E}_2| \leq \sum_j |P_j| \left(\frac{\alpha N}{x_j} + 1 \right) \leq \alpha N S + K.$$

Thus, by Step 3 and (6.9),

$$\left| [N] \setminus \bigcup_j P_j Q_j \right| = |\mathcal{E}_0| + |\mathcal{E}_1| + |\mathcal{E}_2| \leq C_R \alpha N + 3\alpha N S + 2K \leq (C_R + 3C_S)\alpha(1+L)N + 2K.$$

This proves (6.5).

Step 6: The parameter bounds. The identity $\log(1+\alpha) = \int_0^\alpha (1+t)^{-1}\,dt$ gives $\alpha/2 \leq \log(1+\alpha) \leq \alpha$. Using $L > 1$, $\alpha < 1$, and the ceiling bounds, we obtain

$$j_0 \leq \frac{6L}{\alpha} + 1 \leq \frac{7L}{\alpha}, \qquad J \leq j_1 + 1 \leq \frac{j_0}{\alpha} + 2 \leq \frac{9L}{\alpha^2}.$$

Finally, $K = \lceil B \rceil \leq 2B$ since $B > 1$. Hence

$$\log K \leq \log 2 + \log B = \log 2 + (j_1 + 1)\log(1+\alpha) \leq \log 2 + \frac{9L}{\alpha} \leq \frac{10L}{\alpha}.$$

Taking $C \geq \max\{C_R + 3C_S, C_S, 10\}$ proves all the estimates. □

6.2. **Correlated prime pairs.** For $0 < \rho < 1/2$, set $\alpha = (\rho/C_*)^4$, where $C_* \geq 10$ is an absolute constant fixed in the next proof. The symbols P_j, Q_j, x_j, J, S, K now have the values specified in (6.1)–(6.3) for this choice of α.

For $N \geq K$, a one-bounded sequence $a : [N] \to \mathbb{C}$, and distinct primes $p, q \leq K$, define

$$n_{p,q} = \lfloor N/\max(p,q) \rfloor, \qquad R_{p,q} = \left| n_{p,q}^{-1} \sum_{m \leq n_{p,q}} a(pm)\overline{a(qm)} \right|.$$

Proposition 6.2. *There is an absolute constant $C \geq 1$ such that the following holds. Let $0 < \rho < 1/2$, let $N \geq 1$ be an integer, let $a : [N] \to \mathbb{C}$ be one-bounded, and let $\nu : \mathbb{N} \to \mathbb{C}$ be multiplicative with $|\nu(n)| \leq 1$ for every $n \in \mathbb{N}$. If*

$$\left| N^{-1} \sum_{n \leq N} \nu(n) a(n) \right| \geq \rho, \qquad N \geq \exp(C\rho^{-5}), \tag{6.10}$$

then $S > 0$ and some block P_j, with $x = x_j$, satisfies

$$|P_j| \geq \frac{\rho^2 x}{16J} \tag{6.11}$$

and the set of ordered distinct pairs

$$\mathcal{E} = \{(p,q) \in P_j^2 : p \neq q, R_{p,q} \geq \rho^2/(32S)\}$$

satisfies

$$|\mathcal{E}| \geq \frac{\rho^2}{32S}|P_j|^2 \geq \frac{\rho^6}{8192J^2S}x^2. \tag{6.12}$$

The absolute constant C is chosen sufficiently large so that the lower bound in (6.10) implies $N \geq 16K/\rho$; consequently, every $n_{p,q}$ defined above is positive. In particular, there is an absolute constant $c > 0$ such that

$$\begin{aligned} J &\leq C\rho^{-8}\log(2/\rho), \qquad S \leq C\log(2/\rho), \\ |\mathcal{E}| &\geq c\rho^{25}x^2, \qquad\quad R_{p,q} \geq c\rho^3 \quad ((p,q) \in \mathcal{E}). \end{aligned} \tag{6.13}$$

The number x also satisfies $x \leq K \leq \exp(C\rho^{-5})$.

Proof. All sums over j below range from j_0 to j_1. We divide the rest proof into six steps.

Step 1: Choice of constants. Let $C_0 \geq 1$ be an absolute constant valid in Theorem 6.1, and write $L = \log(1/\alpha)$. We first fix $C_* \geq 10$ so that $\alpha = (\rho/C_*)^4$ satisfies

$$C_0\alpha(1+L) \leq \rho/8, \qquad \sqrt{2}\alpha \leq \rho/4 \qquad (0 < \rho < 1/2).$$

This choice is uniform in ρ. Indeed, $\rho^3 \leq 1/8$ and $\log(1/\rho) \leq 1/\rho$ give

$$\frac{C_0\alpha(1+L)}{\rho} \leq \frac{C_0}{C_*^4}\left(\frac{1+4\log C_*}{8}+1\right), \qquad \frac{\sqrt{2}\alpha}{\rho} \leq \frac{\sqrt{2}}{8C_*^4},$$

and both right-hand sides tend to zero as $C_* \to \infty$. With C_* now fixed, (6.6) yields

$$\log\frac{16K}{\rho} \leq 4C_0C_*^4\rho^{-4}\log\frac{C_*}{\rho} + \log\frac{16}{\rho} \leq C_1\rho^{-5} \tag{6.14}$$

for an absolute C_1, again using $\log(1/\rho) \leq 1/\rho$. Choose the constant C in (6.10) at least C_1. Then $N \geq 16K/\rho$, so every $n_{p,q}$ is positive. For the omitted set $D = [N]\setminus\bigcup_j P_jQ_j$, Theorem 6.1 gives

$$|D| \leq C_0\alpha(1+L)N + 2K \leq \rho N/8 + \rho N/8 = \rho N/4. \tag{6.15}$$

Step 2: Decomposition and Cauchy–Schwarz. Extend a by zero outside $[N]$, and define

$$A_j = |P_j||Q_j|, \qquad B_j = \sum_{p\in P_j}\sum_{m\in Q_j}\nu(pm)a(pm),$$

$$r_j = \begin{cases} |P_j|^{-2}\displaystyle\sum_{\substack{p,q\in P_j\\ p\neq q}} R_{p,q}, & P_j \neq \varnothing, \\ 0, & P_j = \varnothing. \end{cases}$$

In particular $0 \leq r_j \leq 1$. Uniqueness of the representations pm and disjointness of the sets P_jQ_j, together with $|\nu(n)a(n)| \leq 1$, imply

$$\sum_j A_j \leq N, \qquad \rho N \leq \left|\sum_{n\leq N}\nu(n)a(n)\right| \leq |D| + \sum_j |B_j|. \tag{6.16}$$

For $p \in P_j$ and $m \in Q_j$, the definition of Q_j gives $p \nmid m$. Thus multiplicativity gives $\nu(pm) = \nu(p)\nu(m)$. Put $M_j = \lfloor N/x_j \rfloor$; then $Q_j \subseteq [M_j]$. Cauchy–Schwarz and $|\nu(m)| \leq 1$ give

$$|B_j|^2 \leq |Q_j|\sum_{m\in Q_j}\left|\sum_{p\in P_j}\nu(p)a(pm)\right|^2 \leq |Q_j|\sum_{m\leq M_j}\left|\sum_{p\in P_j}\nu(p)a(pm)\right|^2. \tag{6.17}$$

The last inequality only enlarges a sum of nonnegative terms.

Step 3: Diagonal and off-diagonal terms. On expanding the last square, the diagonal sum is at most $M_j|P_j| \le (N/x_j)|P_j|$. For $p \ne q$ in P_j, we have $n_{p,q} \le M_j$ because $\max(p,q) \ge x_j$. The zero extension makes the terms with $m > n_{p,q}$ vanish. Since $|\nu(p)\overline{\nu(q)}| \le 1$, the off-diagonal sum is therefore bounded in absolute value by

$$\sum_{\substack{p,q\in P_j\\ p\ne q}} \left| \sum_{m\le M_j} a(pm)\overline{a(qm)} \right| = \sum_{\substack{p,q\in P_j\\ p\ne q}} n_{p,q}R_{p,q} \le \frac{N}{x_j}|P_j|^2 r_j.$$

Using $s_j = |P_j|/x_j$ in (6.17), and then $\sqrt{u+v} \le \sqrt{u} + \sqrt{v}$ for $u, v \ge 0$, we obtain

$$|B_j| \le \sqrt{NA_j/x_j} + \sqrt{NA_j s_j r_j}. \tag{6.18}$$

Step 4: Total correlation. The definition of j_0 gives $x_{j_0} \ge \alpha^{-3}$. Hence

$$\sum_j x_j^{-1} \le \sum_{j\ge j_0} (1+\alpha)^{-j} = \frac{1+\alpha}{\alpha x_{j_0}} \le 2\alpha^2.$$

Applying Cauchy–Schwarz and $\sum_j A_j \le N$ to the first term in (6.18) gives

$$\sum_j \sqrt{A_j N/x_j} \le \left(\sum_j A_j\right)^{1/2} \left(N\sum_j x_j^{-1}\right)^{1/2} \le \sqrt{2}\alpha N \le \rho N/4.$$

The second term satisfies

$$\sum_j \sqrt{NA_j s_j r_j} \le \sqrt{N}\left(\sum_j A_j\right)^{1/2} \left(\sum_j s_j r_j\right)^{1/2} \le N\left(\sum_j s_j r_j\right)^{1/2}.$$

Combining these bounds with (6.15) and (6.16), we obtain

$$\rho N \le \rho N/4 + \rho N/4 + N\left(\sum_j s_j r_j\right)^{1/2}.$$

Consequently,

$$\sum_j s_j r_j \ge \rho^2/4. \tag{6.19}$$

Since $r_j \le 1$, this also gives $S = \sum_j s_j \ge \rho^2/4 > 0$.

Step 5: Selection of an interval and prime pairs. The indices with small s_j or small r_j satisfy

$$\sum_{s_j < \rho^2/(16J)} s_j r_j \le \frac{\rho^2}{16}, \qquad \sum_{r_j < \rho^2/(16S)} s_j r_j \le \frac{\rho^2}{16S}\sum_j s_j = \frac{\rho^2}{16}.$$

Here the first estimate uses $r_j \le 1$ and the fact that there are J indices. Removing both sets of indices from (6.19) leaves a sum at least $\rho^2/8 > 0$. Thus some j satisfies

$$s_j \ge \frac{\rho^2}{16J}, \qquad r_j \ge \frac{\rho^2}{16S}.$$

Set $x = x_j$ and $M = |P_j|$. The first inequality gives (6.11). Put $t = \rho^2/(32S)$. Let $\mathcal{E}$ be the set of ordered distinct pairs in P_j^2 with $R_{p,q} \ge t$, and put $u = |\mathcal{E}|/M^2$. Since $r_j \ge 2t > 0$, we have $M \ge 2$. Each summand defining r_j is at most 1, and it is less than t for ordered distinct pairs outside $\mathcal{E}$. Therefore

$$2t \le r_j \le \frac{|\mathcal{E}| + t\big(M(M-1) - |\mathcal{E}|\big)}{M^2} \le u + t.$$

It follows that $u \ge t$, and hence

$$|\mathcal{E}| \ge \frac{\rho^2}{32S}M^2 \ge \frac{\rho^2}{32S}\left(\frac{\rho^2 x}{16J}\right)^2 = \frac{\rho^6}{8192J^2S}x^2.$$

This proves (6.12).

Step 6: Bounds in terms of ρ. Write $\ell = \log(2/\rho)$. Since C_* is fixed, $L = 4\log(C_*/\rho) \le C_2\ell$ for an absolute C_2. The bounds for J and S in Theorem 6.1 therefore give

$$J \le C_0 C_*^8 \rho^{-8} L \le C_3 \rho^{-8}\ell, \qquad S \le C_0(1+L) \le C_4\ell,$$

where C_3, C_4 are absolute. Using $\ell \le 2/\rho$ in (6.12) and the definition of $\mathcal{E}$, we obtain

$$|\mathcal{E}| \ge \frac{\rho^{22}}{8192C_3^2C_4\ell^3}x^2 \ge \frac{\rho^{25}}{65536C_3^2C_4}x^2,$$

$$R_{p,q} \ge \frac{\rho^2}{32C_4\ell} \ge \frac{\rho^3}{64C_4} \qquad ((p,q)\in\mathcal{E}).$$

Finally, $x = x_j \le K$ by definition, and (6.14) gives $K \le \exp(C_1\rho^{-5})$. Enlarging C to at least $\max\{1, C_1, C_3, C_4\}$ and taking $c = \min\{(65536C_3^2C_4)^{-1}, (64C_4)^{-1}\}$ proves (6.13) and the asserted bound for K. Both constants are absolute. □

For the progression case, let D, N be positive integers, let $b \in \mathbb{Z}$ and $\xi, \beta \in \mathbb{R}$, and let $I \subseteq \mathbb{R}$ be an interval. Suppose $\mathcal{P} = I \cap \{n \in \mathbb{Z} : n \equiv b \pmod D\} \subseteq [N]$ is nonempty, and set $a(n) = \mathbf{1}_{\mathcal{P}}(n)e(\xi n + \beta n^2)$. The difference phase is

$$Q_{p,q}(m) = (p-q)\xi m + (p^2-q^2)\beta m^2.$$

Lemma 6.3. *Let $0 < \rho < 1/2$, and let $\nu : \mathbb{N} \to \mathbb{C}$ be a one-bounded multiplicative function. Suppose (6.10) holds for the sequence a above. Set $\alpha = (\rho/C_*)^4$ and $S = \sum_{j=j_0}^{j_1} |P_j|/x_j$ as in (6.2). Choose x and $\mathcal{E}$ by Theorem 6.2. Then $D \le 2/\rho$. For each $(p,q) \in \mathcal{E}$, there exist $a_0 \in \mathbb{Z}$ and positive integers d, M such that*

$$\{m \in [n_{p,q}] : pm, qm \in \mathcal{P}\} = \{a_0 + d\ell : 1 \le \ell \le M\}, \tag{6.20}$$

with

$$d \le D^2, \qquad |a_0| \le N/x + D^2, \qquad M \ge \frac{\rho^2 N}{128Sx}.$$

Moreover,

$$\sum_{m \le n_{p,q}} a(pm)\overline{a(qm)} = \sum_{\ell=1}^{M} e\big(Q_{p,q}(a_0 + d\ell)\big),$$

$$\left|\frac{1}{M}\sum_{\ell=1}^{M} e\big(Q_{p,q}(a_0 + d\ell)\big)\right| \ge \frac{\rho^2}{32S}. \tag{6.21}$$

Proof. We divide the proof into four steps.

Step 1: Bound for the modulus. The set $\mathcal{P}$ contains at most $N/D + 1$ integers, so the correlation hypothesis in (6.10) gives

$$\rho \le \frac{1}{N}\left|\sum_{n\le N} \nu(n)a(n)\right| \le \frac{|\mathcal{P}|}{N} \le \frac{1}{D} + \frac{1}{N}.$$

Here $|\nu(n)a(n)| \le 1$ and $a(n) = 0$ outside $\mathcal{P}$, which justifies the middle inequality. The lower bound for N implies $N \ge 2/\rho$, and hence $1/N \le \rho/2$. Therefore $1/D \ge \rho/2$, which proves $D \le 2/\rho$.

Step 2: The arithmetic progression. Fix $(p,q) \in \mathcal{E}$. Put $g_p = \gcd(p, D)$ and $g_q = \gcd(q, D)$, and define

$$d = \operatorname{lcm}\left(\frac{D}{g_p}, \frac{D}{g_q}\right) \le D^2.$$

By Theorem 6.2, $S > 0$, $n_{p,q} \ge 1$, and

$$\left|\sum_{m \le n_{p,q}} a(pm)\overline{a(qm)}\right| = n_{p,q}R_{p,q} \ge \frac{\rho^2}{32S} n_{p,q} > 0. \tag{6.22}$$

Hence some $m \in [n_{p,q}]$ satisfies $pm, qm \in \mathcal{P}$. Let m_1 be the least such integer.

Since $pm_1, qm_1 \equiv b \pmod D$, the congruences $pm \equiv b \pmod D$ and $qm \equiv b \pmod D$ are equivalent to $D \mid p(m - m_1)$ and $D \mid q(m - m_1)$. For each $r \in \{p, q\}$, put $g_r = \gcd(r, D)$. Then $\gcd(r/g_r, D/g_r) = 1$, so dividing by g_r and applying Euclid's lemma gives

$$D \mid r(m - m_1) \iff \frac{D}{g_r} \mid \frac{r}{g_r}(m - m_1) \iff \frac{D}{g_r} \mid m - m_1.$$

Thus the two congruences hold if and only if both D/g_p and D/g_q divide $m - m_1$. By the definition of their least common multiple d, this is equivalent to $d \mid m - m_1$, that is, $m \equiv m_1 \pmod d$.

Because $p, q > 0$, the restrictions $pm, qm \in I$ and $1 \le m \le n_{p,q}$ define an interval. The integers in this interval congruent to m_1 modulo d are therefore $m_1, m_1 + d, \dots, m_1 + (M-1)d$ for some $M \ge 1$. Setting $a_0 = m_1 - d$ proves (6.20).

Step 3: Bounds for a_0 and M. Since $1 \le m_1 \le n_{p,q}$,

$$|a_0| = |m_1 - d| \le n_{p,q} + d \le \frac{N}{x} + D^2,$$

where $n_{p,q} \le N/x$ because $p, q \ge x$. Recall that $K = \lceil x_{j_1+1} \rceil$ in (6.1). Since $x = x_j \le K$, Theorem 6.2 gives $N \ge 16K/\rho \ge 4x$. Also $p, q \in P_j \subset [x, 2x)$, so $N/\max(p,q) > N/(2x) \ge 2$. Applying $\lfloor y \rfloor \ge y/2$ for $y \ge 2$, we obtain

$$n_{p,q} = \left\lfloor \frac{N}{\max(p,q)} \right\rfloor \ge \frac{N}{2\max(p,q)} \ge \frac{N}{4x}.$$

The product $a(pm)\overline{a(qm)}$ has absolute value 1 for the M integers in (6.20) and is zero for all other $m \le n_{p,q}$. The triangle inequality and (6.22) therefore give

$$M \ge \left|\sum_{m \le n_{p,q}} a(pm)\overline{a(qm)}\right| = n_{p,q}R_{p,q} \ge \frac{\rho^2}{32S} n_{p,q} \ge \frac{\rho^2 N}{128Sx}.$$

Step 4: Phase identity and average. For every m in the progression, $pm, qm \in \mathcal{P}$, and therefore

$$a(pm)\overline{a(qm)} = e\left(\xi(pm - qm) + \beta\big((pm)^2 - (qm)^2\big)\right) = e\big(Q_{p,q}(m)\big).$$

All other terms with $m \le n_{p,q}$ vanish. Consequently, (6.21) gives the exact sum identity. Finally, the progression is contained in $[n_{p,q}]$, so $M \le n_{p,q}$. Using (6.22) once more,

$$\left|\frac{1}{M}\sum_{\ell=1}^{M} e\big(Q_{p,q}(a_0 + d\ell)\big)\right| = \frac{n_{p,q}R_{p,q}}{M} \ge R_{p,q} \ge \frac{\rho^2}{32S}.$$

This proves the final assertion in (6.21). □

6.3. **Simultaneous rational approximation.** For a positive integer R, let $\tau(R)$ be the number of its positive divisors. In this subsection, the letters p and q denote arbitrary integer variables; no primality assumption is made. We abbreviate $\|x\|_{\mathbb{R}/\mathbb{Z}}$ to $\|x\|$ within this subsection.

Lemma 6.4. *For every real $X \geq 1$ and integer $R \geq 1$,*

$$\#\{(p,q) \in ([X,2X] \cap \mathbb{Z})^2 : p \neq q,\ R \mid p^2 - q^2\} \leq \frac{8X^2\tau(R)}{R} \leq \frac{16X^2}{\sqrt{R}}. \tag{6.23}$$

Similarly,

$$\#\{(p,q) \in ([X,2X] \cap \mathbb{Z})^2 : p \neq q,\ R \mid p - q\} \leq \frac{4X^2}{R}. \tag{6.24}$$

Proof. We divide the proof into two steps.

Step 1. Count pairs with $R \mid p^2 - q^2$. Put $I = [X,2X] \cap \mathbb{Z}$. We first count the pairs in (6.23). For $(p,q) \in I^2$ with $p \neq q$, set

$$A = p - q, \qquad B = p + q.$$

Then

$$0 < |A| \leq X, \qquad 2X \leq B \leq 4X, \qquad p^2 - q^2 = AB.$$

The identities

$$p = \frac{A+B}{2}, \qquad q = \frac{B-A}{2}$$

show that $(p,q) \mapsto (A,B)$ is injective.

Suppose that $R \mid AB$. Define

$$g = \gcd(|A|, R), \qquad A_0 = \frac{A}{g}, \qquad R_0 = \frac{R}{g}.$$

Then $g \mid R$ and $\gcd(|A_0|, R_0) = 1$. The divisibility $R \mid AB$ implies $R_0 \mid A_0 B$. Euclid's lemma therefore gives

$$\frac{R}{g} = R_0 \mid B.$$

Thus each admissible pair is assigned its unique exact-gcd value $g = \gcd(|A|, R) \mid R$; for that value, $g \mid A$ and $(R/g) \mid B$.

Fix $g \mid R$. If we discard the exact condition $\gcd(|A|, R) = g$, the number of possible nonzero A is at most

$$2\left\lfloor \frac{X}{g} \right\rfloor \leq \frac{2X}{g}.$$

If we also discard the lower bound $B \geq 2X$, then the number of positive $B \leq 4X$ divisible by R/g is at most

$$\left\lfloor \frac{4X}{R/g} \right\rfloor = \left\lfloor \frac{4Xg}{R} \right\rfloor \leq \frac{4Xg}{R}.$$

Dropping these restrictions, as well as the parity and interval conditions, only enlarges the set being counted. Hence

$$\#\{(p,q) \in I^2 : p \neq q,\ R \mid p^2 - q^2\} \leq \sum_{g \mid R} 2\left\lfloor \frac{X}{g} \right\rfloor \left\lfloor \frac{4Xg}{R} \right\rfloor \leq \sum_{g \mid R} \frac{2X}{g}\frac{4Xg}{R} = \frac{8X^2\tau(R)}{R}.$$

Every positive divisor d of R pairs with R/d. At most one member of each pair is smaller than $\sqrt{R}$; if R is a square, the divisor $\sqrt{R}$ is counted only once. Therefore

$$\tau(R) \leq 2\sqrt{R},$$

and so

$$\frac{8X^2\tau(R)}{R} \le \frac{16X^2}{\sqrt{R}}.$$

This proves (6.23).

Step 2. Count pairs with $R \mid p-q$. For (6.24), put $h = p-q$. The conditions $p,q \in I$, $p \neq q$, and $R \mid h$ imply that h is a nonzero multiple of R with $|h| \le X$. There are at most

$$2\left\lfloor \frac{X}{R} \right\rfloor$$

such values of h. Once h is fixed, choosing q determines $p = q+h$, so there are at most $|I|$ corresponding pairs. Since

$$|I| \le \lfloor X \rfloor + 1 \le 2X \qquad (X \ge 1),$$

we obtain

$$\#\{(p,q) \in I^2 : p \neq q,\ R \mid p-q\} \le 2\left\lfloor \frac{X}{R} \right\rfloor |I| \le \frac{4X^2}{R}.$$

This proves (6.24). □

For the next proposition, fix an integer $H \ge 2$, a real $X \ge 1$, a positive integer N, and coefficients $\alpha, \beta \in \mathbb{R}$. Consider a set of pairs $\mathcal{E}$[7] satisfying

$$\mathcal{E} \subseteq \{(p,q) \in ([X,2X] \cap \mathbb{Z})^2 : p \neq q\}, \qquad |\mathcal{E}| \ge X^2/H, \tag{6.25}$$

with the following property. For each $(p,q) \in \mathcal{E}$, there are integers $1 \le r,d \le H$, an integer a, and a real $M > 0$ such that

$$|a| \le N/X + H, \qquad M \ge N/(HX), \tag{6.26}$$

and

$$\|rd^2(p^2-q^2)\beta\| \le H/M^2, \tag{6.27}$$

$$\|r\{d(p-q)\alpha + 2ad(p^2-q^2)\beta\}\| \le H/M. \tag{6.28}$$

The choices of r,d,a,M may depend on the pair (p,q).

Proposition 6.5. *Assume the pair and coefficient conditions (6.25)–(6.28). If $X \ge 8H$ and*

$$N \ge 10^9 H^{30} X^2,$$

then there are integers u,v and positive integers S,R such that

$$R \le 256H^9, \qquad \left|\beta - \frac{v}{R}\right| \le \frac{4H^4}{N^2},$$

$$S \le 1024H^{14}, \qquad \left|\alpha - \frac{u}{S}\right| \le \frac{520H^7}{N}.$$

Set $Q = RS \le 2^{18}H^{23}$. Then both approximating fractions can be written with denominator Q:

$$\frac{u}{S} = \frac{uR}{Q}, \qquad \frac{v}{R} = \frac{vS}{Q}.$$

Proof. For each $(p,q) \in \mathcal{E}$, fix parameters r,d,a,M satisfying (6.26)–(6.28). Put $I = [X,2X] \cap \mathbb{Z}$ and $\Delta_{p,q} = p^2 - q^2$. We divide the rest proof into three steps.

[7] In applying Theorem 6.12, set $X = x$, $\alpha = \xi$, and $a = a_0$; use the particular correlation-selected set $\mathcal{E}$ supplied by Theorem 6.2, with H chosen as in (6.42). Theorems 6.3 and 6.9 then verify the remaining hypotheses.

Step 1. Find a pair with large separation. Fix $q \in I$. The condition $|p-q| < X/(8H)$ restricts p to an interval of length $X/(4H)$. This interval contains at most $X/(4H)+1$ integers. Since $|I| \le X+1$, the number of such pairs in I^2 is at most

$$(X+1)\left(\frac{X}{4H}+1\right) \le \frac{X^2}{2H} + 2X \le \frac{3X^2}{4H}.$$

The last inequality uses $X \ge 8H$. Since $|\mathcal{E}| \ge X^2/H$, some $(p_0,q_0) \in \mathcal{E}$ satisfies

$$|p_0-q_0| \ge \frac{X}{8H}, \qquad |\Delta_{p_0,q_0}| = |p_0-q_0|(p_0+q_0) \ge \frac{X^2}{4H}. \tag{6.29}$$

For every pair in $\mathcal{E}$ we also have

$$|p-q| \le X, \qquad |\Delta_{p,q}| \le 4X^2.$$

Step 2. Approximate β. Denote the parameters chosen for (p_0,q_0) by r_0, d_0, a_0, M_0. By (6.27) and $M_0 \ge N/(HX)$,

$$\|r_0 d_0^2 \Delta_{p_0,q_0}\beta\| \le \frac{H}{M_0^2} \le \frac{H^3X^2}{N^2}.$$

Put $\Lambda_0 = r_0 d_0^2 \Delta_{p_0,q_0}$. Then $|\Lambda_0| \ge X^2/(4H)$. Choosing an integer m_0 nearest to $\Lambda_0\beta$ gives

$$\left|\beta - \frac{m_0}{\Lambda_0}\right| \le \frac{H^3X^2/N^2}{|\Lambda_0|} \le \frac{4H^4}{N^2}.$$

Write $m_0/\Lambda_0 = v/R$ in lowest terms, with $v \in \mathbb{Z}$, $R \ge 1$, and $\gcd(v,R)=1$. Since $r_0 d_0^2 \le H^3$ and $|\Delta_{p_0,q_0}| \le 4X^2$,

$$R \le 4H^3X^2, \qquad \beta = \frac{v}{R} + \beta_0, \qquad |\beta_0| \le \frac{4H^4}{N^2}. \tag{6.30}$$

For each $(p,q) \in \mathcal{E}$, combining (6.27) with (6.30) gives

$$\left\| \frac{r d^2 \Delta_{p,q} v}{R} \right\| \le \frac{H^3X^2}{N^2} + 4H^3X^2\frac{4H^4}{N^2} \le \frac{17H^7X^2}{N^2}.$$

Multiplication by the preliminary bound $R \le 4H^3X^2$ yields

$$R\frac{17H^7X^2}{N^2} \le \frac{68H^{10}X^4}{N^2} \le \frac{68}{10^{18}H^{50}} < 1,$$

where we used $N \ge 10^9 H^{30} X^2$. The rational number inside the norm has denominator dividing R. Any nonintegral rational with denominator dividing R has distance at least $1/R$ from the nearest integer. Hence the strict inequality forces it to be an integer. Thus $R \mid r d^2 \Delta_{p,q} v$; since $(v,R)=1$, Euclid's lemma gives

$$R \mid r d^2 \Delta_{p,q} \qquad ((p,q) \in \mathcal{E}).$$

Partition $\mathcal{E}$ according to the chosen values of (r,d). There are at most H^2 classes. Hence one class $\mathcal{E}_1$ satisfies

$$|\mathcal{E}_1| \ge \frac{|\mathcal{E}|}{H^2} \ge \frac{X^2}{H^3}.$$

Let (r_1,d_1) be its associated pair and put

$$R_1 = \frac{R}{\gcd(R, r_1 d_1^2)}.$$

The divisibility just proved implies $R_1 \mid \Delta_{p,q}$ for every $(p,q) \in \mathcal{E}_1$. Theorem 6.4 therefore gives

$$\frac{X^2}{H^3} \le |\mathcal{E}_1| \le \frac{16X^2}{\sqrt{R_1}},$$

so $R_1 \le 256H^6$. Consequently,

$$R \le r_1 d_1^2 R_1 \le 256H^9.$$

Together with (6.30), this proves the two assertions concerning β.

Step 3. Approximate α. By (6.28) and $M \ge N/(HX)$, for each $(p,q) \in \mathcal{E}$ we have

$$\|r\{d(p-q)\alpha + 2ad\Delta_{p,q}\beta\}\| \le \frac{H^2X}{N}.$$

Using $\|Rz\| \le R\|z\|$ and substituting $\beta = v/R + \beta_0$ gives

$$\|Rrd(p-q)\alpha\| \le \frac{RH^2X}{N} + 2Rrd|a|\,|\Delta_{p,q}|\,|\beta_0|.$$

The size assumption on N implies $N \ge HX$, so $|a| \le N/X + H \le 2N/X$. Using $rd \le H^2$, $|\Delta_{p,q}| \le 4X^2$, and (6.30), we obtain

$$\|Rrd(p-q)\alpha\| \le \frac{RH^2X}{N} + \frac{64RH^6X}{N} \le \frac{65RH^6X}{N}.$$

Apply this inequality to (p_0,q_0) and put $\Lambda_1 = Rr_0d_0(p_0-q_0)$. By (6.29),

$$\frac{RX}{8H} \le |\Lambda_1| \le RH^2X.$$

Choose an integer m_1 nearest to $\Lambda_1\alpha$. Then

$$\left|\alpha - \frac{m_1}{\Lambda_1}\right| = \frac{\|\Lambda_1\alpha\|}{|\Lambda_1|} \le \frac{65RH^6X/N}{RX/(8H)} = \frac{520H^7}{N}.$$

Write $m_1/\Lambda_1 = u/S$ in lowest terms, with $u \in \mathbb{Z}$, $S \ge 1$, and $\gcd(u,S) = 1$. Since $S \le |\Lambda_1|$,

$$S \le RH^2X \le 256H^{11}X, \qquad \alpha = \frac{u}{S} + \alpha_0, \qquad |\alpha_0| \le \frac{520H^7}{N}. \tag{6.31}$$

For each $(p,q) \in \mathcal{E}$, substitute $\alpha = u/S + \alpha_0$ into the bound for $\|Rrd(p-q)\alpha\|$. Using (6.31) and $rd|p-q| \le H^2X$ gives

$$\left\|\frac{Rrd(p-q)u}{S}\right\| \le \frac{65RH^6X}{N} + \frac{520RH^9X}{N} \le \frac{585RH^9X}{N}.$$

Using $R \le 256H^9$ and $S \le 256H^{11}X$,

$$S\frac{585RH^9X}{N} \le \frac{585\cdot 256^2H^{29}X^2}{N} < 1,$$

by $N \ge 10^9H^{30}X^2$. The rational number inside the norm has denominator dividing S. Any nonintegral rational with denominator dividing S has distance at least $1/S$ from the nearest integer. Hence the strict inequality forces it to be an integer. Thus $S \mid Rrd(p-q)u$; since $(u,S) = 1$, Euclid's lemma gives

$$S \mid Rrd(p-q) \qquad ((p,q) \in \mathcal{E}).$$

Again partition $\mathcal{E}$ according to the chosen values of (r,d). Choose a class $\mathcal{E}_2$ satisfying

$$|\mathcal{E}_2| \ge \frac{|\mathcal{E}|}{H^2} \ge \frac{X^2}{H^3}.$$

Let (r_2,d_2) be its associated pair and set

$$S_1 = \frac{S}{\gcd(S, Rr_2d_2)}.$$

The divisibility just proved gives $S_1 \mid p-q$ throughout $\mathcal{E}_2$. Theorem 6.4 now yields

$$\frac{X^2}{H^3} \le |\mathcal{E}_2| \le \frac{4X^2}{S_1},$$

so $S_1 \le 4H^3$. Therefore

$$S \le Rr_2d_2S_1 \le (256H^9)(H^2)(4H^3) = 1024H^{14}.$$

Together with (6.31), this proves the two assertions concerning α.

Finally, with $Q = RS$,

$$Q \le (256H^9)(1024H^{14}) = 2^{18}H^{23},$$

and

$$\frac{u}{S} = \frac{uR}{Q}, \qquad \frac{v}{R} = \frac{vS}{Q}.$$

Thus both rational approximations have the common denominator Q. □

Lemma 6.6. *Let $R \ge 1$ be an integer and let $a, b \in \mathbb{Z}$ satisfy $\gcd(a, b, R) = 1$. For real $X \ge 1$,*

$$\#\{(p, q) \in ([X, 2X] \cap \mathbb{Z})^2 : R \mid ap + bq\} \le \frac{X^2}{R} + 2X + 1.$$

Proof. Put

$$I = [X, 2X] \cap \mathbb{Z}, \qquad \mathcal{S} = \{(p, q) \in I^2 : R \mid ap + bq\}.$$

Set

$$g = \gcd(a, R).$$

Then g is a positive integer with $1 \le g \le R$. Every common divisor of b and g also divides a, b, and R. Hence $\gcd(b, g) = 1$.

Take $(p, q) \in \mathcal{S}$. Since $g \mid R$, the divisibility $R \mid ap + bq$ implies $g \mid ap + bq$. Since $g \mid a$, we also have $g \mid ap$, and therefore $g \mid bq$. As $\gcd(b, g) = 1$, Euclid's lemma gives $g \mid q$.

Thus only multiples of g in I can occur as the second coordinate. The interval $[X, 2X]$ has length X, so it contains at most $X/g + 1$ multiples of g.

Fix one such q and write $q = gq'$. Dividing $R \mid ap + bq$ by g gives

$$\frac{a}{g}\, p + bq' \equiv 0 \pmod{R/g}.$$

Since

$$\gcd\left(\frac{a}{g}, \frac{R}{g}\right) = 1,$$

the congruence places p in one residue class modulo R/g.

An interval of length X contains at most $X/m + 1$ integers in one residue class modulo a positive integer m. With $m = R/g$, at most $Xg/R + 1$ values of p are possible. Consequently,

$$|\mathcal{S}| \le \left(\frac{X}{g} + 1\right)\left(\frac{Xg}{R} + 1\right) = \frac{X^2}{R} + \frac{X}{g} + \frac{Xg}{R} + 1 \le \frac{X^2}{R} + 2X + 1,$$

because $1 \le g \le R$, so $X/g \le X$ and $Xg/R \le X$.

The argument includes $a = 0$ and $R = 1$. If $a = 0$, then $g = \gcd(0, R) = R$ and the reduced modulus R/g is 1. If $R = 1$, then $g = 1$ and again the reduced modulus is 1. □

Lemma 6.7. *Let $H \ge 2$ be an integer, let $X \ge 32H$ be a real number, and let N be a positive integer satisfying*

$$N \ge 10^6 H^3 X^3.$$

Set $I = [X, 2X] \cap \mathbb{Z}$. Let $A, B \in \mathbb{R}$ and suppose that $\mathcal{E} \subseteq I^2$ satisfies

$$|\mathcal{E}| \ge X^2/H, \qquad \|pA + qB\|_{\mathbb{R}/\mathbb{Z}} \le HX/N \quad ((p, q) \in \mathcal{E}).$$

Then there exist integers a, b and a positive integer R satisfying $\gcd(a, b, R) = 1$ and $R \le 2H$ such that

$$\left|A - \frac{a}{R}\right| \le \frac{64H^2}{N}, \qquad \left|B - \frac{b}{R}\right| \le \frac{64H^2}{N}. \tag{6.32}$$

Proof. We divide the proof into four steps.

Step 1. Select two pairs with a nonzero determinant. The lower bound on $|\mathcal{E}|$ makes $\mathcal{E}$ nonempty. Choose $(p_1, q_1) \in \mathcal{E}$. For an integer $p \in I$, the real numbers q satisfying

$$|p_1 q - q_1 p| < \frac{X^2}{16H}$$

form an interval of length

$$\frac{2X^2}{16Hp_1} = \frac{X^2}{8Hp_1} \le \frac{X}{8H},$$

because $p_1 \ge X$. An interval of real length L contains at most $L+1$ integers, and $|I| \le X+1$. Therefore the strip

$$\mathcal{S} = \left\{(p,q) \in I^2 : |p_1 q - q_1 p| < \frac{X^2}{16H}\right\}$$

satisfies

$$|\mathcal{S}| \le (X+1)\left(\frac{X}{8H}+1\right) \le \frac{X^2}{4H} + 2X < \frac{X^2}{H}.$$

The first $\le$ follows from $(X-1)(X/(8H)+1) \ge 0$, and the final strict inequality from $X \ge 32H$. Since $|\mathcal{E}| \ge X^2/H$, choose $(p_2, q_2) \in \mathcal{E} \setminus \mathcal{S}$. Its determinant

$$\Delta = p_1 q_2 - p_2 q_1$$

then satisfies

$$\frac{X^2}{16H} \le |\Delta| \le 4X^2.$$

The upper bound follows from $p_i, q_i \in [X, 2X]$ and $|p_1 q_2 - p_2 q_1| \le \max(p_1 q_2, p_2 q_1) \le 4X^2$.

Step 2. Solve the two approximate linear equations. For each $i = 1, 2$, the hypothesis supplies an integer k_i with

$$|p_i A + q_i B - k_i| \le \frac{HX}{N}.$$

Put

$$\varepsilon_i = p_i A + q_i B - k_i.$$

Let (A_0, B_0) be the exact solution of

$$p_i A_0 + q_i B_0 = k_i \qquad (i = 1, 2).$$

Since $\Delta \ne 0$, Cramer's rule gives

$$A_0 = \frac{q_2 k_1 - q_1 k_2}{\Delta}, \qquad B_0 = \frac{-p_2 k_1 + p_1 k_2}{\Delta}.$$

It also gives

$$A - A_0 = \frac{q_2 \varepsilon_1 - q_1 \varepsilon_2}{\Delta}, \qquad B - B_0 = \frac{-p_2 \varepsilon_1 + p_1 \varepsilon_2}{\Delta}.$$

Since $p_i, q_i \le 2X$ and $|\Delta| \ge X^2/(16H)$,

$$|A - A_0|,\ |B - B_0| \le \frac{4X(HX/N)}{|\Delta|} \le \frac{64H^2}{N}. \tag{6.33}$$

Step 3. Put the exact solution in reduced rational form. To make the common denominator explicit, set

$$D = |\Delta|, \qquad \alpha = q_2 k_1 - q_1 k_2, \qquad \beta = p_1 k_2 - p_2 k_1, \qquad g = \gcd(|\alpha|, |\beta|, D).$$

Let

$$R = \frac{D}{g}, \qquad a = \operatorname{sgn}(\Delta)\frac{\alpha}{g}, \qquad b = \operatorname{sgn}(\Delta)\frac{\beta}{g}.$$

Then R is a positive integer, $\gcd(a,b,R)=1$, and the Cramer formulas give exactly

$$A_0=\frac{a}{R},\qquad B_0=\frac{b}{R}.$$

Moreover,

$$R\le D=|\Delta|\le 4X^2.$$

Step 4. Force divisibility and bound the denominator. For $(p,q)\in\mathcal{E}$, the triangle inequality on $\mathbb{R}/\mathbb{Z}$ and (6.33) give

$$\left\|\frac{pa+qb}{R}\right\|=\|pA_0+qB_0\|\le\frac{HX}{N}+(p+q)\frac{64H^2}{N}\le\frac{257H^2X}{N},$$

because $p+q\le 4X$ and $H\ge 2$. Put $C=257H^2X/N$. Using $R\le 4X^2$ and the hypothesis on N,

$$RC\le\frac{1028H^2X^3}{N}\le\frac{1028}{10^6H}<1.$$

Hence $C<1/R$. For an integer m, the rational number m/R has distance either 0 or at least $1/R$ from $\mathbb{Z}$: if $R\nmid m$, then $m-Rj$ is a nonzero integer for every $j\in\mathbb{Z}$, so $\|m/R\|=\min_j|m-Rj|/R\ge 1/R$. Applying this with $m=pa+qb$ forces

$$R\mid ap+bq\qquad((p,q)\in\mathcal{E}).$$

Theorem 6.6, applied to the primitive triple (a,b,R), now gives

$$\frac{X^2}{H}\le|\mathcal{E}|\le\frac{X^2}{R}+2X+1\le\frac{X^2}{R}+3X.$$

Since $X\ge 32H$, $3X\le 3X^2/(32H)$, and therefore

$$\frac{1}{R}\ge\frac{1}{H}-\frac{3}{32H}=\frac{29}{32H}.$$

Thus

$$R\le\frac{32H}{29}<2H.$$

Together with (6.33), this proves (6.32). □

6.4. **Quadratic exponential sums.** We record a quantitative equidistribution theorem. For the related notions, see [23].

Theorem 6.8 (Leng–Sah–Sawhney, [23, Theorem 5.2]). *Fix integers $\ell,k,s\ge 1$, $\delta\in(0,1/10)$, and positive integers M and d, together with $F:G/\Gamma\to\mathbb{C}$. Suppose that G is a dimension-d, at most s-step connected, simply connected nilpotent Lie group with a given degree-k filtration, and the nilmanifold G/Γ is complexity at most M with respect to this filtration. Let g be a polynomial sequence on G with respect to this filtration. Furthermore suppose that $\|F\|_{\mathrm{Lip}}\le 1$ and F has $G^{(s)}$-vertical frequency ξ[8] such that the height of ξ is bounded by M/δ. Suppose that*

$$N\ge(M/\delta)^{O_{k,\ell}(d^{O_{k,\ell}(1)})},\qquad\left|\mathbb{E}_{\widetilde{n}\in[N]^\ell}F(g(\widetilde{n})\Gamma)\right|\ge\delta.$$

There exists an integer $0\le r\le\dim(G/[G,G])$ such that:

- *there are horizontal characters $\eta_1,\dots,\eta_r:G\to\mathbb{R}$ whose heights are bounded by $(M/\delta)^{O_{k,\ell}(d^{O_{k,\ell}(1)})}$;*
- *for every $1\le i\le r$,*

$$\|\eta_i\circ g\|_{C^\infty[N]}\le(M/\delta)^{O_{k,\ell}(d^{O_{k,\ell}(1)})};$$

[8] Here $G^{(s)}$ denotes the s-th term of the lower central series: $G^{(1)}=G$ and $G^{(j+1)}=[G,G^{(j)}]$.

- *for any $w_1, \dots, w_s \in G/[G,G]$ such that the w_i are orthogonal to all of $\eta_1, \dots, \eta_r$, one has*
$$\xi([[[w_1, w_2], w_3], \dots, w_s]) = 0.$$

Now, we use the above theorem to imply the following lemma.

Lemma 6.9. *There is an absolute constant $C_{\mathrm{W}} \geq 1$ with the following property. Let $0 < \tau < 1/10$, let $M \in \mathbb{N}$ satisfy $M \geq \tau^{-C_{\mathrm{W}}}$, and let $A, B \in \mathbb{R}$. If $Q(n) = An + Bn^2$ satisfies*
$$\left| M^{-1} \sum_{n=1}^{M} e(Q(n)) \right| \geq \tau,$$
then there is an integer $1 \leq r \leq \tau^{-C_{\mathrm{W}}}$ with
$$M\|r(A+B)\|_{\mathbb{R}/\mathbb{Z}} \leq \tau^{-C_{\mathrm{W}}}, \qquad M^2\|2rB\|_{\mathbb{R}/\mathbb{Z}} \leq \tau^{-C_{\mathrm{W}}}. \tag{6.34}$$

Proof. We divide the proof into five steps.

Step 1. Realize the scalar phase on the circle nilmanifold. Set
$$G = \mathbb{R}, \qquad \Gamma = \mathbb{Z}, \qquad G_0 = G_1 = G_2 = \mathbb{R}, \qquad G_3 = \{0\}.$$
Choose an absolute positive integer $\mathfrak{M}_0$ that bounds the complexity of this filtered $\mathbb{R}/\mathbb{Z}$. We apply Theorem 6.8 with $\ell = 1$, $k = 2$, $s = 1$, $d = 1$, complexity bound $\mathfrak{M}_0$, and $N = M$. The sequence $g(n) = Q(n)$ is polynomial for the displayed degree-two filtration.

Step 2. Verify the vertical-frequency and correlation hypotheses. Choose $L_0 \geq 1$ so that $\|x \mapsto e(x)\|_{\mathrm{Lip}} \leq L_0$ on $\mathbb{R}/\mathbb{Z}$, and set
$$F(x+\Gamma) = \frac{e(x)}{L_0}, \qquad \delta = \frac{\tau}{L_0}, \qquad 0 < \delta < 1/10.$$
Then $\|F\|_{\mathrm{Lip}} \leq 1$, and
$$\left|\mathbb{E}_{n\in[M]} F(g(n)+\Gamma)\right| = \frac{1}{L_0}\left|\mathbb{E}_{n\in[M]} e(Q(n))\right| \geq \delta.$$
In this specialization, $G^{(1)} = G_2 = \mathbb{R}$. Define $\xi(z) = z$. It is integer-valued on $G \cap \Gamma = \mathbb{Z}$, has height one, and is nonzero. Moreover,
$$F(z + x + \Gamma) = e(z)F(x+\Gamma) = e(\xi(z))F(x+\Gamma).$$
Thus the vertical-frequency hypothesis of Theorem 6.8 is satisfied, and $1 \leq \mathfrak{M}_0/\delta$.

Step 3. Apply the quantitative inverse theorem. For the fixed values $(k, \ell, d) = (2, 1, 1)$, let C_N be an absolute exponent for the lower bound on N in Theorem 6.8. Let C_η be an absolute exponent for the bounds on the heights and the $C^\infty[M]$ norms of the horizontal characters in that theorem. Choose C_{W} so large that, for every $0 < \tau < 1/10$,
$$\left(\frac{\mathfrak{M}_0 L_0}{\tau}\right)^{\max\{C_N, C_\eta\}} \leq \tau^{-C_{\mathrm{W}}}.$$
Since $M \geq \tau^{-C_{\mathrm{W}}}$, the lower bound for N required by Theorem 6.8 holds. Theorem 6.8 then yields an integer $0 \leq q \leq 1$ and horizontal characters $\eta_1, \dots, \eta_q$ whose heights and the norms $\|\eta_j \circ g\|_{C^\infty[M]}$ are at most $\tau^{-C_{\mathrm{W}}}$. Because $s = 1$, the final conclusion of Theorem 6.8 says that ξ vanishes on
$$W := \bigcap_{j=1}^{q} \ker_{\mathbb{R}} \eta_j,$$
with $W = G$ when $q = 0$.

Step 4. Extract a nonzero integer frequency. If all of these horizontal characters were zero, including the case $q = 0$, then $W = G = \mathbb{R}$. This contradicts $\xi(1) = 1$ and the preceding

vanishing conclusion. Therefore $q = 1$ and η_1 is nonzero. Every horizontal character on $G = \mathbb{R}$ is linear, and integer-valuedness on $\Gamma = \mathbb{Z}$ gives $\eta_1(x) = rx$ for some $r \in \mathbb{Z} \setminus \{0\}$. Replacing r by $|r|$ preserves its height and all norms, so

$$1 \le r \le \tau^{-C_{\mathrm{W}}}.$$

Step 5. Read off the coefficient bounds. Finally,

$$n^2 = n + 2\binom{n}{2}, \qquad Q(n) = (A+B)\binom{n}{1} + 2B\binom{n}{2},$$

and hence

$$rQ(n) = r(A+B)\binom{n}{1} + 2rB\binom{n}{2}.$$

For the scalar polynomial $P(n) = \alpha_0 + \alpha_1\binom{n}{1} + \alpha_2\binom{n}{2}$, use the norm

$$\|P\|_{C^\infty[M]} = \max\left\{M\|\alpha_1\|_{\mathbb{R}/\mathbb{Z}}, M^2\|\alpha_2\|_{\mathbb{R}/\mathbb{Z}}\right\}.$$

Applying this norm to rQ and using the bound supplied by Theorem 6.8 gives

$$M\|r(A+B)\|_{\mathbb{R}/\mathbb{Z}} \le \tau^{-C_{\mathrm{W}}}, \qquad M^2\|2rB\|_{\mathbb{R}/\mathbb{Z}} \le \tau^{-C_{\mathrm{W}}},$$

which is exactly (6.34). □

We shall also need a bound for the linear coefficient A of $Q(n) = An + Bn^2$. If the two quantities in (6.34) are at most L, the triangle inequality on $\mathbb{R}/\mathbb{Z}$ gives

$$\|2rA\|_{\mathbb{R}/\mathbb{Z}} \le 2\|r(A+B)\|_{\mathbb{R}/\mathbb{Z}} + \|2rB\|_{\mathbb{R}/\mathbb{Z}} \le \frac{2L}{M} + \frac{L}{M^2} \le \frac{3L}{M}.$$

Hence the same multiplier $2r$ controls both A and B:

$$M\|2rA\|_{\mathbb{R}/\mathbb{Z}} \le 3L, \qquad M^2\|2rB\|_{\mathbb{R}/\mathbb{Z}} \le L. \tag{6.35}$$

6.5. **Means on residue classes.** For a multiplicative $u : \mathbb{N} \to \mathbb{C}$ with $|u| \le 1$ and real $V \ge 3$, $T \ge 1$, write

$$m(u, V, T) = \min_{|t| \le T} \sum_{\substack{p \le V \\ p \text{ prime}}} \frac{1 - \Re(u(p)p^{-it})}{p}.$$

Theorem 6.10 (Halász–Montgomery–Tenenbaum, [17, p. 1192]). *There is an absolute constant $C_H \ge 1$ such that, for every u, V, T just specified,*[9]

$$\left|V^{-1}\sum_{n \le V} u(n)\right| \le C_H\big((1 + m(u,V,T))e^{-m(u,V,T)} + T^{-1/2}\big).$$

Lemma 6.11. *There is an absolute constant $C \ge 1$ with the following property. Assume $0 < \kappa < 1/2$, $Q_0, N \in \mathbb{N}$, and $N \ge (2Q_0/\kappa)^C$. Let*

$$f : \mathbb{N} \longrightarrow \mathbb{S}^1$$

[9] The cited theorem is unnumbered. With $f = u$ and $x = V$, its quantity $M(V, S)$ is defined in [17, (1.2), p. 1192] by minimizing over $|t| \le 2S$. For $T \ge 2$, take $S = T/2$. Then $M(V, S) = m(u, V, T)$ and $S^{-1/2} = \sqrt{2}\,T^{-1/2}$. For $1 \le T < 2$, the trivial estimate $|V^{-1}\sum_{n \le V} u(n)| \le 1 \le \sqrt{2}\,T^{-1/2}$ suffices. Both cases therefore satisfy the displayed bound for a suitable absolute C_H.

be a completely multiplicative function. Suppose that $I \subseteq [N]$ is an interval and that b_0 is a residue class modulo Q_0 satisfying

$$\left| \sum_{n \in I,\ n \equiv b_0 \pmod{Q_0}} f(n) \right| \geq \kappa N.$$

Then there are a primitive character χ of conductor dividing Q_0 and $t \in \mathbb{R}$ such that

$$|t| \leq (Q_0/\kappa)^C, \qquad \sum_{\substack{p \leq N \\ p \text{ prime}}} \frac{1 - \Re(f(p)\overline{\chi(p)}p^{-it})}{p} \leq C \log(2Q_0/\kappa).$$

Proof. We divide the proof into seven steps.

Step 1. Reduce to a large prefix sum. For an integer $0 \leq Y \leq N$, write

$$S(Y) := \sum_{\substack{n \leq Y \\ n \equiv b_0 \pmod{Q_0}}} f(n).$$

Write the interval as $I = \{L+1, \dots, U_1\}$, where $0 \leq L < U_1 \leq N$. The assumed sum is $S(U_1) - S(L)$. Hence one of $S(U_1)$ and $S(L)$ has modulus at least $\kappa N/2$. Denote its endpoint by U. Since $|f(n)| = 1$, $|S(U)| \leq U$, and therefore

$$U \geq \frac{\kappa N}{2}.$$

Thus $U \leq N$ and

$$|S(U)| \geq \frac{\kappa N}{2} \geq \frac{\kappa U}{2}.$$

Step 2. Select an additive Fourier frequency. By additive-character orthogonality modulo Q_0,

$$S(U) = \frac{1}{Q_0} \sum_{b=0}^{Q_0-1} e(-bb_0/Q_0) \sum_{n \leq U} f(n) e(bn/Q_0).$$

The modulus of this average is at most the largest modulus of its inner sums. We may therefore fix $b \pmod{Q_0}$ such that

$$\left| \sum_{n \leq U} f(n) e(bn/Q_0) \right| \geq \frac{\kappa N}{2} \geq \frac{\kappa U}{2}. \tag{6.36}$$

Step 3. Remove the contribution of large Q_0-smooth factors. Every positive integer n has a unique factorization $n = dm$ in which every prime divisor of d divides Q_0 and $(m, Q_0) = 1$. Using complete multiplicativity of f, the sum in (6.36) is

$$\sum_{\substack{d \leq U \\ p \text{ prime},\ p \mid d \Rightarrow p \mid Q_0}} f(d) \sum_{\substack{m \leq U/d \\ (m, Q_0) = 1}} f(m) e(bdm/Q_0).$$

Put

$$C_{Q_0} := \prod_{\substack{p \mid Q_0 \\ p \text{ prime}}} (1 - p^{-1/2})^{-1}, \qquad D := \left\lceil \left(\frac{8 C_{Q_0}}{\kappa} \right)^2 \right\rceil.$$

For $d > D$ the inner sum has modulus at most U/d. Since $d^{-1} \leq D^{-1/2} d^{-1/2}$,

$$U \sum_{\substack{d > D \\ p \text{ prime},\ p \mid d \Rightarrow p \mid Q_0}} \frac{1}{d} \leq U D^{-1/2} \sum_{\substack{d \geq 1 \\ p \text{ prime},\ p \mid d \Rightarrow p \mid Q_0}} d^{-1/2} = U D^{-1/2} C_{Q_0} \leq \frac{\kappa U}{8}.$$

Subtracting the contribution of $d > D$ from (6.36), the terms with $d \le D$ have combined modulus at least $3\kappa U/8$. There are at most D such values of d, so for one of them, which we now fix,

$$\left| \sum_{\substack{m \le U/d \\ (m,Q_0)=1}} f(m) e(bdm/Q_0) \right| \ge \frac{\kappa U}{4D}.$$

Set $V = \lfloor U/d \rfloor$. Since $V \le U$, division by V gives

$$\left| V^{-1} \sum_{\substack{m \le V \\ (m,Q_0)=1}} f(m) e(bdm/Q_0) \right| \ge \frac{\kappa}{4D}.$$

Step 4. Choose a long coprime average. The Euler product satisfies $C_{Q_0} \le Q_0^{C_0}$ for an absolute constant C_0, and hence $D \le (2Q_0/\kappa)^{C_1}$ for an absolute C_1. Choose the absolute exponent C in the lemma large enough that $N \ge 12D/\kappa$ follows from $N \ge (2Q_0/\kappa)^C$. Since $U \ge \kappa N/2$ and $d \le D$,

$$\frac{U}{d} \ge \frac{\kappa N}{2D} \ge 6, \qquad V = \left\lfloor \frac{U}{d} \right\rfloor \ge \frac{U}{2d} \ge \frac{\kappa N}{4D} \ge 3.$$

Thus V lies in the range required by Theorem 6.10.

Step 5. Expand the coprime phase in characters. Let $\mathcal{X}(Q_0)$ be the set of Dirichlet characters modulo Q_0. Then $|\mathcal{X}(Q_0)| = \varphi(Q_0)$, where φ denotes Euler's totient function (and $\varphi(1) = 1$). For $\psi \in \mathcal{X}(Q_0)$, define

$$c_\psi = \frac{1}{\varphi(Q_0)} \sum_{a \in (\mathbb{Z}/Q_0\mathbb{Z})^\times} e(bda/Q_0)\overline{\psi(a)}.$$

Character orthogonality gives

$$\mathbf{1}_{(m,Q_0)=1} e(bdm/Q_0) = \sum_{\psi \in \mathcal{X}(Q_0)} c_\psi \psi(m), \qquad |c_\psi| \le 1.$$

Since $|\mathcal{X}(Q_0)| = \varphi(Q_0)$, the triangle inequality applied to the preceding average gives one $\psi \in \mathcal{X}(Q_0)$ for which

$$\left| V^{-1} \sum_{m \le V} f(m)\psi(m) \right| \ge r := \frac{\kappa}{4D\varphi(Q_0)} \ge (2Q_0/\kappa)^{-C_2} \tag{6.37}$$

for an absolute C_2.

Step 6. Apply the Halász–Montgomery–Tenenbaum estimate. The function $u = f\psi$ is multiplicative and satisfies $|u(n)| \le 1$; its possible zero values occur only when $(n, Q_0) > 1$. Apply Theorem 6.10 with

$$T := \left(\frac{4C_H}{r} \right)^2.$$

Let

$$m_* := \min_{|t| \le T} \sum_{\substack{p \le V \\ p \text{ prime}}} \frac{1 - \Re(f(p)\psi(p)p^{-it})}{p}.$$

The minimum exists because the sum is continuous in t and $[-T, T]$ is compact. Also $m_* \ge 0$. By (6.37) and Theorem 6.10,

$$r \le C_H\big((1 + m_*)e^{-m_*} + T^{-1/2}\big) = C_H(1 + m_*)e^{-m_*} + \frac{r}{4}.$$

Therefore

$$\frac{3r}{4} \le C_H(1+m_*)e^{-m_*} \le 2C_He^{-m_*/2},$$

where $(1+x)e^{-x} \le 2e^{-x/2}$ for $x \ge 0$. Taking logarithms gives

$$m_* \le 2\log\left(\frac{8C_H}{3r}\right) \le 2\log\left(\frac{4C_H}{r}\right).$$

Let t be a minimizing point. Then $|t| \le T$, and (6.37) implies, after enlarging the absolute exponent if necessary,

$$|t| \le \left(\frac{Q_0}{\kappa}\right)^C.$$

Moreover,

$$\sum_{\substack{p\le V\\ p \text{ prime}}} \frac{1-\Re(f(p)\psi(p)p^{-it})}{p} \le 2\log\left(\frac{4C_H}{r}\right).$$

Step 7. Pass to a primitive character and extend the prime range. Let χ be the unique primitive character of conductor $q_\chi \mid Q_0$ that induces $\overline{\psi}$; thus

$$\psi(n) = \overline{\chi(n)} \qquad \text{whenever } (n,Q_0)=1.$$

In particular, for $p \nmid Q_0$ the prime summands above agree. For $p \mid Q_0$, the summand involving $\psi(p)=0$ equals $1/p$, whereas the summand involving $\overline{\chi(p)}$ lies between 0 and $2/p$. Replacing ψ by $\overline{\chi}$ therefore increases the sum by at most

$$\sum_{\substack{p\mid Q_0\\ p \text{ prime}}} \frac{1}{p} \le \log(2Q_0).$$

It remains to extend the sum from $p \le V$ to $p \le N$. Each added summand is at most $2/p$, so

$$2\sum_{\substack{V<p\le N\\ p \text{ prime}}} \frac{1}{p} \le 2\sum_{V<n\le N} \frac{1}{n} \le 2\log(N/V) + \frac{2}{V}.$$

Since $V \ge \kappa N/(4D)$ and $V \ge 3$,

$$\log(N/V) \le \log(4D/\kappa) \ll \log(2Q_0/\kappa).$$

The lower bound for r in (6.37) also gives

$$\log(4C_H/r) \ll \log(2Q_0/\kappa).$$

Adding this estimate, the prime-divisor correction, and the contribution of primes $V < p \le N$ gives

$$\sum_{\substack{p\le N\\ p \text{ prime}}} \frac{1-\Re(f(p)\overline{\chi(p)}p^{-it})}{p} \le C\log(2Q_0/\kappa).$$

This is the required conclusion. □

6.6. **Scalar inverse theorem.** We now state the following theorem. Fix $0 < \rho < 1/2$ and $N \in \mathbb{N}$. Let $f : \mathbb{N} \to \mathbb{S}^1$ be completely multiplicative, let $\xi, \beta \in \mathbb{R}$, and let $\mathcal{P} \subseteq [N]$ be an arithmetic progression. Throughout this subsection, $\|z\| = \|z\|_{\mathbb{R}/\mathbb{Z}}$ for real z.

Theorem 6.12. *There is an absolute constant $C \geq 1$, independent of ρ, N, f, ξ, β, and $\mathcal{P}$, such that, if*

$$N \geq \exp(C\rho^{-5}), \qquad \left| N^{-1} \sum_{n \in \mathcal{P}} f(n) e(\xi n + \beta n^2) \right| \geq \rho,$$

then there are positive integers $Q, Q_0 \leq \rho^{-C}$, integers u, v, an interval $I \subseteq [N]$, and an integer b_0 satisfying

$$|\xi - u/Q| \leq \rho^{-C}/N, \qquad |\beta - v/Q| \leq \rho^{-C}/N^2, \tag{6.38}$$

$$\left| \sum_{\substack{n \in I \\ n \equiv b_0 \pmod{Q_0}}} f(n) \right| \geq \rho^C N. \tag{6.39}$$

There are also a primitive Dirichlet character χ and a real t with

$$q_\chi,\ |t| \leq \rho^{-C}, \qquad \sum_{\substack{p \leq N \\ p \text{ prime}}} \frac{1 - \Re(f(p)\overline{\chi(p)}p^{-it})}{p} \leq C \log(2/\rho). \tag{6.40}$$

Proof. The constant C in the statement is fixed at the end of the proof. We record below the finitely many absolute lower bounds it must satisfy. We divide the rest proof into six steps.

Step 1. Find many correlated prime pairs. Let D be the positive common difference of $\mathcal{P}$, with $D = 1$ if $\mathcal{P}$ is a singleton. Let $b_* = \min \mathcal{P}$, and let $I_* = [\min \mathcal{P}, \max \mathcal{P}] \cap \mathbb{Z}$ be the smallest integer interval containing $\mathcal{P}$. Then

$$\mathcal{P} = \{n \in I_* : n \equiv b_* \pmod{D}\}, \qquad a(n) = \mathbf{1}_{\mathcal{P}}(n) e(\xi n + \beta n^2) \quad (n \in [N]).$$

The hypothesis ensures that $\mathcal{P}$ is nonempty. Also $|a(n)| \leq 1$, and f is multiplicative with $|f(n)| = 1$.

We recall the parameters of Theorem 6.2. Let $C_* \geq 10$ be the fixed absolute constant used there, and define

$$\alpha = (\rho/C_*)^4, \qquad j_0 = \left\lceil \frac{3\log(1/\alpha)}{\log(1+\alpha)} \right\rceil, \qquad j_1 = \lceil j_0/\alpha \rceil,$$

$$x_i = (1+\alpha)^i, \qquad P_i = \{p \text{ prime} : x_i \leq p < x_{i+1}\} \quad (j_0 \leq i \leq j_1),$$

$$J = j_1 - j_0 + 1, \qquad S = \sum_{i=j_0}^{j_1} \frac{|P_i|}{x_i}, \qquad K = \lceil x_{j_1+1} \rceil.$$

Here j_0, j_1, J, K are positive integers. The integer J counts the prime blocks, S is the displayed sum of their sizes divided by their left endpoints, and K bounds every prime in these blocks. All these quantities depend only on ρ.

Let $C_{\mathrm{BSZ}} \geq 1$ be an absolute constant valid for both the size hypothesis and the parameter bounds in Theorem 6.2. Require $C \geq C_{\mathrm{BSZ}}$. Applying that proposition with $\nu = f$ gives $S > 0$ and an index j with $j_0 \leq j \leq j_1$. Put $x = x_j$; then $P_j \subseteq [x, 2x]$ and $x \leq K$. For distinct $p, q \in P_j$, define

$$n_{p,q} = \left\lfloor \frac{N}{\max(p,q)} \right\rfloor, \qquad R_{p,q} = \frac{1}{n_{p,q}} \left| \sum_{m=1}^{n_{p,q}} a(pm)\overline{a(qm)} \right|,$$

$$\mathcal{E} = \{(p,q) \in P_j^2 : p \neq q,\ R_{p,q} \geq \rho^2/(32S)\}.$$

The integer $n_{p,q} \ge 1$ is the largest m for which both pm and qm lie in $[N]$; $R_{p,q}$ measures the correlation of the two sequences on this interval. The proposition gives

$$|\mathcal{E}| \ge \frac{\rho^6 x^2}{8192 J^2 S}, \qquad R_{p,q} \ge \frac{\rho^2}{32S} \quad ((p,q) \in \mathcal{E}). \tag{6.41}$$

In particular, $\mathcal{E}$ is nonempty.

By Theorem 6.3, $D \le 2/\rho$. Let C_{W} be the absolute constant in Theorem 6.9. Choose a real τ and a positive integer H by

$$\begin{aligned} \tau &= \min\{\rho^2/(32S), 1/100\}, \\ H &= \left\lceil \max\left\{2, \frac{4}{\rho^2}, \frac{8192 J^2 S}{\rho^6}, \frac{128S}{\rho^2}, 3\tau^{-C_{\mathrm{W}}}\right\}\right\rceil. \end{aligned} \tag{6.42}$$

The number τ is a positive lower bound for the correlations in (6.41), with $\tau < 1/10$ as required by Theorem 6.9. The integer H combines the bounds for D^2, $x^2/|\mathcal{E}|$, $128S/\rho^2$, and $3\tau^{-C_{\mathrm{W}}}$ that will be used below. The estimates

$$J \le C_{\mathrm{BSZ}} \rho^{-8} \log(2/\rho), \qquad S \le C_{\mathrm{BSZ}} \log(2/\rho), \qquad \tau^{-1} \le 100 + 32S/\rho^2$$

give fixed absolute constants $A_*, C_2 \ge 1$ such that

$$H \le \rho^{-A_*}, \qquad K \le \exp(C_2 \rho^{-5}). \tag{6.43}$$

In particular, $D^2 \le H$ and $|\mathcal{E}| \ge x^2/H$.

Fix $(p,q) \in \mathcal{E}$. By Theorem 6.3, the integers $m \in [n_{p,q}]$ with $pm, qm \in \mathcal{P}$ form an arithmetic progression. Write M for its number of terms, d for its positive common difference, and $a_0 + d$ for its first term. Here a_0 is an integer and d, M are positive integers; they may depend on (p,q). We have

$$\begin{aligned} &\{m \in [n_{p,q}] : pm, qm \in \mathcal{P}\} = \{a_0 + d\ell : 1 \le \ell \le M\}, \\ &d \le D^2 \le H, \qquad |a_0| \le N/x + H, \qquad M \ge \frac{\rho^2 N}{128 S x} \ge \frac{N}{Hx}. \end{aligned} \tag{6.44}$$

Define the real quadratic polynomial $Q_{p,q}(y) = (p-q)\xi y + (p^2 - q^2)\beta y^2$. The sum identity in Theorem 6.3 gives

$$\left| \frac{1}{M} \sum_{\ell=1}^{M} e(Q_{p,q}(a_0 + d\ell)) \right| = \frac{n_{p,q}}{M} R_{p,q} \ge R_{p,q} \ge \frac{\rho^2}{32S} \ge \tau, \tag{6.45}$$

where $M \le n_{p,q}$ because the progression lies in $[n_{p,q}]$.

Step 2. Bound the polynomial coefficients. Require $C \ge C_3$, where $C_3 = \log(10^9) + 30A_* + 2C_2$. Then

$$N \ge 10^9 H^{30} x^2. \tag{6.46}$$

Indeed, (6.43) and $x \le K$ give

$$\log(10^9 H^{30} x^2) \le \log(10^9) + 30A_* \log(1/\rho) + 2C_2 \rho^{-5} \le C_3 \rho^{-5}$$

using $\log(1/\rho) \le \rho^{-5}$. Consequently $M \ge N/(Hx) \ge H \ge \tau^{-C_{\mathrm{W}}}$.

Fix any $(p,q) \in \mathcal{E}$, and set $A_1 = p - q$ and $A_2 = p^2 - q^2$. Expanding $Q_{p,q}$ gives

$$Q_{p,q}(a_0 + d\ell) = Q_{p,q}(a_0) + A\ell + B\ell^2, \quad A = dA_1\xi + 2a_0 d A_2 \beta, \quad B = d^2 A_2 \beta.$$

The constant term does not change the modulus in (6.45). Since $0 < \tau \le 1/100$ and $M \ge \tau^{-C_{\mathrm{W}}}$, Theorem 6.9 applies. It gives an integer $1 \le r' \le \tau^{-C_{\mathrm{W}}}$. Set $r = 2r'$. By (6.35) and $H \ge 3\tau^{-C_{\mathrm{W}}}$, we have $r \le H$ and

$$\|r d^2 (p^2 - q^2)\beta\| \le H/M^2, \tag{6.47}$$

$$\|r\{d(p-q)\xi + 2a_0 d(p^2 - q^2)\beta\}\| \le H/M. \tag{6.48}$$

These inequalities hold for every pair in $\mathcal{E}$; the integers r, d, a_0, M may depend on the pair.

Step 3. Obtain a common rational denominator. First suppose $x \geq 8H$. The set $\mathcal{E}$, the bounds in (6.44), and (6.46) verify the hypotheses of Theorem 6.5, with $X = x$ and $\alpha = \xi$. Hence there are integers u_1, v_1 and positive integers S_1, R_1 such that

$$R_1 \leq 256H^9, \qquad \left|\beta - \frac{v_1}{R_1}\right| \leq \frac{4H^4}{N^2}, \qquad S_1 \leq 1024H^{14}, \qquad \left|\xi - \frac{u_1}{S_1}\right| \leq \frac{520H^7}{N}.$$

Both fractions therefore have a common denominator $R_1 S_1 \leq 2^{18}H^{23}$.

Now suppose $x < 8H$, and choose one pair in $\mathcal{E}$. Since $p \neq q$ are positive, $A_1, A_2 \neq 0$. Put $R_0 = rd^2|A_2|$. We have $|A_1| \leq x < 8H$, $|A_2| \leq 4x^2 < 256H^2$, and

$$\|R_0\beta\| \leq \frac{H}{M^2} \leq \frac{H^3x^2}{N^2} \leq \frac{64H^5}{N^2}.$$

Choose $v_0 \in \mathbb{Z}$ so that $|R_0\beta - v_0| = \|R_0\beta\|$. Then

$$R_0 \leq 256H^5, \qquad \beta = v_0/R_0 + \beta_0, \qquad |\beta_0| = \frac{\|R_0\beta\|}{R_0} \leq \frac{64H^5}{N^2}. \tag{6.49}$$

The size bound (6.46) implies $N \geq Hx$, so $|a_0| \leq 2N/x$. Multiply (6.48) by R_0 and substitute (6.49). The term $2ra_0dA_2v_0$ is an integer. Thus

$$\begin{aligned}\|R_0rdA_1\xi\| &\leq \frac{R_0H}{M} + 2R_0rd|a_0|\,|A_2|\,|\beta_0| \\ &\leq \frac{R_0H^2x}{N} + 2R_0H^2\frac{2N}{x}(4x^2)\frac{64H^5}{N^2} \\ &\leq \frac{1025R_0H^7x}{N} \leq \frac{8200R_0H^8}{N}.\end{aligned}$$

Put $S_0 = R_0rd|A_1|$. Then $R_0 \leq S_0 \leq 2048H^8$. Choosing an integer u_0 nearest to $S_0\xi$ gives

$$\left|\xi - \frac{u_0}{S_0}\right| = \frac{\|S_0\xi\|}{S_0} \leq \frac{8200H^8}{N}, \qquad R_0S_0 \leq 2^{19}H^{13}.$$

In either case, set $B_\rho = 2^{20}H^{23}$ and use the product of the two denominators. After rescaling the numerators, we obtain integers u, v and a positive integer $Q \leq B_\rho$ such that

$$\xi = u/Q + \xi_0, \qquad \beta = v/Q + \beta_0, \qquad |\xi_0| \leq B_\rho/N, \quad |\beta_0| \leq B_\rho/N^2.$$

Set $C_4 = 20 + 23A_*$. Since $2^{20} \leq \rho^{-20}$, (6.43) gives $B_\rho \leq \rho^{-C_4}$. Thus (6.38) holds whenever $C \geq C_4$.

Step 4. Control the remaining exponential factor. For real $y \in [1, N]$, put $P_0(y) = \xi_0 y + \beta_0 y^2$. Then

$$|P_0'(y)| \leq |\xi_0| + 2N|\beta_0| \leq 3B_\rho/N.$$

Require $C \geq \log 2000 + C_4 + 1$. Then

$$\log(2000B_\rho/\rho) \leq \log 2000 + (C_4 + 1)\log(1/\rho) \leq C\rho^{-5},$$

so $N \geq 2000B_\rho/\rho$. Set the positive integer $L = \lfloor \rho N/(1000B_\rho)\rfloor$. Divide $[N]$ into consecutive intervals $I_1, \dots, I_{m_0}$ of length L, except possibly the last. Then

$$L \geq \frac{\rho N}{2000B_\rho} \geq 1, \qquad m_0 = \lceil N/L\rceil \leq N/L + 1 \leq \frac{2000B_\rho}{\rho} + 1 \leq \frac{2001B_\rho}{\rho}.$$

The ceiling accounts for the last interval when its length is less than L. Let n_j be the first point of I_j. The mean value theorem and $|e(s) - e(t)| \leq 2\pi|s - t|$ give

$$|e(P_0(n)) - e(P_0(n_j))| \leq 6\pi B_\rho L/N < \rho/10 \qquad (n \in I_j). \tag{6.50}$$

Step 5. Find a large sum on one residue class. The function $n \mapsto e((un+vn^2)/Q)$ is Q-periodic. Put

$$Q_0 = \operatorname{lcm}(D, Q) \le DQ \le 2B_\rho/\rho.$$

Split each $\mathcal{P} \cap I_j$ into residue classes modulo Q_0. For $1 \le j \le m_0$ and $0 \le b < Q_0$, write

$$\mathcal{C}_{j,b} = \{n \in \mathcal{P} \cap I_j : n \equiv b \pmod{Q_0}\}.$$

Discard the empty sets. The remaining sets partition $\mathcal{P}$, and there are at most $m_0 Q_0 \le 4002 B_\rho^2/\rho^2$ of them. All sums over (j,b) below run over these nonempty sets. Since $Q \mid Q_0$, the function $e((un+vn^2)/Q)$ is constant on each $\mathcal{C}_{j,b}$. Define $z_{j,b} = e((ub+vb^2)/Q)e(P_0(n_j))$, so $|z_{j,b}| = 1$. Using (6.50) and $|f(n)| = 1$, we get

$$\left| \sum_{n \in \mathcal{P}} f(n) e(\xi n + \beta n^2) - \sum_{j,b} z_{j,b} \sum_{n \in \mathcal{C}_{j,b}} f(n) \right| \le \frac{\rho |\mathcal{P}|}{10} \le \frac{\rho N}{10}.$$

The assumed correlation and the triangle inequality therefore give

$$\sum_{j,b} \left| \sum_{n \in \mathcal{C}_{j,b}} f(n) \right| \ge \frac{9\rho N}{10}.$$

At least one pair (j,b) satisfies

$$\left| \sum_{n \in \mathcal{C}_{j,b}} f(n) \right| \ge \frac{9\rho^3 N}{40020 B_\rho^2} \ge \kappa_\rho N, \qquad \kappa_\rho := \frac{\rho^3}{5000 B_\rho^2}. \tag{6.51}$$

For this pair, take $I = I_j \cap I_*$ and $b_0 = b$. The set $\mathcal{C}_{j,b}$ is nonempty, so $b \equiv b_* \pmod D$. As $D \mid Q_0$, it follows that

$$\mathcal{C}_{j,b} = \{n \in I : n \equiv b_0 \pmod{Q_0}\}.$$

The estimate $B_\rho \le \rho^{-C_4}$ gives

$$Q_0 \le 2\rho^{-(C_4+1)} \le \rho^{-(C_4+2)}, \qquad \kappa_\rho^{-1} \le 5000 \rho^{-(2C_4+3)} \le \rho^{-(2C_4+16)}. \tag{6.52}$$

Here $2 \le \rho^{-1}$ and $5000 \le \rho^{-13}$. Thus $C \ge 2C_4 + 16$ ensures both $Q_0 \le \rho^{-C}$ and $\kappa_\rho \ge \rho^C$, proving (6.39).

Step 6. Apply the residue-class lemma. We have $0 < \kappa_\rho < 1/2$. Let $C_{\mathrm{res}} \ge 1$ be the absolute constant in Theorem 6.11. Put $E = 3C_4 + 19$. By (6.52),

$$\frac{2Q_0}{\kappa_\rho} \le \rho^{-E}, \qquad \log\left((2Q_0/\kappa_\rho)^{C_{\mathrm{res}}}\right) \le C_{\mathrm{res}} E \log(1/\rho) \le C_{\mathrm{res}} E \rho^{-5}. \tag{6.53}$$

Requiring $C \ge C_{\mathrm{res}} E$ therefore gives $N \ge (2Q_0/\kappa_\rho)^{C_{\mathrm{res}}}$, as needed.

Applying Theorem 6.11 to (6.51) gives a primitive Dirichlet character χ and a real t with

$$q_\chi \mid Q_0, \qquad |t| \le (Q_0/\kappa_\rho)^{C_{\mathrm{res}}},$$

$$\sum_{\substack{p \le N \\ p \text{ prime}}} \frac{1 - \Re(f(p)\overline{\chi(p)} p^{-it})}{p} \le C_{\mathrm{res}} \log(2Q_0/\kappa_\rho).$$

The estimates (6.52) and (6.53) now give

$$q_\chi \le Q_0 \le \rho^{-(C_4+2)}, \qquad |t| \le \rho^{-C_{\mathrm{res}} E},$$

$$\sum_{\substack{p \le N \\ p \text{ prime}}} \frac{1 - \Re(f(p)\overline{\chi(p)} p^{-it})}{p} \le C_{\mathrm{res}} E \log(1/\rho).$$

We may therefore choose the single constant in the theorem to be

$$C = \max\{C_{\mathrm{BSZ}},\, C_3,\, \log 2000 + C_4 + 1,\, 2C_4 + 16,\, C_{\mathrm{res}}(3C_4 + 19)\}.$$

Every entry in this maximum is an absolute constant. This choice satisfies all size requirements above and gives (6.38)–(6.40). □

7. A multiplicative inverse theorem

This section proves the multiplicative inverse theorem, Theorem 7.10. The proof first extracts a structured quadratic sequence from a large correlation and then converts that structure into a character description of the multiplicative function. Throughout this section, $C, c > 0$ denote absolute constants, which may change from line to line. For a real or integer vector v, $|v|$ denotes its maximum norm, and $\|v\|_2$ its Euclidean norm. The height of an integer matrix is its maximum entry norm.

7.1. Bohr sets and polynomial sequences.

In this subsection, we collect some notation used later.

Finite-group notation. Let A be a finite abelian group and write $\mathbb{T} = \mathbb{R}/\mathbb{Z}$. Its dual group $\widehat{A}$ consists of the homomorphisms $A \to \mathbb{T}$. We use normalized averages and

$$\widehat{u}(\xi) = \mathbb{E}_{x\in A} u(x)e(-\xi(x)), \qquad (u * v)(x) = \mathbb{E}_{y\in A} u(y)v(x-y), \qquad e(t) = e^{2\pi i t}.$$

For a nonempty finite set $S \subseteq \widehat{A}$ and $0 < \rho < 1/2$, define

$$B(S,\rho) = \{x \in A : \max_{\xi\in S} \|\xi(x)\|_{\mathbb{R}/\mathbb{Z}} < \rho\}.$$

Its rank is $|S|$. We call it regular when

$$\left|\frac{|B(S,(1+\kappa)\rho)|}{|B(S,\rho)|} - 1\right| \le 100|S||\kappa| \qquad (|\kappa| \le (100|S|)^{-1}). \tag{7.1}$$

We use the standard estimates

$$|B(S,\rho)| \ge (c\rho)^{|S|}|A|, \qquad |B(S,2\rho)| \le 4^{|S|}|B(S,\rho)|.$$

There is also a $\rho' \in [\rho/2, \rho]$ for which $B(S,\rho')$ is regular in the sense of (7.1). [10]

Local quadratic functions and progressions. Let $D \subseteq A$. A function $\phi : D \to \mathbb{T}$ is locally quadratic if

$$\sum_{\omega\in\{0,1\}^3} (-1)^{|\omega|}\phi(x + \omega\cdot h) = 0$$

whenever all eight arguments belong to D, where $h = (h_1,h_2,h_3)$ and $\omega\cdot h = \omega_1h_1+\omega_2h_2+\omega_3h_3$. Local linearity is defined by the analogous vanishing of second differences.

A symmetric coset progression of rank r is

$$P = H + \left\{\sum_{j=1}^{r} n_j v_j : |n_j| \le L_j\right\}, \qquad H \le A, \quad v_j \in A,$$

where $L_1,\dots,L_r$ are nonnegative integers. Let $B_L = \{(n_1,\dots,n_r) \in \mathbb{Z}^r : |n_j| \le L_j\}$. We say that P is proper when the map

$$B_L \to A/H, (n_1,\dots,n_r) \mapsto H + \sum_j n_j v_j,$$

is injective. Its density is $|P|/|A|$. For a set $E \subseteq A$ and an integer $k \ge 1$, write

$$kE = \{e_1 + \cdots + e_k : e_1,\dots,e_k \in E\}.$$

[10] The size estimates and the regular-radius assertion are given in [20, Section 2, p. 7, immediately before Lemma 2.1]. Our regularity convention agrees with [19, Definition 1.4(ii)].

Degree-two filtrations and polynomial sequences. Let G be a connected, simply connected nilpotent Lie group. A degree-two filtration on G is a sequence

$$G = G_0 = G_1 \supseteq G_2 \supseteq G_3 = \{1\},$$

where G_2 is closed, connected, and central and $[G, G] \subseteq G_2$. We call G_2 the central level of the filtration: every element of G_2 commutes with every element of G. The quotient map $\pi : G \to G/G_2$ records the horizontal coordinate, while a central coordinate is a coordinate along G_2. In the Heisenberg coordinates introduced below, $x = (\theta, \lambda, c)$ has horizontal coordinate (θ, λ) and central coordinate c; in a product of factors, these coordinates are read component by component. Thus the phrase "degree-two central filtration" refers precisely to the chain $G = G_1 \supseteq G_2 \supseteq G_3 = \{1\}$ with $G_2 \subseteq Z(G)$. Fix a lattice Γ in G; that is, Γ is a discrete subgroup and G/Γ is compact. Assume that $G_2/(\Gamma \cap G_2)$ is compact. The image of Γ in G/G_2 is required to be discrete and cocompact. Write $G^{(1)} = G$ and $G^{(j+1)} = [G, G^{(j)}]$ for the lower central series.

A polynomial sequence adapted to this filtration has the form

$$g(n) = g_0 u^n v^{\binom{n}{2}}, \qquad g_0, u \in G, \quad v \in G_2. \tag{7.2}$$

With $\partial_h g(n) = g(n+h)g(n)^{-1}$, one computes

$$\partial_h g(n) = g_0 u^h g_0^{-1} v^{hn+\binom{h}{2}}, \qquad \partial_k \partial_h g(n) = v^{hk}, \qquad \partial_l \partial_k \partial_h g(n) = 1.$$

Thus the successive differences take values in G_1, G_2, G_3. Conversely, if the second differences take values in G_2 and the third differences are trivial, then $\partial_1 \partial_1 g(n) = v \in G_2$ is independent of n. Hence $\partial_1 g(n) = av^n$ for some $a \in G$, and induction gives $g(n) = a^n v^{\binom{n}{2}} g(0)$. Writing $a = g(0)ug(0)^{-1}$ recovers (7.2).

The explicit Heisenberg models. We use the two-step groups

$$\mathcal{H}(r) = \mathbb{R}^r \times \mathbb{R}^r \times \mathbb{R}, \quad (\theta, \lambda, c)(\theta', \lambda', c') = (\theta + \theta', \lambda + \lambda', c + c' + \lambda \cdot \theta'), \tag{7.3}$$

with lattice $\Gamma(r) = \mathbb{Z}^r \times \mathbb{Z}^r \times \mathbb{Z}$ and filtration subgroup $\mathcal{H}(r)_2 = \{(0, 0, c) : c \in \mathbb{R}\}$. We also use products of at most two such groups, with the product lattice and filtration. The dimension of $\mathcal{H}(r)$ is $2r + 1$. The Lie bracket[11] is

$$[(\theta, \lambda, c), (\theta', \lambda', c')]_{\mathrm{Lie}} = (0, 0, \lambda \cdot \theta' - \lambda' \cdot \theta), \qquad \exp(\theta, \lambda, c) = (\theta, \lambda, c + \tfrac{1}{2}\lambda \cdot \theta).$$

In these coordinates, every coefficient in the group law, the Lie bracket, and the exponential map has absolute value at most one.

Metrics, smooth sequences, and Lipschitz norms. For the rest of this subsection, let G be a Heisenberg group $\mathcal{H}(r)$ or a product of at most two such groups, and let Γ be its integer lattice. Use the Euclidean inner product in the displayed Lie algebra coordinates and extend it by right translations. Denote the resulting Riemannian distance by d_G.

Let $Q \geq 1$ be real and $N \geq 1$ an integer. A sequence $\varepsilon : \{0, \ldots, N\} \to G$ is (Q, N)-smooth if

$$d_G(\varepsilon(n), 1) \leq Q \quad (0 \leq n \leq N), \qquad d_G(\varepsilon(n+1), \varepsilon(n)) \leq Q/N \quad (0 \leq n < N).$$

Thus Q bounds the distance from the identity, and Q/N bounds the distance between consecutive terms.

The quotient metric on G/Γ is

$$d_{G/\Gamma}(x\Gamma, y\Gamma) = \inf_{\gamma \in \Gamma} d_G(x, y\gamma).$$

[11] The Lie bracket is the commutator bracket of the left-invariant vector fields generated by the corresponding tangent vectors. In the coordinates above, it is the displayed vector in the c-coordinate direction.

For $F : G/\Gamma \to \mathbb{C}$, define

$$\|F\|_{\mathrm{Lip}} = \|F\|_\infty + \sup_{x\Gamma \neq y\Gamma} \frac{|F(x\Gamma) - F(y\Gamma)|}{d_{G/\Gamma}(x\Gamma, y\Gamma)}.$$

Characters and vertical frequencies. For the same group G and lattice Γ, a horizontal character is a continuous homomorphism $\eta : G \to \mathbb{R}$ with $\eta(\Gamma) \subseteq \mathbb{Z}$. It annihilates $[G, G]$ and therefore descends to a homomorphism on $G/[G, G]$. For $\mathcal{H}(r)$,

$$\eta(\theta, \lambda, c) = a \cdot \theta + b \cdot \lambda, \qquad a, b \in \mathbb{Z}^r,$$

and its height is $\max(|a|, |b|)$; on a product, use the maximum over the factors. If $w \in G/[G, G]$, we say that w is orthogonal to η when the descended homomorphism satisfies $\eta(w) = 0$. For $G = \mathbb{R}$ and $\Gamma = \mathbb{Z}$, horizontal characters are $x \mapsto mx$ with $m \in \mathbb{Z}$.

For a positive integer s, a vertical character of level s is a continuous homomorphism $\xi : G^{(s)} \to \mathbb{R}$ with $\xi(G^{(s)} \cap \Gamma) \subseteq \mathbb{Z}$. A function $F : G/\Gamma \to \mathbb{C}$ has vertical frequency ξ at level s when

$$F(zx\Gamma) = e(\xi(z))F(x\Gamma) \qquad (z \in G^{(s)},\ x \in G).$$

Polynomial norms. Let $N \geq 1$ be an integer. For an $\mathbb{R}/\mathbb{Z}$-valued polynomial in binomial coordinates

$$P(n) = a_0 + a_1\binom{n}{1} + a_2\binom{n}{2}, \qquad a_0, a_1, a_2 \in \mathbb{R}/\mathbb{Z},$$

define

$$\|P\|_{C^\infty[N]} = \max\{N\|a_1\|_{\mathbb{R}/\mathbb{Z}}, N^2\|a_2\|_{\mathbb{R}/\mathbb{Z}}\}.$$

If g is a degree-two polynomial sequence and η is horizontal, then $\eta \circ g$ is reduced modulo $\mathbb{Z}$ and is measured by this norm.

7.2. **Initial quadratic correlation.** We first prove the progression-to-Bohr estimate needed for Theorem 7.2.

Lemma 7.1 (Progression-to-Bohr estimate). *Let A be a finite abelian group. Let $P \subseteq A$ be a symmetric coset progression of rank $r \geq 1$ and density*

$$\alpha := \frac{|P|}{|A|} > 0.$$

Set

$$S := r + \log(2/\alpha).$$

Then there is a nonempty finite set $\Psi \subseteq \widehat{A}$ such that

$$B\Big(\Psi, \frac{1}{4|\Psi|}\Big) \subseteq P, \qquad |\Psi| \ll S\log(2S).$$

Proof. Write the progression in the form

$$P = H + \left\{\sum_{j=1}^{r} n_j v_j : |n_j| \leq L_j\right\}, \qquad H \leq A.$$

We divide the rest proof into five steps.

Step 1. A smaller progression. Choose an even integer $k \geq 4$, put

$$q_j := \left\lfloor \frac{L_j}{k} \right\rfloor, \qquad Q := H + \left\{\sum_{j=1}^{r} n_j v_j : |n_j| \leq q_j\right\}.$$

Since $kq_j \leq L_j$ and the sum of k elements of the subgroup H again lies in H, we have

$$kQ \subseteq P.$$

Since $L_j < k(q_j+1)$, we have $2L_j+1 \le 2k(2q_j+1)$. Divide $[-L_j, L_j]\cap\mathbb{Z}$ into at most $2k$ consecutive blocks, each with at most $2q_j+1$ terms. Each block lies in a translate of $[-q_j,q_j]\cap\mathbb{Z}$. Thus there is a set $T_j \subseteq \mathbb{Z}$ with $|T_j| \le 2k$ and

$$[-L_j, L_j]\cap\mathbb{Z} \subseteq \bigcup_{t\in T_j} \big(t + ([-q_j,q_j]\cap\mathbb{Z})\big).$$

Put $E = \left\{\sum_{j=1}^{r} t_j v_j : t_j \in T_j\right\} \subseteq A$. Then

$$P \subseteq \bigcup_{t\in E}(t+Q), \qquad |P| \le |E|\,|Q| \le (2k)^r |Q|.$$

Consequently, with

$$\delta := \frac{|Q|}{|A|},$$

we have

$$\delta \ge \alpha(2k)^{-r}. \tag{7.4}$$

Step 2. The large spectrum of Q. Let

$$a := \frac{1}{2}\,\delta^{\,1+1/(k-2)}, \qquad \mathcal{L} := \{\gamma \in \widehat{A} : |\widehat{\mathbf{1}_Q}(\gamma)| \ge a\}.$$

Parseval gives

$$\sum_{\gamma\in\widehat{A}} |\widehat{\mathbf{1}_Q}(\gamma)|^2 = \mathbb{E}_{x\in A}\mathbf{1}_Q(x)^2 = \delta.$$

Therefore

$$\sum_{\gamma\notin\mathcal{L}} |\widehat{\mathbf{1}_Q}(\gamma)|^k \le a^{\,k-2} \sum_{\gamma\in\widehat{A}} |\widehat{\mathbf{1}_Q}(\gamma)|^2 = 2^{-(k-2)}\delta^k. \tag{7.5}$$

Because $Q=-Q$, every Fourier coefficient $\widehat{\mathbf{1}_Q}(\gamma)$ is real. Since k is even, $\widehat{\mathbf{1}_Q}(\gamma)^k \ge 0$.

Step 3. Positivity on a Bohr set. Write $\mathbf{1}_Q^{*k}$ for the normalized k-fold convolution of $\mathbf{1}_Q$. Explicitly,

$$\mathbf{1}_Q^{*k}(x) = \frac{1}{|A|^{k-1}} \#\{(y_1,\dots,y_k)\in Q^k : y_1+\cdots+y_k = x\}.$$

In particular, $\mathbf{1}_Q^{*k}(x) > 0$ if and only if $x \in kQ$. Fourier inversion gives

$$\mathbf{1}_Q^{*k}(x) = \sum_{\gamma\in\widehat{A}} \widehat{\mathbf{1}_Q}(\gamma)^k e(\gamma(x)).$$

If $x \in B(\mathcal{L}, 1/4)$, then $\Re e(\gamma(x)) \ge 0$ for every $\gamma\in\mathcal{L}$. The zero frequency contributes δ^k. The retained nonzero frequencies have nonnegative real part, while the total absolute value of the discarded frequencies is bounded by (7.5). Hence

$$\Re\big(\mathbf{1}_Q^{*k}(x)\big) \ge \delta^k - 2^{-(k-2)}\delta^k > 0.$$

Thus $\mathbf{1}_Q^{*k}(x) > 0$, so x is a sum of k elements of Q. Therefore

$$B(\mathcal{L}, 1/4) \subseteq kQ \subseteq P.$$

Step 4. Compressing the large spectrum. We use the spectral form of Chang's lemma appearing in the proof of [25, Proposition 28]: for $0 < b \le 1$, the set

$$\{\gamma \in \widehat{A} : |\widehat{\mathbf{1}_Q}(\gamma)| \ge b\delta\}$$

is contained in the $\{-1,0,1\}$-span of a set of at most

$$Cb^{-2}\log(2/\delta)$$

characters, where C is absolute. Apply this with

$$b := \frac{a}{\delta} = \frac{1}{2}\,\delta^{1/(k-2)}.$$

By Chang's lemma, after including the trivial character (which changes the bound by at most one, absorbed because $b^{-2}\log(2/\delta) \geq 4\log 2 > 1$), choose a set $\Psi \subseteq \widehat{A}$ such that every element of $\mathcal{L}$ is a sum $\sum_{\psi\in\Psi} \varepsilon_\psi \psi$ with $\varepsilon_\psi \in \{-1, 0, 1\}$ and

$$|\Psi| \ll \delta^{-2/(k-2)} \log(2/\delta). \tag{7.6}$$

If $x \in B(\Psi, 1/(4|\Psi|))$, then

$$\|\gamma(x)\|_{\mathbb{R}/\mathbb{Z}} \leq \sum_{\psi\in\Psi} \|\psi(x)\|_{\mathbb{R}/\mathbb{Z}} < \frac{1}{4} \qquad (\gamma \in \mathcal{L}).$$

Thus

$$B\Big(\Psi, \frac{1}{4|\Psi|}\Big) \subseteq B(\mathcal{L}, 1/4) \subseteq P. \tag{7.7}$$

Step 5. Choosing k. Set $T = \log(2S)$. Since $S \geq 1$, we have $T \geq \log 2$. Choose

$$k = 2\lceil 8ST\rceil, \qquad 16ST \leq k \leq 20ST, \qquad k - 2 \geq 8ST. \tag{7.8}$$

These bounds follow from $ST \geq \log 2$ and $16ST \leq k < 16ST + 2$. Also, $\log S \leq T$, $\log T \leq T$, and $\log 40 \leq 6T$, so $\log(2k) \leq \log 40 + \log S + \log T \leq 8T$. Since $r \leq S$ and $\log(2/\alpha) \leq S$, (7.4) gives

$$\log(2/\delta) \leq \log(2/\alpha) + r\log(2k) \leq S + 8ST \leq 10ST. \tag{7.9}$$

Together with (7.8), this yields

$$\frac{\log(2/\delta)}{k-2} \leq \frac{10ST}{8ST} = \frac{5}{4}.$$

Consequently,

$$\delta^{-2/(k-2)} = \exp\Big(\frac{2\log(1/\delta)}{k-2}\Big) \leq e^{5/2}. \tag{7.10}$$

All constants in these three estimates are numerical; in particular, they are independent of A, r, and α. Substituting (7.9) and (7.10) into (7.6) yields

$$|\Psi| \ll S\log(2S).$$

Together with (7.7), this proves the lemma. □

We now use Theorem 7.1 to establish the local quadratic inverse estimate.

Proposition 7.2. *Let L be an odd prime and put*

$$G := \mathbb{Z}/L\mathbb{Z}.$$

Let $0 < \epsilon < 1/2$ and put $\ell := 2 + \log(2/\epsilon)$. Suppose that $u : G \to \mathbb{C}$ satisfies

$$|u(x)| \leq 1 \quad (x \in G), \qquad \|u\|_{U^3(G)} \geq \epsilon.$$

Then there exist an absolute constant $C \geq 1$, a nonempty set $S \subseteq \widehat{G}$ with $r := |S| \geq 1$, a regular Bohr set

$$B_G(S, \rho) := \{x \in G : \max_{\xi\in S} \|\xi(x)\|_{\mathbb{R}/\mathbb{Z}} < \rho\}, \qquad 0 < \rho < 1/2000,$$

a point $x_0 \in G$, a locally quadratic function $\phi : B_G(S, 1000\rho) \to \mathbb{T}$, and a real number $\tau > 0$ such that

$$\left|\mathbb{E}_{h \in B_G(S,\rho)} u(x_0 + h) e(-\phi(h))\right| \geq \tau,$$
$$r \leq C\ell^{43},$$
$$\log(2/\rho),\ \log(2/\tau) \leq C\ell.$$

Proof. We divide the proof into seven steps.

Step 1. Extract a graph. The pointwise bound on u gives

$$\|u\|_{L^{1024}(G)} \leq 1.$$[12]

By $\|u\|_{U^3(G)} \geq \epsilon$ and the above L^{1024} bound, the Cauchy–Schwarz, Balog–Szemerédi–Gowers, and graph-selection steps in the finite-abelian-group form of [20, Lemmas A.6–A.7] supply a set $H' \subseteq G$ and a map $\omega : H' \to \widehat{G}$ with

$$|H'| \geq \epsilon^{C_0}|G|$$

for an absolute C_0. Put

$$\Gamma' := \{(h, \omega(h)) : h \in H'\}.$$

For each $h \in G$, the set $\{\eta \in \widehat{G} : (h, \eta) \in 4\Gamma' - 4\Gamma'\}$ has at most one element. Put $\delta_1 := |H'|/|G|$. Then, with $C_1 := C_0$,

$$\delta_1 \geq \epsilon^{C_1}.$$

Step 2. Obtain the first progression. We use [20, Lemma A.1]. Its parameter $K \geq 1$ measures the additive enlargement of the smaller summand: if $|A_1 + A_2| \leq K \min(|A_1|, |A_2|)$, then $(A_1 - A_1) + (A_2 - A_2)$ contains a proper symmetric coset progression of rank at most $d(K)$ and size at least $\exp(-h(K))|A_1 + A_2|$, with

$$d(K) \ll \log^6(2K), \qquad h(K) \ll \log^6(2K) \log\log(4K).$$

Apply this with $A_1 = A_2 = H'$. Since

$$|H' + H'| \leq |G| = \delta_1^{-1}|H'|,$$

the admissible expansion parameter is

$$K_1 := \delta_1^{-1}.$$

The cited sumset theorem gives a proper symmetric coset progression $P_1 \subseteq 2H' - 2H'$ with

$$|P_1| \geq \exp(-h(K_1))|H' + H'|.$$

Moreover,

$$\log(2K_1) = \log(2/\delta_1) \leq C_1 \log(2/\epsilon) \ll \ell.$$

Substitution into the bounds for d and h, together with $\alpha_1 := |P_1|/|G|$, gives

$$r_1 := \operatorname{rank}(P_1) \ll \ell^6, \qquad \log(2/\alpha_1) \ll \ell^6 \log(2\ell). \tag{7.11}$$

Step 3. Apply Theorem 7.1 once. If $r_1 = 0$, add one zero generator of length zero; this leaves P_1 and α_1 unchanged. Apply Theorem 7.1 in the ambient group G. Put

$$T_1 := r_1 + \log(2/\alpha_1).$$

The lemma gives a Bohr set

$$B_1 = B_G(S_1, \rho_1) \subseteq P_1$$

[12] Here $\|v\|_{L^{1024}(G)} := (\mathbb{E}_{x \in G}|v(x)|^{1024})^{1/1024}$ is the normalized counting norm.

with a frequency set S_1 of rank

$$d_1 := |S_1| \ll T_1 \log(2T_1) \ll \ell^7.$$

The last inequality follows from (7.11); the same lemma gives $\rho_1^{-1} \ll \ell^7$.

Step 4. Linearize on a smaller Bohr set. The graph property from Step 1 is the hypothesis for the linearization argument in [20, Appendix A.2]. It produces a regular Bohr set

$$B_2 = B_G(S_2, \rho_2) \subseteq B_1, \qquad |S_2| \le d_1,$$

a locally linear map

$$M : B_G(S_2, 2000\rho_2) \to \widehat{G},$$

and a translate $x_1 \in G$. Put

$$E_0 := \{h \in B_2 : x_1 + h \in H'\}, \qquad \delta_0 := \frac{|E_0|}{|B_2|}.$$

The quantitative part of the linearization argument gives

$$\rho_2 \ge \epsilon^{C_2}, \qquad \delta_0 \ge \epsilon^{C_2}.$$

The linearization estimate [20, Lemma A.5] gives functions $a : B_2 \to \mathbb{C}$ and $b : G \to \mathbb{C}$, with $|a|, |b| \le 1$, such that

$$|\mathbb{E}_{h \in B_2} \mathbb{E}_{x \in G} a(h) b(x) u(x+h) e(-2\langle M(h), x\rangle)| \ge \epsilon^{C_2}. \tag{7.12}$$

Applying the Cauchy–Schwarz and pigeonhole steps in [20, Appendix A.3] to this correlation gives a set $E_1 \subseteq B_2$ and an absolute C_3 such that

$$\delta_3 := \frac{|E_1|}{|B_2|} \ge \epsilon^{C_3}, \qquad \|2\{x,z\}\|_{\mathbb{R}/\mathbb{Z}} \le \epsilon^{-C_3} \max_{\xi \in S_2} \|\xi(x)\|_{\mathbb{R}/\mathbb{Z}} \quad (x \in B_2, z \in 2E_1 - 2E_1), \tag{7.13}$$

where

$$\{x, z\} := \langle M(x), z\rangle - \langle M(z), x\rangle.$$

This is the only property of E_1 used in the next step.

Step 5. Obtain the second progression. Since $E_1 + E_1 \subseteq B_G(S_2, 2\rho_2)$, the Bohr-volume estimate gives

$$|E_1 + E_1| \le 4^{d_1} |B_2| = 4^{d_1} \delta_3^{-1} |E_1|.$$

Set

$$K_2 := 4^{d_1} \delta_3^{-1}.$$

Thus $|E_1 + E_1| \le K_2 |E_1|$, so [20, Lemma A.1] supplies a proper symmetric coset progression $P_3 \subseteq 2E_1 - 2E_1$ with

$$|P_3| \ge e^{-h(K_2)} |E_1 + E_1| \ge e^{-h(K_2)} \delta_3 |B_2|.$$

Moreover,

$$\log(2K_2) \le \log 2 + d_1 \log 4 + \log(1/\delta_3) \ll \ell^7,$$
$$\operatorname{rank}(P_3) \ll (\log(2K_2))^6 \ll \ell^{42}.$$

The Bohr-volume estimate also gives

$$\log \frac{|G|}{|B_2|} \ll d_1 \log(2/\rho_2) \ll \ell^8.$$

Consequently, with $\alpha_3 := |P_3|/|G|$,

$$\log(2/\alpha_3) \le h(K_2) + \log(2/\delta_3) + \log(|G|/|B_2|) + O(1) \ll \ell^{42} \log(2\ell),$$

so $\alpha_3 \ge \exp(-C\ell^{42} \log(2\ell))$.

Step 6. Convert the second progression into a symmetric frequency form. Apply Theorem 7.1 to P_3 in G. If $r_3 := \operatorname{rank}(P_3)$, we obtain a regular Bohr set

$$B_3 = B_G(S_3, \rho_3) \subseteq P_3$$

with

$$|S_3| \ll (r_3 + \log(2/\alpha_3)) \log(2r_3 + 2\log(2/\alpha_3)) \ll \ell^{43}, \qquad \rho_3^{-1} \ll \ell^{43}.$$

Since $\log \ell \ll \log(2/\epsilon)$, the second bound also gives $\rho_3 \geq \epsilon^C$ after changing the absolute constant. For $\xi \in \widehat{G}$, let $\xi/2$ denote the unique character whose double is ξ; it exists because L is odd. Set

$$S_4 := S_2 \cup S_3 \cup \{\xi/2 : \xi \in S_2\}.$$

Thus $|S_4| \leq 2d_1 + |S_3| \ll \ell^{43}$. Choose a regular radius $\overline{\rho}_4 \in [\min(\rho_2, \rho_3)/2, \min(\rho_2, \rho_3)]$ and put $\overline{B}_4 := B_G(S_4, \overline{\rho}_4)$. For $x, z \in \overline{B}_4$, we have $x/2 \in B_2$ and $z \in B_3 \subseteq 2E_1 - 2E_1$. Local linearity gives $\{x, z\} = 2\{x/2, z\}$. Hence (7.13) gives

$$\begin{aligned} &\overline{\rho}_4 \geq \epsilon^{a_4}, \\ &\|\{x, z\}\|_{\mathbb{R}/\mathbb{Z}} \leq \epsilon^{-b_4} \max_{\xi \in S_2} \|(\xi/2)(x)\|_{\mathbb{R}/\mathbb{Z}} \leq \epsilon^{-b_4} \max_{\eta \in S_4} \|\eta(x)\|_{\mathbb{R}/\mathbb{Z}}, \end{aligned} \tag{7.14}$$

where $a_4 > 0$ is absolute and $b_4 = C_3$. This is the division by 2 used in [20, Appendix A.3]. Write $c_* := \epsilon^{C_*}$, where $C_* > 0$ is chosen so that $2\pi c_* \leq \frac{1}{2}\epsilon^{C_2}$. Put

$$t := \min\{\overline{\rho}_4, c_* \epsilon^{b_4}\}.$$

Choose a regular radius $\rho_4 \in [t/2, t]$ and set $B_4 := B_G(S_4, \rho_4)$. If $A_4 := \max\{a_4, C_* + b_4\}$, then $t \geq \epsilon^{A_4}$ and $\rho_4 \geq \frac{1}{2}\epsilon^{A_4} \geq \epsilon^{A_4+1}$ because $0 < \epsilon < 1/2$. Thus, with $C_4 := A_4 + 1$,

$$\epsilon^{C_4} \leq \rho_4 \leq c_* \epsilon^{b_4}. \tag{7.15}$$

The defect estimate in (7.14) now gives

$$\|\{x, z\}\|_{\mathbb{R}/\mathbb{Z}} \leq c_* \qquad (x, z \in B_4).$$

Define

$$\mathcal{B}(x, z) := \langle M(x), z\rangle + \langle M(z), x\rangle, \qquad \Phi(x) := \langle M(x), x\rangle.$$

The local linearity of M gives, for $h, x \in B_4$,

$$\mathcal{B}(h, x) = \Phi(x + h) - \Phi(x) - \Phi(h). \tag{7.16}$$

Step 7. Localize the correlation and select one phase. We spell out the two operations that are implicit in the localization argument of [20, Appendix A.4].

7a. Symmetrization of the bilinear phase. For $h, x \in B_4$, the definitions in Step 6 give

$$2\langle M(h), x\rangle = \mathcal{B}(h, x) + \{h, x\}.$$

Consequently, by (7.15),

$$|e(-2\langle M(h), x\rangle) - e(-\mathcal{B}(h, x))| \leq 2\pi \|\{h, x\}\|_{\mathbb{R}/\mathbb{Z}} \leq 2\pi c_*. \tag{7.17}$$

Choose C_* in Step 6 so that $2\pi c_* \leq \frac{1}{2}\epsilon^{C_2}$. Thus replacing the original phase in (7.12) by the symmetric phase loses at most one half of its lower bound. Moreover, (7.16) gives

$$e(-\mathcal{B}(h, x)) = e(-\Phi(x + h))\, e(\Phi(x))\, e(\Phi(h))\,. \tag{7.18}$$

The first and third factors in (7.18) depend on $x + h$ and h, respectively, and can therefore be absorbed into the two one-bounded coefficients produced by the localization argument. The only phase that remains attached to the variable on which u is tested is $e(\Phi(x))$.

7b. Quantitative localization. Apply the quantitative conclusion of [20, Appendix A.4] to the correlation after the replacement in (7.17). It gives a regular Bohr set

$$B_5 = B_G(S, \rho), \qquad B_G(S, 1000\rho) \subseteq B_4, \qquad |S| \ll \ell^{43}, \qquad \epsilon^{C_5} \leq \rho < 1/2000.$$

and, for each $y \in G$, functions $a_y, b_y : G \to \mathbb{C}$ with $|a_y|, |b_y| \le 1$, such that

$$\mathbb{E}_{y\in G}\,|\mathbb{E}_{h\in B_4}\mathbb{E}_{x\in B_5} a_y(h)b_y(h+x)u(y+x)e(\Phi(x))| \ge \epsilon^{C_6}. \tag{7.19}$$

The same localization chooses B_5 so that

$$|B_4 + B_5| \le 2|B_4|. \tag{7.20}$$

It also chooses ρ small enough that $B_4 + B_5 \subseteq B_G(S_2, 2000\rho_2)$, the domain of M from Step 4. Thus $\Phi(h+x)$ and the coefficient $b_y(h+x)$ below are defined wherever they are used. The displayed inclusion ensures that all phases in (7.18) are evaluated inside the domain where M is locally linear.

7c. Fourier extraction of a single linear character. Fix $y \in G$ and set

$$v_y(x) := u(y+x)e(\Phi(x)), \quad f_y := a_y\mathbf{1}_{B_4}, \quad q_y := b_y\mathbf{1}_{B_4+B_5}, \quad w_y := v_y\mathbf{1}_{B_5}.$$

All three functions are regarded as functions on G, extended by zero outside their displayed supports. With the normalized Fourier transform, Fourier inversion gives

$$\begin{aligned}&\mathbb{E}_{h\in B_4}\mathbb{E}_{x\in B_5} a_y(h)b_y(h+x)v_y(x)\\ &= \frac{|G|^2}{|B_4||B_5|}\sum_{\xi\in\widehat{G}} \widehat{f_y}(-\xi)\widehat{q_y}(\xi)\widehat{w_y}(-\xi).\end{aligned} \tag{7.21}$$

By Cauchy–Schwarz and Parseval,

$$|\mathbb{E}_{h\in B_4}\mathbb{E}_{x\in B_5} a_y(h)b_y(h+x)v_y(x)| \le \left(\frac{|B_4+B_5|}{|B_4|}\right)^{1/2} \sup_{\xi\in\widehat{G}} |\mathbb{E}_{x\in B_5} v_y(x)e(\xi(x))|\,. \tag{7.22}$$

Indeed, $\|f_y\|_{L^2(G)}^2 \le |B_4|/|G|$, $\|q_y\|_{L^2(G)}^2 \le |B_4+B_5|/|G|$, and

$$\widehat{w_y}(\xi) = \frac{|B_5|}{|G|}\mathbb{E}_{x\in B_5} v_y(x)e(-\xi(x));$$

substitution into (7.21) yields (7.22). By (7.20), the prefactor is at most $\sqrt{2}$. Averaging over y in (7.19) therefore gives, after increasing the absolute exponent,

$$\mathbb{E}_{y\in G}\sup_{\xi\in\widehat{G}} |\mathbb{E}_{h\in B_5} u(y+h)e(\Phi(h)+\xi(h))| \ge \epsilon^{C_7}. \tag{7.23}$$

7d. Pigeonhole and conclusion. Choose $y_0 \in G$ and $\xi_0 \in \widehat{G}$ for which the quantity in (7.23) is at least ϵ^{C_7}. Put

$$x_0 := y_0, \qquad \phi(h) := -\Phi(h) - \xi_0(h), \qquad \tau := \epsilon^{C_7}.$$

The function ξ_0 is linear. The identity (7.16), together with the local linearity of M, shows that Φ has vanishing third differences on $B_G(S, 1000\rho)$; hence ϕ is locally quadratic there. Therefore

$$\left|\mathbb{E}_{h\in B_G(S,\rho)} u(x_0+h)e(-\phi(h))\right| \ge \tau.$$

Finally, the displayed Bohr-set bounds give $r = |S| \ll \ell^{43}$ and $\log(2/\rho) \ll \ell$, while $\tau = \epsilon^{C_7}$ gives $\log(2/\tau) \ll \ell$. Increasing the absolute constant C once proves all the asserted bounds and completes the proof. □

Theorem 7.2 gives a correlation with a locally quadratic phase on a Bohr set in

$$A := \mathbb{Z}/L\mathbb{Z},$$

where L is an odd prime. For $S \subseteq \widehat{A}$ write

$$B_A(S,t) := \{a \in A : \max_{\xi\in S} \|\xi(a)\|_{\mathbb{R}/\mathbb{Z}} < t\}.$$

We write $[n]_L$ for the residue class of n in A. Let $u : A \to \mathbb{C}$ satisfy $|u(a)| \le 1$ for all $a \in A$. The next lemma converts this local correlation into a correlation with a degree-two nilsequence.

Under these assumptions, decrease τ if necessary so that $0 < \tau < 1/2$. We have $0 < \rho < 1/2000$, $S \subseteq \widehat{A}$ is nonempty with $r := |S|$, $B_A(S,\rho)$ is regular, $x_0 \in A$, and $\phi : B_A(S, 1000\rho) \to \mathbb{T}$ is locally quadratic with

$$\left|\mathbb{E}_{h \in B_A(S,\rho)} u(x_0 + h) e(-\phi(h))\right| \geq \tau.$$

Put

$$\Lambda := r \log(2r) + \log(2/\rho) + \log(2/\tau).$$

If S contains a nontrivial character, let

$$(G, \Gamma) := (\mathcal{H}(r), \Gamma(r)),$$

with the standard metric and degree-two filtration from (7.3). If every member of S is trivial, let $(G,\Gamma) = (\mathbb{R}, \mathbb{Z})$ with its degree-two filtration. In both cases $\dim G \leq 2r + 1$.

Lemma 7.3. *Under the above assumptions, there exist a degree-two polynomial sequence $g : \mathbb{Z} \to G$, a function $F : G/\Gamma \to \mathbb{C}$, and a real number $M \geq 2$ such that*

$$|F| \leq 1, \qquad \|F\|_{\mathrm{Lip}} \leq M,$$

and F has vertical frequency one at level 2, namely

$$F((0,0,t)x\Gamma) = e(t)F(x\Gamma) \quad (t \in \mathbb{R},\ x \in G)$$

in the Heisenberg case (with the analogous formula $F(t + x + \mathbb{Z}) = e(t)F(x + \mathbb{Z})$ in the circle case), and

$$\left|\frac{1}{L}\sum_{n=1}^{L} u([n]_L) F(g(n)\Gamma)\right| \geq \exp(-Cr\Lambda), \tag{7.24}$$

$$\log(2M) \leq C\Lambda. \tag{7.25}$$

Proof. Write

$$B_t := B_A(S,t) = \{x \in A : \max_{\xi \in S} \|\xi(x)\|_{\mathbb{R}/\mathbb{Z}} < t\}.$$

We divide the rest proof into eight steps.

Step 1. Dispose of the all-trivial-frequency case. If every character in S is trivial, then $B_\rho = A$. The vanishing third differences of the locally quadratic function ϕ on A give, for $n = 0, \dots, L-1$,

$$\phi([n]_L) = c + an + b\binom{n}{2} \pmod 1.$$

Let n_0 represent x_0 and set

$$G = \mathbb{R}, \qquad \Gamma = \mathbb{Z}, \qquad F(t + \Gamma) = e(t),$$

$$g(n) := -c - a(n - n_0) - b\binom{n - n_0}{2}.$$

Then g is a degree-two polynomial sequence and $F(g(n)+\Gamma) = e(-\phi([n-n_0]_L))$. Consequently the assumed local correlation becomes

$$\left|\frac{1}{L}\sum_{n=1}^{L} u([n]_L) F(g(n)\Gamma)\right| \geq \tau.$$

The function F has vertical frequency one, $|F| = 1$, and its Lipschitz norm is absolute. Thus the lemma follows in this case.

From now on assume that S contains a non-trivial character. First we prove the lattice extension fact needed in Step 3.

Step 2. Lift the Bohr set to a lattice. Assume that S contains a non-trivial character. Identify

$$\mathcal{L} := \{\theta \in \mathbb{R}^r : \theta_\xi = \xi(x) \pmod 1 \text{ for some } x \in A\} \subset \mathbb{R}^r, \qquad r = |S|.$$

The projection from $\mathcal{L}$ to A is well-defined and injective modulo $\mathbb{Z}^r$: indeed a non-trivial character of A is injective because L is prime. Hence $\mathcal{L}/\mathbb{Z}^r \simeq A$, so $\mathcal{L}$ is a rank-r lattice containing $\mathbb{Z}^r$ and

$$[\mathcal{L} : \mathbb{Z}^r] = L, \qquad \det \mathcal{L} = L^{-1}. \tag{7.26}$$

Let $\pi : \mathcal{L} \to A$ denote the induced projection. Since $\rho < 1/2000$, every element of B_t with $t \le 2\rho$ has a unique representative $\theta \in \mathcal{L}$ with $|\theta|_\infty < t$; conversely, such a representative projects to an element of B_t. For each $t > 0$, the discrete John theorem of Tao and Vu [26, Theorem 1.6] gives fixed real numbers $N_i > 0$ and the lattice basis $w_1, \dots, w_r$ such that, with $\kappa = (Cr)^{3r/2}$,

$$[-t\kappa^{-1}, t\kappa^{-1}]^r \cap \mathcal{L} \subseteq \Big\{ \sum_i n_i w_i : n_i \in \mathbb{Z}, |n_i| < tN_i \Big\} \subseteq [-t,t]^r \cap \mathcal{L}. \tag{7.27}$$

For later use, put

$$\mathcal{C}_t := \Big\{ \sum_i n_i w_i : n_i \in \mathbb{Z},\ |n_i| < tN_i \Big\}.$$

The lattice-point asymptotic for a full-rank lattice gives

$$\frac{|\mathcal{L} \cap [-t\kappa^{-1}, t\kappa^{-1}]^r|}{(2t)^r} \longrightarrow \frac{\kappa^{-r}}{\det \mathcal{L}} = L\kappa^{-r}, \qquad \frac{|\mathcal{C}_t|}{(2t)^r} \longrightarrow \prod_i N_i \quad (t \to \infty).$$

Taking cardinalities in the first inclusion of (7.27) and passing to the limit, and using (7.26), yields

$$\prod_i N_i \ge L\kappa^{-r} = L(Cr)^{-3r^2/2}. \tag{7.28}$$

We will use only the schematic consequence

$$\prod_i N_i \ge L \exp(-Cr^2 \log(2r)).$$

Step 3. Extend the local quadratic phase. Let $y \in \mathcal{L}$ and let

$$\psi : \mathcal{L} \cap \big(y + [-2\rho, 2\rho]^r\big) \longrightarrow \mathbb{T}$$

be locally quadratic. Put

$$\sigma_* := \rho\kappa^{-1}/1000 = \rho(Cr)^{-3r/2}/1000.$$

Define

$$D := \Big\{ n \in \mathbb{Z}^r : |n_i| < \rho N_i/8 \text{ for every } i \Big\}, \qquad \Psi : D \to \mathbb{T}, n \mapsto \psi\Big(y + \sum_{i=1}^r n_i w_i \Big).$$

The second inclusion in (7.27), used with $t = \rho/8$, shows that all points used in Ψ lie in $y + [-\rho/8, \rho/8]^r$ and hence in the domain of ψ. Write $\Delta_i \Psi(n) := \Psi(n + e_i) - \Psi(n)$ whenever both terms are defined. Local quadraticity says that every admissible third difference $\Delta_i\Delta_j\Delta_k\Psi$ vanishes. Therefore each second difference $\Delta_i\Delta_j\Psi$ is constant on the coordinate-connected interior of D.

Let $J := \{i : \rho N_i > 32\}$. For $i \in J$, the points $0, e_i, 2e_i$ lie in D; for distinct $i, j \in J$, the points $0, e_i, e_j, e_i + e_j$ lie in D. Define, in $\mathbb{T}$,

$$a_i := \Delta_i \Psi(0), \qquad b_i := \Delta_i^2 \Psi(0), \qquad c_{ij} := \Delta_i\Delta_j\Psi(0) \quad (i < j, i, j \in J).$$

On the subbox $D_J := \{n \in D : n_i = 0 \text{ for } i \notin J\}$, the Newton identity is

$$\Psi(n) = \psi(y) + \sum_{i\in J} a_i n_i + \sum_{i\in J} b_i \binom{n_i}{2} + \sum_{i<j,\ i,j\in J} c_{ij} n_i n_j. \tag{7.29}$$

We now extend this formula to the whole lattice. Every $\theta \in \mathcal{L}$ has a unique representation $\theta = y + \sum_i n_i w_i$. Define

$$\widetilde{\psi}\Big(y + \sum_i n_i w_i\Big) := \psi(y) + \sum_{i\in J} a_i n_i + \sum_{i\in J} b_i \binom{n_i}{2} + \sum_{i<j,\ i,j\in J} c_{ij} n_i n_j \qquad (\mathrm{mod}\ 1).$$

This is a global quadratic map on $\mathcal{L}$. If $\theta - y \in [-\sigma_*, \sigma_*]^r \cap \mathcal{L}$, the first inclusion in (7.27), with $t = \rho/1000$, gives unique integers n_i satisfying $|n_i| < \rho N_i/1000$. For $i \notin J$ this is $\le 32/1000 < 1$, so $n_i = 0$; for $i \in J$ it is $< \rho N_i/8$, so $n \in D_J$. Hence (7.29) gives

$$\widetilde{\psi}(\theta) = \psi(\theta) \qquad \big(\theta \in \mathcal{L} \cap (y + [-\sigma_*, \sigma_*]^r)\big).$$

The construction is translation-invariant in y.

Step 4. Smooth the correlation and choose a good translate. We now treat the non-trivial-frequency case. Let

$$C_0 := \big|\mathbb{E}_{h\in B_\rho} u(x_0 + h) e(-\phi(h))\big| \ge \tau.$$

Choose a regular radius

$$0 < \sigma \le \min\{\sigma_*, \tau\rho/(1000r)\}, \qquad \sigma \ge \tfrac{1}{2}\min\{\sigma_*, \tau\rho/(1000r)\}. \tag{7.30}$$

For $k \in B_\sigma$, regularity gives

$$B_{\rho-\sigma} \subseteq B_\rho \cap (B_\rho - k), \qquad B_\rho \cup (B_\rho - k) \subseteq B_{\rho+\sigma},$$

and consequently

$$\big|\mathbb{E}_{h\in B_\rho} f(h) - \mathbb{E}_{h\in B_\rho}\mathbb{E}_{k\in B_\sigma} f(h+k)\big| \le Cr\sigma/\rho \le \tau/4,$$

where $f(h) = u(x_0+h)e(-\phi(h))$ on $B_{1000\rho}$ (and $f = 0$ outside that set). Thus there is $h_0 \in B_\rho$ with

$$|\mathbb{E}_{k\in B_\sigma} u(x_0 + h_0 + k) e(-\phi(h_0 + k))| \ge \tau/2. \tag{7.31}$$

Step 5. Recenter the phase and build the Heisenberg orbit. The translated phase $k \mapsto \phi(h_0 + k)$ is locally quadratic on $B_{2\rho}$. Choose a lift $y \in \mathcal{L}$ with $|y|_\infty < \rho$ whose image in A is h_0. Apply the construction in Step 3 to $\psi(\vartheta) := \phi(\pi(\vartheta))$ on the lattice points near y. It supplies a global quadratic $\widetilde{q} : \mathcal{L} \to \mathbb{R}/\mathbb{Z}$ agreeing with $\phi(\pi(\vartheta))$ for $\vartheta \in y + [-\sigma, \sigma]^r$. Define the recentered phase $q_0(\theta) := \widetilde{q}(y + \theta)$; then $q_0(\theta) = \phi(h_0 + \pi(\theta))$ for every θ projecting to B_σ. The extension scale in (7.30) ensures this agreement.

Choose a real quadratic polynomial Q on $\mathbb{R}^r$ with

$$Q(\theta) \equiv -q_0(\theta) \pmod 1 \qquad (\theta \in \mathcal{L}).$$

In the integer-valued basis $1, z_i, \binom{z_i}{2}, z_i z_j$, subtract an integer-valued polynomial $Q_\mathbb{Z}$ so that

$$Q = Q_0 + Q_\mathbb{Z}, \quad \text{all coefficients of } Q_0 \text{ belonging to } [-1, 1]. \tag{7.32}$$

Choose $a \in \mathcal{L}$ whose image in A is $[1]_L$; then $na \in \mathcal{L}$ has image $[n]_L$ for every $n \in \mathbb{Z}$. Write

$$Q_\mathbb{Z}(z + na) - Q_\mathbb{Z}(z) = \lambda(n) \cdot z + c(n).$$

Because $Q_\mathbb{Z}$ is quadratic, $\lambda(n) = n\lambda_1$ is linear and $c(n) = nc_1 + \binom{n}{2}c_2$ is quadratic. Define, in the coordinates of $\mathcal{H}(r)$,

$$h(n) := (na, \lambda(n), c(n)).$$

If $u_1 = (a, \lambda_1, c_1)$ and $v_1 = (0, 0, c_2 - \lambda_1 \cdot a)$, then the Heisenberg product gives

$$h(n) = u_1^n v_1^{\binom{n}{2}};$$

thus h is a degree-two polynomial sequence.

Step 6. Construct the nilsequence function. We now turn the real quadratic polynomial into a Lipschitz function on the Heisenberg nilmanifold. Let $w = c\tau/r$, with $c > 0$ an absolute sufficiently small constant. Let $\varphi : \mathbb{R}^r \to [0, 1]$ be 1 on $[-(1-w)\sigma, (1-w)\sigma]^r$, supported on $[-\sigma, \sigma]^r$, and linear in the maximum norm on the intervening annulus. Then

$$\|\varphi\|_{\mathrm{Lip}} \leq (w\sigma)^{-1}.$$

Regularity at σ gives

$$\frac{|B_\sigma \setminus B_{(1-w)\sigma}|}{|B_\sigma|} \leq Crw \leq \tau/4. \tag{7.33}$$

Define on $\mathcal{H}(r)$

$$F_0(\theta, \lambda, c') := \sum_{m \in \mathbb{Z}^r} \varphi(\theta + m) e\big(Q_0(\theta + m) + c' + \lambda \cdot m\big). \tag{7.34}$$

The support has radius $< 1/4$, so at most one summand is non-zero and $|F_0| \leq 1$. Let $\gamma = (m_0, b, k) \in \Gamma(r)$, where $m_0, b \in \mathbb{Z}^r$ and $k \in \mathbb{Z}$. The group law gives

$$\begin{aligned} F_0((\theta, \lambda, c')\gamma) &= \sum_m \varphi(\theta + m_0 + m) e\big(Q_0(\theta + m_0 + m) + c' + k + \lambda \cdot m_0 + (\lambda + b) \cdot m\big) \\ &= \sum_j \varphi(\theta + j) e\big(Q_0(\theta + j) + c' + \lambda \cdot j + k + b \cdot (j - m_0)\big) = F_0(\theta, \lambda, c'), \end{aligned}$$

because $k + b \cdot (j - m_0) \in \mathbb{Z}$. Hence F_0 descends to $\mathcal{H}(r)/\Gamma(r)$. Denote the descended function by F. Moreover,

$$F_0((0, 0, t)x\Gamma) = e(t)F_0(x\Gamma).$$

Step 7. Transfer the Bohr correlation to an average on A. Let $k \in \mathbb{Z}$ satisfy $[k]_L \in B_\sigma$. Since $\pi(a) = [1]_L$ and $\ker \pi = \mathbb{Z}^r$, every lift of $[k]_L$ to $\mathcal{L}$ has the form $ka + m$ with $m \in \mathbb{Z}^r$. The condition $[k]_L \in B_\sigma$ guarantees the existence of $m = m(k) \in \mathbb{Z}^r$ such that

$$\theta_k := ka + m \in (-\sigma, \sigma)^r.$$

This lift is unique in $[-\sigma, \sigma]^r$: two such lifts would differ by an integer vector of maximum norm at most $2\sigma < 1$, so their difference must be zero. Since φ is supported in $[-\sigma, \sigma]^r$, only this term can contribute to the sum defining $F_0(h(k))$. Consequently,

$$F_0(h(k)) = \varphi(ka + m) e\big(Q(ka + m)\big). \tag{7.35}$$

The equality is modulo one: the exponent in (7.34) is

$$Q_0(ka + m) + Q_{\mathbb{Z}}(ka + m) - Q_{\mathbb{Z}}(m) \equiv Q(ka + m) \pmod 1,$$

because $Q_{\mathbb{Z}}(m) \in \mathbb{Z}$. On $B_{(1-w)\sigma}$, (7.35) equals $e(-q(k))$. Combining (7.31), (7.33), and $|u| \leq 1$ gives

$$|\mathbb{E}_{k \in B_\sigma} u(x_0 + h_0 + k) F_0(h(k))| \geq \tau/4. \tag{7.36}$$

Step 8. Normalize the correlation and bound the Lipschitz norm. For completeness, the lattice estimate needed to normalize (7.36) is

$$\frac{|B_\sigma|}{L} \geq \exp(-Cr^2 \log(2r))\sigma^r.$$

Indeed, the second inclusion in (7.27) gives at least $\prod_i(2\lceil\sigma N_i\rceil-1)$ points in B_σ. Since $2\lceil x\rceil-1\ge x$ for $x\ge 0$, this is bounded below by $\sigma^r\prod_i N_i$. Using (7.28) therefore gives the displayed bound, after enlarging C. Therefore (7.36) implies

$$|\mathbb{E}_{k\in A}u(x_0+h_0+k)F_0(h(k))|\ge\exp(-Cr\Lambda). \tag{7.37}$$

Choose $n_0\in\{0,\dots,L-1\}$ representing x_0+h_0 and put $g(n):=h(n-n_0)$. Translation of a degree-two polynomial sequence is again degree two, and the change of variable $k=[n-n_0]_L$ converts (7.37) into the required correlation with $u([n]_L)$.

Finally, use the right-invariant coordinate fields ∂_{θ_i}, $\partial_{\lambda_i}+\theta_i\partial_{c'}$, and $\partial_{c'}$. On every non-zero summand of (7.34), $|\theta_i+m_i|\le\sigma$; hence the middle field differentiates its phase by $m_i+\theta_i$ of size at most σ. The first fields differentiate φ and Q_0; the coefficients of Q_0 in (7.32) give $\|\nabla Q_0\|\le Cr^2$ on the support. Consequently

$$\|F_0\|_{\mathrm{Lip}}\le Cr^2\bigl(1+(w\sigma)^{-1}\bigr).$$

By (7.30), $w=c\tau/r$, and the definition of σ_*,

$$\log(2\sigma^{-1})\le C\Lambda,\qquad \log(2\|F_0\|_{\mathrm{Lip}})\le C\Lambda.$$

Set $M=\max\{2,\|F_0\|_{\mathrm{Lip}}\}$. With the descended function F and this value of M, the bounds in the lemma are exactly (7.24) and (7.25).

This completes the non-trivial-frequency case. Together with Step 1, this proves the lemma. □

Corollary 7.4. *Let $N\ge 1$ be an integer. Let $f:[N]\to\mathbb{C}$ satisfy $|f(n)|\le 1$ for $n\in[N]$ and*

$$\|f\|_{U^3[N]}\ge\epsilon,\qquad 0<\epsilon<1/2.$$

Put $\ell:=2+\log(2/\epsilon)$. Then there exist an integer $d\ge 1$, real $M\ge 2$ and $0<\eta<1/2$, a filtered nilmanifold G/Γ of dimension d, a degree-two polynomial sequence $g:\mathbb{Z}\to G$, and a function $F:G/\Gamma\to\mathbb{C}$ such that

$$|F(x\Gamma)|\le 1,\qquad \|F\|_{\mathrm{Lip}}\le M,$$

where (G,Γ) is either $(\mathbb{R},\mathbb{Z})$ or $(\mathcal{H}(r),\Gamma(r))$ for an integer $r\ge 1$. The function F has central frequency one in the following sense: in the Heisenberg case,

$$F((0,0,t)x\Gamma)=e(t)F(x\Gamma)\qquad(t\in\mathbb{R},\ x\in G),$$

while in the circle case,

$$F(t+x+\mathbb{Z})=e(t)F(x+\mathbb{Z})\qquad(t,x\in\mathbb{R}).$$

Moreover,

$$\left|\frac{1}{N}\sum_{n=1}^{N}f(n)F(g(n)\Gamma)\right|\ge\eta,$$

and, moreover,

$$d\ll\ell^{43},\qquad\log(2M/\eta)\ll\ell^{87}.$$

Proof. We divide the proof into three steps.

Step 1. Embed the interval into a cyclic group. By Bertrand's postulate choose a prime L with $8N<L<16N$. Let $A:=\mathbb{Z}/L\mathbb{Z}$ and define the zero extension $f_0:A\to\mathbb{C}$ by

$$f_0([n]_L):=\begin{cases}f(n), & n\in[N],\\ 0, & n\in\{0,\dots,L-1\}\setminus[N].\end{cases}$$

The residue classes $[1]_L,\dots,[N]_L$ are distinct. By the interval-embedding identity (3.3),

$$\|f_0\|_{U^3(A)}=\|f\|_{U^3[N]}\|\mathbf{1}_{[N]}\|_{U^3(A)}.$$

There is an absolute $c_0 > 0$ with $\|\mathbf{1}_{[N]}\|_{U^3(A)} \geq c_0$. Indeed, if $N \geq 64$, take

$$x \in [3N/8, 5N/8] \cap \mathbb{Z}, \qquad h_1, h_2, h_3 \in [-\lfloor N/32 \rfloor, \lfloor N/32 \rfloor] \cap \mathbb{Z}.$$

Every one of the eight vertices $x + \omega \cdot h$ lies in $[N]$, so there are $\gg N^4$ supported cubes. Since $L < 16N$, their proportion among all L^4 cyclic cubes is bounded below by an absolute constant. If $N < 64$, the N constant cubes already give the same conclusion after reducing c_0. Hence

$$\|f_0\|_{U^3(A)} \geq c_0\epsilon.$$

Write $\epsilon_0 := c_0\epsilon$ and $\ell_0 := 2 + \log(2/\epsilon_0) \ll \ell$.

Step 2. Extract a locally quadratic phase. Apply Theorem 7.2 to $u = f_0$ with parameter ϵ_0. It gives a nonempty $S \subseteq \widehat{A}$, with $r := |S| \ll \ell_0^{43}$, a regular radius $0 < \rho < 1/2000$, a point $x_0 \in A$, a locally quadratic function $\phi : B_A(S, 1000\rho) \to \mathbb{T}$, and $\tau_0 > 0$ such that

$$\left|\mathbb{E}_{h \in B_A(S,\rho)} f_0(x_0 + h) e(-\phi(h))\right| \geq \tau_0,$$

with $\log(2/\rho), \log(2/\tau_0) \ll \ell_0$. Replace τ_0 by $\tau := \min\{\tau_0, 1/4\}$; the correlation remains true and $\log(2/\tau) \ll \ell$.

Step 3. Realize the phase and transfer the parameters. Apply Theorem 7.3 to $u = f_0$ and the above data. It gives $g : \mathbb{Z} \to G$, $F : G/\Gamma \to \mathbb{C}$, and $M \geq 2$ with the properties in the statement, together with

$$\left|\frac{1}{L}\sum_{n=1}^{L} f_0([n]_L) F(g(n)\Gamma)\right| \geq \eta_0, \qquad \eta_0 \geq \exp(-Cr\Lambda),$$

where

$$\Lambda := r\log(2r) + \log(2/\rho) + \log(2/\tau), \qquad \log(2M) \leq C\Lambda.$$

In the circle case $d = 1$, and in the Heisenberg case $d = 2r + 1$; hence $d \ll \ell^{43}$. Because $f_0([n]_L) = 0$ for $N < n < L$, the cyclic average is exactly

$$\left|\frac{1}{L}\sum_{n=1}^{N} f(n) F(g(n)\Gamma)\right| \geq \eta_0.$$

Since $L/N > 1$, replacing L^{-1} by N^{-1} only increases its magnitude. Set $\eta := \min\{\eta_0, 1/4\}$ to ensure $0 < \eta < 1/2$. Finally,

$$r\log(2r) \ll \ell^{44}, \qquad \Lambda \ll \ell^{44}, \qquad r\Lambda \ll \ell^{87},$$

so

$$\log(2M/\eta) \ll \Lambda + r\Lambda \ll \ell^{87}.$$

This proves the corollary. □

7.3. **Fourier and bracket estimates.** We first approximate Lipschitz functions on a torus by finite Fourier sums, with explicit dependence on the dimension.

Lemma 7.5. *Let $m \geq 1$ be an integer, let $L_{\mathrm{Lip}} \in [1, \infty)$ be a real number, and let $\tau \in \mathbb{R}$ satisfy $0 < \tau < 1/2$. Give $\mathbb{T}^m := \mathbb{R}^m/\mathbb{Z}^m$ its Euclidean quotient metric. If $F : \mathbb{T}^m \to \mathbb{C}$ satisfies $|F(x)| \leq 1$ and $\|F\|_{\mathrm{Lip}} \leq L_{\mathrm{Lip}}$, then there are an integer $K \geq 2$ and coefficients c_k such that*

$$F_K(x) := \sum_{k \in [-K,K]^m \cap \mathbb{Z}^m} c_k e(k \cdot x), \qquad |c_k| \leq 1,$$

$$\|F - F_K\|_\infty \leq \tau, \qquad K \leq C\left(\frac{mL_{\mathrm{Lip}}}{\tau}\right)^4.$$

In particular,

$$\sum_{k \in [-K,K]^m \cap \mathbb{Z}^m} |c_k| \leq (2K+1)^m.$$

Proof. We divide the proof into four steps.

Step 1. Establish the one-dimensional Fej'er-kernel bounds. For an integer $K \geq 2$, define

$$\mathcal{F}_K(t) := \frac{1}{K+1}\left|\sum_{j=0}^{K} e(jt)\right|^2, \qquad t \in \mathbb{T}.$$

Orthogonality gives $\int_{\mathbb{T}} \mathcal{F}_K(t)\,dt = 1$. The geometric-sum formula and $|\sin(\pi t)| \gg \|t\|_{\mathbb{R}/\mathbb{Z}}$ give

$$0 \leq \mathcal{F}_K(t) \leq K+1, \qquad \mathcal{F}_K(t) \leq \frac{C}{(K+1)\|t\|_{\mathbb{R}/\mathbb{Z}}^2}.$$

Writing $d(t) := \|t\|_{\mathbb{R}/\mathbb{Z}}$ and splitting at $d(t) = 1/(K+1)$ gives

$$\mu_K := \int_{\mathbb{T}} d(t)\mathcal{F}_K(t)\,dt \leq 2(K+1)\int_0^{1/(K+1)} t\,dt + \frac{C}{K+1}\int_{1/(K+1)}^{1/2} \frac{dt}{t} \leq C\frac{\log(2K)}{K+1}.$$

Step 2. Smooth on the torus and estimate the approximation error. Let

$$\mathcal{K}_K(y) := \prod_{j=1}^{m} \mathcal{F}_K(y_j), \qquad y = (y_1, \ldots, y_m) \in \mathbb{T}^m,$$

and define

$$F_K(x) := \int_{\mathbb{T}^m} F(x-y)\mathcal{K}_K(y)\,dy.$$

The quotient metric is

$$d_{\mathbb{T}^m}(y,0) = \left(\sum_{j=1}^{m} \|y_j\|_{\mathbb{R}/\mathbb{Z}}^2\right)^{1/2} \leq \sum_{j=1}^{m} \|y_j\|_{\mathbb{R}/\mathbb{Z}}.$$

Since $\mathcal{K}_K$ is a product probability kernel,

$$|F_K(x) - F(x)| \leq L_{\mathrm{Lip}} \int d_{\mathbb{T}^m}(y,0)\mathcal{K}_K(y)\,dy \leq L_{\mathrm{Lip}}\, m\mu_K \leq C\frac{mL_{\mathrm{Lip}}\log(2K)}{K+1}.$$

Step 3. Choose the truncation scale. Put

$$A := \frac{mL_{\mathrm{Lip}}}{\tau} \geq 2, \qquad K := \left\lceil C_0 A^2 \log(2A) \right\rceil,$$

where C_0 is a sufficiently large absolute constant. Then $K \geq 2$, $\log(2K) \ll \log(2A)$, and the preceding estimate gives $\|F - F_K\|_\infty \leq \tau$. Since $A \geq 2$ and $\log(2A) \leq A$,

$$K \leq CA^4 = C\left(\frac{mL_{\mathrm{Lip}}}{\tau}\right)^4.$$

Step 4. Identify the Fourier coefficients and finish. The one-dimensional Fourier expansion

$$\mathcal{F}_K(t) = \sum_{|j|\leq K} \left(1 - \frac{|j|}{K+1}\right) e(jt)$$

shows that

$$c_k = \widehat{F}(k)\prod_{j=1}^{m}\left(1 - \frac{|k_j|}{K+1}\right), \qquad \widehat{F}(k) := \int_{\mathbb{T}^m} F(x)e(-k\cdot x)\,dx.$$

Thus $|c_k| \leq |\widehat{F}(k)| \leq 1$, and there are $(2K+1)^m$ possible frequencies. Therefore

$$\sum_k |c_k| \leq (2K+1)^m.$$

This proves the lemma. □

The next lemma is a one-parameter bracket estimate. It converts the assumption that a bracket polynomial is close to an integer on a positive density of h into two complementary systems of integer relations. The notation must distinguish centered representatives from distance to the nearest integer: for $t \in \mathbb{R}$ put

$$\|t\|_{\mathbb{R}/\mathbb{Z}} := \min_{q\in\mathbb{Z}} |t-q|, \qquad \{t\} \in [-1/2, 1/2) \text{ with } t - \{t\} \in \mathbb{Z},$$

and for $z = (z_1, \dots, z_m) \in \mathbb{R}^m$ put $\{z\} := (\{z_1\}, \dots, \{z_m\})$. Thus $a \cdot \{h\alpha\}$ is an ordinary real number.

Lemma 7.6. *Let $m, N \in \mathbb{Z}_{\geq 1}$ with $N \geq 2$, let $0 < \rho < 1/10$ and $M \geq 2$ be real numbers, and let $K \geq 1$ be real. Let $a, \alpha \in \mathbb{R}^m$ and $\beta, \gamma \in \mathbb{R}$, with $|a| \leq M$. Suppose that*

$$\|\beta + \gamma h + a \cdot \{h\alpha\}\|_{\mathbb{R}/\mathbb{Z}} \leq K/N$$

for at least ρN integers $h \in [N]$. If

$$N \geq \left(\frac{mM}{\rho}\right)^{C(m+1)^2} K,$$

then there are an integer t with $0 \leq t \leq m$ and two families

$$y_1, \dots, y_t \in \mathbb{Z}^m, \qquad x_1, \dots, x_{m-t} \in \mathbb{Z}^m.$$

The vectors $y_1, \dots, y_t$ are linearly independent over $\mathbb{R}$. The vectors $x_1, \dots, x_{m-t}$ are also linearly independent over $\mathbb{R}$. They satisfy

$$\operatorname{span}_{\mathbb{R}}(y_1, \dots, y_t) = \bigcap_{j=1}^{m-t} x_j^{\perp}, \tag{7.38}$$

$$|y_i|, |x_j| \leq \left(\frac{mM}{\rho}\right)^{C(m+1)^2}, \tag{7.39}$$

$$\|y_i \cdot a\|_{\mathbb{R}/\mathbb{Z}},\ \|x_j \cdot \alpha\|_{\mathbb{R}/\mathbb{Z}} \leq \left(\frac{mM}{\rho}\right)^{C(m+1)^2} K/N. \tag{7.40}$$

Here the span of an empty list is $\{0\}$ and the intersection of an empty family of hyperplanes is $\mathbb{R}^m$.

Proof. We divide the proof into four steps.

Step 1. Apply the external lemma with one fixed bound. In this proof, the constant C in the statement is chosen once and then kept fixed. We use Leng's one-parameter refined bracket-polynomial lemma [21, Lemma 3.1]. Its quantitative form is as follows. Suppose that $D, N \geq 1$ are integers with $N \geq 2$, that $0 < \delta < 1/10$, $M' \geq 2$, $K' \geq 2$, and that $b, \theta \in \mathbb{R}^D$, $\beta_0 \in \mathbb{R}$, with $|b| \leq M'$. If

$$\|\beta_0 + b \cdot \{h\theta\}\|_{\mathbb{R}/\mathbb{Z}} < K'/N \quad \text{for at least } \delta N \text{ integers } h \in [N], \tag{7.41}$$

then either

$$N \ll (D^D M'/\delta)^{O(1)} \quad \text{or} \quad K'/N > 1/100,$$

or there are linearly independent integer families $w_1, \dots, w_s$ and $\eta_1, \dots, \eta_{D-s}$ such that

$$\operatorname{span}_{\mathbb{R}}(w_1, \dots, w_s) = \bigcap_{j=1}^{D-s} \eta_j^{\perp},$$

$$|w_i|,\ |\eta_j| \leq (D^D M'/\delta)^{O(1)},$$

$$|w_i \cdot b|,\ \|\eta_j \cdot \theta\|_{\mathbb{R}/\mathbb{Z}} \leq (M'/\delta)^{O(1)} D^{O(D)} K'/N.$$

Set

$$D := m+1, \qquad \widetilde{a} := (a,1), \qquad \widetilde{\alpha} := (\alpha,\gamma), \qquad K_* := 2K.$$

Then $|\widetilde{a}| \le 2M$, and

$$\widetilde{a}\cdot\{h\widetilde{\alpha}\} = a\cdot\{h\alpha\} + \{h\gamma\} \equiv a\cdot\{h\alpha\} + \gamma h \pmod 1.$$

The hypothesis therefore gives (7.41) with $b = \widetilde{a}$, $\theta = \widetilde{\alpha}$, $\beta_0 = \beta$, $\delta = \rho$, $M' = 2M$, and $K' = K_*$.

Define the single auxiliary bound

$$R := \frac{mM}{\rho}, \qquad T := R^{C(m+1)^2/4}.$$

Choose $C \ge 1$ large enough that T dominates the exceptional threshold and every height or error factor in the Leng's lemma. This choice is uniform in all the parameters. Indeed, $R > 20m$, $D = m+1 \le 2m < R/10$, and

$$\frac{2D^D M}{\rho} = \frac{2R}{m} D^D \le 2R(R/10)^D \le R^{D+1}.$$

Consequently all these upper bounds are $R^{O(D)}$ and are at most $R^{CD^2/4} = T$ for one sufficiently large absolute C. Since $T \ge R > 20$, our hypothesis now reads

$$N \ge T^4 K > T, \qquad \frac{K_*}{N} \le \frac{2}{T^4} < \frac{1}{100}.$$

Neither exceptional alternative can occur. We obtain the above orthogonal families in $\mathbb{Z}^D$ with

$$|w_i|,\ |\eta_j| \le T, \qquad |w_i\cdot\widetilde{a}|,\ \|\eta_j\cdot\widetilde{\alpha}\|_{\mathbb{R}/\mathbb{Z}} \le TK_*/N. \tag{7.42}$$

Step 2. Project the w-relations. Write $w_i = (w_i', q_i)$ with $w_i' \in \mathbb{Z}^m$ and $q_i \in \mathbb{Z}$. Let $Y := \operatorname{span}_{\mathbb{R}}\{w_1', \dots, w_s'\}$ and $t := \dim Y$. Choose a maximal linearly independent sublist of the w_i' and call it $y_1, \dots, y_t$. Then $|y_i| \le T$ and the y_i span Y. If $y_i = w_{\nu(i)}'$, then the last coordinate $q_{\nu(i)}$ is an integer, so

$$\|y_i\cdot a\|_{\mathbb{R}/\mathbb{Z}} = \|w_{\nu(i)}\cdot\widetilde{a} - q_{\nu(i)}\|_{\mathbb{R}/\mathbb{Z}} \le |w_{\nu(i)}\cdot\widetilde{a}| \le TK_*/N.$$

Step 3. Construct an integer basis of $Y^\perp$. Put $p := D - s$ and

$$W := \operatorname{span}_{\mathbb{R}}(w_1,\dots,w_s), \qquad E := \operatorname{span}_{\mathbb{R}}(\eta_1,\dots,\eta_p) = W^\perp, \qquad U := Y^\perp.$$

Here a basis of U means a basis over $\mathbb{R}$ whose vectors have integer coordinates. We construct it directly from the η_j.

3a. Identify the vectors whose last coordinate is zero. For $u \in \mathbb{R}^m$, the identity $(u,0)\cdot w_i = u\cdot w_i'$ gives

$$(u,0) \in E \iff u\cdot w_i' = 0 \quad (1 \le i \le s) \iff u \in Y^\perp = U. \tag{7.43}$$

Thus deleting the last coordinate identifies $E \cap (\mathbb{R}^m \times \{0\})$ with U. Write

$$\eta_j = (v_j, b_j), \qquad v_j \in \mathbb{Z}^m, \quad b_j \in \mathbb{Z} \quad (1 \le j \le p).$$

3b. Choose a basis of that intersection. If $b_j = 0$ for every j, all the η_j already lie in $\mathbb{R}^m \times \{0\}$. Therefore $v_1, \dots, v_p$ form a basis of U. In this case $p = \dim U = m - t$, and we set

$$x_j := v_j, \qquad (x_j, 0) = \eta_j \quad (1 \le j \le p). \tag{7.44}$$

This also covers $p = 0$, when $U = \{0\}$ and the x-family is empty.

Otherwise, reorder the η_j so that $b_1 \ne 0$. For $2 \le j \le p$, set

$$x_{j-1} := b_1 v_j - b_j v_1 \in \mathbb{Z}^m, \qquad (x_{j-1}, 0) = b_1\eta_j - b_j\eta_1 \in E. \tag{7.45}$$

By (7.43), every x_{j-1} lies in U. To prove independence, suppose that $\sum_{j=2}^{p} d_j x_{j-1} = 0$ with $d_j \in \mathbb{R}$. Then

$$\sum_{j=2}^{p} b_1 d_j \eta_j - \left(\sum_{j=2}^{p} b_j d_j \right) \eta_1 = 0.$$

The η_j are linearly independent and $b_1 \neq 0$, so every $d_j = 0$.

To prove spanning, take $u \in U$. By (7.43), there are real numbers $d_1, \dots, d_p$ such that

$$(u, 0) = \sum_{j=1}^{p} d_j \eta_j, \qquad \sum_{j=1}^{p} b_j d_j = 0, \qquad d_1 = -\frac{1}{b_1} \sum_{j=2}^{p} b_j d_j.$$

Substitution yields

$$(u, 0) = \sum_{j=2}^{p} \frac{d_j}{b_1} (b_1 \eta_j - b_j \eta_1), \qquad u = \sum_{j=2}^{p} \frac{d_j}{b_1} x_{j-1}.$$

Hence these $p - 1$ vectors form a basis of U, and $p - 1 = m - t$. If $p = 1$, the sum is empty and the same argument gives $U = \{0\}$.

Step 4. Verify orthogonality and the quantitative bounds. In both cases, $x_1, \dots, x_{m-t}$ form a basis of $U = Y^{\perp}$. Consequently,

$$\bigcap_{j=1}^{m-t} x_j^{\perp} = U^{\perp} = Y = \operatorname{span}_{\mathbb{R}}(y_1, \dots, y_t).$$

This identity also holds when $t = 0$ or $t = m$, with the stated empty-family conventions.

By (7.42), $|v_j|, |b_j| \leq T$ and $T \geq 1$. In the first case of Step 3b, (7.44) gives

$$|x_j| \leq T, \qquad \|x_j \cdot \alpha\|_{\mathbb{R}/\mathbb{Z}} = \|\eta_j \cdot \widetilde{\alpha}\|_{\mathbb{R}/\mathbb{Z}} \leq T K_* / N.$$

In the second case, (7.45) gives

$$|x_{j-1}| \leq |b_1||v_j| + |b_j||v_1| \leq 2T^2.$$

Since b_1, b_j are integers, the triangle inequality on $\mathbb{R}/\mathbb{Z}$ gives

$$\|x_{j-1} \cdot \alpha\|_{\mathbb{R}/\mathbb{Z}} = \|b_1 \eta_j \cdot \widetilde{\alpha} - b_j \eta_1 \cdot \widetilde{\alpha}\|_{\mathbb{R}/\mathbb{Z}} \leq |b_1| \|\eta_j \cdot \widetilde{\alpha}\|_{\mathbb{R}/\mathbb{Z}} + |b_j| \|\eta_1 \cdot \widetilde{\alpha}\|_{\mathbb{R}/\mathbb{Z}} \leq 2T^2 K_* / N.$$

Together with Step 2 and $K_* = 2K$, these estimates imply, for all applicable indices,

$$|y_i|, |x_j| \leq 2T^2, \qquad \|y_i \cdot a\|_{\mathbb{R}/\mathbb{Z}}, \quad \|x_j \cdot \alpha\|_{\mathbb{R}/\mathbb{Z}} \leq 4T^2 K / N. \tag{7.46}$$

Since $T \geq 2$, the same bound T chosen in Step 1 gives

$$2T^2 \leq T^4, \qquad 4T^2 \leq T^4 = R^{C(m+1)^2}.$$

Thus (7.46) gives exactly (7.39) and (7.40). □

7.4. **Horizontal relations.** For the next theorem, fix $q \in \{1, 2\}$ and integers $r_\nu \geq 1$, and let

$$G := \prod_{\nu=1}^{q} \mathcal{H}(r_\nu)$$

with the explicit lattice

$$\Gamma := \prod_{\nu=1}^{q} \left(\mathbb{Z}^{r_\nu} \times \mathbb{Z}^{r_\nu} \times \mathbb{Z} \right).$$

Set $G_1 := G$, $G_2 := [G, G]$, and $G_3 := \{1\}$. In the present coordinates,

$$G_2 = \{z(c) : c \in \mathbb{R}^q\}, \qquad [G, G_2] = \{1\};$$

these are the only subgroup levels used below. Put $d := \dim G$. Write

$$z(c) := ((0, 0, c_1), \dots, (0, 0, c_q)), \qquad c = (c_1, \dots, c_q) \in \mathbb{R}^q$$

for a central element. Fix $M \geq 2$ and an integer vector $\ell = (\ell_1, \dots, \ell_q) \in \mathbb{Z}^q$ with $\max_\nu |\ell_\nu| \leq M$, and define

$$\xi : G_2 \longrightarrow \mathbb{R}, \qquad \xi(z(c)) := \ell \cdot c.$$

Thus ξ is integer-valued on $\Gamma \cap G_2$. Let $F : G/\Gamma \to \mathbb{C}$ satisfy $|F| \leq 1$ and $\|F\|_{\mathrm{Lip}} \leq M$, and assume that

$$F(z(c)x\Gamma) = e(\ell \cdot c)F(x\Gamma) \qquad (c \in \mathbb{R}^q,\ x \in G). \tag{7.47}$$

The displayed identity is the exact assumption used below: $z(c) \in G_2$ acts on F by the scalar $e(\xi(z(c)))$. The coordinate height of ξ is $\max_\nu |\ell_\nu|$. Since G is two-step nilpotent, a polynomial sequence for $G_1 \supseteq G_2 \supseteq G_3$ is exactly the representation

$$g(n) = g_0 u^n v^{\binom{n}{2}}, \qquad g_0, u \in G, \quad v \in G_2.$$

Theorem 7.7. *With this setup, let $N \geq 2$ be an integer and suppose*

$$\left| N^{-1} \sum_{n=1}^{N} F(g(n)\Gamma) \right| \geq \delta, \qquad 0 < \delta < 1/10, \qquad Z := 10Md/\delta.$$

If $N \geq Z^{C(d+1)^4}$, there are an integer s with $0 \leq s \leq 2d$ and horizontal characters $\lambda_1, \dots, \lambda_s : G \to \mathbb{R}$ which are integer-valued on Γ and vanish on G_2. Put

$$W := \bigcap_{i=1}^{s} \ker \lambda_i, \qquad \ker \lambda_i := \{w \in G : \lambda_i(w) = 0 \text{ in } \mathbb{R}\}.$$

Then

$$|\lambda_i|,\ \|\lambda_i \circ g\|_{C^\infty[N]} \leq Z^{C(d+1)^4}, \qquad \xi([W, W]) = 0.$$

Here $[W, W]$ denotes the subgroup generated by all commutators $[w_1, w_2] = w_1 w_2 w_1^{-1} w_2^{-1}$ with $w_1, w_2 \in W$.

Proof. We use the convention $[x, y] = xyx^{-1}y^{-1}$ throughout the proof. Put

$$m := \dim(G/G_2) = 2 \sum_{\nu=1}^{q} r_\nu$$

and identify G/G_2 with $\mathbb{R}^m$ by the horizontal coordinates. Let $\pi : G \to \mathbb{R}^m$ be the quotient homomorphism. We divide the rest proof into seven steps.

Step 1. Choose bounded representatives and normalize the polynomial sequence. We first make the bounded representatives used in the proof explicit. For one Heisenberg factor, write $x = (\theta, \lambda, c)$ with $\theta, \lambda \in \mathbb{R}^r$ and $c \in \mathbb{R}$. Choose $p, q \in \mathbb{Z}^r$ so that $\theta_0 := \theta - p$ and $\lambda_0 := \lambda - q$ belong to $[-1/2, 1/2)^r$, and then choose $k \in \mathbb{Z}$ so that $c_0 := c - \lambda_0 \cdot p - k$ belongs to $[-1/2, 1/2)$. The group law then gives

$$x = (\theta_0, \lambda_0, c_0)(p, q, k).$$

For the product G, let $\mathcal{B}$ be the product of these centered half-open boxes and apply this construction in every factor. Thus every $x \in G$ can be written as $x = b\gamma$ with $\gamma \in \Gamma$ and with $b \in \mathcal{B}$. In particular, if $u^h = \zeta_h \gamma_h$, then $\pi(\zeta_h) = \{h\pi(u)\}$, where $\{\cdot\}$ denotes the representative in $[-1/2, 1/2)^m$.

Write $g_0 = b\gamma_0$ by this construction. Since v is central,

$$g_0 u^n v^{\binom{n}{2}} = b(\gamma_0 u \gamma_0^{-1})^n v^{\binom{n}{2}} \gamma_0. \tag{7.48}$$

Define $\widetilde{F}(x\Gamma) := F(bx\Gamma)$ and replace u by $\gamma_0 u \gamma_0^{-1}$. The final factor $\gamma_0 \in \Gamma$ is invisible on G/Γ, so this replacement leaves the nilsequence unchanged. It remains to check that the translation by b preserves a controlled Lipschitz norm.

We use on G the right-invariant metric associated with the Euclidean norm in the displayed coordinates, and write $d_{G/\Gamma}$ for the induced quotient metric. If $x(t) = ((\theta_\nu(t), \lambda_\nu(t), c_\nu(t)))_\nu$ is

a differentiable path, write $\mathrm{speed}(x(t))$ for its metric speed.[13] Differentiating $x(t+s)x(t)^{-1}$ at $s=0$ gives

$$\mathrm{speed}(x(t))^2 = \sum_{\nu=1}^{q} \left(\|\dot\theta_\nu\|_2^2 + \|\dot\lambda_\nu\|_2^2 + |\dot c_\nu - \dot\lambda_\nu \cdot \theta_\nu|^2 \right). \tag{7.49}$$

If $b_\nu = (p_\nu, q_\nu, r_\nu)$, applying the same formula to the path $bx(t)$ changes the last term in factor ν to

$$\dot c_\nu - \dot\lambda_\nu \cdot \theta_\nu + q_\nu \cdot \dot\theta_\nu - p_\nu \cdot \dot\lambda_\nu.$$

Put $E_\nu := q_\nu \cdot \dot\theta_\nu - p_\nu \cdot \dot\lambda_\nu$. Since each coordinate of p_ν and q_ν has absolute value at most $1/2$, Cauchy–Schwarz gives

$$\left(\sum_{\nu=1}^{q} |E_\nu|^2 \right)^{1/2} \le \sqrt{d} \left(\sum_{\nu=1}^{q} (\|\dot\theta_\nu\|_2^2 + \|\dot\lambda_\nu\|_2^2) \right)^{1/2} \le \sqrt{d}\, \mathrm{speed}(x(t)).$$

The Euclidean triangle inequality applied to the central coordinates now gives

$$\mathrm{speed}(bx(t)) \le (1+\sqrt{d})\, \mathrm{speed}(x(t)). \tag{7.50}$$

The quotient distance is the infimum of the lengths of lifted paths, with lattice translates allowed at the endpoints. Applying (7.50) to every such path and then taking the two infima gives

$$d_{G/\Gamma}(bx\Gamma, by\Gamma) \le (1+\sqrt{d}) d_{G/\Gamma}(x\Gamma, y\Gamma), \qquad \|\widetilde F\|_{\mathrm{Lip}} \le M(1+\sqrt{d}). \tag{7.51}$$

We now rename $\widetilde F$ as F.

Next normalize the linear coefficient. Write $u = u_0\gamma$ with $\gamma \in \Gamma$ and with $u_0 \in \mathcal{B}$. Since commutators are central,

$$(u_0\gamma)^n = u_0^n \gamma^n [\gamma, u_0]^{\binom{n}{2}}. \tag{7.52}$$

The commutator factor is central, so it can be moved past γ^n; the factor γ^n then disappears on the quotient, and the remaining central factor is absorbed into v. We may therefore replace u by u_0 and v by $v[\gamma, u_0]$, obtaining

$$g(n) = u^n v^{\binom{n}{2}}, \qquad \alpha := \pi(u) \in [-1/2, 1/2)^m, \qquad \omega := \xi(v).$$

The real number ω is not subject to a quantitative bound. For a horizontal character λ, let P_{old} and P_{new} denote the scalar polynomials before and after the two normalizations, viewed in $\mathbb{R}/\mathbb{Z}$. The calculation above gives the exact identity

$$P_{\mathrm{old}}(n) - P_{\mathrm{new}}(n) = \lambda(g_0) + n\lambda(\gamma) \pmod 1, \qquad \lambda(\gamma) \in \mathbb{Z}.$$

Thus the second term is zero in $\mathbb{R}/\mathbb{Z}$, and only the constant coefficient changes. Since the $C^\infty[N]$ norm ignores the constant coefficient and measures the remaining coefficients modulo $\mathbb{Z}$, $\|\lambda \circ g\|_{C^\infty[N]}$ is unchanged.

Step 2. Find many large derivative sums. Set

$$A_n := F(g(n)\Gamma), \qquad S_h := \sum_{n=1}^{N-h} A_{n+h}\overline{A_n} \quad (1 \le h < N).$$

The hypothesis and $|A_n| \le 1$ imply

$$\delta^2 N^2 \le \left| \sum_{n=1}^{N} A_n \right|^2 \le N + 2\sum_{h=1}^{N-1} |S_h|.$$

[13] For a differentiable path $x : I \to G$, the metric speed at t is $\lim_{s\to 0} d_G(x(t+s), x(t))/|s|$, when this limit exists; the length of the path is the integral of this speed.

Since the theorem's lower bound on N implies $N \ge 4\delta^{-2}$, the sum on the right gives

$$\sum_h |S_h| \ge 3\delta^2 N^2/8.$$

Shifts with $|S_h| < \delta^2 N/8$ contribute less than $\delta^2 N^2/8$. Consequently, for

$$\mathcal{H} := \{h \in [N-1] : |S_h| \ge \delta^2 N/8\},$$

the remaining shifts contribute at least $\delta^2 N^2/4$. Since $|S_h| \le N - h \le N$, we obtain

$$|\mathcal{H}| \ge \delta^2 N/4, \qquad |S_h| \ge \delta^2 N/8, \qquad N - h \ge \delta^2 N/8 \quad (h \in \mathcal{H}).$$

Step 3. Rewrite each derivative on a horizontal torus.
3a. The torus function. For $h \in \mathcal{H}$, write $u^h = \zeta_h \gamma_h$ as in Step 1 and define

$$H_h(y) := F(\zeta_h y \Gamma)\overline{F(y\Gamma)}.$$

Because Γ is the quotient lattice, $F(x\gamma\Gamma) = F(x\Gamma)$ for $\gamma \in \Gamma$; hence H_h is right-Γ-invariant. For $z \in G_2$, (7.47) gives

$$H_h(zy) = e(\xi(z))F(\zeta_h y\Gamma)\, e(-\xi(z))\overline{F(y\Gamma)} = H_h(y). \tag{7.53}$$

If $\pi(y') - \pi(y) \in \mathbb{Z}^m$, choose $\gamma \in \Gamma$ with this horizontal coordinate. Then $y' = zy\gamma$ for some $z \in G_2$, so (7.53) and lattice invariance show that $H_h(y)$ depends only on $\pi(y) \bmod \mathbb{Z}^m$. Thus there is a well-defined $\Phi_h : \mathbb{T}^m \to \mathbb{C}$ with

$$\Phi_h(\pi(y) \bmod \mathbb{Z}^m) = H_h(y), \qquad |\Phi_h| \le 1. \tag{7.54}$$

3b. Lipschitz bound. Use the horizontal section

$$\sigma(t) := ((\theta_\nu, \lambda_\nu, 0))_{\nu=1}^q, \qquad t = ((\theta_\nu, \lambda_\nu))_\nu \in \mathbb{R}^m.$$

Choose lifts $t, t' \in [-1,1]^m$ of two torus points for which $\|t - t'\|_2$ equals their Euclidean quotient distance. Along $t_s = t + s(t'-t)$, (7.49) gives

$$\operatorname{speed}(\sigma(t_s))^2 \le (1+d)\|t'-t\|_2^2.$$

Hence (7.51) and (7.50) imply

$$d_{G/\Gamma}(\sigma(t)\Gamma, \sigma(t')\Gamma) \le (1+\sqrt{d})\|t-t'\|_2,$$
$$d_{G/\Gamma}(\zeta_h\sigma(t)\Gamma, \zeta_h\sigma(t')\Gamma) \le (1+\sqrt{d})^2\|t-t'\|_2.$$

The elementary inequality $|z_1\overline{z}_2 - w_1\overline{w}_2| \le |z_1 - w_1| + |z_2 - w_2|$ for $|z_i|, |w_i| \le 1$, together with (7.51), therefore gives

$$\|\Phi_h\|_{\mathrm{Lip}} \le 1 + M(1+\sqrt{d})^2(2+\sqrt{d}) \le L, \qquad L := 100M(d+1)^3. \tag{7.55}$$

The periodicity in (7.54) permits the same segment argument across the boundaries of the coordinate cube.
3c. The bilinear phase. For horizontal vectors s, t, define

$$B_\ell(s,t) := \sum_{\nu=1}^q \ell_\nu\big(s_{\lambda,\nu}\cdot t_{\theta,\nu} - t_{\lambda,\nu}\cdot s_{\theta,\nu}\big).$$

The group law gives

$$\xi([x,y]) = B_\ell(\pi(x), \pi(y)). \tag{7.56}$$

Define $a \in \mathbb{R}^m$ by $a \cdot t = B_\ell(\alpha, t)$. Its coordinates are $(\ell_\nu\alpha_{\lambda,\nu}, -\ell_\nu\alpha_{\theta,\nu})_\nu$, so, in the maximum norm used for character heights, $|a| \le M/2 \le L$.

For $h \in \mathcal{H}$, the identity

$$\binom{n+h}{2} - \binom{n}{2} = hn + \binom{h}{2}$$

and (7.52) give

$$\Delta_{h,n} := \big(\zeta_h^{-1} g(n+h)\gamma_h^{-1}\big) g(n)^{-1} = [\gamma_h, u]^n v^{hn+\binom{h}{2}} = [u, \zeta_h]^n v^{hn+\binom{h}{2}} \in G_2.$$

Equivalently,

$$g(n+h)\Gamma = \Delta_{h,n}\, \zeta_h g(n)\Gamma.$$

Applying (7.47), then multiplying by $\overline{F(g(n)\Gamma)}$, gives

$$A_{n+h}\overline{A_n} = e(\xi(\Delta_{h,n}))\, H_h(g(n)). \tag{7.57}$$

By (7.56)–(7.57),

$$\xi(\Delta_{h,n}) = n\, a \cdot \{h\alpha\} + \left(hn + \binom{h}{2}\right)\omega.$$

Since $\pi(g(n)) = n\alpha$, (7.54) now yields the exact formula

$$A_{n+h}\overline{A_n} = e\left(\binom{h}{2}\omega\right) \Phi_h(n\alpha) e\left(n\big(a \cdot \{h\alpha\} + h\omega\big)\right). \tag{7.58}$$

Step 4. Obtain one bracket relation for many shifts. Apply Theorem 7.5 with $\tau := \delta^2/32$ and the common Lipschitz bound L in (7.55). The resulting Fourier frequency bound K_0 is an integer independent of h. For each $h \in \mathcal{H}$,

$$P_h(t) := \sum_{k \in [-K_0, K_0]^m \cap \mathbb{Z}^m} c_{h,k} e(k \cdot t), \qquad |c_{h,k}| \le 1, \qquad \|\Phi_h - P_h\|_\infty \le \tau,$$

where $K_0 \ll (mL/\tau)^4$. Put $J := (2K_0+1)^m$. Replacing Φ_h in (7.58) changes the sum by at most $\tau(N-h) \le \delta^2 N/32$. Therefore

$$\left|\sum_k c_{h,k} \sum_{n=1}^{N-h} e\big(n\vartheta_{h,k}\big)\right| \ge \frac{3\delta^2 N}{32}, \qquad \vartheta_{h,k} := k \cdot \alpha + a \cdot \{h\alpha\} + h\omega.$$

By the triangle inequality,

$$\frac{3\delta^2 N}{32} \le J \max_k \left|\sum_{n=1}^{N-h} e(n\vartheta_{h,k})\right|.$$

Thus, for some $k_h \in [-K_0, K_0]^m \cap \mathbb{Z}^m$,

$$\left|\sum_{n=1}^{N-h} e(n\vartheta_{h,k_h})\right| \ge \frac{3\delta^2 N}{32J}.$$

The geometric-sum bound

$$\left|\sum_{n=1}^{H} e(n\theta)\right| \le \frac{1}{2\|\theta\|_{\mathbb{R}/\mathbb{Z}}}$$

implies

$$\|\vartheta_{h,k_h}\|_{\mathbb{R}/\mathbb{Z}} \le \frac{16J}{3\delta^2 N}.$$

Set

$$\rho := \frac{\delta^2}{8J}, \qquad K_{\mathrm{b}} := \rho^{-1} = \frac{8J}{\delta^2}.$$

Then

$$\|k_h \cdot \alpha + a \cdot \{h\alpha\} + h\omega\|_{\mathbb{R}/\mathbb{Z}} \le \frac{K_{\mathrm{b}}}{N}. \tag{7.59}$$

Partition $\mathcal{H}$ according to the value of k_h. There are at most J classes, so one class contains at least $|\mathcal{H}|/J \ge 2\rho N$ shifts. Fix its frequency k. Then (7.59) holds with this same k for at least ρN integers $h \in [N]$.

Step 5. Apply Lemma 7.6 and check its parameters. We apply Theorem 7.6 with

$$\beta := k \cdot \alpha, \qquad \gamma := \omega, \qquad M_{\mathrm{a}} := L, \qquad K := K_{\mathrm{b}} = \rho^{-1}.$$

Let C_0 be a fixed absolute constant large enough for the constants in Lemmas Theorem 7.5 and Theorem 7.6. Since $m \le Z$, $L \le Z^3$, and $\tau^{-1} = 32\delta^{-2} \le Z^2$, Theorem 7.5 gives $K_0 \ll Z^{24}$. Increasing C_0 once, we may assume

$$L \le Z^{C_0}, \qquad 2K_0 + 1 \le Z^{C_0}.$$

Consequently,

$$J \le Z^{C_0 d}, \qquad \rho^{-1} = 8J\delta^{-2} \le Z^{C_0(d+1)}.$$

Since $mL \le Z^4$,

$$\frac{mL}{\rho} \le Z^{(C_0+4)(d+1)}.$$

Define

$$B := \left(\frac{mL}{\rho}\right)^{C_0(m+1)^2}.$$

Then

$$B \le Z^{C_0(C_0+4)(d+1)^3}, \qquad \frac{MB}{\rho} \le Z^{(C_0(C_0+4)+C_0+1)(d+1)^4}$$

Choose the theorem's absolute constant $C \ge \max\{C_0(C_0+4)+C_0+1, 3\}$ once and for all. Then the hypothesis $N \ge Z^{C(d+1)^4}$ implies both $N \ge 4\delta^{-2}$ and the sample-size inequality $N \ge B/\rho$ required by Theorem 7.6.

Theorem 7.6 now gives integer vectors $y_1, \dots, y_t$ and $x_1, \dots, x_{m-t}$ such that

$$\begin{aligned} Y := \operatorname{span}_{\mathbb{R}}(y_1, \dots, y_t) = \bigcap_{j=1}^{m-t} x_j^{\perp}, \qquad |x_j|,\ |y_i| \le B, \\ \|x_j \cdot \alpha\|_{\mathbb{R}/\mathbb{Z}},\ \|y_i \cdot a\|_{\mathbb{R}/\mathbb{Z}} \le \frac{B}{\rho N}. \end{aligned} \tag{7.60}$$

Each family is linearly independent over $\mathbb{R}$.

Step 6. Construct the horizontal characters. Define

$$\chi_j(z) := x_j \cdot \pi(z), \qquad \mu_i(z) := B_\ell(y_i, \pi(z)).$$

Both maps are homomorphisms because π is a homomorphism. They vanish on G_2 and are integer-valued on Γ because x_j, y_i, ℓ have integer coordinates. More explicitly,

$$\mu_i(z) = \sum_{\nu=1}^{q} \ell_\nu\big(y_{i,\lambda,\nu} \cdot \theta_\nu(z) - y_{i,\theta,\nu} \cdot \lambda_\nu(z)\big).$$

Hence

$$|\chi_j| \le B, \qquad |\mu_i| \le MB, \qquad \chi_j(u) = x_j \cdot \alpha, \qquad \mu_i(u) = -y_i \cdot a.$$

Since $v \in G_2$, $\lambda(g(n)) = n\lambda(u)$ for every one of these characters λ. Therefore

$$\|\lambda \circ g\|_{C^\infty[N]} = N\|\lambda(u)\|_{\mathbb{R}/\mathbb{Z}} \le \frac{B}{\rho}.$$

The normalization in Step 1 shows that the same estimate holds for the original sequence. There are $m \le d \le 2d$ characters in total.

Step 7. Prove that the kernel is isotropic for ξ. Let W be the common real kernel of all the characters in Step 6. The equations $\chi_j(w) = 0$ imply $\pi(w) \in Y$ by (7.60), while $\mu_i(w) = 0$ for every $w \in W$. For $w_1, w_2 \in W$, write $\pi(w_1) = \sum_{i=1}^{t} s_i y_i$ with $s_i \in \mathbb{R}$. Then

$$\xi([w_1, w_2]) = B_\ell(\pi(w_1), \pi(w_2)) = \sum_{i=1}^{t} s_i B_\ell(y_i, \pi(w_2)) = \sum_{i=1}^{t} s_i \mu_i(w_2) = 0.$$

This is an equality in $\mathbb{R}$, because the kernels defining W are real kernels. Since $[W,W] \subseteq G_2$ and ξ is additive on G_2, it vanishes on the subgroup generated by these commutators. Thus $\xi([W,W]) = 0$. □

7.5. **Reduction to quadratic phases.** We record the quantitative forms of the two results used next. Theorem 6.2 applies to $|a(n)| \le 1$ and a multiplicative function f with $|f| \le 1$. If

$$\left| N^{-1} \sum_{n \le N} f(n)a(n) \right| \ge \eta, \qquad N \ge \exp(C\eta^{-5}), \qquad 0 < \eta < 1/2,$$

then there are $2 \le x \le \exp(C\eta^{-5})$ and at least $c\eta^{25}x^2$ ordered distinct prime pairs $p, q \in [x, 2x]$ for which

$$\left| \frac{1}{N_{p,q}} \sum_{n \le N_{p,q}} a(pn)\overline{a(qn)} \right| \ge c\eta^3, \qquad N_{p,q} = \left\lfloor \frac{N}{\max(p,q)} \right\rfloor. \tag{7.61}$$

Theorem 6.7 says that if $H \ge 2$ is an integer, $x \ge 32H$ is real, and N is a positive integer satisfying $N \ge 10^6 H^3 x^3$, then any set of at least x^2/H pairs in $([x, 2x] \cap \mathbb{Z})^2$ satisfying $\|pA + qB\|_{\mathbb{R}/\mathbb{Z}} \le Hx/N$ obeys

$$\left| A - \frac{a}{R} \right|, \left| B - \frac{b}{R} \right| \le \frac{64H^2}{N}, \qquad a, b \in \mathbb{Z}, \quad R \text{ a positive integer}, \quad R \le 2H. \tag{7.62}$$

We apply this lemma separately to each of a fixed finite list of characters; the denominator may depend on the character.

The next two results use the following fixed model. Let $r \ge 1$, let $G = \mathcal{H}(r)$ with $\Gamma = \Gamma(r) = \mathbb{Z}^r \times \mathbb{Z}^r \times \mathbb{Z}$, and write

$$G_1 := G, \qquad G_2 := [G,G] = \{z(c) : c \in \mathbb{R}\}, \qquad G_3 := \{1\}, \qquad d := \dim G = 2r + 1,$$

where $z(c) = (0, 0, c)$. Thus G is step two and G_2 is its one-dimensional center. Let $\sigma \in \{-1, 1\}$ and define $\xi : G_2 \to \mathbb{R}$ by $\xi(z(c)) := \sigma c$. Assume that

$$F(z(c)x\Gamma) = e(\sigma c)F(x\Gamma) \qquad (c \in \mathbb{R},\ x \in G).$$

This is the precise vertical-frequency condition used below. Let $f : \mathbb{N} \to \mathbb{S}^1$ be completely multiplicative. For an integer $N \ge 1$, a degree-two polynomial sequence $g : \mathbb{Z} \to G$, and $0 < \eta < 1/2$, assume

$$\left| N^{-1} \sum_{n \le N} f(n)F(g(n)\Gamma) \right| \ge \eta, \qquad Z := 10dM/\eta. \tag{7.63}$$

Let

$$\mathcal{B}_r := \{(\theta, \lambda, c) : \theta, \lambda \in [-1/2, 1/2)^r,\ c \in [-1/2, 1/2)\}$$

be the centered coordinate box, and assume below that $u \in \mathcal{B}_r$.

Lemma 7.8. *Assume (7.63), with*

$$g(n) = u^n v^{\binom{n}{2}}, \qquad u \in \mathcal{B}_r, \qquad v \in G_2.$$

If $N \ge \exp(Z^{C(d+1)^6})$, then there exist an integer s with $0 \le s \le 2r$ and homomorphisms $\lambda_1, \dots, \lambda_s : G \to \mathbb{R}$ such that each λ_i vanishes on G_2, is integer-valued on Γ, and the family is linearly independent over $\mathbb{R}$. Equivalently, $\lambda_i(\theta, \lambda, c) = a_i \cdot \theta + b_i \cdot \lambda$ with $a_i, b_i \in \mathbb{Z}^r$, and the vectors (a_i, b_i) are linearly independent over $\mathbb{R}$. Put

$$H := \bigcap_{i=1}^{s} \ker_{\mathbb{R}} \lambda_i, \qquad \ker_{\mathbb{R}} \lambda_i := \{w \in G : \lambda_i(w) = 0 \text{ in } \mathbb{R}\}.$$

For $s = 0$ we use the convention $H = G$. There is an integer J with

(7.64)
$$Z \le J \le Z^{C(d+1)^6}, \qquad |\lambda_i| \le J, \qquad \|\lambda_i(u)\|_{\mathbb{R}/\mathbb{Z}} \le J/N, \qquad H \supseteq G_2, \qquad \xi([H,H]) = 0.$$

Here $|\lambda_i|$ denotes the maximum absolute value of the integer coefficients of λ_i in the horizontal coordinates. The subgroup H is connected, normal, and abelian.

Proof. We divide the proof into six steps.

Step 1. Produce many correlated prime pairs. Set $a(n) := F(g(n)\Gamma)$. Applying (7.61) to a gives a real number x with $2 \le x \le \exp(Z^C)$ and a set $\mathcal{P}$ of at least $c\eta^{25}x^2$ ordered distinct prime pairs $(p,q) \in [x,2x]^2$ such that

$$\left| \frac{1}{N_{p,q}} \sum_{n \le N_{p,q}} F(g(pn)\Gamma)\overline{F(g(qn)\Gamma)} \right| \ge c\eta^3, \qquad N_{p,q} := \left\lfloor \frac{N}{\max(p,q)} \right\rfloor. \tag{7.65}$$

Because $p, q \le 2x$ and the assumed lower bound for N is enormous, $N_{p,q} \ge N/(4x)$ for all $(p,q) \in \mathcal{P}$.

Step 2. Apply Theorem 7.7 to each pair. Fix $(p,q) \in \mathcal{P}$. Work on $G^\times := G \times G$ with lattice $\Gamma^\times := \Gamma \times \Gamma$ and center $G_2^\times := G_2 \times G_2$. Define

$$\mathcal{F}(x_1,x_2) := F(x_1\Gamma)\overline{F(x_2\Gamma)}, \qquad g_{p,q}(n) := (g(pn), g(qn)).$$

The identity

$$\binom{pn}{2} = p^2\binom{n}{2} + n\binom{p}{2}$$

gives

$$g(pn) = (u^p v^{\binom{p}{2}})^n (v^{p^2})^{\binom{n}{2}},$$

and the analogous formula holds for $g(qn)$. Thus $g_{p,q}$ is again a degree-two polynomial sequence adapted to the step-two filtration. Then the left side of (7.65) is the average of $\mathcal{F}(g_{p,q}(n))$. Moreover,

$$\mathcal{F}(z(c_1)x_1, z(c_2)x_2) = e\big(\sigma(c_1 - c_2)\big)\mathcal{F}(x_1,x_2),$$

so the vertical frequency on the product is $\Xi(c_1,c_2) = \sigma(c_1 - c_2) = (\xi, -\xi)$. The product has dimension $2d$, and the Lipschitz norm of $\mathcal{F}$ is at most a fixed multiple of M. After increasing the absolute constant in the threshold if necessary, Theorem 7.7 applies with length $N_{p,q}$. It yields at most $4d$ linearly independent real characters $\Lambda_{p,q,j}$ on $G^\times$ that are integer-valued on $\Gamma^\times$, vanish on $G_2^\times$, and satisfy

$$|\Lambda_{p,q,j}|, \ \|\Lambda_{p,q,j} \circ g_{p,q}\|_{C^\infty[N_{p,q}]} \le L_0, \qquad L_0 := Z^{C(d+1)^4}. \tag{7.66}$$

The exponent $C(d+1)^4$ is enlarged once to absorb the product dimension $2d$.

Every such character has the form

$$\Lambda_{p,q,j}(x_1,x_2) = \alpha_{p,q,j}(x_1) + \beta_{p,q,j}(x_2),$$

where $\alpha_{p,q,j}$ and $\beta_{p,q,j}$ are horizontal integer characters on G.

Step 3. Fix one tuple of characters. Write $m := \dim(G/G_2) = 2r < d$. Order each family and append zero pairs until exactly $4d$ positions are present. Each pair $(\alpha_{p,q,j}, \beta_{p,q,j})$ has at most $2m \le 2d$ integer coefficients, each of size at most L_0. Consequently the number of resulting $4d$-tuples is at most

$$Q := (2\lfloor L_0 \rfloor + 1)^{8d^2}.$$

Choose an integer H_0 satisfying

$$H_0 \ge \max\{2, 4L_0, Q/(c\eta^{25})\}.$$

By increasing the absolute constant C, we may also assume $H_0 \le Z^{C(d+1)^6}$. Pigeonholing the pairs in $\mathcal{P}$, one resulting $4d$-tuple

$$(\alpha_1,\beta_1),\dots,(\alpha_{4d},\beta_{4d})$$

occurs for at least x^2/H_0 pairs. Fix this tuple and one corresponding pair (p,q) whenever a single pair is needed below.

Because the characters vanish on $G_2^{\times}$,

$$\Lambda_i(g_{p,q}(n)) = n\big(p\alpha_i(u)+q\beta_i(u)\big).$$

Thus (7.66) and $N_{p,q}\ge N/(4x)$ imply, for every retained pair and every i,

$$\|p\alpha_i(u)+q\beta_i(u)\|_{\mathbb{R}/\mathbb{Z}} \le \frac{L_0}{N_{p,q}} \le \frac{4L_0x}{N} \le \frac{H_0x}{N}. \tag{7.67}$$

Step 4. The case $x \ge 32H_0$. Apply (7.62) with $H = H_0$ to each of the $4d$ fixed linear relations in (7.67). The required conditions $x \ge 32H_0$ and $N \ge 10^6 H_0^3 x^3$ follow from the present case and the assumed threshold, after increasing C. We obtain integers $1 \le R_i \le 2H_0$ for which

$$\|R_i\alpha_i(u)\|_{\mathbb{R}/\mathbb{Z}},\ \|R_i\beta_i(u)\|_{\mathbb{R}/\mathbb{Z}} \le \frac{128H_0^3}{N}.$$

Take all nonzero characters $R_i\alpha_i$ and $R_i\beta_i$, and then retain a maximal linearly independent subfamily. Its real common kernel is unchanged. If w_1, w_2 lie in this kernel, then $(w_j, 1)$ lies in the real common kernel of the product characters. Therefore Theorem 7.7 gives

$$0 = \Xi([(w_1,1),(w_2,1)]) = \xi([w_1,w_2]).$$

Step 5. The case $x < 32H_0$. Fix one retained pair (p,q); it has $p \ne q$. Define

$$\lambda_i := p\alpha_i + q\beta_i \qquad (1 \le i \le 4d),$$

and retain a maximal linearly independent subfamily. These are integer-valued horizontal characters. Their heights and values at u satisfy

$$|\lambda_i| \le 4xL_0 \le 128H_0L_0, \qquad \|\lambda_i(u)\|_{\mathbb{R}/\mathbb{Z}} \le \frac{H_0x}{N} \le \frac{32H_0^2}{N}.$$

Let w_1, w_2 lie in their common real kernel and set

$$\widetilde{w}_j := (w_j^p, w_j^q) \in G \times G.$$

Every discarded character is an $\mathbb{R}$-linear combination of the retained ones, so it also vanishes on w_j. Hence all product characters vanish on $\widetilde{w}_j$, because $p\alpha_i(w_j) + q\beta_i(w_j) = \lambda_i(w_j) = 0$. Theorem 7.7, together with bilinearity of the step-two commutator, gives

$$0 = \Xi([\widetilde{w}_1,\widetilde{w}_2]) = \Xi\big([w_1,w_2]^{p^2},[w_1,w_2]^{q^2}\big) = (p^2-q^2)\xi([w_1,w_2]).$$

Since $p \ne q$, we conclude $\xi([w_1,w_2]) = 0$.

Step 6. Finish the quantitative and structural conclusions. In both cases the resulting characters vanish on G_2 and are integer-valued on Γ. After retaining a maximal linearly independent subfamily, their number is at most $2r$, and their heights and N-scaled errors are bounded by a fixed power of H_0L_0. Choose J to be an integer satisfying

$$J \ge Z, \qquad J \ge 128H_0L_0, \qquad J \ge 128H_0^3, \qquad J \le Z^{C(d+1)^6},$$

which is possible after enlarging C once more. Then the bounds in (7.64) hold.

Let $V \subseteq \mathbb{R}^{2r}$ be the intersection of the real kernels of the horizontal coefficient vectors of the retained characters. Since every character vanishes on G_2, the common kernel is $H = \pi^{-1}(V)$, where $\pi : G \to G/G_2 \simeq \mathbb{R}^{2r}$. Thus H is connected and contains G_2. It is normal because

G/G_2 is abelian and $[G,H] \subseteq G_2 \subseteq H$. The two case arguments prove $\xi([H,H]) = 0$. Since $\xi(z(c)) = \sigma c$ is injective on the one-dimensional center, $[H,H] = \{1\}$; hence H is abelian. $\square$

Theorem 7.9. *Assume (7.63) for an arbitrary degree-two polynomial g. There is an absolute C such that, if*

$$N \geq \exp\bigl(Z^{C(d+1)^8}\bigr),$$

there exist real numbers a, b and a nonempty arithmetic progression $\mathcal{P} \subseteq [N]$ (with positive integer common difference) such that

$$\left| N^{-1} \sum_{n \in \mathcal{P}} f(n) e(an + bn^2) \right| \geq Z^{-C(d+1)^8}.$$

The common difference of $\mathcal{P}$ is at most $Z^{C(d+1)^7}$.

Proof. We divide the proof into six steps.

Step 1. Normalize the polynomial sequence. The constant and linear normalizations in (7.48) and (7.52) replace the original data by data with exactly the same correlation and with controlled changes in the Lipschitz norm. In particular, after enlarging the absolute constant in the hypothesis, we may assume

$$g(n) = u^n v^{\binom{n}{2}}, \qquad u \in \mathcal{B}_r, \qquad v \in G_2, \tag{7.68}$$

where $G = \mathcal{H}(r)$ and $d = 2r+1$. The equivariance relation $F(z(c)x\Gamma) = e(\sigma c)F(x\Gamma)$ is preserved by these normalizations. The bound $|F| \leq 1$ is preserved, and its Lipschitz norm is at most a fixed power of dM. The linear normalization changes the central coefficient to $v[\gamma, u_0]$; this new central factor is retained exactly in (7.68). Thus the parameter $Z = 10dM/\eta$ changes by at most a fixed power, which is absorbed into $Z^{C(d+1)^8}$ and $Z^{C(d+1)^7}$ below.

Step 2. Use Theorem 7.8 to obtain an abelian subgroup. Apply Theorem 7.8 to the normalized data in (7.68). We obtain independent horizontal characters $\lambda_1, \dots, \lambda_k$ and

$$H_1 := \bigcap_{i=1}^{k} \ker_{\mathbb{R}} \lambda_i, \qquad H_1 \supseteq G_2,$$

where H_1 is connected, normal, and abelian. Let $m := 2r = \dim(G/G_2)$, let A be the $k \times m$ integer matrix whose rows are the horizontal coefficient vectors of the λ_i, and let $\alpha \in [-1/2, 1/2)^m$ be the horizontal coordinate of u. The lemma gives an integer J such that

$$Z \leq J \leq Z^{C(d+1)^6}, \qquad \|A\alpha\|_{\mathbb{R}^k/\mathbb{Z}^k} \leq J/N, \qquad \text{every entry of } A \text{ has absolute value at most } J.$$

Here the norm on the left is the maximum of the coordinate distances to $\mathbb{Z}$. Since G is non-abelian but H_1 is abelian, $k \geq 1$.

After permuting the horizontal coordinates, choose a nonsingular $k \times k$ column minor B of A and write $A = [B\ C_1]$. Put $\mathfrak{q} := \det B \neq 0$, and choose $b_0 \in \mathbb{Z}^k$ with $\|A\alpha - b_0\|_\infty \leq J/N$. Define vectors supported on the pivot columns by

$$e_B := B^{-1}(A\alpha - b_0), \qquad c_B := B^{-1}b_0, \qquad e_{\text{free}} = c_{\text{free}} := 0, \qquad w := \alpha - e - c.$$

Then

$$Aw = 0, \quad Ae = A\alpha - b_0, \quad Ac = b_0, \quad |e| \leq \frac{(dJ)^{C(d+1)}}{N}, \quad |c| \leq (dJ)^{C(d+1)}, \quad \mathfrak{q}c \in \mathbb{Z}^m.$$

For the bounds, write $B^{-1} = \mathfrak{q}^{-1}\operatorname{adj}(B)$, where $\operatorname{adj}(B)$ is the integer adjugate matrix characterized by $B\operatorname{adj}(B) = \mathfrak{q}I_k$; each entry of this matrix is a signed determinant of a $(k-1)\times(k-1)$ submatrix of B. Since every entry of B is bounded by J, these determinants are bounded by a fixed power of J. Cramer's rule then gives the displayed bound for e and c because each coordinate of b_0 is $O(dJ)$, while $|\mathfrak{q}| \leq (dJ)^{C(d+1)}$.

Embed the vectors $e, c \in \mathbb{R}^m$ into the Lie algebra by giving them zero entries in the c-coordinate direction of every Heisenberg factor. Define

$$\varepsilon(n) := \exp(ne), \qquad \gamma(n) := \exp(nc), \qquad h(n) := \varepsilon(n)^{-1} u^n \gamma(n)^{-1}.$$

The horizontal coordinate of $h(n)$ is $nw \in \ker A$, so $h(n) \in H_1$. The Heisenberg group law adds a bilinear commutator term; therefore the c-coordinate of $h(n)$ is a polynomial of degree at most two. Since $v \in G_2$ is central,

$$g(n) = \varepsilon(n) h_1(n) \gamma(n), \qquad h_1(n) := h(n) v^{\binom{n}{2}} \in H_1.$$

All quadratic commutator terms are contained in $h_1(n)$; no error term is being discarded.

Step 3. Control the slowly varying and periodic factors. Choose a fixed power bound $P_0 = (dJ)^{C(d+1)}$ large enough that $P_0 \geq d, M, \eta^{-1}, 2|\mathfrak{q}|$ and that all estimates below hold. The path $t \mapsto \exp(te)$ has constant speed $\|e\|_2$, so

$$d_G(\varepsilon(n), 1) \leq n\|e\|_2 \leq P_0, \qquad d_G(\varepsilon(n+1), \varepsilon(n)) \leq \|e\|_2 \leq P_0/N. \tag{7.69}$$

Thus $\varepsilon(n)$ varies by at most P_0/N per step. Moreover, $\mathfrak{q}c \in \mathbb{Z}^m$ and

$$\exp(\theta, \lambda, 0) = (\theta, \lambda, \tfrac{1}{2}\lambda \cdot \theta)$$

show that $\exp(2\mathfrak{q}c) \in \Gamma$. Define D_γ to be the least positive integer D for which $\gamma(n+D)\Gamma = \gamma(n)\Gamma$ for every integer n; equivalently, D_γ is the least positive D with $\exp(Dc) \in \Gamma$. The preceding calculation supplies such a period, so

$$D_\gamma \leq 2|\mathfrak{q}| \leq P_0, \qquad P_0 \leq Z^{C(d+1)^7}. \tag{7.70}$$

Step 4. Select one arithmetic cell. Put

$$L_{\mathrm{cell}} := \left\lfloor \frac{\eta N}{20 M P_0} \right\rfloor.$$

The theorem's lower bound on N implies $L_{\mathrm{cell}} \geq \eta N/(40MP_0) \geq 1$. Partition $[N]$ into consecutive intervals of length L_{cell}, and then split each interval according to the residue class of n modulo D_γ. Each resulting nonempty cell is an arithmetic progression, and the number of cells is at most

$$D_\gamma \left(1 + \frac{N}{L_{\mathrm{cell}}}\right) \leq \frac{41 P_0^3}{\eta}.$$

For a cell $\mathcal{C}$ choose $n_{\mathcal{C}} \in \mathcal{C}$ and replace $\varepsilon(n)$ by $\varepsilon(n_{\mathcal{C}})$. By (7.69), the pointwise error in F is at most $MP_0 L_{\mathrm{cell}}/N \leq \eta/20$. Summing over n and using $|f(n)| = 1$, we obtain

$$\sum_{\mathcal{C}} \left| \frac{1}{N} \sum_{n \in \mathcal{C}} f(n) F(\varepsilon(n_{\mathcal{C}}) h_1(n) \gamma(n) \Gamma) \right| \geq \frac{19\eta}{20}.$$

Consequently one cell, denoted by $\mathcal{P}$, satisfies

$$\left| \frac{1}{N} \sum_{n \in \mathcal{P}} f(n) F(\varepsilon(n_0) h_1(n) \gamma(n) \Gamma) \right| \geq R, \qquad R := \frac{\eta^2}{50 P_0^3},$$

where $n_0 \in \mathcal{P}$ is fixed. The progression $\mathcal{P}$ has step D_γ.

Step 5. Put the selected cell on a torus. On $\mathcal{P}$, the coset $\gamma(n)\Gamma$ is constant. Let $\alpha_0 \in G$ be its representative in the centered fundamental box $\mathcal{B}_r$. Define

$$F_{\mathcal{P}}(z) := F(\varepsilon(n_0) \alpha_0 z).$$

Then, for $n \in \mathcal{P}$,

$$F(\varepsilon(n_0) h_1(n) \gamma(n) \Gamma) = F_{\mathcal{P}}(\alpha_0^{-1} h_1(n) \alpha_0 \Gamma).$$

Normality of H_1 implies that $\widetilde{h}(n) := \alpha_0^{-1} h_1(n) \alpha_0$ remains in H_1.

Recall that $\operatorname{adj}(B)$ is the integer matrix made from the signed $(k-1)\times(k-1)$ minors of B, so that $B\operatorname{adj}(B)=\mathfrak{q}I_k$. Let U be the integer matrix

$$U:=\begin{pmatrix}-\operatorname{adj}(B)C_1\\ \mathfrak{q}I\end{pmatrix}.$$

Its columns span $\ker A$ over $\mathbb{R}$. The vectors $2U_j$, together with the lattice vector whose horizontal coordinates are zero and whose c-coordinate is 1, generate a lattice $\Lambda\subseteq H_1\cap\Gamma$. Indeed, the horizontal coordinates of $2U_j$ are even integers, and the Heisenberg c-coordinate $\frac{1}{2}\lambda\cdot\theta$ is then an integer. Thus H_1/Λ is a torus of dimension $s_H=m-k+1\le d$. The sequence $\widetilde{h}(n)\Lambda$ is a degree-two polynomial sequence on this abelian torus.

Let $\widetilde{F}_{\mathcal{P}}$ denote the pullback of $F_{\mathcal{P}}$ to H_1/Λ. It has supremum at most 1. Its Lipschitz norm is at most P_0^C. The vectors $2U_j$ have size at most P_0. To account for the left translation by $a_0:=\varepsilon(n_0)\alpha_0$, let X_0 be the Lie-algebra coordinate vector with $a_0=\exp(X_0)$, and let V_j be the tangent vector associated with $2U_j$ (and likewise for the central lattice direction). In a step-two group, differentiating $\exp(X_0)\exp(tV_j)\exp(-X_0)$ at $t=0$ gives $V_j+[X_0,V_j]$. Here $[X_0,V_j]$ is the Lie bracket, which is central and bilinear in the horizontal coordinates. The bounds on X_0 and V_j therefore imply the claimed fixed power of P_0.

Step 6. Extract one scalar quadratic phase. Apply Theorem 7.5 on the s_H-dimensional torus with error parameter $R/4$. We obtain

$$T(x)=\sum_{k\in[-K,K]^{s_H}\cap\mathbb{Z}^{s_H}}c_k e(k\cdot x),\qquad \|\widetilde{F}_{\mathcal{P}}-T\|_\infty\le R/4,$$

where

$$K\le C\left(\frac{s_H P_0^C}{R}\right)^4\le P_0^C,\qquad \sum_k|c_k|\le(2K+1)^{s_H}.$$

The error in the normalized correlation is at most $R/4$, so

$$\left|\frac{1}{N}\sum_{n\in\mathcal{P}}f(n)T(\widetilde{h}(n)\Lambda)\right|\ge\frac{3R}{4}. \tag{7.71}$$

Write the torus orbit in additive coordinates as

$$\widetilde{h}(n)\Lambda=t_0+nt_1+\binom{n}{2}t_2\pmod{\mathbb{Z}^{s_H}},\qquad t_0,t_1,t_2\in\mathbb{R}^{s_H}.$$

For each k, this gives real numbers a_k,b_k,c_k' such that

$$e(k\cdot\widetilde{h}(n))=e(c_k'+a_kn+b_kn^2).$$

The constant factor $e(c_k')$ has absolute value one and can be absorbed into the Fourier coefficient. By the triangle inequality applied to (7.71), some k satisfies

$$\left|\frac{1}{N}\sum_{n\in\mathcal{P}}f(n)e(a_kn+b_kn^2)\right|\ge\frac{3R}{4(2K+1)^{s_H}}.$$

Finally, R^{-1} and P_0 are bounded by fixed powers of $Z^{C(d+1)^7}$, while $s_H\le d$. Therefore

$$\frac{3R}{4(2K+1)^{s_H}}\ge Z^{-C(d+1)^8}.$$

By (7.70), the step of $\mathcal{P}$ is at most $Z^{C(d+1)^7}$. This proves the scalar correlation and the progression bound in the theorem. □

7.6. **Proof of the inverse theorem.** We now combine the preceding reductions with the scalar inverse theorem.

Theorem 7.10. *There is an absolute constant $C \geq 1$, independent of ϵ, N, and f, such that, for $0 < \epsilon < 1/2$, on writing $\ell = 2+\log(2/\epsilon)$, every completely multiplicative function $f : \mathbb{N} \to \mathbb{C}$ with $|f(n)| = 1$ satisfying*

$$N \geq \exp\exp(C\ell^{431}), \qquad \|f\|_{U^3[N]} \geq \epsilon$$

has a primitive Dirichlet character χ and a real t such that

$$q_\chi,\ |t| \leq \exp(C\ell^{431}), \qquad \sum_{\substack{p \leq N \\ p \text{ prime}}} \frac{1 - \Re(f(p)\overline{\chi(p)}p^{-it})}{p} \leq C\ell^{431}. \tag{7.72}$$

Proof. We divide the proof into four steps.

Step 1. Obtain the structured nilsequence. Apply Theorem 7.4 to the restriction of the given completely multiplicative function f to $[N]$. Its hypotheses hold because $|f(n)| = 1$ and $\|f\|_{U^3[N]} \geq \epsilon$. We obtain an integer $d \geq 1$, a real number $M \geq 2$, a number $0 < \eta < 1/2$, and either the circle nilmanifold $\mathbb{R}/\mathbb{Z}$ or a Heisenberg nilmanifold G/Γ, together with a degree-two polynomial sequence g and a function F satisfying

$$|F| \leq 1, \qquad \|F\|_{\mathrm{Lip}} \leq M, \qquad \left|\frac{1}{N}\sum_{n=1}^{N} f(n)F(g(n)\Gamma)\right| \geq \eta.$$

The same corollary gives

$$d \ll \ell^{43}, \qquad \log(2M/\eta) \ll \ell^{87}, \qquad \ell := 2 + \log(2/\epsilon).$$

Set

$$\eta_0 := \min\{\eta, 1/4\}, \qquad Z := \frac{10dM}{\eta_0}.$$

Because $0 < \eta < 1/2$, we have $\eta_0 \geq \eta/2$. The correlation is still at least η_0. Moreover,

$$\log Z \leq \log(20d) + \log(2M/\eta) \ll \ell^{87}, \qquad \log(\eta_0^{-1}) \ll \ell^{87}. \tag{7.73}$$

Here and below all implicit constants are absolute and may be enlarged.

Step 2. Dispose of the circle case. Suppose first that $(G, \Gamma) = (\mathbb{R}, \mathbb{Z})$. In additive notation write

$$g(n) = g_0 + nu + \binom{n}{2}v, \qquad g_0, u, v \in \mathbb{R}.$$

The frequency-one identity from Theorem 7.4 gives $F(x + \mathbb{Z}) = F(0 + \mathbb{Z})e(x)$ for every real x. Hence, with $F_0 := F(0 + \mathbb{Z})$,

$$F(g(n) + \mathbb{Z}) = F_0 e(g_0)e\big((u - v/2)n + (v/2)n^2\big).$$

The correlation bound forces $F_0 \neq 0$. Taking absolute values and using $|F_0| \leq 1$, it implies

$$\left|\frac{1}{N}\sum_{n=1}^{N} f(n)e(\xi n + \beta n^2)\right| \geq \eta_0, \qquad \xi := u - v/2, \quad \beta := v/2. \tag{7.74}$$

Apply Theorem 6.12 with $\rho := \eta_0$ and $\mathcal{P} = [N]$. Its correlation hypothesis is (7.74). Its length hypothesis also holds: by (7.73), $\eta_0^{-5} \leq \exp(C\ell^{87})$, so the assumed bound $N \geq \exp\exp(C\ell^{431})$ dominates $N \geq \exp(C_s\eta_0^{-5})$, where C_s is the absolute constant in Theorem 6.12. That theorem therefore supplies a primitive Dirichlet character χ and a real number t with

$$q_\chi,\ |t| \leq \eta_0^{-C_s}, \qquad \sum_{\substack{p \leq N \\ p \text{ prime}}} \frac{1 - \Re(f(p)\overline{\chi(p)}p^{-it})}{p} \leq C_s \log(2/\eta_0).$$

The bounds in (7.73) imply that these quantities satisfy the conclusion of Theorem 7.10 after increasing its absolute constant.

Step 3. Reduce the Heisenberg case to a scalar phase. Now suppose that $(G,\Gamma) = (\mathcal{H}(r),\Gamma(r))$. The data obtained in Step 1 satisfy the input condition (7.63) with the present value of η_0 and with $Z = 10dM/\eta_0$. The dimension and parameter bounds give

$$(d+1)^8 \log Z \ll \ell^{8\cdot 43+87} = \ell^{431}. \tag{7.75}$$

Consequently, after enlarging the absolute constant in the hypothesis on N, we have

$$N \geq \exp\bigl(Z^{C_9(d+1)^8}\bigr),$$

where C_9 is the constant in Theorem 7.9. We may therefore apply that theorem. It gives real numbers a, b and a nonempty arithmetic progression $\mathcal{P} \subseteq [N]$ such that

$$\left|\frac{1}{N}\sum_{n\in\mathcal{P}} f(n)e(an+bn^2)\right| \geq \rho_0, \qquad \rho_0 := Z^{-C_9(d+1)^8}. \tag{7.76}$$

The progression has positive integer common difference, but no further property of that difference is needed here. Put

$$\rho := \min\{\rho_0, 1/4\}.$$

Then $0 < \rho < 1/2$ and the left-hand side of (7.76) is at least ρ. Furthermore, by (7.75),

$$\rho^{-1} \leq 4 + Z^{C_9(d+1)^8} \leq \exp(C'\ell^{431}) \tag{7.77}$$

for an absolute C'. Hence $N \geq \exp(C_s\rho^{-5})$ after another enlargement of the absolute constant in the assumed bound for N.

Step 4. Apply the scalar inverse theorem and finish. Apply Theorem 6.12 to the original completely multiplicative function f, the same ambient interval $[N]$, the progression $\mathcal{P}$, and the phase $an+bn^2$. This is a direct application of its stated hypothesis: the theorem normalizes the sum by N, so no reparametrization of $\mathcal{P}$ is required. By (7.76) and the definition of ρ, the correlation hypothesis holds, and the preceding paragraph verifies its length hypothesis. We obtain a primitive Dirichlet character χ and a real number t satisfying

$$q_\chi,\ |t| \leq \rho^{-C_s}, \qquad \sum_{\substack{p\leq N\\ p \text{ prime}}} \frac{1-\Re(f(p)\overline{\chi(p)}p^{-it})}{p} \leq C_s\log(2/\rho). \tag{7.78}$$

By (7.77), $\rho^{-C_s} \leq \exp(C\ell^{431})$ and $\log(2/\rho) \ll \ell^{431}$. Thus (7.78) is exactly the conclusion (7.72), after choosing the absolute constant in Theorem 7.10 large enough. □

8. Proof of the main theorem

We combine the preceding results with the numerical exponent 431 of Theorem 7.10. All parameters in this section are chosen independently of the multiplicative function.

8.1. **The uniformity estimate.** Throughout this subsection, let $w : \mathbb{R}^2 \to [0,\infty)$ be the fixed C_c^2 probability density from (4.39), so that $\int_{\mathbb{R}^2} w = 1$, and let $C_0 \in \mathbb{N}$ be the support parameter from (4.43). Once a real $\delta \in (0, 1/10)$ is fixed, let $D_0 = D_0(\delta)$ and let $W_0 = W_{0,\delta} : \mathbb{Z}^2 \to [0,\infty)$ be the period and periodic weight constructed in (4.30)–(4.32). Thus

$$\operatorname{supp} w \subseteq \{(s,t) : s > t > 0,\ s+t < C_0 - 1\},$$
$$W_0 \text{ is } D_0\text{-periodic in each variable and } D_0^{-2}\sum_{a,b \bmod D_0} W_0(a,b) = 1.$$

All constants denoted by C_w below depend only on the fixed weight w and the fixed integer C_0; in particular, they are independent of δ, D_0, W_0, X, and g. For each integer $X \geq 1$ and each $g \in \mathcal{M}$, Z_X and $K_X(g)$ are the normalization and normalized kernel defined in (4.44) whenever $Z_X > 0$; explicitly, the kernel averages $g(R(m,n))$ with $R(m,n) = (m^2 - n^2)/(2mn)$.

Proposition 8.1. *For every real δ with $0 < \delta < 1/10$, with $D_0 = D_0(\delta)$ and $W_0 = W_{0,\delta}$ chosen as above, there are real numbers $L_\delta \geq 1$ and $X_\delta \geq 1$, depending only on δ, the fixed data (w, C_0), and the prescribed construction of (D_0, W_0), such that*

$$\log(2L_\delta) + \log(2X_\delta) \leq C_w \delta^{-3}, \tag{8.1}$$

and, for every integer $X \geq X_\delta$ and every $g \in \mathcal{M}$,

$$|K_X(g)| \leq L_\delta \|g\|_{U^3[C_0 X]} + \delta/100. \tag{8.2}$$

Consequently, with

$$\epsilon_\delta = \frac{\delta}{100 L_\delta}, \qquad \mathcal{L}_\delta = 2 + \log(2/\epsilon_\delta),$$

we have $\mathcal{L}_\delta \leq C_w \delta^{-3}$, and $\|g\|_{U^3[C_0 X]} < \epsilon_\delta$ implies $|K_X(g)| \leq \delta/50$.

Proof. We divide the proof into six steps.

Step 1. Approximate the fixed spatial weight. Fix δ and put $W := \|W_0\|_\infty$. Choose an integer

$$J := \lceil C_w(1 + W)/\delta \rceil, \tag{8.3}$$

where C_w depends only on the fixed weight w and the fixed integer C_0, and is large enough for all estimates below. Partition $\mathbb{R}^2$ into half-open squares of side length $1/J$, and retain the squares meeting $\operatorname{supp} w$. Since the support is compact, the number of retained squares is at most $C_w J^2$. The support lies a positive distance from the boundary of $s > t > 0$ and from the line $s + t = C_0 - 1$; after increasing C_w if needed, every retained square is contained in $s > t > 0$ and $s + t < C_0$.

Let w_J equal the value of w at the centre of each retained square and be zero outside their union. The mean-value theorem gives

$$\|w - w_J\|_\infty \leq C_w/J, \qquad w_J = \sum_{\mathcal{R}} c_{\mathcal{R}} \mathbf{1}_{\mathcal{R}}, \quad 0 \leq c_{\mathcal{R}} \leq \|w\|_\infty, \quad \#\{\mathcal{R}\} \leq C_w J^2.$$

Here and below $\mathcal{R}$ ranges over the retained squares.

Step 2. Bound the error caused by replacing w by w_J. Both w and w_J vanish outside a fixed bounded set. Hence, for every integer $X \geq 1$, that set contains at most $C_w X^2$ lattice points after scaling by X. By (4.44), $I_X(g)$ is the unnormalized sum and $Z_X = I_X(1)$ is its value for the constant function 1. Since $|W_0(m,n)| \leq W$, replacing w by w_J in $I_X(g)$ changes it by at most

$$X^{-2} \, (C_w X^2) \, (C_w/J) \, W \leq C_w W/J.$$

Let $\widetilde{I}_X(g)$ denote this modified unnormalized sum, and define $\widetilde{K}_X(g) := \widetilde{I}_X(g)/Z_X$ using the original normalization $Z_X = I_X(1)$. By the normalization estimate (4.46), $Z_X \geq 1/2$ whenever $X \geq C_w D_0$. Consequently,

$$|K_X(g) - \widetilde{K}_X(g)| \leq 2C_w W/J \leq \delta/100$$

by (8.3).

Step 3. Expand the periodic weight. The D_0-periodicity of W_0 gives the finite Fourier expansion

$$W_0(m,n) = \sum_{\alpha,\beta \bmod D_0} \widehat{W_0}(\alpha,\beta) e((\alpha m + \beta n)/D_0),$$
$$\widehat{W_0}(\alpha,\beta) := D_0^{-2} \sum_{a,b \bmod D_0} W_0(a,b) e(-\alpha a/D_0 - \beta b/D_0),$$
$$\sum_{\alpha,\beta \bmod D_0} |\widehat{W_0}(\alpha,\beta)| \le D_0^2 W.$$

The first identity is Fourier inversion on $(\mathbb{Z}/D_0\mathbb{Z})^2$; it follows from the orthogonality relation

$$D_0^{-1} \sum_{j \bmod D_0} e(jr/D_0) = \mathbf{1}_{D_0 | r}.$$

The last inequality follows from the triangle inequality applied to the finite definition of each coefficient.

Step 4. Rewrite each summand as a four-linear form. For $m > n > 0$, complete multiplicativity and $|g(q)| = 1$ give

$$g\Big(\frac{m^2-n^2}{2mn}\Big) = \overline{g(2)}\, g(m-n) g(m+n) \overline{g(m)}\, \overline{g(n)}.$$

Indeed, $(m^2-n^2)/(2mn) = (m-n)(m+n)/(2mn)$, and $g(1/q) = \overline{g(q)}$ for $q > 0$. Set $Z := C_0 X$ and $P := 32Z+1$. For a retained rectangle $\mathcal{R} = I \times J'$ and a Fourier pair (α,β), define functions on $\mathbb{Z}/P\mathbb{Z}$ by zero extension from $[Z]$: Here I and J' are intervals in the scaled variables m/X and n/X. We use the notation

$$XI := \{m \in \mathbb{Z} : m/X \in I\}, \qquad XJ' := \{n \in \mathbb{Z} : n/X \in J'\};$$

thus XI and XJ' are sets of integers, not real intervals.

$$a_Z(u) := g(u)\mathbf{1}_{[Z]}(u), \qquad b_Z(u) := g(u)\mathbf{1}_{[Z]}(u),$$
$$c_Z(u) := \overline{g(u)}\mathbf{1}_{XI}(u) e(\alpha u/D_0), \qquad d_Z(u) := \overline{g(u)}\mathbf{1}_{XJ'}(u) e(\beta u/D_0).$$

The indicators in the last two definitions select integer points. All four functions have modulus at most one. If $c_Z(m) d_Z(n) \neq 0$, then $m > n > 0$ and $m+n < Z$ because every retained square is contained in $s > t > 0$ and $s + t < C_0$. Therefore $m-n, m+n, m, n$ all lie in $[Z]$, and the four-linear expression has no wrap-around in $\mathbb{Z}/P\mathbb{Z}$. The phases in c_Z and d_Z are evaluated only at these selected integer representatives; no periodic extension of the phases is required, because the functions are defined by zero extension.

Step 5. Apply the U^3 four-forms estimate. Let $G = \mathbb{Z}/P\mathbb{Z}$ and write

$$\Lambda := \mathbb{E}_{m,n \in G} a_Z(m-n) b_Z(m+n) c_Z(m) d_Z(n).$$

Because $P = 32Z+1$ is odd and $P > 8Z$, the hypotheses of Theorem 3.2 and Theorem 3.1 hold. The four-forms estimate gives

$$|\Lambda| \le \|a_Z\|_{U^3(G)}.$$

The support observation in Step 4 shows that the cyclic sum defining Λ is exactly the corresponding integer sum: every nonzero term has $m > n > 0$ and $m+n < Z$, so reduction modulo P introduces no wrap-around. The factor $\overline{g(2)}$ in the multiplicative identity has modulus one and therefore disappears after taking absolute values. Since a normalized average on G^2 is P^{-2} times an unnormalized sum, the contribution to $I_X(g)$ from this rectangle and Fourier mode is bounded by

$$\frac{P^2}{X^2}|\Lambda| \le (P/X)^2 \|a_Z\|_{U^3(\mathbb{Z}/P\mathbb{Z})} \le (32C_0+1)^2 \|g\|_{U^3[Z]}. \tag{8.4}$$

Here the last inequality is the interval-embedding estimate applied to the zero extension a_Z of $g|_{[Z]}$, together with $P/X = 32C_0 + 1/X \le 32C_0 + 1$.

Step 6. Sum the finitely many contributions. There are at most $C_w J^2$ retained rectangles, and the Fourier coefficients have total absolute mass at most $D_0^2 W$. Summing (8.4), dividing by $Z_X \ge 1/2$, and absorbing the fixed factors into C_w gives

$$1 \le L_\delta \le C_w J^2 D_0^2 (1+W), \qquad X_\delta := \lceil C_w D_0 \rceil. \tag{8.5}$$

Thus $|\widetilde{K}_X(g)| \le L_\delta \|g\|_{U^3[Z]}$. Combining this with the estimate for $|K_X(g) - \widetilde{K}_X(g)|$ from Step 2 proves (8.2) for every integer $X \ge X_\delta$.

Finally, the weight-construction estimate Theorem 4.4 gives $\log(2D_0), \log(2W) \le C_w \delta^{-3}$. Taking logarithms in (8.3) and (8.5) yields (8.1). From $\epsilon_\delta = \delta/(100 L_\delta)$ we obtain $\mathcal{L}_\delta \le C_w \delta^{-3}$. If $\|g\|_{U^3[C_0 X]} < \epsilon_\delta$, then (8.2) gives

$$|K_X(g)| < L_\delta \epsilon_\delta + \delta/100 = \delta/50.$$

This proves the proposition. □

8.2. **Monotone prime sums.** To compare g with twisted Dirichlet characters at different prime cutoffs, we use one fixed set of admissible parameters. At each cutoff we take the smallest prime-distance among those parameters. Theorem 8.2 records the two facts needed later: this minimum is nondecreasing with the cutoff, and a minimizing pair controls the contribution of the primes between two cutoffs.

Fix an integer $Q \ge 1$ and a real $T_1 \ge 0$. For $g \in \mathcal{M}$ and real $z \ge 1$, set

$$E_g(z) = \min_{\substack{\chi \text{ primitive}, \ q_\chi \le Q \\ |t| \le T_1}} \sum_{\substack{p \le z \\ p \text{ prime}}} \frac{1 - \Re(g(p)\overline{\chi(p)} p^{-it})}{p}. \tag{8.6}$$

For a pair (χ, t) in these ranges and a cutoff $y \ge 1$, let $\widetilde{g}$ be the completely multiplicative function specified by

$$\widetilde{g}(p) = \begin{cases} g(p), & p \le y, \\ \chi(p) p^{it}, & p > y \end{cases}. \tag{8.7}$$

Lemma 8.2. *For every $g \in \mathcal{M}$, the minimum in (8.6) exists and is nonnegative and nondecreasing in z. A minimizing pair at z satisfies*

$$\sum_{\substack{y < p \le z \\ p \text{ prime}}} \frac{1 - \Re(g(p)\overline{\chi(p)} p^{-it})}{p} \le E_g(z) - E_g(y) \qquad (1 \le y \le z). \tag{8.8}$$

If $y \ge Q$, its truncation (8.7) belongs to $\mathcal{M}$.

Proof. We divide the proof into four steps.

Step 1. Existence and nonnegativity of the minimum. There are finitely many primitive Dirichlet characters with conductor at most Q; the set is nonempty because it contains the principal character of conductor one. Fix one such character χ. For real t in $[-T_1, T_1]$, write

$$D_{\chi,t}(z) := \sum_{\substack{p \le z \\ p \text{ prime}}} \frac{1 - \Re(g(p)\overline{\chi(p)} p^{-it})}{p}.$$

This is a finite sum of continuous functions of t, since $p^{-it} = e^{-it \log p}$. It therefore attains its minimum on the compact interval $[-T_1, T_1]$. Taking the minimum over the finite set of characters also attains a minimum. Every summand is nonnegative because $|g(p)| = 1$ and $|\chi(p)| \le 1$, so $E_g(z) \ge 0$.

Step 2. Monotonicity in the cutoff. If $z_2 \geq z_1$, then for every admissible pair (χ, t), $D_{\chi,t}(z_2) \geq D_{\chi,t}(z_1)$, because the second sum contains all terms of the first and the added terms are nonnegative. Taking the minimum over the same admissible set gives

$$E_g(z_2) \geq E_g(z_1).$$

Step 3. A minimizing pair controls the tail. Let (χ, t) minimize $E_g(z)$, and let $D(u) := D_{\chi,t}(u)$. Then $D(z) = E_g(z)$. Since $E_g(y)$ is the minimum over all admissible pairs, $D(y) \geq E_g(y)$. Subtracting gives

$$\sum_{\substack{y<p\leq z\\ p \text{ prime}}} \frac{1 - \Re(g(p)\overline{\chi(p)}p^{-it})}{p} = D(z) - D(y) \leq E_g(z) - E_g(y),$$

which is (8.8).

Step 4. The truncation remains completely multiplicative. Assume $y \geq Q$. Every prime divisor of the conductor q_χ is at most Q, so $p > y$ implies $p \nmid q_\chi$ and hence $|\chi(p)| = 1$. Therefore both values prescribed in (8.7) have modulus one: $|g(p)| = 1$ for $p \leq y$, and $|\chi(p)p^{it}| = 1$ for $p > y$. Extending these prime values multiplicatively defines an element of $\mathcal{M}$. □

The next lemma explains how to average the errors from different cutoffs. Truncation at C_1 permits a uniform estimate even when the original sequence exceeds C_1.

Lemma 8.3. *Let $k \geq 1$ be an integer, and let $C_1, a, b \in \mathbb{R}$ satisfy $C_1 > 0$ and $a, b \geq 0$. Suppose $E_0 \leq E_1 \leq \cdots \leq E_k$ are nonnegative real numbers and $z_1, \dots, z_k \in \mathbb{C}$ satisfy*

$$\begin{cases} \Re z_j \geq -a, & E_j > C_1, \\ \Re z_j \geq -a - b\sqrt{E_j - E_{j-1}}, & E_j \leq C_1. \end{cases} \tag{8.9}$$

Then

$$\Re\left(\frac{1}{k}\sum_{j=1}^{k} z_j\right) \geq -a - b\sqrt{C_1/k}.$$

Proof. Put $F_j = \min(C_1, E_j)$. If $E_j \leq C_1$, monotonicity gives $F_j - F_{j-1} = E_j - E_{j-1}$. If $E_j > C_1$, then $F_j - F_{j-1} \geq 0$, so the first alternative in (8.9) also implies

$$\Re z_j \geq -a - b\sqrt{F_j - F_{j-1}}. \tag{8.10}$$

By Cauchy–Schwarz and telescoping,

$$\begin{aligned} \frac{1}{k}\sum_{j=1}^{k}\sqrt{F_j - F_{j-1}} &\leq \left(\frac{1}{k}\sum_{j=1}^{k}(F_j - F_{j-1})\right)^{1/2} \\ &= \sqrt{(F_k - F_0)/k} \leq \sqrt{C_1/k}. \end{aligned}$$

Averaging (8.10) gives the assertion. □

8.3. **Construction of the measure.** The kernels chosen below depend only on δ. In the proof, the minimizing pair (χ, t) may depend on g and on the index j, but it is used only to estimate the already-fixed kernel K_{X_j}; it does not enter the definition of μ_δ.

Proposition 8.4. *For every $0 < \delta < 1/10$, there is a finitely supported probability measure μ_δ on Pythagorean ratios of height at most H_δ such that*

$$\Re\sum_{r}\mu_\delta(r)g(r) \geq -\delta \qquad (g \in \mathcal{M}). \tag{8.11}$$

and

$$H_\delta \le \mathrm{Exp}_2\left(C\delta^{-1295}\log^2(2/\delta)\right).$$

Here $C \ge 1$ is absolute.

Proof. We divide the proof into six steps.

Step 1. Fix the inverse-theorem parameters. Fix a real $\delta \in (0, 1/10)$ and put $\ell = \log(2/\delta)$. The fixed density w, fixed integer C_0, and prescribed δ-dependent parameters $D_0(\delta), W_{0,\delta}$ determine the kernels K_X. These objects are fixed once δ is fixed and do not depend on g. Choose $L_\delta, X_\delta, \epsilon_\delta$, and $\mathcal{L}_\delta$ as in Theorem 8.1. In particular,

$$\mathcal{L}_\delta \le C\delta^{-3}, \qquad \log(2L_\delta) + \log(2X_\delta) \le C\delta^{-3}. \tag{8.12}$$

Set the integer Q, the real parameter T_1, the real threshold C_1, and the integer threshold N_1 by

$$Q := \left\lceil \exp(A\delta^{-1293}) \right\rceil, \qquad T_1 := Q, \qquad C_1 := A\delta^{-1293},$$
$$N_1 := \left\lceil \mathrm{Exp}_2(A\delta^{-1293}) \right\rceil.$$

Here A is an absolute constant chosen sufficiently large. Since $1293 = 3 \cdot 431$, Theorem 7.10, together with (8.12), gives the following contrapositive for every completely multiplicative $g \in \mathcal{M}$ and every integer $z \ge N_1$:

$$E_g(z) > C_1 \quad \Longrightarrow \quad \|g\|_{U^3[z]} < \epsilon_\delta. \tag{8.13}$$

Indeed, if the norm were at least ϵ_δ, Theorem 7.10 would produce an admissible primitive character and real parameter in the ranges $q_\chi \le Q$ and $|t| \le T_1$ whose prime distance is at most C_1, contradicting $E_g(z) > C_1$.

Step 2. Choose all arithmetic scales independently of g. Choose an absolute constant A' sufficiently large and put

$$M_1 := \left\lceil \mathrm{Exp}_2(A'\delta^{-1293}) \right\rceil. \tag{8.14}$$

Increase A' so that the integer M_1 satisfies

$$M_1 \ge \max\{2, Q, T_1, N_1, X_\delta, D_0, C_w D_0^2, \exp(C_*\delta^{-3})\}.$$

Here C_* is the lower-bound constant in Theorem 4.10, and $C_w D_0^2$ is the lower bound required by Theorem 5.5. Such an A' exists because (4.4) gives $\log(2D_0) + \log(2\|W_0\|_\infty) \ll \delta^{-3}$, whereas $\log\log M_1 = A'\delta^{-1293} + O(1)$. Theorem 4.10 therefore supplies an integer U depending only on δ such that

$$2 \le U \le C\delta^{-3},$$

and its conclusion applies whenever

$$y \ge M_1, \qquad X \ge (M_1 y)^U,$$

with $q_\chi \le M_1$, $|t| \le M_1$.

Let $b_\delta \ge 1$ be the constant in Theorem 5.5. Its bound and $B = (200/\delta)^2$ give

$$b_\delta^2 \le C\log(2B) \le C\ell.$$

Choose

$$k := \left\lceil 10^4 b_\delta^2 C_1 \delta^{-2} \right\rceil$$

so that

$$k \le C\delta^{-1295}\ell. \tag{8.15}$$

Fix $a_0 = 2$ and define integer scales by

$$Z_0 := M_1^{a_0}, \qquad X_j := \left\lceil (M_1 Z_{j-1})^U \right\rceil, \qquad Z_j := C_0 X_j \quad (1 \le j \le k). \tag{8.16}$$

These scales depend only on δ and the fixed, g-independent parameters $D_0(\delta), W_{0,\delta}, w, C_0$. If $y_j := Z_{j-1}$, then $y_j \geq M_1$, $X_j \geq (M_1 y_j)^U$, and $X_j \geq M_1 \geq X_\delta, C_w D_0^2$. Since $M_1 \geq 1$ and $U \geq 2$, $X_j \geq y_j$, so $y_j \leq Z_j = C_0 X_j$. Thus the hypotheses of both Theorem 4.10 and Theorem 5.5 will hold at every scale once the character chosen below satisfies $q_\chi \leq Q \leq M_1$.

Step 3. Estimate one scale by an energy dichotomy. Fix $g \in \mathcal{M}$ and $1 \leq j \leq k$. If $E_g(Z_j) > C_1$, then $Z_j \geq N_1$ and (8.13) gives $\|g\|_{U^3[Z_j]} < \epsilon_\delta$. Since $X_j \geq X_\delta$, Theorem 8.1 yields

$$|K_{X_j}(g)| \leq \delta/50.$$

Suppose instead that $E_g(Z_j) \leq C_1$. Choose an admissible pair (χ, t) minimizing $E_g(Z_j)$ and put $y := Z_{j-1}$. Define the truncated completely multiplicative function

$$\widetilde{g}(p) = \begin{cases} g(p), & p \leq y, \\ \chi(p)p^{it}, & p > y. \end{cases}$$

Since $y \geq M_1 \geq Q$, Theorem 8.2 gives $\widetilde{g} \in \mathcal{M}$. Its hypotheses for Theorem 4.10 are explicit: $q_\chi \leq Q \leq M_1$, $|t| \leq T_1 \leq M_1$, $y \geq M_1$, and $X_j \geq (M_1 y)^U$. Hence

$$\Re K_{X_j}(\widetilde{g}) \geq -\delta/50. \tag{8.17}$$

Step 4. Compare g with the truncated model. Let $h := g/\widetilde{g}$. Then $h(p) = 1$ for every prime $p \leq y$, including the primes dividing D_0, while for $p > y$, $h(p) = g(p)\overline{\chi(p)}p^{-it}$. Theorem 5.5 applies because $y \geq M_1 \geq D_0, q_\chi, 2$, $X_j \geq C_w D_0^2$, and $y \leq Z_j = C_0 X_j$. On the support of the kernel, all four positive linear forms $m-n, m+n, m, n$ are at most Z_j. Theorem 5.5 approximates h on these four forms by the same unit complex number ζ. Their product is $\zeta^2\overline{\zeta}^2 = 1$, so no extra phase remains in the four-linear kernel. Therefore

$$|K_{X_j}(g) - K_{X_j}(\widetilde{g})| \leq b_\delta \left(\sum_{\substack{y < p \leq Z_j \\ p \text{ prime}}} \frac{1 - \Re(g(p)\overline{\chi(p)}p^{-it})}{p} \right)^{1/2}$$
$$\leq b_\delta \sqrt{E_g(Z_j) - E_g(Z_{j-1})}.$$

The final inequality is exactly (8.8): it bounds the prime sum over $(Z_{j-1}, Z_j]$ by $E_g(Z_j) - E_g(Z_{j-1})$ for the pair minimizing at Z_j. Combining this estimate with (8.17) gives

$$\Re K_{X_j}(g) \geq -\delta/50 - b_\delta \sqrt{E_g(Z_j) - E_g(Z_{j-1})}.$$

Step 5. Average over the fixed scales. For $0 \leq j \leq k$ set $E_j := E_g(Z_j)$, and for $1 \leq j \leq k$ set $z_j := K_{X_j}(g)$. Theorem 8.2 gives $E_0 \leq E_1 \leq \cdots \leq E_k$. The large-energy estimate and the last inequality in Step 4 verify the two alternatives of Theorem 8.3 with $a = \delta/50$ and $b = b_\delta$. Since $k \geq 10^4 b_\delta^2 C_1 \delta^{-2}$,

$$b_\delta \sqrt{C_1/k} \leq \delta/100.$$

Therefore

$$\Re\left(\frac{1}{k}\sum_{j=1}^{k} K_{X_j}(g)\right) \geq -\frac{\delta}{50} - \frac{\delta}{100} = -\frac{3\delta}{100} \geq -\delta. \tag{8.18}$$

The scales used here are the same for every g.

Step 6. Define the common measure and bound its height. Define

$$\mu_\delta(r) := \frac{1}{k}\sum_{j=1}^{k} \frac{X_j^{-2}}{Z_{X_j}} \sum_{\substack{m>n>0 \\ R(m,n)=r}} w(m/X_j, n/X_j) W_0(m,n).$$

Each inner sum is finite and nonnegative. Moreover,

$$\sum_r \mu_\delta(r) = \frac{1}{k}\sum_{j=1}^{k} K_{X_j}(1) = 1,$$

so μ_δ is a finitely supported probability measure independent of g. (8.18) is precisely (8.11) for this measure.

For every support pair (m,n),

$$(m^2-n^2)^2 + (2mn)^2 = (m^2+n^2)^2.$$

Thus $R(m,n)$ is a Pythagorean ratio. It is not equal to 1, since $R(m,n)=1$ would imply $(m-n)^2 = 2n^2$, impossible for positive integers m,n. Since $m+n < Z_j \le Z_k$ on the support, both unreduced legs m^2-n^2 and $2mn$ are at most Z_k^2. Reducing the two legs to lowest terms can only decrease them, so we may take

$$H_\delta := Z_k^2.$$

It remains to estimate H_δ. Write $z_j := \log Z_j$ and $m_1 := \log M_1$. Since $\lceil u \rceil \le 2u$ for $u \ge 1$, (8.16) gives

$$z_j \le U z_{j-1} + U m_1 + \log(2C_0).$$

As $U \ge 2$, summing this recurrence yields

$$\begin{aligned} z_k &\le U^k z_0 + (U m_1 + \log(2C_0))\frac{U^k - 1}{U-1} \\ &\le U^k\{z_0 + 2m_1 + 2\log(2C_0)\}. \end{aligned} \tag{8.19}$$

Here $z_0 = 2m_1$. From (8.14),

$$\log m_1 \le A'\delta^{-1293} + C, \qquad \log U \le C\ell. \tag{8.20}$$

Taking logarithms in (8.19), then using (8.15) and (8.20), we obtain

$$\log\log Z_k = \log z_k \le k\log U + O(\log m_1) \le C\delta^{-1295}\ell^2.$$

Finally, $\log\log H_\delta = \log(2z_k) = \log 2 + \log\log Z_k$, so increasing the absolute constant gives

$$H_\delta \le \operatorname{Exp}_2\left(C\delta^{-1295}\log^2(2/\delta)\right).$$

This proves the proposition. □

8.4. **The density bound.** We first combine Theorem 8.4 with Theorem 3.3 to obtain a threshold implication. We then choose δ as a function of N and invert that implication.

Proof of Theorem 1.1. We divide the proof into three steps.

Step 1. Prove the threshold implication. For $0 < \delta < 1/10$, set

$$F_\delta = \delta^{-1295}\log^2(2/\delta).$$

Theorem 8.4 provides a measure satisfying the hypotheses of Theorem 3.3, with $H_\delta \le \operatorname{Exp}_2(C_H F_\delta)$ for an absolute $C_H \ge 1$. Choose an absolute $C_N > C_H + 2$ so large that, for every $F \ge 1$,

$$\frac{2}{\log 2}\exp((C_H+1)F) \le \operatorname{Exp}_2(C_N F),$$

and set $C_{\text{main}} := C_N$. Then

$$N \ge \operatorname{Exp}_2(C_N F_\delta) \tag{8.21}$$

ensures both size conditions in (3.8). We verify them separately. The first one follows from $H_\delta \le \mathrm{Exp}_2(C_H F_\delta) \le \mathrm{Exp}_2(C_N F_\delta) \le N$. For the second one, note that on the stated range of δ, $F_\delta \ge 1$ and $\delta^{-2} \le e^{F_\delta}$. Therefore

$$\frac{2\delta^{-2}\log H_\delta}{\log 2} \le \frac{2}{\log 2}\exp((C_H+1)F_\delta) \le \mathrm{Exp}_2(C_N F_\delta),$$

Thus Theorem 3.3 applies and gives $f(N) \le 2\delta N^N$. Since $C_{\mathrm{main}} = C_N$, this proves (1.1).

Step 2. Choose δ and retain the logarithmic correction. Put $t = \log_2 N$ and $s = 1295$. Choose an absolute constant $A_0 \ge 1$ sufficiently large, and then choose t_0 sufficiently large so that for every $t \ge t_0$ the choice

$$\delta = A_0\left(\frac{(\log t)^2}{t}\right)^{1/s}$$

lies in $(0, 1/10)$ and satisfies

$$\log(2/\delta) = \log(2/A_0) + \frac{\log t - 2\log\log t}{s} \le \frac{2\log t}{s},$$
$$C_N\delta^{-s}\log^2(2/\delta) \le \frac{4C_N}{s^2A_0^s}\,t \le t.$$

where the final inequality follows from fixing $A_0^s \ge 4C_N/s^2$. Since $N = \mathrm{Exp}_2(t)$, the inequality $C_N\delta^{-s}\log^2(2/\delta) \le t$ is exactly the hypothesis in (8.21). Thus

$$\frac{f(N)}{N^N} \le 2A_0\left(\frac{(\log t)^2}{t}\right)^{1/s} = 2A_0\left(\frac{(\log_3 N)^2}{\log_2 N}\right)^{1/1295}. \tag{8.22}$$

Choose an integer N_0 so large that $\log_2 N \ge t_0$ and all logarithms in (8.22) are defined for $N \ge N_0$. After enlarging an absolute constant if necessary, set $C_1 := 2A_0$. Then the first inequality in (1.2) follows.

Step 3. Deduce the pure-power bounds. Let $0 < b < 1/s$ and put $\kappa = 1/s - b > 0$. For $x = \log t > 0$, differentiation shows that the maximum of $x^{2/s}e^{-\kappa x}$ is attained at $x = 2/(s\kappa)$. Consequently,

$$t^{-1/s}(\log t)^{2/s} \le t^{-b}\left(\frac{2}{es\kappa}\right)^{2/s}.$$

Combining this inequality with (8.22), and taking $b_* = 1/(s+1) = 1/1296$, gives

$$\frac{f(N)}{N^N} \le C_2(\log_2 N)^{-1/1296} \qquad (N \ge N_0),$$

after setting $C_2 := C_1\big(2/(es(1/s - b_*))\big)^{2/s}$ and enlarging it if needed. This proves the second inequality in (1.2). For a general fixed $0 < b < 1/1295$, set

$$C_b := C_1\left(\frac{2}{es(1/s-b)}\right)^{2/s}, \qquad N_b := N_0.$$

If necessary enlarge N_b so that the choice of δ in Step 2 lies in $(0, 1/10)$ for every $N \ge N_b$. The same calculation then yields (1.3). □

AI-use disclosure

The author used OpenAI Codex in GPT-6 Astra Ultra Fast mode during the preparation of this manuscript. After obtaining the bound

$$\frac{f(N)}{N^N} \ll \left(\frac{(\log\log\log N)^2}{\log\log N}\right)^{1/s},$$

the author used Codex to calculate an admissible value of s, to reorganize the proofs around the value $s = 1295$, and to polish the proofs. Codex also assisted with mathematical auditing and English-language editing. The author has checked all mathematical statements and takes responsibility for the final manuscript.

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